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Geographic Information
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eurostat

Technical Documents
2022

LUCAS 2022

(Land Use / Cover Area Frame Survey)

Technical reference document C1
Instructions for Surveyors

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DOCUMENT CHANGE RECORD			
ISSUE	DATE	CHANGE AUTHORITY / AUTHOR	REASON FOR CHANGE AND AFFECTED SECTIONS
V0.1			New chapter on Copernicus. New chapter on grassland.
V0.2	2016.12.22	EUROSTAT E4.LUCAS / HR	New chapter for erosion and soil. Integration of the new soil manual. Grassland and Water management moved to respect the order of the field form and workflow.
V0.3	2017.05.05	EUROSTAT E4.LUCAS/BE	Revise structure. Finalise new chapters.
V0.4	2017.12.06	EUROSTAT E4.LUCAS/BE	Fine tuning
V1.0	2018.02.20	EUROSTAT E4.LUCAS/BE	Revision after project manager training
V2.0	2019.02.14	EUROSTAT E4.LUCAS/BE	Integrated revised Copernicus chapter, errata
V2.0	2019.02.22	EUROSTAT E4.LUCAS/BE	Integrated FAQs from LUCAS survey 2018
V3.0	2021.02.15	EUROSTAT E4.LUCAS/BE	Revision, integrated contributions for different modules
V4.0	2021.02.19	EUROSTAT E4.LUCAS/BE	Formatting
V5.0	2021.03.29	EUROSTAT E4.LUCAS/BE/ZS	Comments
V6-11	2021.05.27	EUROSTAT E4.LUCAS	Fine-tuning, clarifications on PI, Copernicus, Grassland
V6-12	2022.01.21	EUROSTAT E4.LUCAS/BE	Copyright information in the text
V20	2023.02.06	EUROSTAT E4.LUCAS	Revision of images

1 Scope

The aim of this document is to provide instructions to the surveyor during the Land Use/Cover Area frame Statistical survey 2022 (LUCAS survey 2022).

The LUCAS survey serves to collect land cover/land use, agro-environmental and soil data by field observation of geographically referenced points. The surveyor documents its observations by filling in the field form, by taking photographs and by taking soil samples. All the data is stored in the Data Management Tool (DMT).

The document therefore covers the following topics:

- Safety
- Quality of the work
- Helpdesk
- Equipment
- Preparatory work
- Filling in the field form: step by step
- Annexes:

Record descriptor, code lists, possible combinations of LC/LC – LU/LU – LC/LU, unlikely changes (LC, LU), EUNIS habitat classification, decision trees

This document is part of a series of technical reference documents defining the framework of the Land Use / Cover Area frame statistical Survey (LUCAS):

- C1: Instructions for surveyors
- C2: Field Form and Ground Document (template)
- C3: Classification
- C4: Quality control procedures
- C5: Plant Identification Guide
- C6: Grassland Identification Guide

Furthermore, document "D1 – DMT user guide" is the guide for the use of the Data Management Tool (DMT) that allows entering data collected in the field into the central database:

2 Introduction

LUCAS stands for Land use and land cover area frame survey. It is a survey that takes place in the field, and in the current form it has been run every 3 years from 2009 to 2018, except in 2022 which is 4 years after the last survey.

LUCAS survey 2022 consists of the following parts:

- LUCAS core, including extended grassland
- LUCAS specific module on soil (Soil module)
- LUCAS specific module on grassland (Grassland module)
- LUCAS specific module on landscape features (Landscape Feature Module)
- LUCAS specific module on Copernicus (Copernicus Module)

LUCAS core includes the identification of the point, different aspects related to land cover and land use, land and water management, some aspects related to habitats and an extended grassland assessment on 40.000 points.

The grassland module is assessed on 20,000 points to collect extensive data about this specific topic.

The soil module, which takes place in 41,000 LUCAS points, consists of different parts. In all soil points a topsoil sample is collected either with the standard sampling procedure (using a spade) or with the soil core sampling procedure for measuring bulk density (4000 points). Furthermore, a sample for soil biodiversity is collected in 2000 points.

The feasibility of the standard LUCAS core part and the standard topsoil sampling have been assessed over the years and a thorough quality control takes place for these parts. Also the grassland module, some smaller additions to the core part (habitats), the Copernicus module and the new parts of the soil module have been successfully tested. The practical feasibility and in some cases the scientific feasibility of these modules and elements have been assessed during the LUCAS survey 2018. Some fine-tuning took place, based on the 2018 results. These elements are now an integral part of the LUCAS survey and it is therefore of uttermost importance that this part of the survey is carefully planned and the assessment is done thoroughly. The extended grassland, while new, is very similar to some of the elements of the grassland module.

In LUCAS 2022 a new module on landscape features has been added. This module is extremely important for the evaluation of the agricultural landscape and a careful assessment is crucial.

However, as this is a new element of the LUCAS Survey, the quality assessment is not as strict as for the other parts and takes into account that some particularly difficult situations might occur in the field. In such a case it is recommended that the helpdesk is contacted immediately.

On the other hand, a request for correction is always possible.

The data collected during the LUCAS survey are used to produce statistical tables on land cover and land use, it is used for modelling agro-environmental aspects and it is used as ground control for satellite images.

More information on land cover/use statistics and on LUCAS in general can be found on the Eurostat website: <https://ec.europa.eu/eurostat/web/lucas>

The LUCAS survey is co-ordinated by the Statistical Office of the European Commission (Eurostat).

3 Safety

Safety is of paramount importance and adequate safety measures need to be taken, as surveyors are sometimes in remote areas. The surveyor shall take into account that even if LUCAS points are chosen to be rather close to the road network, longer walks might be needed.

This will mean that the surveyor may have to go to areas where there is no telephone or radio communication, with little or no traffic, sometimes more than one hour away from medical or emergency support, or where rough terrain, dense vegetation or other features make it difficult to obtain help in case of need.

At least one more person should be informed of the daily planning of the surveyor. This will normally be the role of the supervisor.

If the point can only be reached by foot, and the walking time for reaching the point and coming back to the car is estimated as lasting more than 1 hour (excluding time of the survey), the point is considered as inaccessible and is not to be visited.

The surveyor and his supervisor shall take action to eliminate, minimize, avoid or report any hazards of which they are aware and follow occupational, safety and health instructions.

Safety devices and personal protection equipment shall be used at all times (e.g. reflecting jacket, flashlight, etc.). This includes appropriate dress standards for the type of work and climate conditions (resistant shoes/boots, raincoat, cap, sun protection etc.).

The surveyor shall not place his or other people's safety or health at risk. They must accept appropriate medical advice where relevant and disclose to the supervisor any limitation imposed by their health that may affect their ability to participate safely in any fieldwork activity. This obligation applies both before and during fieldwork. Information provided must be treated as confidential information.

Everyone participating in fieldwork must be adequately insured.

Information on any incident has to be transmitted to the supervisor as soon as possible, so that he can take appropriate measures, including informing other relevant parties about the types of risks encountered. Any relevant information is to be transmitted to the contractor responsible for the quality support, who will then make this information available to all players of the LUCAS 2022 survey.

4 Quality of the work

The elements of the good quality fieldwork are the following:

- correct and continuous use of GPS and orthophotos
- precise location of the point in the field
- correct and complete observation
- compliance with the instructions given in this document
- filling in the questionnaire with readable marks/letters
- adding relevant comments.

Besides the mandatory comments foreseen in the workflow, it is also very important to document every unusual circumstance in the **comments** fields and to justify the decisions of the surveyor.

It is mandatory to use structured comments whenever possible.

No special characters should be used to fill in free text fields. The use of English is mandatory in case of free text comments.

The work is accepted only if the field form is precisely filled in with comments as well, and if all the relevant photos are taken.

The fieldwork must be checked at least once by either the Central Office (CO) or the Regional Offices (ROs) using the photos taken by the surveyor and the recorded GPS tracks.

More details on quality control can be found in document C4.

5 Helpdesk

Support on technical questions is available throughout the survey.

A general helpdesk is managed by the entity responsible for the external quality control together with Eurostat.

Specific helpdesk for questions related to soil, grassland and the DMT software are also available.

The separate helpdesks for soil, grassland and the DMT assure the direct support of experts in these specific fields.

Contact details will be provided at the beginning of the project.

6 Equipment

6.1 Equipment for LUCAS core

The following items are always needed for the survey:

- Accreditation document (Document A1) and ID card
- Surveyors' instructions (Document C1 and Annexes)
- Blank field forms and Ground Document (Document C2)
- Classification (Document C3)
- Appropriate device to access the data management tool (DMT) 2022 (screen size 10 – 30 inches)
- Mobile phone
- Hand-held GPS navigation device for:
 - o locating the point,
 - o saving and exporting the points and the track the surveyor follows daily.
- Digital camera + memory cards
- Extra batteries
- Magnetic compass
- Point marker/flag
- Pencil or water resistant pens + writing pad
- Plain DVDs / CDs / USB sticks
- 12V rechargers
- Backpack
- Safety equipment (e.g. first-aid kit, reflective coat, lantern, whistle, insect repellent, etc.)
- Suitable vehicle(s) to reach the point, including, where relevant, through unpaved roads (e.g. rented 4x4 car, motorbike)
- Car navigator with licensed maps of the survey area
- Road maps for efficient route planning
- Other, useful documents (charts, crop calendars etc.)

The **ID card and the accreditation document** are necessary for the surveyor to explain and justify his mission and to present it to the farmer or landowner, or local authorities.

The present document (Document C1) provides guidance on the survey implementation and gives answers to many technical questions on LUCAS. The **decision tree** (Annex 9.10) provides guidance for the correct survey sequence and a summary of the **information to be collected** can be found in Annex 9.1).

Both, a detailed version (Document C3) and a short version (Annexes 9.2) of the **LUCAS classification** are given to the surveyor. Document C3 describes the land cover and land use categories precisely, while the short version gives only the list of codes and titles of categories.

A **suitable car** (4x4 if necessary) equipped with a **car navigation system** enables the surveyor to approach the point if roads are accessible.

A **hand-held GPS receiver** is to be used to approach the point. It is mandatory to register in the field form the GPS coordinates (Lat/Long) of the position of the car park, the position from which the surveyor observes the point on the ground and to record the GPS tracks.

The **ground document** includes a **topographic map** (the most widely used types of maps with a scale usually between 1:10.000 and 1:200.000) and an **orthophoto** (an orthorectified aerial photograph, that is, free from the distortion caused by inclination angle and relief, with a scale that normally, varies between 1:10.000 and 1:2.000). The ground document is the base for the surveyor to locate the point most exactly and to estimate the area of the parcel where the point is located in.

A traditional **magnetic compass** has to be used to indicate a particular direction. North is indicated on both the topographic maps and the orthophotos. With the help of the compass, the surveyor can correctly orient the map

and the orthophoto before examining them. Moreover, the compass is helpful to correctly take the photos in the cardinal directions (N, E, S, W) and locating the location where to collect soil samples. Note that the GPS compass alone is not suitable for these tasks.

The **digital camera** (minimum 2 mega pixels) is used to take pictures of the point, the crop and the N, E, S, W landscape photos as well as the irrigation, grassland, soil and additional pictures. It is recommended to use digital cameras equipped with GPS to record the position of the photo taken.

A **portable computer** shall be put at disposal of the surveyor to enter the collected data into the LUCAS database using the Data Management Tool (DMT). The DMT integrates automatic controls and checks, enabling the surveyor to deliver complete and formally correct data.

Charts, diagrams, decision keys, crop calendar and classification overviews etc. can be copied into a double-page, which can be plasticized as a visual help for the surveyors.

6.2 Equipment for the LUCAS Grassland module

Equipment needed in addition to the tools listed in chapter 6.1 Equipment for LUCAS core are as follows:

- Blank field forms for grassland
- Measuring stick (1 m)
- 4 poles & tape/cord to delimit the transect (transect size: 20m, 2.5m)
- White linen

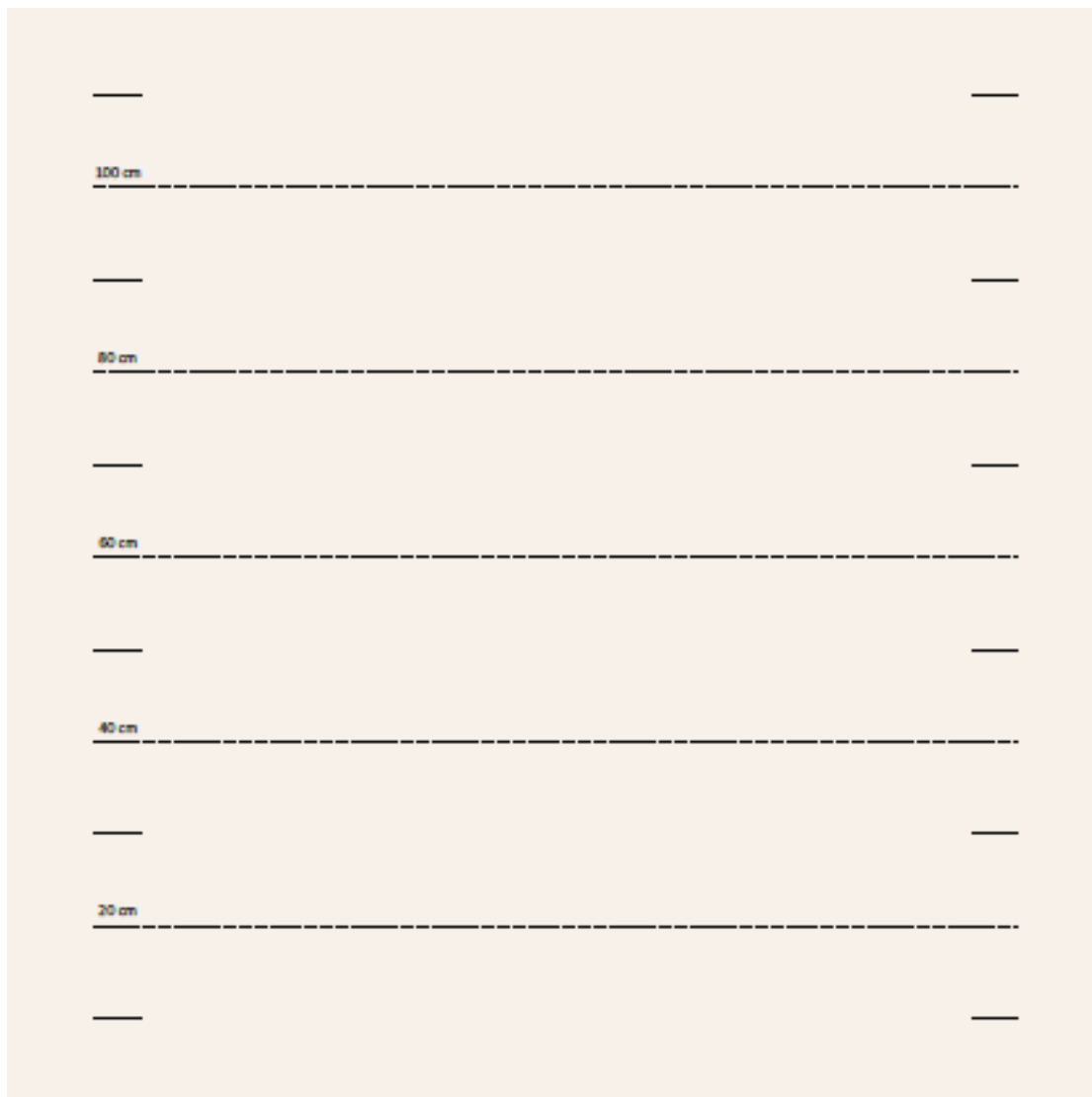
The white linen tool has the following characteristics:

Total weight: 1.5 kg.

Dimensions when packed: 10cm*10cm*130cm.

The following material is needed to produce the white linen tool:

- PVC-banner (120x120 cm) with hemstitch seam at the top and bottom,
- 2 wooden poles (130 cm x 2 cm Ø),
- 2 telescopic fishing pole holders with V-bracket (minimum high 130 cm)
- 6 tent pegs (additional)
- 4 guy lines with runners (additional)



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Figure 6-1: Height gauges on white linen tool

Further recommendations:

- use a bag with shoulder strap for transport in the field
- take water and a cloth to clean the WLT
- take spare tent pegs, guy lines and runners
- attach colourful markers to the tent pegs to avoid them getting lost in the vegetation
- you can use a one-legged camera support to ensure the correct camera height.

6.3 Equipment for the LUCAS Soil module

Equipment needed in addition to the tools listed in chapter 6.1 Equipment for LUCAS core are as follows:

6.3.1 Standard soil sample

For the standard soil sampling, required in 37 000 points, the equipment is as follows:

- a spade with a blade of 30-cm length and approx. 20-cm width
- a trowel (small spade)
- a plastic bucket
- a meter stick

- a marker of indelible ink
- a box to store and transport the samples
- mail boxes to ship the samples to supervisor (cardboard boxes, enough resistant to be transported, and with a maximum weight of 2,5 to 5 kg)
- boxes to transport samples to the JRC or laboratory (cardboard boxes, enough resistant to be transported without crushing, and with a maximum weight of 20 kg)
- garbage bag
- For cropland, grassland, and peatland points
 - 2 printed plastic labels (white colour) per point
 - 2 plastic bags and 2 ties per point (polyethylene transparent bags, 1 small bag of at least 25×40 cm and 1 big bag of at least 40×60 cm, plastic twist ties)
- For woodland points
 - 3 printed plastic labels (white colour) per point
 - 3 plastic bags and 3 ties per point (polyethylene transparent bags, 1 small bag of at least 25×40 cm and 2 big bags of at least 40×60 cm, plastic twist ties)

It is the responsibility of the contractor to buy and provide their surveyors with all this material, except for the printed plastic labels.

Contractors need to indicate the exact size of the blade of the spades that their surveyors will be using.

The JRC will provide contractors with printed labels and contractors will distribute them among their surveyors.

6.3.2 Core sampling for measuring bulk density

In addition to the equipment listed in section 6.3.1, the following equipment is needed for core sampling for measuring bulk density:

- A number of metallic rings (stainless steel rings, diameter 53 x 50 mm, height 51 mm, vol. 100 cc with max. volume deviation of 0.5 %, 1 bevelled edge, 2 polyethylene cover per ring)
- a rubber mallet (face diameter 5-8 cm)
- a flat-bladed knife
- a wood block (of approximately 15 x 7 x 10 cm)
- a balance to weigh samples (portable battery operated balance, digital display, min. capacity 3 kg, max. capacity 5 kg, precision 0.1 g)
- For cropland, grassland, and peatland points
 - 4 printed plastic labels (orange colour) per point
 - 3 small plastic bags and 1 big plastic bag with their ties per point (polyethylene transparent bags, small bags of at least 25×40 cm and big bag of at least 40×60 cm, plastic twist ties)
- For woodland points
 - 5 printed plastic labels (orange colour) per point
 - 3 small plastic bags and 2 big plastic bags with their ties per point (polyethylene transparent bags, small bags of at least 25×40 cm and big bags of at least 40×60 cm, plastic twist ties)

Note that plastic bags and plastic labels listed here replace the material used for the standard soil sampling.

It is the responsibility of the contractor to buy and provide their surveyors with all this material, except for the printed plastic labels, metallic rings, balances and mallets. The JRC will provide contractors with this specific material. Contractors will distribute them among their surveyors.

6.3.3 Soil biodiversity sampling

In addition to the equipment needed for the standard/core sampling, in case a sample for assessing soil biodiversity has to be taken, the following equipment is needed:

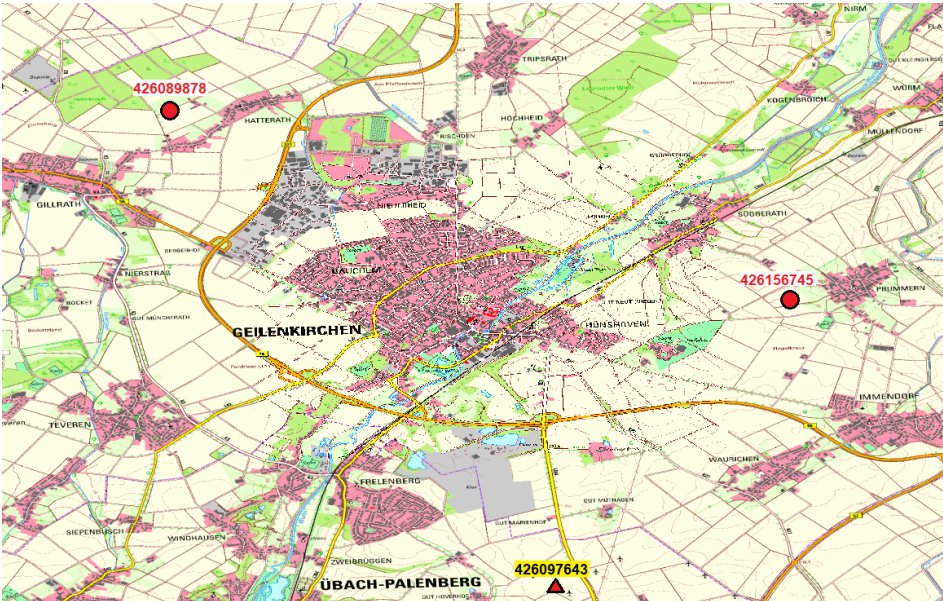
- a 1 L bottle of water per point (tap water is sufficient)
- 2 pairs of latex single use gloves per point
- 2 printed plastic labels per point (green colour)
- 2 small plastic bags per point (for collecting soil biodiversity sample) and plastic twist tie
- 2 freezer pack per point
- 1 polystyrene box/cool box to transport the freezer packs needed for each sampling day and the collected samples
- address label for box (to be sent to the national coordinator)
- alcohol wipes
- a garbage bag

Note that plastic bags and plastic labels listed here replace the material used for the standard/core sampling for bulk density.

It is the responsibility of the contractor to buy and provide their surveyors with all this material, except for the printed plastic labels, polystyrene boxes, plastic bags for soil biodiversity samples and freezer packs. The JRC will provide contractors with this specific material. Contractors will distribute them among their surveyors.

7 Preparatory work

The surveyor has to prepare his itinerary before going out in the field. This preparation is essential for the efficient conduction of the survey. It is recommended to use an up to date road map to find the most suitable access to the point. GPS can be helpful as well. The surveyor needs to identify the different problems or obstacles he will have to deal with during his journey.

<i>Daily route planning increases efficiency:</i>	It is practical to use a small-scale topographic map to plan the travelling. The order of the points to be visited on the same day should be planned in advance to find the optimal route. Moreover, a quick photo analysis of the orthophoto, a look at the results of the previous survey, or additionally looking at the point on free accessible earth observation programs like Google Earth, is helpful before starting the work.
<i>Small scale topographic map including LUCAS points</i>	 © Geobasis NRW, Digitale Topographische Karte 1:25 000 (TIFF), dtk25_32296_5648_4_nw_col.tif, Data licence Germany - Zero - version 2.0 <u>Edited</u>: addition of the 3 plausible LUCAS points Figure 7-1. Example of a topographic map including the LUCAS points to be observed
<i>Be prepared for difficult conditions</i>	The surveyor should be aware that any kind of field conditions might be encountered at the point. Sometimes difficult access to the point is not visible on the orthophoto. Adequate shoes and clothing are necessary. Even if the survey is in the spring and summer, clothing which covers the whole body might be necessary. A proper bag should be taken to protect the equipment and the documents in case of rain.
<i>Know how to handle your GPS and digital camera</i>	The instructions for the camera and the GPS should be studied and their use should be practised before the fieldwork. Spare batteries and memory cards must be supplied to surveyors.
<i>GPS</i>	Before going out into the field, the coordinates of the points have to be uploaded into the GPS (It can be done by the contractor, centrally, or by the surveyors). Do not only rely on the GPS. Orthophotos and previous field forms should be examined before the fieldwork, to become generally familiar with the natural and man-made elements. As orthophotos are taken from the air, the view of the landscape is quite unusual at first sight. Roads and tracks can be recognized easily. Plan the change of the battery in GPS so that no tracks are lost.

8 Filling in the field form: step by step

Filling in the form step by step, focuses on filling in the form in-situ. Fields are presented in the order of appearance in the field form (C2), which sometimes might not correspond to the order of filling, e.g. the end time of the survey is compiled at the very end but in the field form it appears almost at the beginning, next to the start time.

If the point is photointerpreted in the office (point is part of the PI sample or has been classified as ex-ante) or if the point is photointerpreted in the field using orthophotos or satellite images some fields cannot be compiled, others can sometimes be filled in. As a principle during the photointerpretation in the office as much information as possible should be collected.

Please also check the decision tree.

8.1 Comments, Photos and Warnings

The surveyors have the best view on the LUCAS point, on the surrounding landscape, on the way to the point and on many other circumstances and conditions that enable them to assess all the necessary information that is required in the Field form.

Therefore it is essential that this knowledge is transmitted to the other players, the quality control (Regional or Central Offices, the External quality Control or by Eurostat) and also the final users that will analyse the data and generate valuable information from this data, such as statistics, indicators, distribution maps.

To this end Comments and Photos that describe and underline the decisions of the surveyors are crucial elements.

Quality controllers in the office need to understand what led to the specific assessment of the features in the field. Their assessment is based on the elements in their "hands". Comments explaining the conditions in the fields and photo showing the specific site, object... help the quality controllers to do their assessment, to keep a good data flow and prevent unnecessary loops of data due to rejections. For final users it is extremely valuable to be able to go back to specific points and see how they look like and what specific conditions can be found there.

8.1.1 Comments



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When a comment is needed the surveyor always has to select a structured comment. It is possible, and sometimes mandatory to add a "free text" comment.

For free text the use of English is mandatory. Special characters are to be avoided.

A complete list of comments can be found in 9.1.18 Comments.

8.1.2 Photos



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The surveyor has to document his observation by means of several photos.

Landscape photos in the 4 cardinal directions are taken on the point reached by the surveyor in the field.

The format of the landscape photos is 1/6 sky – 5/6 terrain.

If the surveyor gets close enough to the LUCAS point and can see the point, a point photo is taken.

What other photos have to be taken depends e.g. the land cover, presence or absence of water management, the need to collect a soil sample and the need to document conflict cases.

Photographs create a basis for a long-term monitoring of landscape changes. Moreover in the context of earth observation related mapping activities, LUCAS micro data and photos are used for production, verification and validation processes. It is thus important that they meet standard quality requirements.

Basic requirements:

- Photos need to have exif data on coordinates.
- In addition the following exif data should be collected:
Gpslongitude, gpslongituderef, gpslatitude, gpslatituderef, gpsdatestamp, gpstimestamp, datetimeoriginal, datetimedigitized, datetime, gpsimgdirection, gpsimgdirectionref, imagesize, gpslatitudedd, gpslongitudedd, gpsgeom, gpsdatetimestampz.
- For each photo, the photo ID needs to be recorded
- When a photo is required and cannot be taken or if there is a problem with the photo, a structured comment is required.

Further information on how to take photos and on the different photos can be found in 8.19 Photos.

8.1.3 Warnings

A yellow triangle with an exclamation mark is used to draw your attention on particularly important aspects.



© 2014, Clker-Free-Vector-Images, Pixabay license

8.2 Section: Identification

This part of the field form is partly prefilled.

If the point is identified (pre-filled information) as being part of the PI extension (D) or as being ex-ante (E), it is not to be assessed in the field. The point will be assessed in the office through photointerpretation of available orthophotos and/or satellite images.

PI extension points are points that are part of the PI sample.

Ex-ante points are points that are part of the in-situ sample and in principle have to be visited in the field. If however there are evident reasons that make a visit in the field impossible, known a posteriori (e.g. access to the area is forbidden because of the risk of forest fires, area of landmines), then the point can be set to ex-ante and will not be visited in the field. This is to be seen as an exceptional measure.

<i>User ID (A)</i>	The surveyor's ID number corresponds to the number given to the surveyor by the Central office.
<i>Point ID (B)</i>	The point ID: it is printed on the maps and orthophotos and loaded into the tablet and/or GPS. Might be pre-filled
<i>Point altitude (C)</i>	The theoretical elevation of the point, as determined in the LUCAS database (the LUCAS master). Pre-filled

Following appropriate preparation, the surveyor has to get as close to the point as possible by car. Most of the points can be reached only by foot.

When the surveyor reaches the car park he turns on the GPS and starts the GPS tracking (if not already on). The GPS track record should include only the route after leaving the car and not the car trip. Nonetheless, if the memory capacity of the GPS allows it, the all day trip can be recorded.

The survey starts as the surveyor starts to approach the point by foot, so this is the starting time of the survey, which has to be recorded on the field form, together with the date. At the end of the survey, the end time is to be recorded.

Date and time (7, 8, 9)	Date (7), start time (8) and end time (9) are to be registered here.
-------------------------	--



Comment needed:

In case the survey duration is below 15 min or above 1hour and 15 minutes, one of the following comments is needed.

CODE	LABEL
BF04	Survey time < 15' because it was easy to assess the LC/LU
BF06	Survey time > 1h 15' because it was very difficult to assess the LC/LU
BF07	Survey time < 15' because LC/LU photo interpreted
BF08	Survey time > 1h 15' because it was very difficult to reach the point
BF99	Other survey time comments

8.3 Section: Access to point

8.3.1 Fields: Access to point

Register the location of the car park

<i>Car park coordinates</i>	Don't forget to register the car park coordinates (11, 13). This will help surveyors reach an appropriate location for parking the car in a future survey.
-----------------------------	--

After leaving the car, the GPS is at first the most useful tool for orientation.

The surveyor will need to reach the point itself – or get as close as possible - for examining the environmental parameters, like estimation of cover percentages etc.

When the surveyor has gotten as close as possible to the theoretical LUCAS point all relevant information on the location of the LUCAS point has to be filled in: GPS coordinate system (14), precision (15), elevation (16), co-ordinates of the point (17, 19) and distance to point (20) (all these from the GPS device) in the field form.

The DMT will automatically calculate the value for the field "GPS calculated distance to point" (21) based on the GPS coordinates of the point reached, the theoretical LUCAS point and the GPS precision and compare it to the distance to point inserted by the surveyor.

The reading of the GPS coordinates of the point can be done either at the beginning or at the end of the point assessment (so in any case before doing the grassland and/or the soil assessment, as this includes a change of location). As the precision of the GPS coordinates increase over time, a reading at the end is more precise.



Comment needed:

1) Problem with GPS signal

If the GPS cannot receive the signal, select option 2 "Problem with signal" in field 14: GPS coordinate system. In this case neither the coordinates of the parking nor the co-ordinates of the point will be filled in. In this case ("Problems with signal") the surveyors have to note down in the comments field the reason for not having the GPS signal.

The following structured comments can be selected:

CODE	LABEL
BE01	GPS battery discharged - track to the point not saved
BE02	GPS battery discharged - point location not saved
BE03	GPS battery discharged - track and point location not saved
BE04	GPS not working - track to the point not saved
BE05	GPS not working - point location not saved
BE06	GPS not working - track and point location not saved
BE07	GPS signal weak, lost or high positional error (forest, dense urban,...)
BE08	GPS wrong settings (different ellipsoid/coordinate system,...)
BE20	GPS problem : impossible to reach the point
BE21	Ground document quality problem : impossible to reach the point
BE22	GPS & Ground document quality problems : impossible to reach the point
BE99	Other GPS comments

2) Difficult access to the point: if the point can be accessed without any problem, a specific comment on "Description of the way to the point" is not necessary. However, in all cases where the way to the point was challenging or difficult to find, a short comment is needed explaining the difficulties, indicating the easiest access or any other important aspect. This information helps during the next survey and explains

for example the effort required to reach the point (duration of the observation). The structured comments should be used whenever possible.

The following structured comments can be selected:

CODE	LABEL
AA01	Long walk
AA02	Water body between the road and the point
AA03	Easiest access from North
AA04	Easiest access from South
AA05	Easiest access from East
AA06	Easiest access from West
AA07	Information requested by land owner
AA08	Point revisited (insert date of previous visit)
AA99	Other description of the way comments

On the ground document Field F "Way to the point" provides a description of the way to the point from previous campaigns (if available).

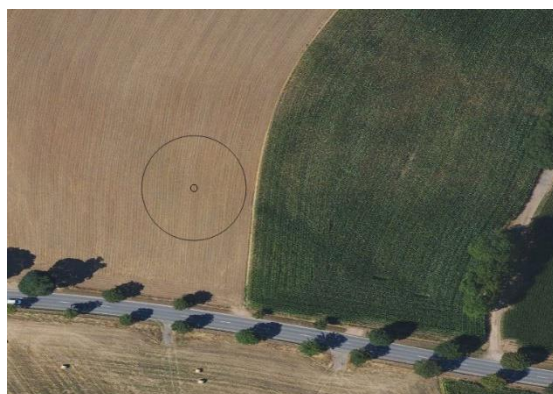


Photo needed:

Take any photo on the access to the point as needed: ACCESS_Other (multiple photos possible)

8.3.2 How to find the exact LUCAS point

Use the orthophoto for approaching the point



(Source of the orthophoto: ortho_2020 WMS, Open Data WMS and WMTS de l'Administration du cadastre et de la topographie du Grand-Duché du Luxembourg (data.public.lu), Creative Commons 1.0 Universal (CC0 1.0).

Edited: addition of the EW of observation

Figure 8-1.

When the surveyor is close enough, the larger-scale topographic map can be used together with the orthophoto and the previous survey photos to locate the point. There are several stable elements marked on a topographic map, like water-towers, bigger roads, rivers, lakes, look-out towers, etc. which can be a good reference for orientation, as well as the point photo. More detailed topographic features can be found on the orthophoto. The points are marked on the orthophoto and the topographic map by a specific sign, i.e. the intersection of two bars resulting in a cross.

To locate a point, the surveyor must give priority to fixed landmarks rather than to uncertain ones, such as limits of parcels. In regions where agriculture is receding, the border of a forest can gain ground over fields or meadows and can't be used as a landmark.

<i>Point shown on the orthophoto is the reference location</i>	The exact location of the point can be identified by using the GPS, the compass, the orthophoto and the topographical map together. The GPS signal might show a relevant deviation from the orthophoto (e.g. in case only very few or bad signals are received). Also, the topographic map might not be up-to-date. The orthophoto has to be taken as main reference. In most instances, it is easy to orientate, especially when the image is a recent one, by comparing the image and the more stable landmarks such as roads or buildings.
<i>Check the enlarged subset for exact location of the point</i>	  (Source of the ortho_2020 WMS, Open Data WMS and WMTS de l'Administration du cadastre et de la topographie du Grand-Duché du Luxembourg (data.public.lu), Creative Commons 1.0 Universal (CCO 1.0). Edited: addition of the EW of observation) Figure 8-2. Figure 8-3. The ground document includes an enlarged subset of the orthophoto enabling the exact location of the LUCAS observation point.
<i>Imagery date and land cover/use changes.</i>	If the orthophoto is an old one, the surveyor has to pay attention to landscape modifications. For instance, an isolated tree or a hedge or a dirt track may have been removed. Buried - and invisible - tanks may have replaced water towers shown on a topographic map. In any case, difficulties to exactly locate the points are to be noted on the field form. In such a case (the old/unclear orthophoto does not show the current land cover/use), the surveyor tries to reach the point as close as possible with the help of the GPS device. A structured comment should be added in the comment field for helping quality check.
<i>Quality</i>	To ensure the highest quality of the survey, the above-mentioned tools for locating the point have to be used together!

8.3.3 Difficult cases in reaching the point

In specific circumstances it can be impossible to reach the point (e.g. excess of time needed to reach the point, access restrictions...).

8.3.3.1 Point is at more than 30 min walking distance

If the point can only be reached by foot, and the walking time for reaching the point and coming back to the car is estimated as lasting clearly more than 1h (excluding time of the survey), the point is considered as inaccessible and not visible. The surveyor will not try to reach the point.



Photo and comment needed:

The surveyor has to take a (conflict) photo to document the impossibility to reach the point in the set time frame.

The relevant structured comment needs to be given and the free text shall be used to provide details on the specific circumstances.

CODE	LABEL
AB09	shortest possible walking time (from car to point and back without time for data collection) more than 1 hour



Photo and comment needed:

If the point can only be reached by foot, and the walking time for reaching the point and coming back to the car is not clear, the surveyor will try to reach the point. If at a certain moment it becomes clear that it is not possible to reach the point within the expected time, the surveyor will not go any further.

Also in this case the surveyor has to take a (conflict) photo to document the impossibility to reach the point in the set time frame: OBS_Conflict

The relevant structured comment needs to be given and the free text shall be used to provide details on the specific circumstances.

CODE	LABEL
AB09	Shortest possible walking time (from car to point and back without time for data collection) more than 1 hour

8.3.3.2 Point not reached: restricted or inaccessible area, other difficulties

If the point is located in a restricted or inaccessible area such as: a large wetland area, an impenetrable natural forest or in a forbidden zone (military camp) or in a fenced pasture for fighting bulls (signs, fences etc.) or privately owned zones with clearly restricted access, the surveyor cannot reach the point. In the last case the surveyor should try to get access permission from the owner.

These points have to be either observed by approaching the point as close as possible so that the surveyor can see the point (if the point is visible from the distance) or they have to be photointerpreted in the field (if the point is not visible). See the chapter 8.4.1 on "Observation of the point" for further information on the photointerpretation in the field.



Photo and comment needed:

If the point cannot be reached, the surveyor has to take a conflict photo (OBS_Conflict) to document the impossibility to reach the point and use the relevant structured comment to describe the situation. The following structured comments can be selected to explain why a point cannot be reached in a restricted or inaccessible area:

CODE	LABEL
AB01	Fence, wall, locked gate/door, natural obstacle
AB02	Access forbidden sign found
AB03	Landowner refused the entrance
AB04	Adverse meteorological conditions (storm, flooding, snow ice etc.)
AB05	Dangerous animals (cows/bulls, dogs, wild animals etc.)
AB06	Impenetrable vegetation / poisonous plants
AB07	Point in water
AB08	Point in wetland
AB10	Military or official area (e.g. naval, airfield)
AB11	Steep slope
AB12	High crop on the ground
AB13	Non-drivable road/track
AB14	Access forbidden due to natural protected area
AB15	Restricted access due to fire hazard
AB16	Restricted access due to hunting
AB17	Restricted access due to fitosanitary treatment
AB18	Restricted access. Payment needed
AB19	Need to cross by boat to access the point
AB20	GPS problem : impossible to reach the point
AB99	Other reasons for not reaching the point comments
AC01	Car damaged

8.3.4 Panel points

Panel points are points that were visited in more than one campaign (be it in consecutive campaigns or not).

These points are used to derive information on land use and land cover change, so it is of utmost importance that they are comparable over time.

For Panel points the ground document contains valuable information to understand which location was assessed by the surveyor in a previous campaign:

- Description of the way to the point from previous campaigns
- Previous photos
- Previous LC/LU
- Information if the Point was located on a border or linear Feature (Use of Look to North rule)



Comment needed:

If the point observed in a previous campaign was located differently, the surveyor has to assess the correct LUCAS point and add a comment.

If the point observed in the current campaign is classified differently as in the previous campaign, a comment has to be made.

The following structured comments can be selected:

CODE	LABEL
BP01	Previous data probably erroneous or wrongly located
BP02	Different location from previous survey/visit: better GD
BP03	Different location from previous survey/visit: point not reachable in present campaign
BP04	Different location from previous survey/visit: closer or better view to the point
BP05	Different Look to N/E decision than in previous survey/visit. Explain why.
BP06	Shift in Orthophotos of different years
BP07	Shift between Orthophoto and GPS
BP08	Different decision on LC: real change
BP09	Different decision on LU: real change
BP11	Different decision than in previous survey/visit: rules changed
BP12	Different decision than in previous survey/visit: PI in previous survey
BP99	Other panel points comments

8.3.5 Care to be taken

<i>Private property</i>	When the surveyor enters fields or forests, he is usually on private property. He must be able to explain his mission and present his accreditation documents if asked by the farmer or owner of the property. If a point is in a building or a parcel next to it and enclosed by a wall, the surveyor will ask permission to enter or stay outside to observe the point.
<i>Don't damage the existing crops</i>	The surveyor must not damage fences or crops. In order to avoid damaging high crops, the surveyor should move along rows between the plants and in tractor tracks for its assessment.

8.4 Section: Point observation

8.4.1 Observation of the point

It needs to be stressed that land use and land cover must be observed at the LUCAS point and from a position as close to the LUCAS point as possible.

The land cover and land use (as well as the other information) to be assessed refer to the exact LUCAS point, which is determined on the orthophoto of the ground document of each point.



Comment needed:

If the duration of the survey is <15min or > 1h15min a comment is needed:

CODE	LABEL
BF04	Survey time < 15' because it was easy to assess the LC/LU
BF06	Survey time > 1h 15' because it was very difficult to assess the LC/LU
BF07	Survey time < 15' because LC/LU photo interpreted
BF08	Survey time > 1h 15' because it was very difficult to reach the point
BF99	Other survey time comments

8.4.2 Field: Type of observation (24)

8.4.2.1 The point is visible (option 1 or 2)

In case the point is visible the distance to the point is to be noted (less or more than 100 m) by ticking the relevant box in the field form.

While the surveyor has to get as close as possible to the theoretical LUCAS point, if this is not possible but the surveyor is able to see the point from a distance, the regular land cover/land use observation can be done and the field form can be filled in. This holds true also if the point is located in a forbidden or inaccessible zone.

If the point is not visible from the closest reachable place to the point but is visible from further away (e.g. due to a very high crop), the observation should be done at the closest place to the point where the point is visible.



Photo needed:

If the point is visible but the distance to the point is >100m the reason for not getting closer needs to be documented through an additional photo: OBS_Distance.



Comment needed:

If the access to the point was difficult and/or the distance is >50 m this needs to be noted in the comments:

CODE	LABEL
AB01	Fence, wall, locked gate/door, natural obstacle
AB02	Access forbidden sign found
AB03	Landowner refused the entrance
AB04	Adverse meteorological conditions (storm, flooding, snow ice etc.)
AB05	Dangerous animals (cows/bulls, dogs, wild animals etc.)
AB06	Impenetrable vegetation / poisonous plants
AB07	Point in water
AB08	Point in wetland
AB09	Shortest possible walking time (from car to point and back without time for data collection) more than 1 hour
AB10	Military or official area (e.g. naval, airfield)
AB11	Steep slope
AB12	High crop on the ground
AB13	Non-drivable road/track
AB14	Access forbidden due to natural protected area
AB15	Restricted access due to fire hazard
AB16	Restricted access due to hunting
AB17	Restricted access due to fitosanitary treatment
AB18	Restricted access. Payment needed
AB19	Need to cross by boat to access the point
AB20	GPS problem : impossible to reach the point
AB99	Other reasons for not reaching the point comments
AC01	Car damaged



Photo needed:

If the LUCAS point is reached put the marker on the point.

Take the point (fields 602, 604) and landscape photos (fields 608 – 619): LCLU_Point, LCLU_North, LCLU_East, LCLU_South, LCLU_West



Comment needed:

If there is a problem with the marker add a comment.

The following comments can be selected:

CODE	LABEL
BJ01	Point marker/flag missing
BJ02	Point marker/flag replaced by other object
BJ03	Point marker/flag placed by mistake or not on point
BJ99	Other point marker comments

8.4.2.2 In situ photointerpretation (option 3)

If the surveyor is not able to see the point from his observation place and the quality of the orthophoto is sufficient to at least apply a simplified land cover and land use classification, the point will be photointerpreted in the field based on the orthophoto. The chapter 8.5 “Photo-Interpretation” provides more details on the use of this observation type.



Photo and comment needed:

Reasons for not observing the point are to be written down in the comments field and a photo (OBS_Conflict) documenting why the LUCAS point was not reached has to be taken from the point reached.

<i>In situ photointerpretation is an exception to the observation rule</i>	LUCAS is an in-situ survey and the points have to be visited in the field. The in-situ photointerpretation can only be used in rare cases if the point is not visible at the closest accessible location nor at a location nearby.
<i>In situ photointerpretation is only allowed in some situations. It requires good documentation</i>	The in situ photointerpretation is only allowed on the condition that the surveyor documents the existence of at least one of the following circumstances by comments and by an additional picture: - Point is located in the water

<i>by comments and pictures.</i>	- Point is located in inaccessible wetland - Impenetrable vegetation blocks the way to the point from all directions (thick shrubs with spikes, impenetrable forest) - Fenced area with “no access” signs or with a no accessible fence (forbidden area) - Walled or hedged private areas (companies' premises, private gardens etc.) - o In this case it is advisable to try to get a permission from the owner to enter if the owner is available (e.g. by contacting the gate guard in companies or if the house owner is available) - Landowner refuses the entrance to the property - Dangerous animals are present close or in the walking way to the point - Shortest possible walking time from the car to the point and back is more than 1 hour (excluding working time for the survey and for walking the transect) Only if the above mention conditions are found, the surveyor registers “photointerpretation” (PI) as type of observation in the field form.
<i>Use the orthophoto to make the photointerpretation</i>	The orthophoto and the visual assessment done from the point reached by the surveyor will possibly allow to identify the land cover on the LUCAS point. This is the case if the surveyor can see the parcel where the point is located (which can be confirmed by photointerpretation) although he cannot see the point itself, e.g. in large field with the point several hundred meters away)
<i>Take landscape pictures from the reached location</i>	Landscape photos have to be taken from the point reached (mark the GPS point coordinates in the field form). Point and crop/coverphotos are not needed in this case.

Possible comments are:

CODE	LABEL
AB01	Fence, wall, locked gate/door, natural obstacle
AB02	Access forbidden sign found
AB03	Landowner refused the entrance
AB04	Adverse meteorological conditions (storm, flooding, snow ice etc.)
AB05	Dangerous animals (cows/bulls, dogs, wild animals etc.)
AB06	Impenetrable vegetation / poisonous plants
AB07	Point in water
AB08	Point in wetland
AB09	Shortest possible walking time (from car to point and back without time for data collection) more than 1 hour
AB10	Military or official area (e.g. naval, airfield)
AB11	Steep slope
AB12	High crop on the ground
AB13	Non-drivable road/track
AB14	Access forbidden due to natural protected area
AB15	Restricted access due to fire hazard
AB16	Restricted access due to hunting
AB17	Restricted access due to fitossanitary treatment

AB18	Restricted access. Payment needed
AB19	Need to cross by boat to access the point
AB20	GPS problem : impossible to reach the point
AB99	Other reasons for not reaching the point comments
AC01	Car damaged

8.4.2.3 Point not visible and photointerpretation not possible (option 4)

In case of the field survey this option is to be selected if the point is located in an inaccessible area or forbidden zone and the point is not visible and the orthophoto doesn't allow photointerpretation to be applied, not even using a simplified classification for land cover and land use (very bad quality of orthophoto, incomplete coverage, cloud coverage).

In the case of the photointerpretation in the office this option is to be selected if the orthophoto doesn't allow photointerpretation to be applied, not even using a simplified classification for land cover and land use (very bad quality of orthophoto, incomplete coverage, cloud coverage).

In the above described cases for the field "Type of observation" option 4 - in the field form should be selected.

In this case the survey ends here.



Photo and comment needed:

The reasons for not observing the point has to be written down in the comments field. In case of the in-situ survey also an additional photo (OBS_Conflict) documenting why the LUCAS point was not reached has to be taken from the point reached in the field.

In addition to the comments available under 8.4.2.2 - In situ photointerpretation (option 3), the following comment:

CODE	LABEL
BA01	Ground documents not correct
BA02	Orthophoto not available
BA03	Not updated orthophoto
BA04	PI reference other than GD ortho
BA99	Other comments on quality of background information

To finalise the survey of the point the end time has to be added (including a comment for the short survey duration).

8.4.2.4 Out of national territory/EU, (options 5, 6)

In case the point falls out of the national territory or into marine sea, the surveyor does not have to reach the point. He selects option 5 or 6. In this case the survey for this point is concluded, once the type of observation has been filled in and the end time has been added.



Comment needed:

Please select the specific comment stating why the photointerpretation is not possible and explain.

CODE	LABEL
AB99	Other reasons for not assessing the point

8.4.2.5 Photointerpretation in the office (option 7)

This option applies if the point has been assigned to photo-interpretation in the office and the photo-interpretation can be done, using a simplified land cover and land use classification if needed. The chapter 8.5 "Photo-Interpretation" provides more details on the use of this observation type.

8.4.3 Field: Homogeneous plot fills extended window (701)

In this field it needs to be indicated if the plot where the point is located on is completely covering the extended window of observation. The extended window of observation expands to a radius of 20 meters of distance (or 40 m diameter) from the point (representing an area of 0.13 ha) and it is drawn on the orthophoto of the ground document.

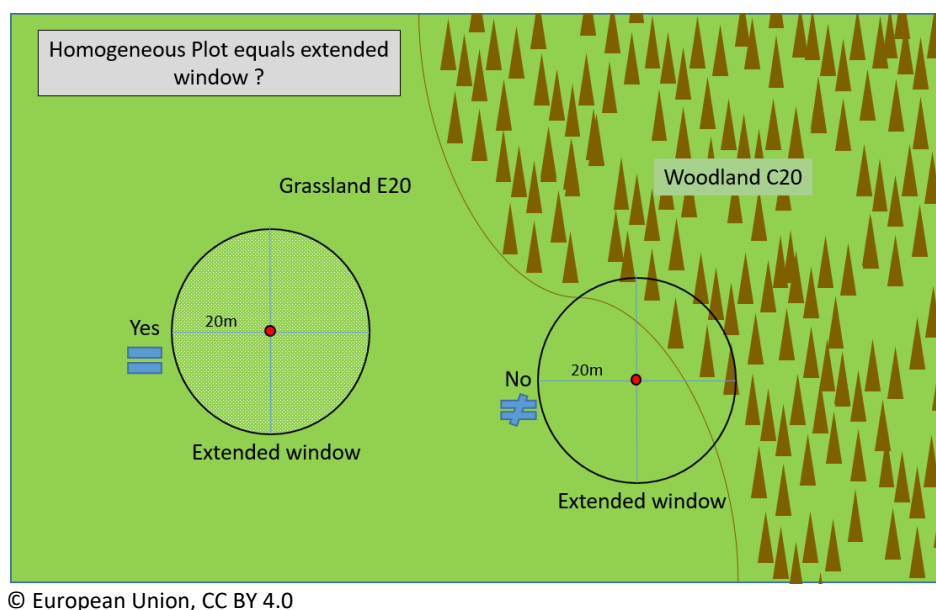


Figure 8-4. Homogenous plot = Extended window

8.4.4 Field: Direction of observation (25)

If:

- the point is located on the boundary of two parcels and it is ambiguous, which parcel has to be observed (Border case)

or

- the point is located on a linear feature < 3 m wide (small hedge, small track, roadside verge) or a small object (e.g. a tool shed < 3 m wide in a garden) (Linear feature case)

(=> the criterion of point definition (minimum width of 3m) is not fulfilled.)

then the "look to the North" rule applies:

Look to the North – Rule:

The surveyor has to observe land cover and use north of the point and if this is not possible to the east (see detailed explanation below).



Photo needed:

If the "Look to the north" or the "Look to the east" rule is applied an additional photo OBS_LTNE is needed to document the decision. The photo needs to show the border or the linear feature.



Photo needed:

An additional photo may be needed to show the land cover assessed (direction North or East): OBS_LTNE_LC



Comment needed:

Only for Panel Points: If the decision on which rule to apply is different from the decision of the last campaign a comment is needed.

CODE	LABEL
BP05	Different Look to N/E decision than in previous survey/visit. Explain why.

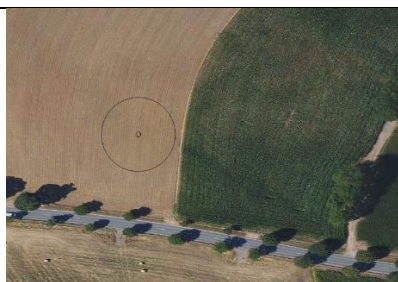


Figure 8-5.

Point location clear, since it is entirely located in an agricultural field.



Figure 8-6.

Point located on a border



Figure 8-7.

Point located on a small linear feature (roadside verge)

Source of the orthophoto: ortho_2020 WMS, Open Data WMS and WMTS de l'Administration du cadastre et de la topographie du Grand-Duché du Luxembourg (data.public.lu), Creative Commons 1.0 Universal (CC0 1.0) Edited: addition of the EW of observation and LTN rule

Look to the North rule and special cases:

<i>Look to the North</i>	If the point is located on a boundary between two land cover types, or the point falls on a small linear feature and the criterion of point definition (minimum width of 3m) is not fulfilled" then the surveyor has to observe the land cover/land use north of the point.
<i>Look to the East if feature is oriented N/S</i>	In case the border between two parcels is exactly in North-South direction (=meridian) and even when applying the "look to the north" rule a clarification cannot be obtained, land cover/land use to the East of the point are to be observed. In the survey field form, the surveyor must note this "virtual" shift of the point. Moreover, he has to explain the decision under the comments field.
<i>Look to the East if more than one linear feature exist to the North</i>	 (Source of the orthophoto: ortho_2020 WMS, Open Data WMS and WMTS de l'Administration du cadastre et de la topographie du Grand-Duché du Luxembourg (data.public.lu), Creative Commons 1.0 Universal (CC0 1.0). <u>Edited</u>: addition of the EW of observation) Figure 8-8. If the point is located on a linear entity (e.g. path) the surveyor applies the rule "look to the north". If when looking to the North a second linear entity is found (e.g. grass margin north of the road), then the surveyor looks to the East.
<i>Where to place the flag/point marker</i>	Following strictly the workflow, the point marker is put in place and the point picture taken BEFORE applying the "Look to the North" rule. Place the point marker on the border, and NOT on the shifted point.
<i>Implications for land use</i>	Land use is to be assessed on the same place as the land cover.
<i>Comparability</i>	If the point has already been observed in a previous survey the same direction of observation should be used, unless the direction of observation was clearly wrong. In this case the correct direction is to be used and a comment on the reasons for changing the direction of observation needs to be added

8.4.5 Field: Reason for looking to North or East (702,703, 704)

In this field the conditions that led to applying the "look to north" or the "look to the East" rule need to be documented. The reasons can be that the point is either falling on a border of two land covers or that the point is falling on a linear feature, which is less than 3 meters wide.

In case the point is falling on a border, the Land Cover Codes on both sides of the border need to be filled in the respective field "Border".

In case the point is falling on a linear feature, it needs to be exactly described in the respective field "Linear feature". The list of linear features can be found in Annex 9.2.7 Code list for linear features.

8.4.6 Point and landscape photos



Photos needed:

Landscape photos in the 4 cardinal directions are to be taken at the point reached by the surveyor: LCLU_North, LCLU_East, LCLU_South, LCLU_West

If the LUCAS point is at max. a 100 m distance (obs type 1 or 2) a point photo is to be taken: LCLU_Point

If the point can be reached, indicate the LUCAS point with an object.

8.4.7 Other photos and comments about point observation



Other comments:

Other comments about the point observation (why observation not at the point, why different decision than last survey, ground document not clear, etc.) should be added as needed (500).

CODE	LABEL
BA01	Ground documents not correct
BA02	Orthophoto not available
BA03	Not updated orthophoto
BA04	PI reference other than GD ortho
BA99	Other comments on quality of background information



Other photos:

Take other photos on the point observation as needed: OBS_Other (multiple photos possible)

8.5 Section: Photo-Interpretation

Unlike a map, features on an aerial photograph are not generalized or symbolized. Aerial photos record all visible features on the Earth's surface from a birds-eye view. Although the features are visible, they are not always easily identifiable. The process of studying and gathering the information required to identify the various cultural and natural features is called photo interpretation.

With careful interpretation, air photos are an excellent source of spatial data for studying the Earth's surface. Compared to a point observation in the field, the photo-interpretation has its limits and some features are not detected. This chapter will clarify these limits, define the fields that can or cannot be filled in.

Generally, all possible features of the LUCAS point that can be identified by the means of photo interpretation (by using ortho-photo and additional sources) shall be coded, but if something cannot be detected clearly and remains questionable, the assessment must be skipped.

If a proper and uncomplicated Photo-Interpretation cannot be done, the following fields shall help to document the reasons and procedures that were used for this observation type.

8.5.1 Field: Poor Ortho-photo conditions? (710)

In this field, the photo-interpreter documents the Ortho-photo quality if it was not good enough to properly assess the point.

<i>Outdated</i>	The orthophoto is clearly outdated, landscape or constructions have changed significantly and the orthophoto does not present the actual surrounding of the LUCAS point anymore.
<i>Detail quality Issue</i>	The orthophoto's level of detail is too low to detect and assess the Land Cover, Land Use and other features correctly.
<i>Outdated and detail quality Issue</i>	Combination of both the before mentioned issues.
<i>No</i>	The orthophoto quality is fine to assess the point.



Other comments:

Additionally comments can be used to explain quality issues with the orthophoto.

CODE	LABEL
BA02	Orthophoto not available
BA03	Orthophoto not updated
BA99	Other quality issue with background information

8.5.2 Field: LC/LU assessment by using additional sources (711)

Here, the photo-interpreter is able to document if additional sources were used to assess the land cover, the land use or both, the land cover and land use at the point.

<i>For LC</i>	Additional sources besides the orthophoto were needed and used to assess the Land Cover.
<i>For LU</i>	Additional sources besides the orthophoto were needed and used to assess the Land Use.
<i>For LC/LU</i>	Additional sources besides the orthophoto were needed and used to assess the Land Cover and Land Use.
<i>Not used</i>	No additional sources were used to assess the LUCAS point. (This can be due to the fact that they were not needed or that there were no additional sources available.)

8.5.3 Field: Additional sources used (712)

In this field, the photo-interpreter is able to document which additional sources were used to assess the point.

<i>Google Earth / Google Maps</i>	As an additional source Google Earth / Google Maps was used
<i>Previous campaign information</i>	As an additional source information from the previous campaign (e.g. previous LC/LU, LTN-decision or previous photos) was used
<i>Other additional sources</i>	A comment explaining the sources is needed.
<i>No other suitable sources available</i>	No additional sources were available, only a simplified photo-interpretation was possible. Please note that any information collected through photointerpretation needs to be reliable. If the orthophoto is so bad that no reliable information can be collected, then the photointerpretation is not possible. In such a case please go back to the field "Type of observation" and select option 4 "Point not visible and photointerpretation not possible".



Comment needed:

If other additional sources have been used, please add a comment listing these sources:

CODE	LABEL
BD08	LC classified with the help of other information
LU03	LU classified with the help of other information

8.5.4 Field: Only assessment on lower detail level possible (713)

If land cover and/or land use are not recognisable in the orthophoto, classes of a lower detail level have to be used, which are only allowed in case the LUCAS point is photointerpreted.

<i>For LC</i>	Land Cover can only be assessed on lower detail level
<i>For LU</i>	Land Use can only be assessed on lower detail level
<i>For LC/LU</i>	Land Cover and Land Use can only be assessed on lower detail level
<i>Not used</i>	No assessment on lower detail level needed

The lists of classes for the lower detail level for Land Cover and Land Use can be found in Annexes 9.2.2 and 9.2.5.

8.6 Section: Land cover and land use

8.6.1 Extended window of observation

Theoretically, a point has neither width nor length. Considering the subject of the survey – land cover and land use – there is a standard definition to be applied for the size of the point: in the LUCAS survey the point corresponds to a circle with a 1.5 m radius (or 3 m diameter), therefore representing an area of about 7m².

If the point falls in a homogenous area (e.g. arable land, top of a house etc.) the above-mentioned point definition can be easily applied and the location of the point can be identified without any doubt.

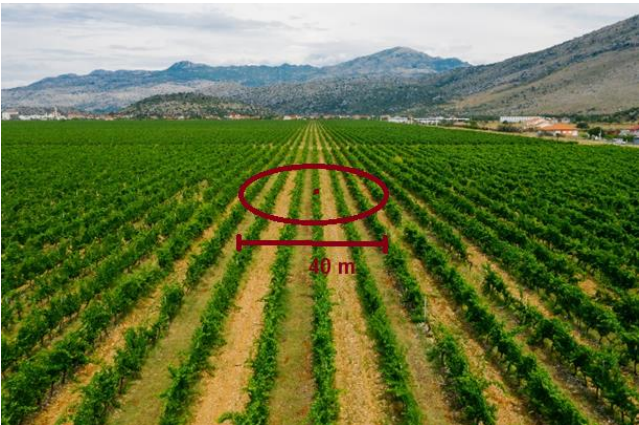
However, some cases might occur, where the location of the point and the observation of the land cover/land use are ambiguous:

- if the point falls on a boundary or small linear feature
- if there is heterogeneity in the land cover

The case where the point falls on a boundary or on a small linear feature has been discussed in chapter 8.4.4.

When the land cover is not homogeneous, for example when it is composed of trees or shrubs interspersed with grass, the scale of observation has to be changed to classify it. In these cases a systematic observation of the “environment” near the point, which in LUCAS is called the extended window of observation, has to be adopted.

The extended window of observation expands to a radius of 20 meters of distance (or 40 m diameter) from the point (representing an area of 0.13 ha).



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Edited: addition of the EW of observation)

Figure 8-9.

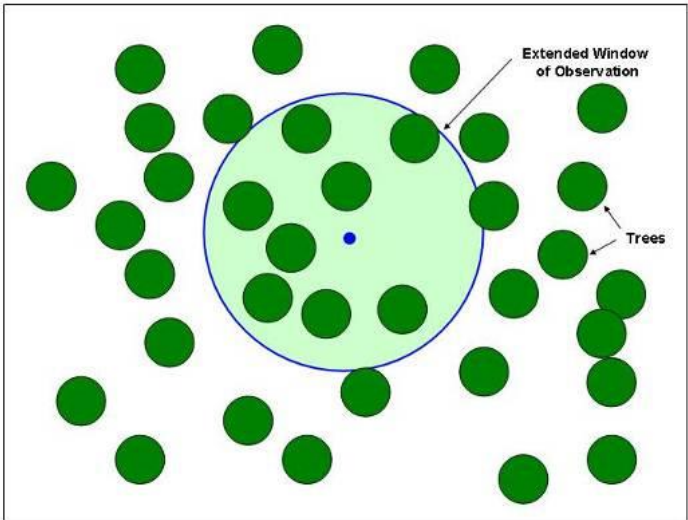


© 2018, Carlos Aguilar (@zenderista), Pixabay License.
Edited: addition of the EW of observation)

Figure 8-10.

Extended window of observation

Using the extended window



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Figure 8-11.

In the example above, the point is located in an area with a sparse tree coverage and grass. By means of the extended window of observation, the area share of the tree canopy can be estimated.

If the extended window is not used and only the point with its 7m² is considered, it is not possible to obtain a reasonable estimation of the tree canopy.

In this example, the area share of the tree’s canopy is approximately 30%, so the point has to be recorded as woodland (minimum coverage of the tree crown 10%). If the tree coverage would range between 5 –10%, the corresponding land cover would be grassland with sparse tree coverage (depending on the land cover).

When to use the

The window of observation has to be extended whenever the land cover at the point is

<i>extended window</i>	heterogeneous. This occurs regularly in areas such as: - permanent crops (B7X, B8X, except nurseries B83): parcels of permanent crops where the trees or other plants alternate with bare soils and/or grassland or another crop - woodland (CXX) - shrub land (DXX): where a mix of e.g. shrubs and trees might occur - grassland (EXX), where land features may alternate (e.g. grassland with trees) - bare land (FXX) - wetland (HXX)
<i>Example: an orchard</i>	(Source: ortho_2020 WMS, Open Data WMS and WMTS de l'Administration du cadastre et de la topographie du Grand-Duché du Luxembourg (data.public.lu), Creative Commons 1.0 Universal (CC0 1.0)) <u>Edited</u>: addition of the EW of observation) Figure 8-12. The orthophoto above shows a heterogeneous area with grassland and fruit trees (B7X). This is a situation where the extended window of observation is to be applied.

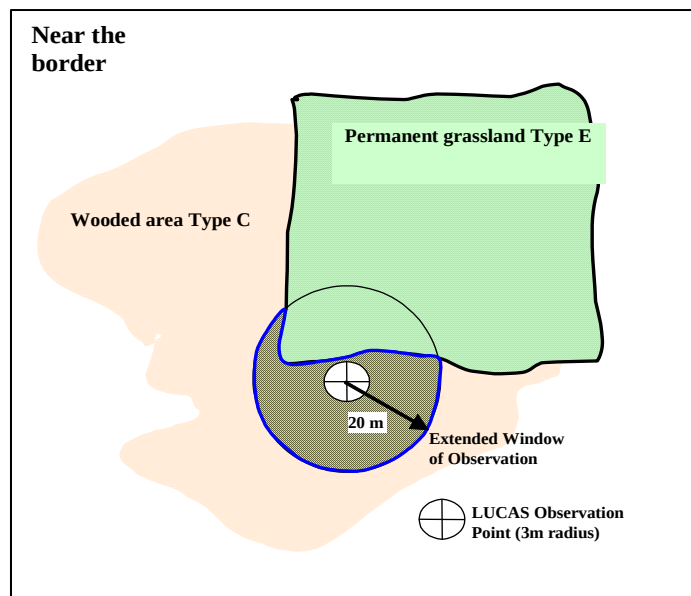
8.6.2 The homogeneous plot

Note that when the extended window of observation is applied, the surveyor has to observe the land cover **within the borders of the parcel where the point is located**. This means that in many cases not all the extended window will be observed, but only the **homogeneous plot** within it.

Generally speaking, for the LUCAS survey purposes, the extended window covers the maximum area within the 20 m radius circle having the same land cover and use.

Three specific examples are presented below for clarification of the concept.

The extended window of a point spanning two adjacent parcels



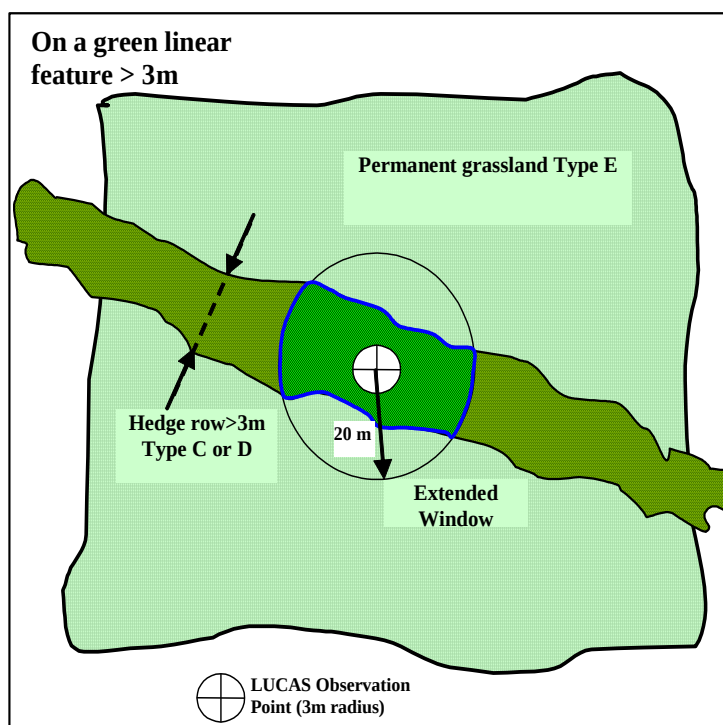
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Figure 8-13.

At the point, the cover being a wooded area, the observation has to be done **within** the extended window. The extended window has to be considered only **within the parcel** it belong to, so in this case the parcel defined by the border of the wooded area and the grassland, not considering the zone inside the grassland (which is a different parcel).

The density of trees and or shrub is to be assessed in the hatched part of the circle to decide whether it is a shrub land (DXX) or a wooded area (CXX), and in the latter case whether it is coniferous (C2X), broad-leaved (C10) or mixed (C3X), and which forest type (CXX1 - CXXE).

The extended window of a point falling on a linear feature wider than 3 m



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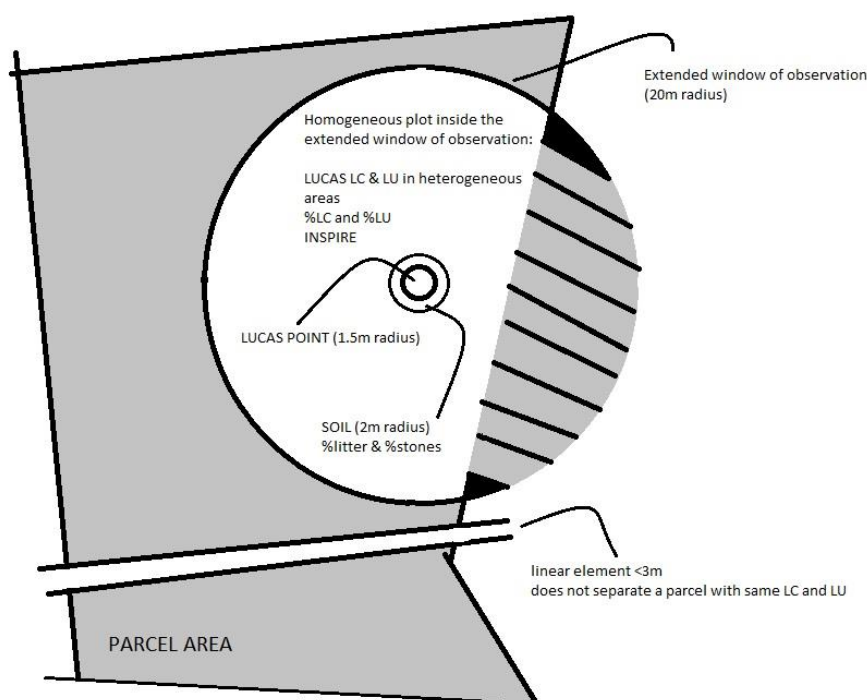
Figure 8-14.

A linear feature wider than 3 m is to be treated like any other parcel. The point is assigned

	to the existing land cover of the specific linear feature (road, river, wooded area). The assessment has to be made within the area defined by the linear feature (the parcel) in order to assess the density of trees and/or shrubs (but it is not extended into the adjacent parcels). In the example the density is to be assessed in the hatched part of the circle to define whether it is a shrub land (DXX) or a wooded area (CXX), and in the latter case whether it is coniferous or broad-leaved.
<i>The extended window of a point falling on the grassy area of a garden belonging to a residential building</i>	© European Union, CC BY 4.0 Figure 8-15. Once again, the surveyor has to observe the parcel in which the point is located. In this example the grass zone enclosed by a wall (dashed square in the figure) defines the plot to be observed within the extended window of observation. As within this area shrubs and trees are growing, the point is to be classified as E10 because the tree crown (within the homogeneous plot) is below 10 %.

8.6.3 Schema: window of observation and homogenous plot

See below an image summarizing the concepts mentioned in this chapter.



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Figure 8-16.: Window of observation and homogeneous plot

8.6.4 Field: LC1 land cover 1 (28)

Land cover refers to the bio-physical coverage of land (e.g. crops, grass, broad-leaved forest, or built-up area). Land cover (LC) will be noted for each point according to the classification mentioned in LUCAS reference document C3-Classification.

For every point one or 2 land covers are to be recorded. The dominant land cover is recorded as first land cover (LC1).

Examples and special cases:

- A road covered with grass has heavily compacted soil (else it would be impossible to drive on), this is therefore the main characteristic: LC1=A22, LC2=E20,
- Wooded land is always considered a dominant land cover CXX=LC1)

In case of layered, competing (no clear dominance) land covers the classification starts from the top.

Examples and special cases:

- A bridge over a river is recorded as LC1=A30, LC2=G21).
- If both land covers belong to artificial (XX) only one land cover is recorded as it is in any case sealed (e.g. bridge over road LC1=A30, no LC2; bridge over square LC1=A30, no LC2).
- If a powerline over wooded land (or over shrubland or grassland) is <3m the powerline is not recorded: LC1=CXX

There is a difference between features superimposed (both are be registered, unless both are artificial) and features juxtaposed, the feature nearby is not registered.

Examples and special cases:

- When trees are near the road with the crowns covering the road but the stems of the trees being in another plot , the land cover of the road is only LC1=A22).

For structures with multiple layers (e.g. buildings), the rule is to attribute 100% land use to each layer (e.g. floor).

Examples and special cases:

- A building has a museum on the ground floor and bank offices in all the other 4 floors: LC1= A12, LU1 = U342 (400%), LU2 = U361 (100%)



Photo needed:

If the point is visible and LC1 is B, C, D or E take a crop/cover photo (fields 605/607): LCLU_CropLC1.



Comment needed:

If the decision on LC1 is different from the decision in the last campaign, add a comment explaining why.

CODE	LABEL
BP08	Different decision on LC: real change
BP11	Different decision than in previous survey/visit: rules changed
BP12	Different decision than in previous survey/visit: PI in previous survey
BP13	No real change. Reason not clear
BP14	Different decision on LC/LU because of shift in orthophotos of different years
BP99	Other panel points comments



Other comments if relevant:

CODE	LABEL
BP01	Previous data probably erroneous or wrongly located
BP02	Different location from previous survey/visit: better GD
BP03	Different location from previous survey/visit: point not reachable in present campaign
BP04	Different location from previous survey/visit: closer or better view to the point
BP06	Shift in Orthophotos of different years
BP07	Shift between Orthophoto and GPS
BP99	Other panel points comments
BD01	Dehesa

BD02	Layered crop
BD03	Local knowledge
BD04	Tilled arable land not sown during current season
BD05	Management of migratory animals
BD99	Other land cover comments



Other photos as relevant: LCLU_Other (multiple photos possible)

8.6.5 Field: LC2 land cover 2 - Multiple registration of land cover (31)

When more than one land cover co-exists in the same area, a second land cover may be needed to describe the point. A detailed crosstable containing all the possible combinations of two different land covers can be found in Annexes (9.3).

It is possible to record 2 land covers and 1 land use.

Exception:

- For wooded land (LC1=CXX) a second land cover is recorded only if there is a second land use linked to it.

This is common in specific landscapes such as for example in agro-forestry areas, where wooded land is combined with crops, but it can also happen in arable land, when different crops are intrinsically mixed in the same field.

For possible combinations of LC1 and LC2 check the cross tables in the Annexes.

Note that the information recorded under (LC1 and LU1) and (LC2 and LU2) must correspond.



Photo needed:

If the point is visible and LC2 is B, C, D or E take a crop/cover photo: LCLU_CropLC2.



Comment needed:

If the decision on LC2 is different from the decision in the last campaign, add a comment explaining why.

Same comments as in section 8.6.4 Field: LC1 land cover 1 (28)



Other comments if relevant:

Same comments as in section 8.6.4 Field: LC1 land cover 1 (28)



<i>A second land cover LC2 may be needed to describe the point</i>	The need for a second land cover is normally the case in greenhouses, association of crops and crops under cover (e.g. under trees).
<i>Some classes include already a second land cover</i>	For complex or heterogeneous areas, particularly in semi-natural areas, the double land cover classes are already coded in the main classification, as the classification is already referring to a mix of components. For example, grassland with sparse shrub/tree cover (E10) or shrub land with sparse tree cover (D10). For these classes LC2 should not be recorded if there is no additional LC class. Likewise, if grass or shrubs appear in combination with permanent tree crops (B7X, B8X), or woodland (CXX) take into account that the tree component has already been assessed as LC1, therefore D20 and E20 are to be used as LC2.
<i>Example: an orchard with grass between the rows</i>	 © 2014, Gary Fisher (@gafisher), Pixabay License Figure 8-17. In the photo a mix of fruit trees (e.g. apple trees B71) and permanent grass can be seen. It is a heterogeneous area and the point falling in the orchard is evaluated using the extended window of observation. The main land cover consists of the fruit trees, as they determine the main use of the area. The resulting classification is LC1= B71 and LC2 = E20 In case of spontaneous grassland under the trees classify LC2 by using the corresponding code E30 (trees have been assessed in LC1).
<i>Orchards with mixed species</i>	Some orchards have different species (maybe in alternate rows or mixed within a given row). If two species have a significant presence in the orchard (at least 33% each), two types of land cover are noted. Otherwise, the major one is noted and the others are ignored.
<i>Orchards and annual crops</i>	Some orchards are organised in association with annual crops between the rows. If the annual crop has a significant presence (at least 33%) within the extended window of observation, two types of land cover are noted, the first is the permanent crop and the second is the annual crop.
<i>LC2 in wooded areas (CXX) only if there is a different use</i>	If the main land cover is wooded land (CXX), and the use is forestry (U120), a second land cover should only be given if it refers to another land use (e.g. U111 agriculture or U361 leisure).

<i>Nurseries</i>	For nurseries (B83) other than forest nurseries (B83f) the main type of plant/tree has to be noted as second land cover. The exception for forest nurseries (B83f) is due to the fact that CXX cannot be used as LC2.
<i>Priorities</i>	In case of multiple land cover combinations, there is some degree of prioritization of LC1 against LC2. For example, land cover with trees should always be recorded as LC1.
<i>Be aware of impossible combinations of LC1 and LC2</i>	For more details on impossible or unlikely combinations of LC1 and LC2, see the annexes to this document.

8.6.6 Field: LU1 land use 1 (34) and coverage % (36)

Land use indicates the socio-economic use of land (e.g. agriculture, forestry, recreation or residential use). Land use (LU) will be noted for each point according to the classification mentioned in LUCAS reference document C3-Classification.

To determine the main land use it is important to evaluate the context and to determine if the area analysed is associated to a neighbouring area that determines its use. Grassland for example can be part of a residential area, but it can also be part of a commercial building or to the road network (e.g. a green island in the middle of a highway crossing).

The LC1 and LU1 information need to correspond to each other. If the land use of two land cover types is the same, only one land use class is to be assigned.

Possible combinations of LC and LU are presented in the annexes.

The dominant land use is recorded as first land use (LU1)

Examples and special cases:

- The first land use LU1 for roads is always transport (U312)).

Associated areas get the land use from the main area they belong to

Examples and special cases:

- A gas station on the road side belongs to the road infrastructure: U312
- A lawn belonging to an industrial establishment has land use: U22X)

Abandoned and unused areas cannot have another land use.



Comment needed:

If the decision on LU1 is different from the decision in the last campaign, add a comment explaining why.

CODE	LABEL
BP09	Different decision on LU: real change
BP11	Different decision than in previous survey/visit: rules changed
BP12	Different decision than in previous survey/visit: PI in previous survey
BP13	No real change. Reason not clear
BP14	Different decision on LC/LU because of shift in orthophotos of different years
BP99	Other panel points comments



Other comments if relevant:

CODE	LABEL
BP01	Previous data probably erroneous or wrongly located
BP02	Different location from previous survey/visit: better GD
BP03	Different location from previous survey/visit: point not reachable in present campaign
BP04	Different location from previous survey/visit: closer or better view to the point
BP06	Shift in Orthophotos of different years
BP07	Shift between Orthophoto and GPS
BP99	Other panel points comments
LL99	Other comment on land use



Other photos as relevant: LCLU_Other (multiple photos possible)

8.6.7 Field: LU2 land use 2 and coverage (%) - Multiple registration of land use (37, 39)

As for land cover, also for land use a second land use is to be registered, if relevant. A detailed cross table containing all the possible combinations for LU1 and LU2 can be found in Annexes (9.4).

In case of more than one land use being present it is also possible that the sum of land uses in the observation unit is more than 100% (e.g. a building with several floors holding residential and commercial use).

In general the following rules apply:

- 1) If there are two Land Covers at the point, as far as possible, LU1 shall relate to LC1 and LU2 to LC2
- 2) If there is only one Land Cover then LU1 shall be the dominant and main land use of the Lucas point and LU2 a secondary land use.



Comment needed:

If the decision on LU2 is different from the decision in the last campaign, add a comment explaining why.

See comments in 8.6.6 Field: LU1 land use 1 (34)



Other comments if relevant:

See comments in 8.6.6 Field: LU1 land use 1 (34)



Other photos as relevant: LCLU_Other (multiple photos possible)

Examples of multiple land use:

<i>Example 1: Multiple uses in urban area</i>	Buildings are often used for multi-functional purposes. The dominant land use – that is the most important land use - is to be recorded as LU1. The second more dominant land use has to be assessed as LU2, since it is impossible to assess all the LU a multi-functional building may have. The point falls on an office building with 20 floors and there is a museum downstairs (included in the 20 floors, since the ground floor counts as a floor). For buildings, the rule is to attribute 100% land use to each floor. The dominant land use relates to the offices (U342). LC1= A12 (Buildings with more than 3 floors) LU1=U342 (Business); LU1%=1900% LU2=U361 (Museums); LU2%=100%
<i>Example 2: Multiple use in roads</i>	In case of roads, the first land use has to be always transport (U312). A second land use might be added for final socio-economic use of the road, e.g. agriculture (U111) if it is for agricultural machines, forestry (U120) if it is used for forestry and so on.
<i>Example 3: Multiple use (a gas station with retail in an area associated to a road)</i>	The point falls on the building of a gas station. Inside there is a travel agency. LC1= A11 (Buildings with one to three floors) LU1=U312 (Road transport is the first land use, since road transport includes the relevant associated areas) LU2=U341 (The second most important land use is the gas station, so a retail shop. As there is no third land cover recorded, the travel agency is not registered)
<i>Example 4: Grazing in forest areas</i>	Shrubland or wooded land may be used for forestry and for the feeding of cattle (agriculture). In this case 2 land uses are to be registered. LU1=U120 (forestry) LU2=U111 (agriculture)

<i>Example 5:</i> <i>Water areas can have many uses</i>	If a point is located in the water body or a reservoir its LUs have to be classified with the main uses this area may have (fishing, leisure, sports, etc.). Whether a dam has been built for the production of electricity, for water consumption, etc. has not impact on the LU of the water it stores. e.g. LC1=G21 LU1=U361 (leisure) But if the LUCAS point is located on the dam or any infrastructure around it used to produce energy (hydroelectric), then it is assessed as follows: LC1=A21 (dam) LU1=U210
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Special case:

<i>Abandoned and unused areas cannot have another land use.</i>	The land use classes abandoned (U410) and unused (U420) cannot be registered in combination with another land use codes, neither can both of these classes (U410, U420) be recorded at the same time).
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8.6.8 Land cover x Land use

If for two land covers (LC1 and LC2) the land use is the same, (e.g. "U111" Agriculture for different species of fruit trees), only one land use is to be registered.

It is also possible to record one land cover and two land use categories (e.g. river: transport, fishing). A list of possible LC1 & LC2 and LC & LU combinations is provided in the annexes.

Check the Annexes for possible, impossible (or unlikely) combinations of land cover and land use.

Within the possibilities, LU codes are to be associated with the respective LC code, thus LC1 -> LU1 and LC2 -> LU2 if the LC and LU are related.

Example:

<i>Wooded land (cover) and energy production (use)</i>	Trees can be found below power lines. The correct classification is LC1=A30 LC2=CXX LU1=U319 LU2=U120 The classification is made from the highest layer to the lowest layer.
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8.6.9 Changes in land cover between campaigns

LUCAS sample points may be visited in more than one campaign (see also 0). This allows identifying whether changes in land cover or land use have occurred at the point. However, a thorough review of point data collected in the 2009 and 2012 campaigns revealed that in some points changes had been registered but no real change had occurred at the point. This could be found especially in case of unlikely changes registered.

Unlikely changes are changes that do not occur very often, such as e.g. a building becoming a forest.

The causes of changes registered that where not real changes were:

- Shifts in points surveyed on the ground:
 - Due to a shift of the point in the ground document,
 - Ground document in previous campaign not detailed enough to locate the point with precision,
 - The new structure is not on the orthophoto and therefore it is difficult to say exactly where the LUCAS point is,
 - A wrong positioning of the point in the field in one of the campaigns;
- Differences in classifying the same point:
 - Linear feature rules not applied consistently (e.g. a small road is considered as small linear feature in one campaign and not in the other),
 - Linear feature rule applied inconsistently according to the feature type (e.g. a grass strip is considered a linear feature, but a similar sized side walk is not),
 - The same road is classified as A22 in one year and F00 in the other;
- Border problem:
 - The point is at the border between 2 areas and the look to the north rule has not been applied in the same way in different years.

The evaluation of land cover changes is an important result of the LUCAS survey, therefore the surveyor **should carefully document changes** (with structured comments, BP code) between the present and past campaigns, and avoid classifying the point differently if no real change is present.

While errors that may be found in the data of the previous campaigns cannot be corrected by the surveyor, it is of **utmost importance** that the surveyor **inserts a structured comment** to document the fact that the classification of the previous campaign differs from his classification but there is not real change at the point. The surveyor also has to insert a comment to support his/her decision on how to correctly classify that point.

These rules apply also to evaluation of Land Use. See a list of unlikely changes in Annex 9.

8.6.10 Examples of classification by land cover

8.6.10.1 Artificial land (A00)

Built-up areas (A10) include roofed constructions (buildings) distinguished in 3 categories

- ≤ 3 floors or less than 10m height (A11)
- > 3 floors or higher than 10m (A12)
- Greenhouses (A13)

Non-built-up areas with artificial cover (A20) include area features (A21) or linear features wider than 3 meters (A22).

In 2015 a new class (A30) was introduced to cover for other artificial areas.

For this class it is very important to identify the correct position of the point in residential areas, especially those where houses are mixed with small parcels of gardens, yards and parkings. Remember to use the “Look to the North/East” rule on linear features (those less than 3m wide) and to apply correctly the extended window if needed. With very small constructions (e.g. a tool shed less than 3m wide) the look to the North/East rule is also to be used.

See below some examples of classification

<i>Parking lot</i>	The point is located on a parking area of a supermarket. LC1 = A21 (non built-up area feature)
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	LU1 = U341 (commerce)
<i>Building (>10m)</i>	The point is located in a building hosting offices with more than 3 floors, and hosts a textile products' manufacturer. LC1 = A12 (buildings with more than three floors) LU1 = U222X (textile products can be coded as U222R (raw) or U222L (light end industry, depending on the specific activity))
<i>Road (>3m)</i>	The point is located on a sealed road wider than 3m. LC1 = A22 (non-built-up linear feature) LU1 = U312 (road transport)
<i>Road (<3m)</i>	The point is located on a sealed road not wider than 3m. The point is located on a linear feature (<3m) and the surveyor has to apply the "Look to the North/East" rule, i.e. observe the land cover and land use in the northern direction (or eastern direction if the limit or the linear feature follows the N-S direction).
<i>Agricultural road, sealed</i>	The point is located on an agricultural road more than 3 m wide. The road is sealed. LC1 = A22 (non-built-up linear feature) LU1 = U312 (road transport) LU2 = U111 (agriculture)
<i>Unsealed roads and parking lots</i>	If a parcel can be recognized as a road or parking lot, it is an artificial construction so LC1 should reflect the fact, and these areas are classified as LC1 = A2X. If the roads or parking lots is not sealed, it is important to register the secondary land cover, which can be sand (F20), gravel (F40) or even permanent (E20) or spontaneous grass (E30). The first land use of a road is always U312. In case of agricultural unsealed road with permanent grass coverage: LC1 = A22 (non-built-up linear feature) LC2 = E20 LU1 = U312 (road transport) LU2 = U111 (agriculture)
<i>Unsealed roads with only some vegetation</i>	Such roads are also commonly a mix of bare and vegetated patches. To decide on whether to code the LC2 as bare (FXX) or as grass (EXX), use the 10% rule: if the amount of vegetation is negligible (less than 10%) then an A22+FXX is to be selected. It is more important though, that even if we are speaking of a natural area (the road was not strictly "built" but has been formed by the successive passage of vehicles and the soil has been compacted) that it is recognized as an A22 and the use is U312 (road transport). The secondary land cover is additional information.
<i>Greenhouse</i>	The point is located in a greenhouse (an installation of glass, plastic or any material which is translucent but impervious to water) dedicated to agricultural production. The type of

	crop inside the greenhouse will be noted as second land cover. LC1 = A13 LC2 = BXX (code of the crop) LU1 = U111 (agriculture) (If the greenhouses are not dedicated to agricultural production, the land use may be different)
<i>Greenhouses are raised structures</i>	If only the ground is covered by plastic for protecting the plants against frost without a raised structure, this is not considered a greenhouse. The land cover to be assessed is the crop. Therefore only the crop code has to be provided.
<i>Small constructions</i>	If the point falls on a small tool shed (<3m wide) in a garden, use the Look to the North/East rule.
<i>Bridges and other superimposed features</i>	For <i>artificial</i> superimposed features take the feature on top only. This means that if there is a bridge over a dam, only the bridge is classified. However, in case the second land cover is not artificial land (e.g. a bridge over a river, or overlapping a crop, the second land cover is to be taken. A point falling on a bridge over a field of sunflower is to be classified: LC1 = A30 (other artificial area) LC2 = B31 (sunflower) LU1 = U312 (road transport) LU2 = U111 (agriculture)
<i>Special Case: Electric Lines</i>	If a point is falling on an electric line, which is a linear feature it is nevertheless considered an artificial construction. Instead of using the Look to North rule the surveyor shall note LC1 and LU1 as A30 and U319. In addition it is important to register as secondary land cover and use what is situated below the electric line(s). In case of an electric line above an agricultural field: LC1 = A30 LC2 = E20 LU1 = U319 (Electricity, gas and thermal power distribution) LU2 = U111 (agriculture) All Land Cover and Land Use Percentages must be set to 100% in this special case.
<i>Be aware of exceptions: superimposed is different from juxtaposed</i>	There is a difference between the treatment of features that are superimposed and features that are juxtaposed. In case of a point falling on a road that is covered by tree canopy, do NOT take the tree into consideration (unless, of course, we are on a road less than 3m wide, where the Look to the North rule would have to be observed). This is a case of a juxtaposed feature, where the trees are “next to” the road, not “on top” of the road.
<i>Be aware of exceptions: power lines and other "non"</i>	In case of a point falling on a road (or parking lot, or building...) that is crossed by a power line, do not take the power line and give priority to the feature which is sealing the soil.

<i>sealed" artificial features over (e.g.) buildings</i>	Check the cross tables for impossible combinations of LC1 x LC2 LC1 = A22 (or A21 or A1X)
<i>Be aware of exceptions: green rooftops and bridges for animal crossings</i>	 © 2017, René Boinski (@ReneBoinski), Pixabay License Figure 8-18. In these cases, the grass (or other vegetation) is used as part of the construction, so the AXX code prevails (as these are sealed areas). Do not use a second land cover (LC2) in these cases.
<i>Artificial grass</i>	Artificial grass (e.g. football field with artificial grass) is to be classified as A21 or A22, depending on the shape.

8.6.10.2 Cropland (B00)

Cropland means there is crop production on a parcel. The most recent crop must be recorded according to the LUCAS LC 2022 classification. Note that if the surveyor is visiting the point too soon and the crop cannot be recognized, a second visit is needed at a later date.

See below some examples of classification

<i>Cropland</i>	The point is located in a common wheat field. LC1=B11 (wheat) LU1=U111 (agriculture)
<i>Crop not yet emerged in a tilled and sown field but seeds are visible</i>	A point falls in arable land that does not yet have a growing crop at the time of the survey. If seeds are visible, and the crop can be identified, take a crop photo of the seeds on the ground. The point does not need to be revisited when the crop can be identified. If only seeds are visible on the field, mark the percentage of land cover as less than 5% as it is the current status of the field that has to be described not the coverage which is expected at harvest time.
<i>Revisit is needed</i>	A point falls in arable land (e.g. tilled) that does not have a growing crop at the time of the

<i>when it is impossible to identify a crop</i>	survey, or the land is covered with snow. The point can be fallow (U112), and then the existing cover (E30, F40...) has to be taken. But if the point is not fallow (U112) then the point has to be revisited in order to identify the crop.
<i>Usage of Bx1 and Bx2</i>	When photo interpreting a point, if the crop is not recognisable in the orthophoto, the two main classes BX1 "Temporary crops" covering the classes from B11 to B55 and BX2 "Permanent crops" covering the classes from B71 to B84 have to be used. BX1 and BX2 are only allowed in the photointerpretation. In case the crop (e.g. an olive grove (B81) or a vineyard (B82)) can be identified by photointerpretation, the specific codes can be collected by the surveyor.
<i>Crop already harvested but identifiable by residuals</i>	The point falls in a field which is already harvested. The surveyor has to try to recognize the crop from crop residuals in the field and use the code for that crop. Special comment "Harvested field" cannot be used in this case, as the crop is recognisable.
<i>Crop already harvested not identifiable by residuals</i>	The point falls in a field which is already harvested and it is not possible to identify the crop by the residuals. LC1=F40 (bare soil) or E30 (spontaneous vegetation) LU1=U111 (agriculture) Special comments (harvested field) must be used, as the crop cannot be identified.
<i>Inclusion of plant species</i>	The point is located in a jute field. LC1=B35 (other fibre and oleaginous crops) LU1=U111 (agriculture) LC1 plant species = B35h
<i>Mixed crops</i>	A point falls in a field with alternating rows of potatoes and sugar beet LC1=B21 (potatoes) LC2=B22 (sugar beet) LU1=U111 (agriculture)
<i>Mix of two cereals</i>	A point falls in a field with common wheat and rye. LC1=B11 (wheat) LC2=B14 (rye) LU1=U111 (agriculture)
<i>Mixes of more than two cereals</i>	A point falls in a field of mixed cereals with more than 2 species. LC1=B54 (mix of cereals)

	LU1=U111 (agriculture)
<i>Bare areas between the rows of an orchard</i>	A point falls on a field with stripes of bare ground between and around the rows of apple-trees. These areas are considered as belonging to the orchard (the extended window of observation was used). LC1=B71 (apple) LC2=F40 (bare soil) LU1=U111 (agriculture) The second land cover is used here to provide environmentally important information. In case of grass instead of bare land, LC2 could be E30 (spontaneous vegetation).
<i>Grassy areas between the rows of a vineyard</i>	A point falls on a vineyard with stripes of grass between and around the rows. These areas are considered as belonging to the vineyard (the extended window of observation was used). LC1=B82 (vineyards) LC2=E30 (spontaneous vegetation) LU1=U111 (agriculture) The second land cover (E20 or E30) is used here to provide environmentally important information.
<i>Fruit trees in kitchen gardens</i>	A point falls in a kitchen garden with fruit trees from several (3 or more) species. LC1=B7X (according to the dominant species) LU1=U113 (kitchen garden)
<i>Isolated fruit trees</i>	A point falls on the crown of an isolated fruit tree. The zone is considered heterogeneous and the extended window has to be used. The land cover of the surroundings is assessed as grassland. The tree covers between 5 and 10% of the extended window. Note that an isolated fruit tree is not an orchard, but the use can still be agriculture (U111). LC1=E10 (grassland with trees)
<i>Chestnuts, cherries, walnuts, and other forest trees that also bear fruit</i>	It may happen that trees that are known as forest trees can also be grown as an orchard. This is common for chestnuts, walnuts or cherry trees. Note however that the tree density is expected to be different in both cases. In the case of orchards the land cover will be B7X and not CXX and the use will be agriculture (U111) and not forestry (U120).
<i>Abandoned crop areas</i>	A point falls on an abandoned olive grove. If the olive trees are the dominant land cover: LC1=B81 (olive) LU1=U410 (abandoned) Crop areas, which are no longer cultivated, must be classified by observing the present cover.

8.6.10.3 Woodland (C00)

Woodland refers to areas covered by trees with a canopy of at least 10% of the extended window of observation.

When broadleaved and coniferous trees are mixed, the criterion to respect is that 75% or more of the surface is occupied by the canopy of one group. Otherwise the forest is considered as mixed. The canopy is the aerial volume occupied by branches and leaves.

In the past the following errors occurred frequently:

- on one hand, it was common to forget to assess the 10% cover of the canopy in shrub land with trees or grassland with trees on the extended window
- on the other hand the surveyor used the CXX class when the canopy was less than 10% due to ambiguous instructions on the LUCAS 2012 documentation

See below some examples to help clarify the correct classification of woodland areas

<i>Example 1:</i> <i>Young stands and plantations yet to reach a density of trees of 10%</i>	The “FAO forest” definition includes areas of young stands and plantations yet to reach 10% crown coverage. However, in LUCAS land cover assessment these areas should be coded with the existing land cover, and not a potential future land cover. This means that such areas are coded with D10 (shrub land with trees) or E10 (grassland with trees), and in some extreme cases even E20 (grassland without trees) or FXX. For young stands and plantations, even if less than 10% crown cover, land use will always be forestry. Do not forget to assess the height of trees at maturity. A forest type is not added in these areas. LC1=D10 (shrub land with trees) or E10 (grassland with trees) LU1=U120
<i>Example 2:</i> <i>Mixed stands of uneven age</i>	The breakdown between broadleaved and coniferous may be difficult when broadleaved trees have been planted under the coniferous trees they will replace. During the first few years the point will be coded coniferous until the broadleaved trees dominate the canopy.
<i>Example 3:</i> <i>Urban parks</i>	A point is located in a big urban park with mainly broadleaved trees (e.g. a beech grove), used for recreation. The extended window of observation is to be used. The point is classified as C10, therefore a forest type has to be added in the field LC1 plant species. LC1=C10 (broadleaved woodland) LU1=U361 (leisure) LC1 plant species=C10-6 (beech forest)
<i>Example 4:</i> <i>Trees bordering roads or railways.</i>	Roadside (or rail side) trees that can be considered as associated areas to the road or railway will still be classified as woodland (CXX) if the density is above 10%. Accordingly, on what concerns use, they have to be classified as U312 (or 311) as LU1. LC1=CXX (woodland) LU1=U312 (roads and associated areas)
<i>Example 5:</i> <i>Burnt areas</i>	A point is located in a burnt forest area. There are still more than 10% (living) trees in the homogeneous plot within the extended window, so it can be classified as woodland. LC1=CXX (woodland) LU1=U120 (forestry) Special comments=burnt areas Note that burnt areas are coded according to their present cover (e.g. shrub land (DXX) or

	bare land (FXX)). Also land use will be coded according to the actual land use (e.g. U111 (agriculture) or U120 (forestry) or U4XX (abandoned/unused)).
<i>Example 6:</i> <i>Clear cuts</i>	A point is located in a zone where a clear cut was made. The point will be classified according to the actual existing land cover at the moment of the survey. LC1=EXX or DXX or FXX LU1=U120 (forestry) Special comment=Clear cut
<i>Example 7: Fire breaks</i>	A point falls in a fire break completely bare, within a forest area (wider than 3m; otherwise the look to the north rule would have to be used). LC1=F40 (bare soil) LU1=U120 (forestry) Note that following the general rule, codification is made according to the existing land cover (CXX, DXX, EXX or FXX). In areas with frequent forest fires, fire breaks are often found in forests, but they can also be found on agricultural areas or other. Use is to be registered accordingly.
<i>Example 8:</i> <i>Grassland x Woodland</i>	A point falls on a forest area dominated by pine trees (density of trees is above 10% on the extended window of observation, more than 75% of the trees are pines). There is grass under the trees, but according to the rules this is only to be coded as LC2 if there is a different use (e.g. animals are grazing). Use E20 as LC2, since the trees have already been assessed at LC1 LC1=C22 (pine dominated forest) LC2=E20 (as LC2 it provides environmental interest information) LU1=U120 (forestry) LU2=U111 (agriculture) See also example on Cropland x Woodland below
<i>Example 9:</i> <i>Cropland x Woodland</i>	A point falls on a “dehesa” or “montado”, with more than 10% of cork oak trees, but with a crop (e.g. oats) underneath (it is common to find this landscape in southern countries, where cereals, dry pulses, potatoes or even fresh vegetables may grow under the trees). LC1=C10 (broadleaved forest) LC2=B15 (oats) LU1=U120 (forestry) LU2=U111 (agriculture)
<i>Example 10:</i> <i>Hedges and Coppices</i>	A point falls on a hedge composed of broadleaved tree species. The hedge is wider than 3 m (else the look to the North rule would have to be applied), and the tree coverage is above 10% on the homogeneous plot within the extended window. The area has no signs of use. LC1=C10 LU1=U420 Areas which are composed by young or coppiced trees, which can reach more than 5 m of

	height are not to be classified under shrub land (DXX) but as woodland (CXX) if the tree cover is >10%.
<i>Example 11: Wet forests</i>	Regardless of the water content of the soil, wooded areas that comply with the general rule of having a canopy of at least 10% are to be coded as CXX and not as wetland.

8.6.10.4 Shrubland (D00)

Areas dominated (more than 10% of the surface) by shrubs and low woody plants are classified as shrub land (DXX).

Be aware that for shrub land included in woodland areas, or surrounded by trees, it is not always easy to define if a point falls on a separate parcel covered only by shrubs or if the parcel includes the trees. This occurs mainly in the Mediterranean area where the cover is a mix of shrubs and trees. If a point was visited in the past assess carefully whether a change is due to a real evolution of the landscape or simply a case of different interpretation of a situation (and therefore not a real change), an information that should be noted down on the field form .

<i>Hedges</i>	Hedges composed only of shrubby species (e.g. by ornamental plants or dwarf shrub, heathland or Mediterranean shrub species) are considered as shrub land (DX0).
<i>Grazing on shrub land</i>	If the point falls in a shrub area (without trees) that is used for pasture. LC1=D20 (shrubs without trees) LU1=U111 (agriculture) for regular grazing, U420 for very occasional grazing
<i>Shrubland with trees</i>	Shrubland may include sparsely occurring trees within a limit of a tree-crown density of 10% (D10). This can be encountered in two cases: - Permanent vegetation in regions where soil and climate conditions are poor; - Temporary covers evolving from a cultivated state to woodland in regions with better conditions. For the LC class D10, the height of trees (below or above 5m) at the moment of the survey and at maturity as well as the width of the feature (larger or smaller than 20m) have to be assessed in the field form.

8.6.10.5 Grassland (E00)

Land predominantly covered by grass or communities of grass like plants is classified as grassland (EXX). In case of existence of trees and/or shrubs, the density of tree-crown is less than 10% and the density of tree+shrub-crown is less than 18%. Note however that LUCAS also separates from this class the communities of grass corresponding to wetlands that are classified under (HXX).

<i>Pastures</i>	The point falls in a pasture under trees. The density of trees is less than 10%, or else the point would have to be classified as woodland (CXX). LC1=E10 LU1=U111
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<i>Grassland under permanent crops</i>	As mentioned before, grass under permanent crops like fruit trees or olive trees is to be classified as LC2=E20. The usage of EXX as land cover 2 is restricted to E20 or E30, since normally the tree component is assessed at LC1.
<i>Spontaneous herbaceous vegetation</i>	Spontaneous, herb or tall fringe vegetation is coded under E30.
<i>Campsite</i>	The point is located in the lawn of a campsite with less than 10% of trees. LC1=E10 LU1=U341 (holiday camp)

8.6.10.6 Bare land and lichens/moss (F00)

This class includes areas covered by lichens or moss or with hardly any vegetation cover (where at least 90% of the ground or more is bare).

Note that to classify an area as lichens or moss these can cover more than 10% of the ground, but no other vegetation can be present in a density higher than 10%.

<i>Crop not yet emerged</i>	If the survey is being carried out before the planting period of certain crops, a second visit has to be organised for those fields found bare in arable land, if no seeds or young plants could be found which unambiguously refer to the sown crop. Special comment "Crop not recognisable – Tilled and/or sowed (To be revisited)" must be used.
<i>Fallow land</i>	In case of fallow land (U112) that is not sown during the current season, the land cover is to be noted (e.g. F40 Other bare soil, E30 Spontaneous vegetation, etc.) and a comment needs to be provided.
<i>Use of FXX as LC2 in unsealed roads and parking lots</i>	It is common to find roads or parking lots which are not sealed but have clearly compacted soil. If a parcel can be recognized as a road or parking lot, it is an artificial construction so LC1 should reflect the fact, and these areas are classified as LC1 = A2X. In an unsealed area, it is important to register also the secondary land cover, which can be sand (F20), gravel (F40) or even permanent (E20) or spontaneous grass (E30). The first land use of a road is always U312.

8.6.10.7 Water (G00)

This class includes Inland or coastal areas without vegetation and covered by water and flooded surfaces, or likely to be so over a large part of the year.. The average water level of a river defines the delimitation of the inland running water. The definition of lakes refers to permanent water surfaces. Islands in lakes are to be included in the land surface area.

Man-made lakes and lakes regulated at a dam may have water levels shifting very much up and down depending on the need to store or use water e.g. for hydroelectric purposes. The area of such reservoirs should be defined at

the maximum filling. On the other hand specific meteorological conditions can cause a considerable increase of the water level of a lake and cause flooding.

The LUCAS definition takes these fluctuations into account:

- 1) by defining the water areas as "areas without vegetation and covered by water and flooded surfaces or likely to be so for a large part of the year".
- 2) by providing a comment on "currently dry" areas that is to be used in case a water feature is observed at a lower level than usual
- 3) by providing a comment on "currently flooded" for areas exceptionally covered with water.

In the past, the problems identified were mainly related to the correct classification of sandbanks in the river bed and irrigation canals in arable land.

<i>Separating fresh and salt water</i>	Due to the need of mapping LUCAS classes to INSPIRE, a separation of fresh and salty water features was needed. Although normally expected at the coastal areas, salty water lakes can also occur inland in volcanic regions or where evaporation is important.
<i>Rice fields are not GXX</i>	This class excludes rice fields which are classified as cropland (B17).
<i>Stagnating water with vegetation is not GXX</i>	This class also excludes brackish or stagnating water bearing vegetation cover of grass, shrubs or trees. The appropriate classes will either be wetland (HXX), shrub land (DXX) or woodland (CXX).
<i>Varying water levels in sandbanks, margins of artificial lakes and islands within water features.</i>	Temporarily submerged areas and sandbanks are to be assigned in FXX classes. On the other hand, a temporarily exposed sandbank inside the mean bed of the river is to be considered as water (GXX). A similar rule is to be used for margins of artificial lakes (where the water level can vary considerably): if the point falls on the temporarily dry area (normally bare land), code it anyway as water (GXX) and use the relevant comment "temporarily dry". Always use the relevant comment temporarily flooded/temporarily dry to complement the information collected.
<i>Tidal areas</i>	In the rare case a point falls on a tidal area, use the normal high tide mark as a reference to use the "temporarily flooded/temporarily dry" comment.
<i>Non-permanent snow cover is not G50</i>	The areas permanently covered by snow (G50) do not include those under snow for only a certain period of the year. It is therefore necessary to dig a little to find the bare soil or the grass, or to organise a second visit for these points that were found under snow cover during the survey to assess the actual land cover.
<i>Fishing</i>	A distinction may be needed between professional and recreational fishing. In case an area of a river, lake or even a coast is used for professional fishing or

	aquaculture, LU=U130 is to be assessed in these cases. But if an area is just used for recreational fishing, it has to be coded as LU=U361 (Leisure)
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8.6.10.8 Wetland (H00)

Wetlands are areas that fall between land and water. They are areas that are wet for long enough periods that the plants and animals living in or near them are adapted to, and often dependent on, wet conditions for at least part of their life cycle.

Wetlands are defined as land that is inundated on a temporary or permanent basis, with water usually slow moving or stationary and that can be shallow, fresh, brackish or saline.

8.6.11 Examples of classification by land use

In general, land use is easy to define, if signs of use can be seen. It is more difficult when there is no visible proof of the use.

See below some examples.

<i>Agricultural use</i>	An orchard (B7X) is trimmed, traces of work are visible in the fields. In such case there is no ambiguity about the use. LU1=U111 (agriculture)
<i>Why is landscape care and maintenance no longer under agricultural activities (U111)?</i>	Landscape care and maintenance includes the planting, care and maintenance of parks and gardens, greenery (e.g. sport grounds) and plants for protection against noise, wind, erosion, visibility and dazzling. It is seen as a service, not as primary production.
<i>Set-aside</i>	Agricultural land which has been set-aside, will be classified as fallow land (U112).
<i>Can fallow land be ploughed?</i>	In some southern countries fallow land (U112) is ploughed in summer (to kill the weeds). If the surveyor find a ploughed field and his local knowledge allows him to determine that this field is fallow land he has to add the relevant comment (Local Knowledge) else quality control will ask to revisit to determine what is growing on the field.
<i>How to distinguish between fallow and abandoned agricultural land?</i>	In principle abandoned agricultural land is expected to have more woody growth than a fallow land, which has been out of use for a short time. Local knowledge and contact with the land owner may be needed to determine the use.
<i>Forests (also known as LUCAS woodland or CXX) can have many uses</i>	Parts of forest may also be exploited for the leisure of people during weekends (recreation, leisure, sport), or be part of a nature protection area (indicated by signs). It can happen that a woodland area is not used for forestry: a camping site in a wooded area could be classified as LC1=CXX (woodland)

	LU1=U341 (commercial activities)
<i>How to distinguish between professional hunting or sport?</i>	Professional hunting is defined under NACE activity A01.70 as "hunting and trapping on a commercial basis" as opposed to the recreational or sport activity. It also includes "taking of animals dead or alive for food, fur, skin or for use in research, zoos or pets" and "production of fur skins, reptile or bird skins from hunting and trapping activities", "land based catching of sea mammals as walrus or seal". It excludes the same type of activities for sport or recreation (and related service activities) and service activities that promote hunting and trapping.
<i>Is it other primary production (U150) when the hunter pays for the hunting activity?</i>	If there is a company profiting from the hunting or the selling of the hunted meat and this is the main use of the land, then U150 is LU1. Else it is sport (U362) – see also NACE 93.2
<i>Military zones</i>	Military zones are to be included in community services (U350).
<i>Use of grassland in urban areas</i>	For the grassland in urban areas, the assigned land use needs to be in relation to the use of the area. Grassland may belong to a hotel (U341) or to a residential house (U370), or to a museum or botanical garden (U361).
<i>Abandoned areas</i>	An area is considered abandoned when it is not used but traces of their former use are still evident.
<i>Abandoned vs. non abandoned agricultural areas</i>	Agricultural areas can be out of use as part of the field rotation or because of a long-term set aside. This area is to be classified as U112 Fallow land. It might not be possible to distinguish a recently abandoned agricultural area from a fallow land. In case of doubt it is to be classified as U112 Fallow Land.
<i>Abandoned or unused</i> <i>"natural areas"</i>	Forest that has no signs of intervention (no recent tracks, no cuttings) can be abandoned (U410) or a protected area subject to a non-intervention management regime (U420). If a reed-bed does not have any signs of being regularly cut, use will either be unused (U420) or abandoned (U410) (if signs of former use are visible).

8.6.12 Fields: LC crop species (29, 32)

This field allows to register more detailed information on the crops previously assessed:

For crops (LC1=BXX) classified as "other", for nurseries (B83) and for permanent industrial crops (B84); more specifically if:

LCx is B19 or B23 or B35 or B37 or B43 or B53 or B75 or B77 or B83 or B84

the species is to be registered.

The relevant species are listed in the document C3 – Classification under the relevant class.

In case of PI this assessment might not be possible.

8.6.13 Fields: LU land use type (35, 38)

This field is used to further differentiate some activities to allow to compare the information collected to the INSPIRE land use classification.

Industrial activities are further classified according to INSPIRE in:

- Raw industry (R)
- Heavy end industry (H)
- Light end industry (L)

Transport via pipelines is subdivided in:

- Transport of water via pipelines (W)
- Transport of other material via pipelines (O)

The full overview of codes can be found in chapter 9.2.6 Code list for land use types

8.6.14 Fields: Crop cover (%) of LC1 land cover 1 and LC2 land cover 2 (30, 33)

If LC = BXX, the % of crops covering the ground has to be assessed on the whole area of reference (observation window).

8.6.15 Field: Parcel area (ha) (40)

The surveyor needs to roughly estimate the area of the parcel (in ha) to which the point belongs. This has to be done for all points regardless of their land cover. The size is graded according to area classes.

The area calculation takes into account the whole parcel having the same land cover and land use as the observation point, except for wooded land.

PLEASE NOTE: As an exception, for wooded land the total extent of wooded land (C00) is to be assessed:

This means that the total extend of wooded land (C00) is to be assessed regardless of the fact that there might be a part of broadleaved woodland (C10), coniferous woodland (C2X) and or mixed woodland (C3X) and the same use. This because the total C00 area is needed to calculate the FAO Forest area.

Note that this is not the “homogeneous plot” mentioned in chapter 8.6.2, The homogeneous plot.

<i>Evaluating parcel areas in the field</i>	In order to assess an area of 0.1 ha in the field, the surveyor has to have in mind that a circle of 18 m radius (a bit less than the extended window of observation) around the point represents an area of 0.1 ha. In order to assess an area of 0.5 ha in the field, the surveyor has to have in mind that a circle of 40 m radius (twice the extended window of observation) around the point represents an area of 0.5027 ha. A circle of around 56 m radius represents an area of almost 1 ha. In case of quite large parcels (greater than 1 ha), the assessment of the size should be
---	---

	done with the help of the orthophoto.
<i>Parcels cut by linear features less than 3m wide</i>	If land is cut by linear features (fences, ditches, roads, railways, rivers etc.) <u>narrower than 3m</u> , consider all the area with the same LC & LU as homogeneous for assessing the area of the parcel.
<i>“Linear parcels”</i>	For points in water courses, roads, rivers and railways wider than 3 m, mark the area of the parcel always as bigger than 10 ha. For hedges wider than 3m, the orthophoto can be used as base for the area estimation.
<i>Different uses indicate different parcels, except for wooded land</i>	For example, if two vineyards are adjacent to each other, one is in use and the other is abandoned, consider only the one where the point is located for the area estimation.
<i>Parcel area for wooded land</i>	For wooded land the total extent of the wooded land – broadleaved, coniferous and mixed - is assessed. This assessment does not take into account the lower level classification (CXX).

8.7 Section: INSPIRE



Take photos as relevant: INSPIRE Other (multiple photos possible)

8.7.1 Fields: INSPIRE PLCC

Data on INSPIRE Pure Land Cover Components (PLCC) will be collected for all LC classes (based on bird-eye view) **within the whole extended window of observation (20 m radius):**

The following list of INSPIRE classes provides a quick overview, a detailed descriptions of the classes can be found in Annex 9.11 INSPIRE Pure Land Cover Components:

- Arable Land
- Artificial Constructions
- Coniferous forest trees
- Broadleaved forest trees
- Shrubs
- Herbaceous plants
- Lichens and mosses
- Consolidated (bare) surface

- Unconsolidated (bare) surface
- Permanent woody and shrubby crops
- Wetlands and marshes
- Organic deposits (peatland)
- Chemical deposits
- Intertidal flats
- Fresh water course

Unlike what happens in LUCAS classes, where the sum of percentage of combined land cover can be more than 100%, the sum of PLCC must be 100%. Assessment of these percentages is made using the “birds-eye” view.



Source of the orthophoto: ortho_2020 WMS, Open Data WMS and WMTS de l'Administration du cadastre et de la topographie du Grand-Duché du Luxembourg (data.public.lu), Creative Commons Zero (CC0)

Figure 8-19.

For Photo-Interpretation:

Only code what can be clearly identified on the orthophoto.

Depending also on the quality of the orthophotos, it will not be possible to assess the presence of certain classes (e.g. lichens/mosses).

The assessment is to be based only on the available orthophotos and not on local knowledge or other sources of information.

8.7.2 Field: Point in urban area? (80)

In order to assure the mapping to Corine Land Cover it is important to determine if the LUCAS point is located in an urban area or not.

A good way to identify the “urban area” is to use the orthophoto, checking that it corresponds to reality. In general, the boundaries of an “urban area” are easily recognisable on an orthophoto and are marked by, for example, fences or walls around kitchen or allotment gardens, graveyards, residential areas or dwellings.

8.7.3 Field: Unvegetated area (%) (81)

The unvegetated area is assessed in order to support the first steps in the production process of the Copernicus HRL “Imperviousness”. The percentage of the area that is not vegetated is to be assessed on the LUCAS point inside the extended window of observation (20m radius) and is based on birds eye view. **The observation is not restricted to the homogenous plot.**

The degree of unvegetated area is the percentage (0 – 100%) of the area within the extended window that does not show vegetation.

The assessment is based on the in-situ survey and on the orthophoto.



Comment needed:

If the unvegetated area has been assessed using partly PI a comment is needed (only for obs type 1 and 2).

CODE	LABEL
TA01	Unvegetated area partly assessed through PI

Examples:

E.g. bare soil = non-vegetated = 100% “unvegetated area”;

E.g. water = non-vegetated = 100% “unvegetated area”;

8.8 Section: FAO parameters

Parameters collected under this heading are needed to determine:

- points that fall under the definition of “FAO Forest” and “FAO Other Wooded Land”, as specified in “FRA 2015 – Terms and Definitions, 2012” (Forest Resources Assessment Working Paper 180) from FAO (<http://www.fao.org/docrep/017/ap862e/ap862e00.pdf>),
- the height of permanent crops (B70 and B80).



Take photos as relevant: FAO_Other (multiple photos possible)

8.8.1 Fields: Height of trees/crops at survey and at maturity (42, 43)

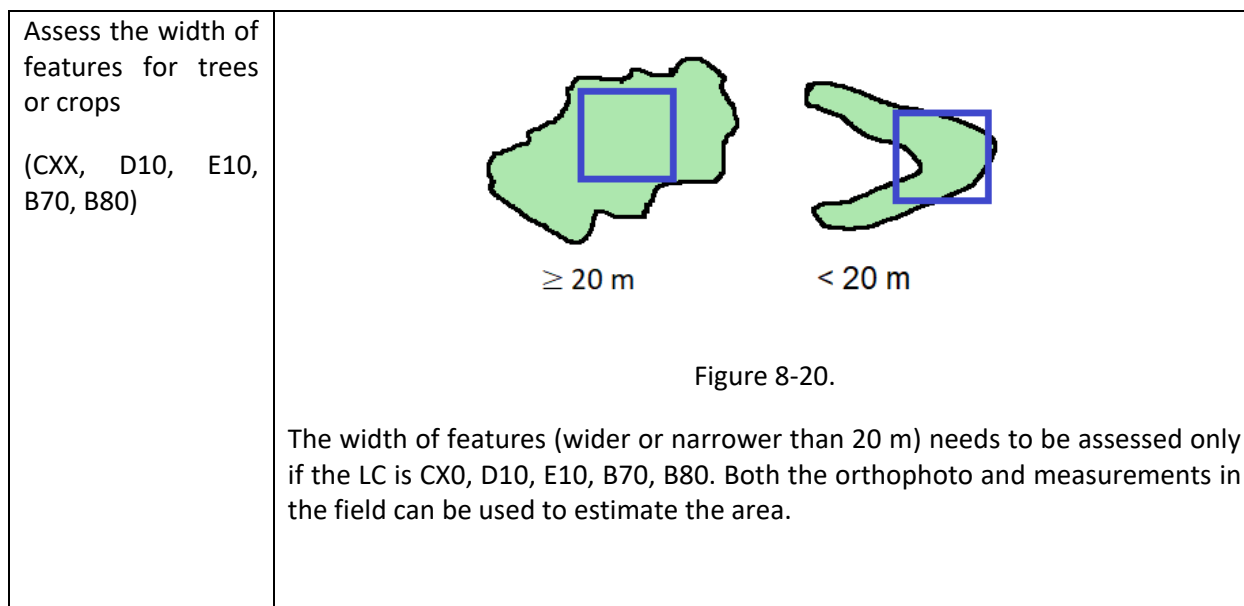
For points with trees in CX0, D10, E10 and for points with crops in B70 or B80 (whatever the size of the parcel), the height of the trees/crops (below or above 5m) at the moment of the survey and at maturity has to be assessed and noted in the field form.

The height to be taken is the height of the highest plants.

8.8.2 Field: Width of feature (44)

For points with a tree cover (CX0, D10, E10) as well as for points with crops (B70 or B80) the width of the feature has to be assessed to provide information on the shape of the assessed parcel. When within the whole parcel (covered by trees or relevant crop) is an area, which covers a quadrant of 20*20 m², ≥ 20 m is to be ticked. If a quadrant of 20*20 m² does not fit completely into the parcel, <20 m is to be ticked.

- **Area covers a quadrant of 20*20 m² = ≥ 20 m** is to be ticked
- **Quadrant of 20*20 m² does not fit completely** into the parcel = **<20 m** is to be ticked



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8.9 Section: Land management

Different land management activities can have an impact – among others - on erosion and are to be registered. This information is to be collected for points with the following land cover: BXX, CXX, DXX, EXX, FXX, HXX, except for crop residues and standing vegetation, which are to be assessed only in case the land cover is BXX.

The reference area is the field, as far as visible from the point reached by the surveyor. There is no need for major walks to assess these parameters.

8.9.1 Field: Stone walls (49)

This field allows information to be registered on the presence of stone walls and their maintenance status land. These are masonry constructions, often built without the use of mortar from local rocks, generally excavated from the ground nearby.

See example photographs in subsequent pages.

As a general guide, well maintained stone walls would act as a barrier to the passage of livestock.

8.9.2 Field: Hedgerows (730)

This field allows information to be registered on the presence of hedgerows. These are lines of closely spaced shrubs and sometimes trees, planted and managed to form a barrier or to mark the boundary of an area (such as between neighbouring properties or roads and fields).

See example photographs in subsequent pages.

8.9.3 Field: Grass margins (51)

This field allows information to be registered on the presence and width of grass margins. These are herbaceous areas (generally 0.5 - 3 m wide) between the main crop and the field boundary> They are deliberately managed to (a) provide habitat for wild flora and fauna, (b) support biodiversity and ecosystem services, (c) protect water quality, and (d), provide weed control.

See example photographs in subsequent pages.

If present, the average width in the field containing the LUCAS point should be recorded.

8.9.4 Field: Grass margins and hedgerows(731)

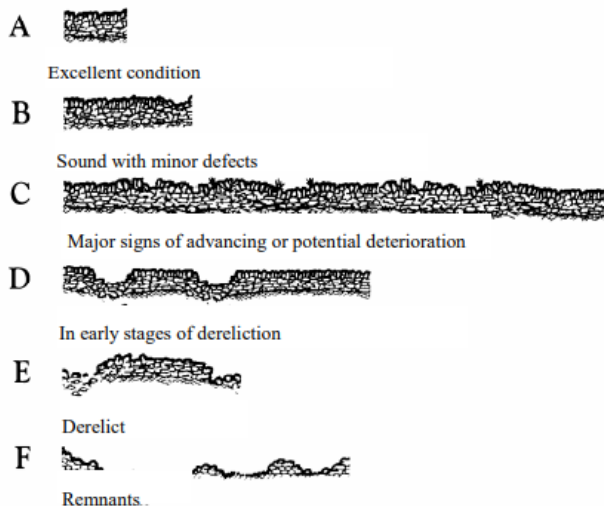
This field allows information to be registered on the presence of both grass margins and hedgerows. See previous points for definitions.

8.9.5 Field: Soil surface cover: standing vegetation and crop residues (732, 733,734, 735)

Theses field are only assessed in BXX classes, from the information collected concerns standing vegetation and crop residues, generally following harvest.

The following pictures illustrate the concepts described.

1 Stone walls – not maintained



© Centre for Ecology and Hydrology, "Condition of dry stone walls (adapted from Countryside Commission leaflet CCP 482)" in Centre for Ecology and Hydrology, *CS Technical Report No.1/07 – Field Mapping Handbook*, November 2008, p. 114 (<https://nora.nerc.ac.uk/id/eprint/5194/1/N005194CR.pdf>)

Figure 8-21.



© 2016, Roger Templeman, CC BY-SA 2.0

Figure 8-22.



© 2007, John S Turner, CC BY-SA 2.0

Figure 8-23.



© 2020, Pijarn Jangsawang Pxhere.com, CC0 1.0 Universal

Figure 8-24.



© 2009, Andy Stott, CC BY-SA 2.0

Figure 8-25.



© 2009, Trevor Littlewood, CC BY-SA 2.0

Figure 8-26.

2 *Stone walls –
well maintained*



© 2007, John S Turner, CC BY-SA 2.0

Figure 8-27.



© 2004, Pollinator, CC BY-SA 3.0

Figure 8-28.



© 2019, Pacopac, CC BY-SA 4.0 Int.

Figure 8-29.



© 2019, Jacqueline Macou (@Jackmac34), Pixabay License

Figure 8-30.



© 2018, Peter K Burian, CC BY-SA 4.0 Int.

Figure 8-31.

3 Grass margins



© 2007, Keith Edkins, CC BY-SA 2.0

Figure 8-32.



© 2011, Michael Trolove, CC BY-SA 2.0

Figure 8-33.



© 2011, Nigel Mykura, CC BY-SA 2.0

Figure 8-34.



© 2013, Michael Trolove, CC BY-SA 2.0

Figure 8-35.



© 2015, Richard Humphrey, CC BY-SA 2.0

Figure 8-36.

4 Hedgerows



© 2005. Caronna (Fotograf: Steffen Heinz).CC BY-SA 3.0

Figure 8-37.



© 2019, Franck Barske (@braskefranck), Pixabay License
Figure 8-38.

**5 Combined
grass margin +
hedgerows**



© 2006, John H Darch, CC BY-SA 2.0

Figure 8-39.



© 2016, Roger Jones, CC BY-SA 2.0

Figure 8-40.

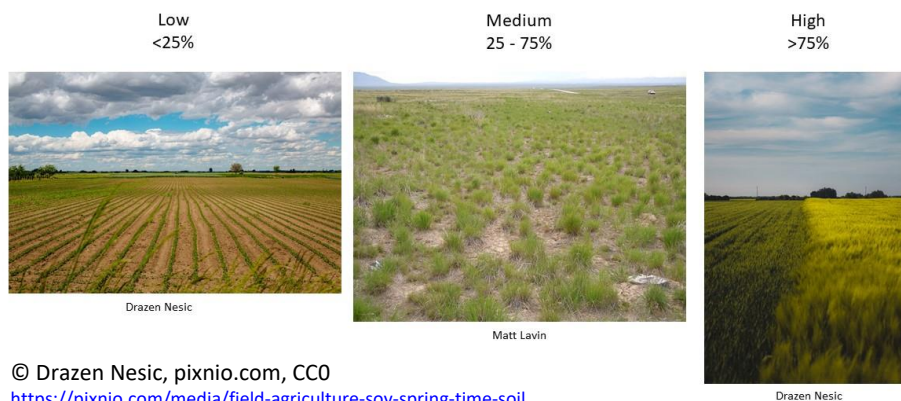


© 2017, Roger Jones, CC BY-SA 2.0

Figure 8-41.

6 Presence of surface cover

Photographs of Standing Vegetation



© Drazen Nestic, pixnio.com, CC0

<https://pixnio.com/media/field-agriculture-soy-spring-time-soil>

<https://pixnio.com/media/wheatfield-plain-field-agriculture-rural>

© 2011, Matt Lavin, flickr.com, CC-BY-SA-2.0

https://www.flickr.com/photos/plant_diversity/5881275315

Figure 8-42.

Photographs of Corn Residue Cover



Adapted from Shelton, David P. and Jasa, Paul J., "G95-1134 Estimating Percent Residue Cover Using the PhotoComparison Method" (1995). Historical Materials from University of Nebraska-Lincoln Extension. 782. <https://digitalcommons.unl.edu/extensionhist/782>

<https://digitalcommons.unl.edu/cgi/viewcontent.cgi?article=1780&context=extensionhist>

Figure 8-43.

8.9.6 Comments and photos on land management



Comments on land management

CODE	LABEL
QA99	Other land management comments



Take photos as relevant: LANDMNG_Other (multiple photos possible)

8.10 Section: Special remarks



Photo needed:

If signs for the presence of any of the following features are present please document with an additional photo "Other" (fields 669 – 695).



Take photos as relevant: REMARKS_Other (multiple photos possible)

8.10.1 Field: Signs of grazing in the field (53)

Information on grazing is to be collected for points with the following land cover: BXX, CXX, DXX, EXX, FXX, HXX. The surveyor has to observe whether the field to which the point belongs shows signs of grazing.

The reference area is the field, or if there are no field borders the area up to a distance of some 500 m. There is no need for major walks, only the area visible from the point reached in the field is to be assessed.

In the past it was assumed that signs of grazing could be found only on U111 "Agriculture". Reality shows that grazing can happen in a variety of different areas. Therefore, the grazing rule has been extended and grazing can be assessed also on other land covers.

This concerns mostly:

U112 + Grazing: Grazing on revegetated fallow land (or crop residues from previous years)

U420/415 + Grazing: Wild grazing in remote semi-natural or abandoned areas

Intensive grazing is expected to happen on agricultural land: U111.

Practically this rule can cause LU changes (U111 -> U420/U415) for some points that must be commented with BP11 ("Different decision than in previous survey/visit: rules changed").

For photo interpretation:

Often grazing can be identified quite well by means of photo interpretation, through the presence of grazing animals visible on the orthophoto. In addition, it is often possible to identify the typical structures for grazed areas, such as drinking and feeding spots for animals, shelter and surrounding fences.

<i>Signs of grazing</i>	Grazed areas are mostly on grassland, with visible signs for grazing management: pastures, where cattle are out at feed and/or infrastructure is seen (fences, stables, drinking troughs) dung and/or cattle trampling can be observed. If there is nothing to be grazed, select "Not relevant".
<i>Check for visible impacts</i>	The impacts of poaching by livestock/trampling (destruction of grass layer, compaction etc.) might be visible. As grazed grassland is more susceptible to erosion than mown areas, various measures might be in place, which are controlling or minimising erosion (fencing off trampled areas, re-seeding bare ground, prevention of bank erosion by livestock etc.).
<i>Take location into account</i>	Grazed areas are also found on hilly or mountainous land, which is difficult to access and to manage (steep slopes), or on poor soils, and might bear grass species of lower nutritional values.
<i>Transhumance areas are covered too</i>	Areas grazed during summer (transhumance) are covered by this land management category as well. Its land cover is mostly marked by grassland with trees or shrub, bare land (if with rock outcrops or covered by lichens only), shrub land or young forestland.
<i>Crop areas can be grazed</i>	Cereal fields might be grazed after harvesting (e.g. grazing of residuals), if the harvest is not expected to be satisfying or if cereal mixtures have been sown as fodder crops.
<i>Forest and shrub areas can be grazed</i>	In some regions, forests are traditionally grazed (permanently or not). This can be observed by the special species composition of the forest, its structure or simply by the presence of cattle and their residues or fences and stables.
<i>Suitable areas might not be grazed</i>	If the land cover is suitable for grazing but no signs of grazing are visible, the surveyor should mark " <u>no signs of grazing</u> ".
<i>Artificial (AXX) and water (GXX) are not grazed</i>	If the LUCAS point has a LC = AXX or GXX, the land management does not apply (not relevant in the field form).

8.10.2 Field: Special status (54)

<i>Check for signs of special status or use local knowledge</i>	The surveyor shall note if the point is in a protected area (meaning inside an area for nature conservation) or a hunting area (e.g. hunting signs, hunting towers). Application of local knowledge may be needed.
<i>Is the special status "hunting" needed when use is U150 Other primary production?</i>	Yes, if it is relevant. Hunting is a part of U150, but U150 also includes activities other than hunting.

Both statuses – hunting and protected area - can exist simultaneously. If none exists, select "no special status".

8.10.3 Field: Special remarks on land cover / land use (55)

The surveyor shall note special remarks concerning land cover/use as follows:

<i>Crop not recognizable - Field harvested (no re-visit)</i> (Land covers not Bx1)	If the field has been harvested during the current season and the crop is not recognisable by residuals then the option “Crop not recognizable - Field harvested (no re-visit)” has to be selected. LC1: F40 (or E30 if spontaneous vegetation is present) LU1: U111 This comment is only possible if the crop cannot be recognized
<i>Crop not recognizable - Tilled and/or sowed</i> (Land Cover is not Bx1)	The remark “Crop not recognizable - Tilled and/or sowed” is selected to inform the Central/Regional office that the point has to be re-visited, because the crop is not yet recognisable, but will be so soon. Use of local knowledge may be needed to identify that a tilled field is to be left fallow.
<i>Clear cut (and windthrows)</i>	Clear cut in forest areas is considered to have forestry use: If most of trees have been cut down uniformly the point should be classified according to the cover present on the ground: LC1: DXX or EXX or FXX LU1: U120  For windthrows please select "clear cut" and add a structured comment BD07 “Windthrow”.
<i>Burnt area</i>	A burnt area can occur in any LC, but signs of fire need to be observed. The point will be classified according to the land cover observed.
<i>Fire break</i>	Firebreaks are manmade gaps in vegetation in order to stop fires. They can occur in cropland, woodland/forests, grassland or shrub land.
<i>Nursery</i>	A distinction needs to be made between Nurseries (trees are to be transplanted) and Re- or Afforestation (trees are planned to stay at the location). Indications for a nursery that is no longer in use, and therefore to be considered as ordinary forest (LU1: U120), is the size of the trees and the absence of signs of management. As trees over approx. 5m can no longer be transplanted, the presence of a majority of trees over 5 m is an indication for abandonment of the nursery. Nursery is used for forest nurseries (wildings): This special remark is used for nurseries having forestry use (i.e. to produce forest trees). Unlike the specialized forest nurseries (B83f) this remark refers to forest nurseries

	normally found inside the forest (to differentiate these areas from plantations). The point will also be classified as: LC1: CXX LU1: U120
<i>Dump site</i>	This special remarks is used in the case the surveyor observes an area, which is used as a dump site. Open dump site LC1: A30
<i>Currently dry river bed/lake</i>	This special remark is used in case the surveyor observes an area, which is part of a riverbed or lake, but is currently dry. Land cover would still be classified as water. Note that the limits of the lake/river are normally visible. LC1: GXX
<i>Currently flooded</i>	This special remarks is used in case the surveyor observes an area, which is currently flooded. A currently flooded sandbank, which normally would be above the mean river level, should still be classified as FXX.
<i>No remark</i>	If there is no remark, check the option "No remark"

8.11Section: Gully Erosion

Gully erosion occurs when running water is channelled across land and washes away soil where it passes. In the following Gullies are classified depending on their permanent or ephemeral (temporary or seasonal) nature. **Permanent gullies** are deep channels that cannot be eliminated by tillage ⁽¹⁾. They can range from 0.5 to 30 m depth on deep soils ⁽²⁾. **Ephemeral gullies** are more shallow (< 0.5 m deep) channels formed in cropland that can be relatively easily obliterated by tillage. They form repeatedly at or near the same location on a yearly basis ⁽³⁾. **Badlands** are heavily gullied landscapes, generally found in drylands. They are characterized by steep sided gullies, steep bottoms and a high drainage density. Badlands often develop on soft or unconsolidated geological materials with low permeability (e.g. marls), which reduces infiltration. Badlands account for some of the highest erosion rates worldwide and are associated with poor soil development and scarce vegetation cover.

Gully erosion in general can cause serious difficulties of land management and loss of productivity, reduction of surface water quality, sedimentation in drainage networks and the release of greenhouse gases.

For all points visited in the field, gully erosion is to be assessed.

8.11.1 Field: Can you see signs of gully erosion? (740)

Are there any signs of gully erosion within the field or within a distance of 500 m from the LUCAS point?

If so, indicated this in the field form. In this case also the further fields with type of erosion are to be filled in (260 - 331). Else they are to be skipped.

The reference area is the field (plot), or if there are no field borders the area up to a distance of some 500 m. There is no need for major walks, only the area visible from the point reached in the field is to be assessed.



Photo needed:

If gully erosion is present, make a photo of the biggest erosion channel(s): GULLY_Biggest

If the signs of erosion are near the point reached please include a scale (e.g. spade, car, ...).

8.11.2 Fields: What type of gully erosion do you observe? (741)

In this instance, we are asking surveyors to notice if there is any evidence of ephemeral gullies, permanent gullies and/or badlands.

Gully erosion occurs when running water is channelled across land and washes away soil where it passes. Gullies are classified depending on their permanent or ephemeral (temporary or seasonal) nature.

Permanent gullies are deep channels that cannot be eliminated by tillage ⁽⁴⁾. They can range from 0.5 to 30 m depth on deep soils ⁽⁵⁾.

Ephemeral gullies are more shallow (< 0.5 m deep) channels formed in cropland that can be relatively easily obliterated by tillage. They form repeatedly at or near the same location on a yearly basis ⁽⁶⁾.

⁽¹⁾ Poesen, J. and Govers, G., 1990. Gully erosion in the loam belt of Belgium: typology and control measures. In: Boardman, J., Foster, I.D.L. and Dearing, J.A. (eds), *Soil erosion on agricultural land*. Wiley, Chichester. pp 513-530.

⁽²⁾ Soil Science Society of America., 2001. *Glossary of Soil Science Terms*, Soil Science Society of America, Madison, WI.

⁽³⁾ Casali, J., López, J. and Giráldez, J., 1999. Ephemeral gully erosion in southern Navarra (Spain). *Catena*, 36(1): 65-84.

⁽⁴⁾ Poesen, J. and Govers, G., 1990. Gully erosion in the loam belt of Belgium: typology and control measures. In: Boardman, J., Foster, I.D.L. and Dearing, J.A. (eds), *Soil erosion on agricultural land*. Wiley, Chichester. pp 513-530.

⁽⁵⁾ Soil Science Society of America., 2001. *Glossary of Soil Science Terms*, Soil Science Society of America, Madison, WI.

⁽⁶⁾ Casali, J., López, J. and Giráldez, J., 1999. Ephemeral gully erosion in southern Navarra (Spain). *Catena*, 36(1): 65-84.

Badlands are heavily gullied landscapes, generally found in drylands. They are characterized by steep sided gullies, steep bottoms and a high drainage density.

Gully erosion is assessed on the point and in the four cardinal directions, up to a distance of 500 m (if visible from the point). Tick the relevant fields if a specific type of erosion is present.

8.11.3 Section: Details of the largest gully (742-745)

Details on the largest gully only need to be assessed for ephemeral or permanent gullies and not for badlands.

The surveyor should estimate the direction, the average length, width and depth of the largest gully in meters.

The details are also assessed in the same field within a distance of 500 m from the LUCAS point (limited to the area visible from the point the surveyor).

8.11.4 Comments and other photos on gully erosion



Comments on gully erosion

CODE	LABEL
RA99	Other erosion comment



Take photos as relevant: GULLY_Other (multiple photos possible)

8.11.5 Examples of gully erosion

Examples of the types of gully erosion are shown below.

1 Ephemeral gullies



Ephemeral gully formed in a bare potato field with ridges and furrows
Note person for scale.

© 2018, Jean Poesen, Neuville en Condroz, Belgium

Figure 8-44.



© 2016, Jean Poesen, Cassel, France
Ephemeral gully formed in a potato field

Figure 8-45.

2 Permanent gullies



© 2011, Jean Poesen, Moldova Province, Romania
Permanent gully that formed in valley bottom

Figure 8-46.



Bank gully formed in agricultural terrace bank
© 2019, Jean Poesen, Rambla Honda, Tabernas, Spain

Figure 8-47.

3 Badlands

Badlands are gullied landscapes with some of the highest erosion rates worldwide. Badlands often develop on soft or unconsolidated geological materials with low permeability (e.g. marls), which reduces infiltration. Badlands are associated with poor soil development and scarce vegetation cover.



© 2015, Jean Poesen, Badlands, Atri, Italy

Figure 8-48.



© 2017, Jean Poesen, Badlands, Puebla de Valles, Spain

Figure 8-49.

8.12 Section: Water management (251 – 254, 748)

Data for water management is collected if a point falls in an area where the use is agriculture or fallow land (LU=U111 or U112) or if the grassland module has been assessed.

Determine the presence of water management in the field (field 251) you are in, as far as visible from your point of observation. If present, determine the type of irrigation (field 252), the source and the delivery system.

<i>Record source and delivery if visible</i>	The source of the water and the delivery from the source (only relevant for irrigation) should be recorded if visible.
<i>Mark the dominant source or delivery</i>	If more than one source or delivery systems are present, mark the dominant one.
<i>It is not always possible to identify a source</i>	Sometimes it will not be possible to see the source of irrigation, as the water is distributed over a long distance.

<i>Identify watering devices in the field</i>	The surveyor should take the supplying devices in the visible surroundings of the point, either on the same field parcel or at the edge of it, or if within walking distance (≤ 100 m), of the neighbouring field parcel if the water device is covering the survey point as well.
---	--

Source (field 253) and delivery systems (field 254) can be found in various combinations. See below some examples

<i>Type: gravity irrigation</i>	Irrigation water is delivered to the field through an irrigation ditch. Water is siphoned from the ditch and allowed to gravity flow down the furrows.
<i>Type: gravity / pressure</i>	Irrigation water is delivered to the field through an irrigation ditch. A side roll sprinkler irrigation system picks up water from the ditch and irrigates the field.
<i>Type: pressure (sprinkle irrigation)</i>	Irrigation water is delivered to the field through a pressure pipeline. The field is irrigated with a centre pivot irrigation system.
<i>Type: pressure: (micro-irrigation)</i>	Irrigation water is delivered to the field through a pressure pipeline. The field is irrigated with a drop-by-drop irrigation system, which is partly buried in the ground.
<i>Source: Well</i>	Examination of surroundings reveals a well house with photovoltaic panels or power lines entering it.
<i>Delivery: canal</i>	Examination of surroundings reveals a canal along the side of the field.

Signs of use of reclaimed water (field 748):

Reclaimed water is often delivered in dedicated pipes, but not only, in some other cases the reclaimed water is blended in ponds or reservoirs. Normally the use of reclaimed water is indicated through panels or by using specially coloured pipes.



Photo needed:

Take photos of features present: „WATERMNG_Other“

If features are present (at least one field has been ticket) then at least one photo is needed (multiple photos possible).

8.13 Specific module: Copernicus

The LUCAS core data can also be an important source of in-situ data for Earth Observation products. The LUCAS Copernicus module was introduced in 2018 and is specifically tailored to meet Earth Observation requirements.

The LUCAS Copernicus in-situ data is used for the Copernicus products, be it for the calibration and the training during the production process, be it for the validation of the products. To allow for a better applicability of LUCAS for Earth Observation in the context of the Copernicus programme, the information on the extent of the land cover from the point into the 4 cardinal directions is particularly relevant. This allows recreating polygons with homogenous land cover (see d'Andrimont et al., 2021). Therefore, it is also of particularly relevant that exif information is included with the photo.

8.13.1 Field: Can you do the Copernicus survey [57]?

The Copernicus data is to be assessed if:

- the point is part of the Copernicus module (Copernicus point, G) (pre-filled),
- if the observation type (field 24) is 1, 2 or 3
- if the point reached by the surveyor is not a linear feature < 3 m width or a small object with a diameter < 3m

If at least one of the above requirements is not satisfied, the Copernicus assessment cannot be done. Select "No" for field 57: "Can you do the Copernicus module?" and skip the remaining questions of this module.



Comment needed:

In case Copernicus cannot be assessed, the surveyor has to add a structured comment (PA04) explaining the reason why it was not possible.

In 2022 survey, the Copernicus module has to be assessed with no minimum extent to the next LC. The 2018 survey rule "the extent of the Copernicus LC is at least 5m in any possible direction" it is not relevant anymore.

8.13.2 Fields: Copernicus LC (58), Extend of Copernicus LC, (in 4 cardinal directions) (60 – 63)

There are 2 key observations of the Copernicus module to be done:

Copernicus land cover:

The land cover of the point reached by the surveyor in the field is registered using the LUCAS LC classification at level 3 (Field 58).

Extent of the Copernicus land cover:

The extent is registered for the 4 cardinal directions (Fields 60 – 63), up to the first land cover change encountered:

- North (covering 90 degrees: north west – north east);
- East (covering 90 degrees: north east – south east);
- South (covering 90 degrees: south east – south west);
- West (covering 90 degrees: south west – north west).

For each sector the visible extent (or spatial dimension) of the Copernicus land cover (LUCAS LC classification level 3) is recorded as the radial distance in meters [fields 60 – 63] from the observation position up to the first land cover change encountered within 50 m. If the land cover extends beyond 50 m, a value of "51" is to be inserted.

The assessment is to be based on the in-situ observation assisted by the orthophoto.



Photo needed:

A photo of the point, where the surveyor stands when taking the landscape photos, needs to be taken: COPERNICUS_LC.

DO NOT indicate the point with an object. Step back some 5m, possibly to the south of the point, and make a photo of the point (not a landscape photo). The photo should show as much as possible the land cover on the point.

Especially for this photo, the surveyor should make sure to include the exif metadata information with the photo (location, azimuthal direction of the photo, etc.).

SPECIAL CASE I

If in one or more sectors a feature like a small hedge (<3m wide) or the terrain completely hides the landscape behind, the assessment is done as usual (based on the in-situ assessment, supported by the orthophoto), but a comment needs to be added:



Comment needed:

- Comment code: PA01 "LC change hidden by a feature <3 m or the terrain"
- Comment detail: Free text comment: "XX YYm; xx yym" (multiple entries to be separated by semicolons)
Where XX = relevant direction (N, NE, E, SE, S, SW, W, NW), YY = distance from standing point.

In case that there are several land cover features not visible in different sectors, the comment shall look for example: "NE 10m A1; SW 20m B1; NW 20m C2".

The order of registration is based on the order of the directions N, NE, W, SE, S, SW, W, NW.

CODE	LABEL
PA01	Other Copernicus comment

SPECIAL CASE II

If there is a LC change (>3m width) within 50 m not located in the field of view of the camera (because the angle covered by the photo is <90 degrees), the assessment is done as usual, but a comment is needed:



Comment needed:

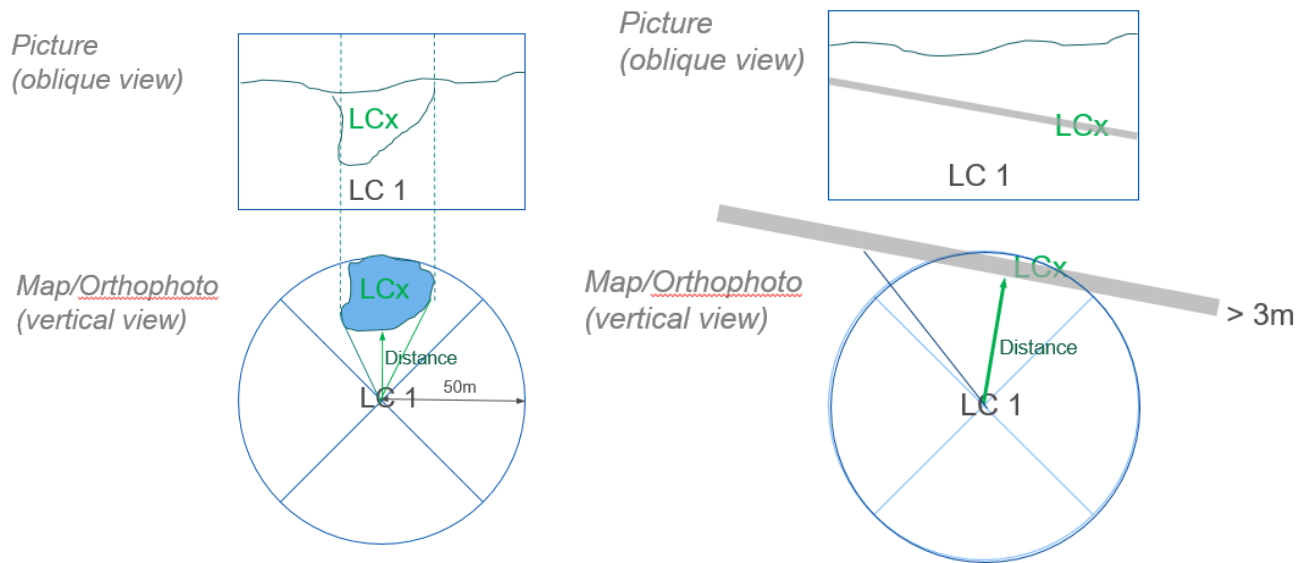
- Comment code: PA02 "LC change existing but not located within the area covered by the photo"
- Comment detail: Free text comment: "XX YYm; xx yym" (multiple entries to be separated by semicolons)

Where XX = relevant direction (N, NE, E, SE, S, SW, W, NW), YY = distance from standing point.

CODE	LABEL
PA02	Other Copernicus comment

In case that there are several land cover features not visible in different sectors, the comment shall look for example: "NE 10m A1; SW 20m B1; NW 20m C2".

The order of registration is based on the order of the directions N, NE, W, SE, S, SW, W, NW.



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Figure 8-50.



Any other comment on Copernicus assessment:

E.g. If there is a complex landscape but you can do the assessment, add a structured comment explaining the situation.

Add any other comment as needed.

CODE	LABEL
PA99	Other Copernicus comment



Take other photos as relevant: COPERNICUS_Other (multiple photos possible)

8.13.3 Some examples

Example 1: Copernicus point is inside a plot with E20



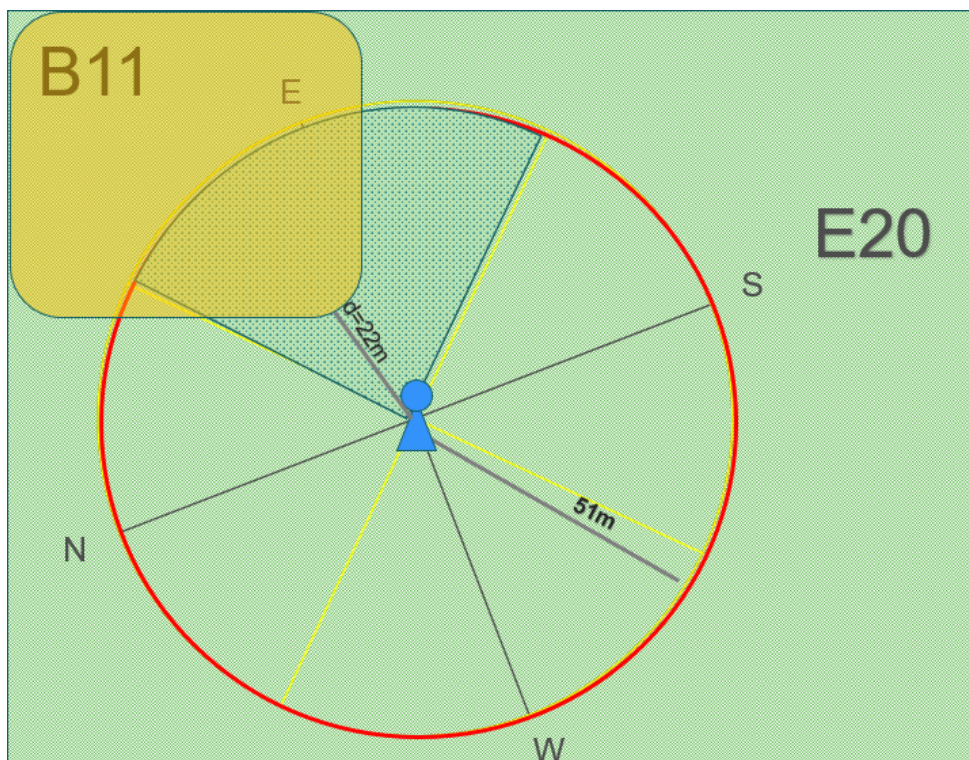
© European Union, CC BY 4.0

Figure 8-51.

There is no LC change within the 50 meters within any of the sectors/cardinal directions.

	Copernicus LC: E20
	Extension (d) of Copernicus LC in cardinal direction in m
N	51
E	51
S	51
W	51

Example 2: LC change within 50m from the position of observation, distance and LUCAS LC level 3 code of the LC in the relevant sector has to be recorded.



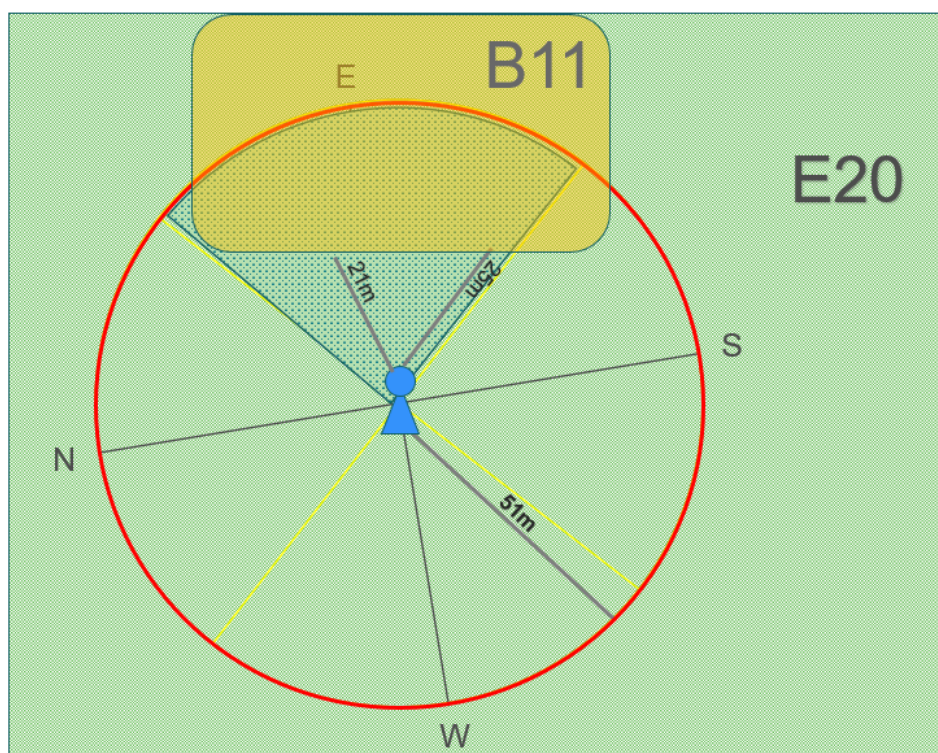
© European Union, CC BY 4.0

Figure 8-52.

Identify the distance (d) towards the next different LC feature within the 50 m in each sector, in this case 25m towards B11 in the sector to the East and 40m in the sector to the North.

	Copernicus LC: E20
	Extension (d) of Copernicus LC in cardinal directions in m
N	40
E	22
S	51
W	51

Example 3: Another example with LC change within 50 meters from the position of observation, where the distance has to be recorded.



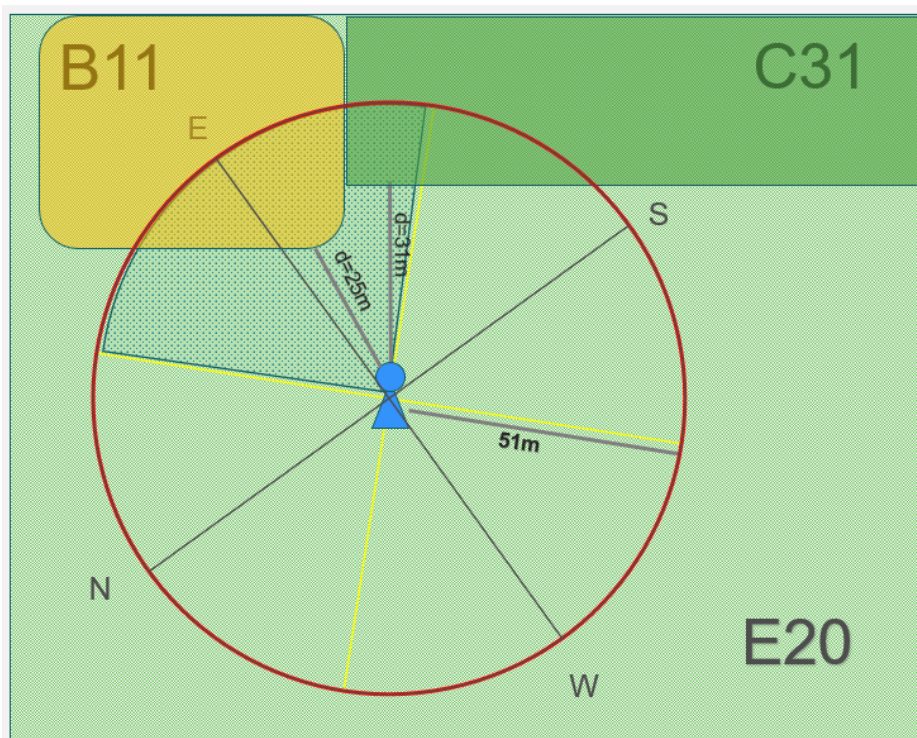
© European Union, CC BY 4.0

Figure 8-53.

In this case the extent from the Copernicus point towards the B11 in the East and South sector is recorded in each sector.

	Copernicus LC: E20
	Extension (d) of Copernicus LC in cardinal directions in m
N	51
E	21
S	25
W	51

Example 4: Two different LC changes within one cardinal sector

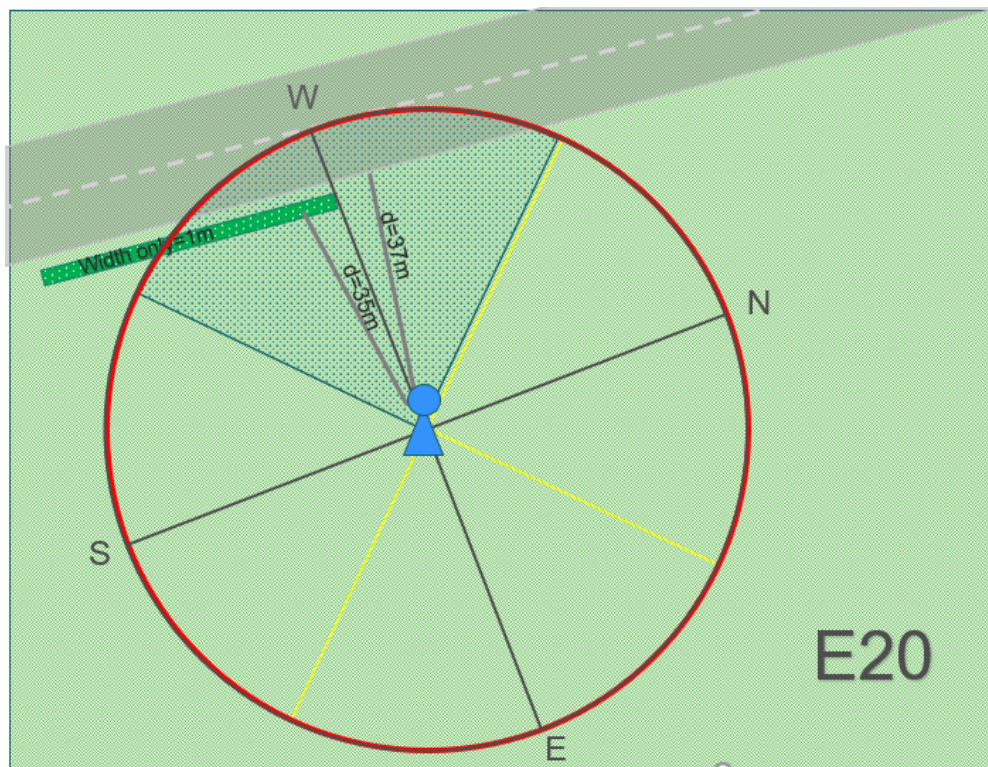


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Figure 8-54.

The closest LC change in the cardinal sector has to be registered. In this example only the extension to B11 is considered in the East sector, the C31 is ignored in this section. In the South sector the extension to C31 is recorded.

	Copernicus LC: E20
	Extension (d) of Copernicus LC in cardinal directions in m
N	51
E	25
S	31
W	51



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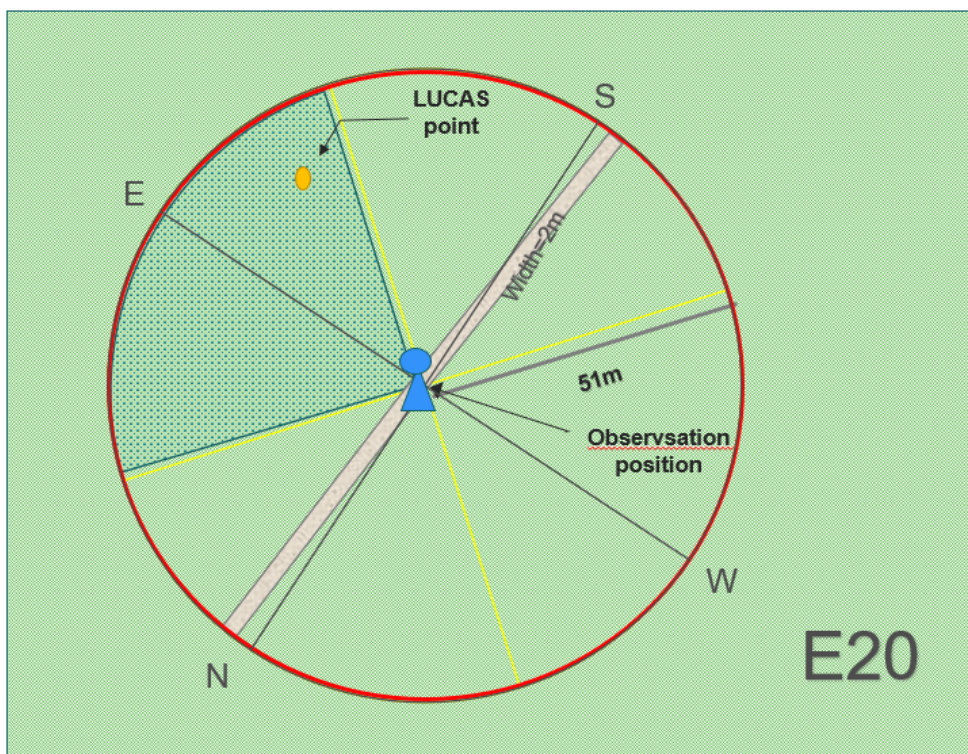
Example 5: Land cover change is more important than a small linear feature

Figure 8-55.

Linear features with width < 3m are not considered a land cover change. A land cover change is visible next to it, therefore the extension up to the next visible land cover is recorded.

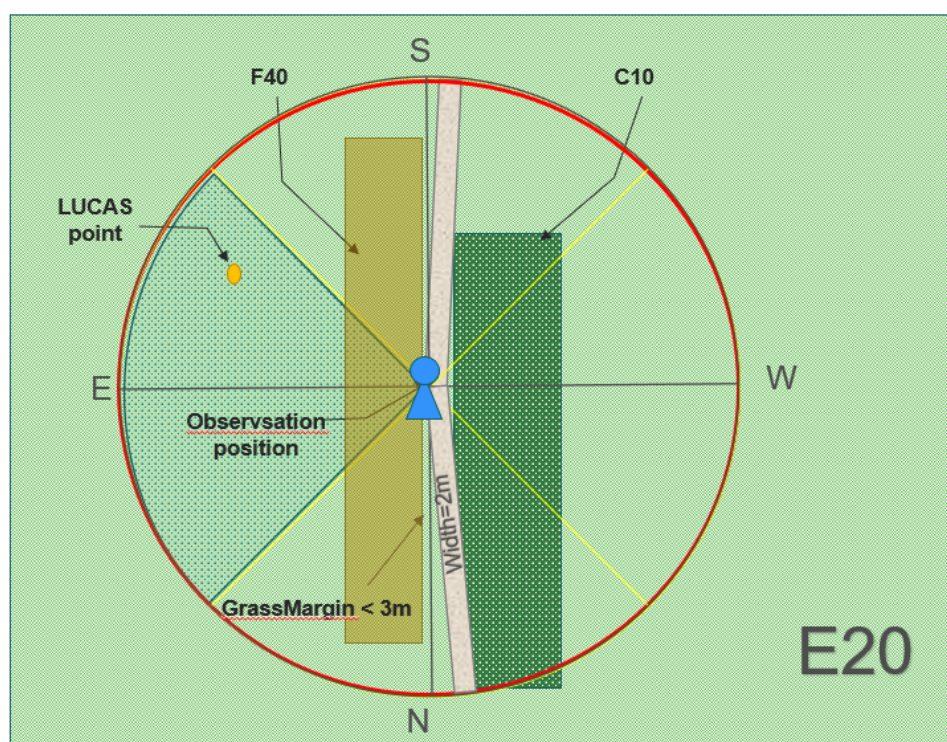
	Copernicus LC: E20
	Extension (d) of Copernicus LC in cardinal directions in m
N	51
E	51
S	51
W	37

Example 6: Point of observation is on a border or feature with less than 3m width



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Figure 8-56.



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Figure 8-57.

In such complex cases, the Copernicus survey module is not applicable.

8.14Section: Trees with secondary crops, meadow or pasture (92)

For points with trees (permanent crops included) (CXX, BX2, E10, D10) the interest here is to assess if there is agro-forestry. To this end evaluate the homogenous plot within the extended window to assess if one of the 2 habitat complexes described below is present (92).

Agro-forestry: "A land-use system that involves deliberate retention, introduction, or mixture of trees or other woody perennials in crop and animal production systems to take advantage of economic or ecological interactions among the components" (Dictionary of Forestry, Helms, 1998, Society of American Foresters, quoted from Vol 4, Forest land, 2006 IPCC Guidelines for national GHG inventories).

Therefore the presence of two specific habitat complexes is to be assessed here:

- Cultivated trees with secondary crops, meadow or pasture
- Forest trees with grazed grassland, heathland and/or woodland flora

If none of them is present select NR.

Strictly speaking X06 includes combinations that are not agro-forestry (when the trees are not forest trees). However both types of trees are of interest and the land cover information of the point allows to separate forest trees from non-forest trees. So no further information needs to be collected here.

Cultivated trees with secondary crops, meadow or pasture (X06):

Crops, meadows or pastures developed under orchards or other cultivated tree plantations. The component habitat types may include elements of I1, E2.6 and FB. Strictly speaking X06 includes combinations that are not agro-forestry (when the trees are not forest trees).

Source: Revision of habitat descriptions originating from Davies, Moss & Hill, M.O. (2004)

Dehesas:

Dehesas are to be assigned X06, as the trees are cultivated. So even if it is forest species, as they are cultivated the habitat complex is X06.

Forest trees with grazed grassland, heathland and/or woodland flora (X09):

Typically this structure consists of large, open-grown or high forest trees (often pollards) at various densities, in a matrix of grazed grassland, heathland and/or woodland floras. This habitat is most common in southern Britain, but scattered examples occur throughout the UK. Outgrown wood-pasture and mature high forest remnants occur in northern and central Europe.

Source: EUNIS habitat classification descriptions from Hill, M.O., Moss, D. & Davies, C.E. (2004)

Not relevant (N.R.):

This is selected when clearly none of the two agro-forestry uses mentioned before (X06 and X09) can be detected.



If the use of Agro-Forestry is not clearly visible on the LUCAS point and landscape photos, the surveyor shall use a comment to explain the reasons for detecting signs of this use at the LUCAS point.

The same applies vice versa:

In case the LUCAS point and landscape photos could lead to the conclusion that Agro-Forestry is taking place at the point but the surveyor is sure of the opposite then he has to give an explanation by using this comment and a free text.

If you would like to provide additional information add a structured comment.

In all cases the comment code is:

CODE	LABEL
HA99	Other Trees with secondary crops, meadow or pasture comment



Take photos as relevant: TREEWITH_Other (multiple photos possible)

8.15 Section: Extended Grassland

Grassland is a key European land cover type and natural capital that significantly contributes to biodiversity and ecosystems conditions with socio-economic importance for the European citizens. In opposition to this, it came under enormous pressure and is among Europe's most threatened ecosystems (Owen, 2008). In this respect major European policies, such as the Habitats and Birds directives, the EU Biodiversity Strategy 2020, the 7th Environmental Action Programme, or others have been formulated with objectives, targets and actions that are directly relevant to the conservation and wise use of grasslands.

The LUCAS extended grassland component is aiming to contribute more detailed qualitative information on European grassland in order to monitor effectiveness of protective measures and to support and strengthen relevant policies. This module is aimed to bridge between the traditional LUCAS core grassland information and the specific LUCAS grassland module.

8.15.1 Field: Assessment possible? (752)

If the point is part of the extended grassland module [extended grassland point – YES] and has been reached (same field, less than 100 m distance = observation type = 1) the parameters of this section have to be observed.

“Can you do the extended grassland point” = yes.

The assessment will be done on the extended window within the homogenous plot.

If these conditions are not fulfilled, the extended grassland survey is NOT to be done and “Can you do the extended grassland point” = no.



Comment needed:

CODE	LABEL
GE01	Extended grassland not possible

8.15.2 Field: Grassland composition (753)

In the extended grassland assessment in addition to the primary LC class (E10, E20, E30), information on the dominating components is to be collected.

Grassland is generally formed by two major components. One component is made of graminoids (also called grass-like plants or herbaceous plants), which includes grasses, sedges, rushes and reeds. Graminoids can be recognized through their parallel leaf "nerve" imprints (those are only grass species, orchids and palms).

The other grassland component are the herbaceous forbs, mostly plants that will be flowering sometime during the year. These plants have net-structured leaf patterns in common.

Grassland layers are generally formed by a composition of both components, however, in some grasslands either grasses/graminoids or forbs /herbaceous plants dominate.

The assessment will be done on the extended window within the homogenous plot.

LUCAS class (extended window + homogenous plot)		Ex1	Ex2	Ex3
E10	Grassland with sparse tree/shrub cover	Graminoids	Forbs	Mixed: Graminoids / forbs
E20	Grassland without tree/shrub cover	(grass-like)* Dominated > 75%	(herbacious)** Dominated > 75%	(grass/forbs) Comprising both >25 %
E30	Spontaneously re- vegetated surfaces	☐	☐	☐

All the following 8.15 points need not to be done, if the point is part of the grassland module (→ then skip the following and go directly to section 8.16)

8.15.3 Fields: Vegetation layers

8.15.3.1 Field: Grass-Forb ratio (754)

To be done only if the point is NOT part of the Grassland module.

Within the herb layer, the ratio between the coverage of graminoids and forbs should be estimated, so that the sum of these two elements is 100%. The aim is to determine which element dominates, or if the coverage is approximately equal.

- graminoid layer = all graminoid species including grasses, sedges, rushes and reeds;
- forb layer: all forb species (i.e. non-woody plants that are not grasses, sedges, rushes or reeds), as well as fern species;

Thus for a pure graminoid vegetation the ratio is 100:0;

For a pure forb vegetation (e.g. stinging nettles, also pure fern) the ratio is 0:100.

For most meadows and pastures there will be a mix of e.g. 90:10, 50:50, 40:60 or 20:80. The sum is always 100, also if the total coverage of this layer is only 50 % (and 50 % of bare layer).

8.15.3.2 Layer components (755-757)

To be done only if the point is NOT part of the Grassland module, otherwise it will be done within the grassland module. Grassland types often are characterised by different types of vegetation height and density or structural pattern. Indicate the importance of the main types of vegetation.

The assessment will be done on the extended window within the homogenous plot.

Herbaceous layer + Bare layer + Woody layer = 100%

xx	Herbaceous layer* (%) : 0 ☐ 0 % 1 ☐ >0 - 3 % 2 ☐ >3 - 8 % 3 ☐ >8 - 15 % 4 ☐ >15 - 25 % 5 ☐ >25 - 40 % 6 ☐ >40 - 60 % 7 ☐ >60 - 80 % 8 ☐ >80 - 100 % *herbs and graminoids, also includes mosses, lichens and mushrooms),	144	Bare layer** (%) : 0 ☐ 0 % 1 ☐ 0 - < 3 % 2 ☐ 3 - < 8 % 3 ☐ 8 - < 15 % 4 ☐ 15 - < 25 % 5 ☐ 25 - < 40 % 6 ☐ 40 - < 60 % 7 ☐ 60 - < 80 % 8 ☐ 80 - 100 % **pure soil, sand, stones, rocks	145	Woody layer*** (%) : (only features that touch the ground) 0 ☐ 0 % 1 ☐ 0 - < 3 % 2 ☐ 3 - < 8 % 3 ☐ 8 - < 15 % 4 ☐ 15 - < 25 % 5 ☐ 25 - < 40 % 6 ☐ 40 - < 60 % 7 ☐ 60 - < 80 % 8 ☐ 80 - 100 % *** features that touch the ground and small bushes, trees, dwarf shrub etc. ≤ 1.0m height)
----	--	-----	---	-----	--

The previous layer components refer to the ground layer (≤ 1.0m height) – the following woody layer components are assessed from bird's eye view and on the extended window; the sum of all components can be maximum 100%.

	Woody layer components (crown coverage) Woody layer from bird's eye perspective (extended window + homogenous plot) (Sum of percentages max. 100)	Percentage (Total ≤ 100) 0 %, >0 % - 3 %, >3 - 8 %, >8 - 15 %, >15 - 25 %, >25 - 40 %, >40 - 60 %, >60 - 80 %, >80 - 100 %
758	Orchard	
759	Old trees	
760	Shrub	
761	Dwarf shrub	
762	Other wooded	

8.15.4 Field: Grassland type (765)

To be done only if the point is NOT part of the Grassland module.

Assess more detailed grassland types according to the nomenclature of the “European Monitoring of Biodiversity in Agricultural Landscapes” (EMBAL) project.

The assessment will be done on the extended window within the homogenous plot.

For explanation see section 8.16.5.

☐ eC11 – Very intensive meadow
☐ eC12 – Intensive meadow
☐ eC13- Medium intensive meadow
☐ eC14 – Extensive meadow
☐ eC15 – Very extensive meadow
☐ eC16 - Meadow or pasture (or both uses)
☐ eC21 - Intensive pasture
☐ eC22 - Medium intensive pasture
☐ eC23 - Extensive pasture
☐ eC24 - Only occasionally grazed (very extensive)
☐ eC25 - Pastured woodland
☐ eC31 - Ruderal / fallow grassland
☐ eC41 - Amenity grassland
☐ eC51 -Unclear / Other kinds of grassland

8.15.5 Field: Growth status (766)

To be done only if the point is NOT part of the Grassland module.

The current growth status of the plots at the time of recording is an important factor in the evaluation of the grassland productivity, an aspect that is also monitored through earth observation.

The growth status assessed in the field can be used a ground reference data for this type of earth observation activities.

Growth status is assessed with regard to mowing or grazing events in three major classes.

☐ Grassland not yet mown or pastured
☐ Grassland mown or pastured a few days ago - (Cut /pastured grass / herbs, either lying on the ground or having been removed from plot, and vegetation ~ <20 cm – no inflorescences or blossoms)
☐ Grassland in growth after mowing / grazing – (Vegetation > 20 cm)

The assessment will be done within the extended window and the homogenous plot.

8.15.6 Field : EUNIS Grassland habitat type (767)

To be done only if the point is NOT part of the Grassland module.

EUNIS is an EU-wide habitat classification system. Please indicate the level 2 EUNIS habitat type according to the habitat identification key given in Annex 9.8 (98).

The most important habitat types for the grassland survey are E1 – E7, for their identification and description see Annex 9.8.

- E1 Dry grasslands
- E2 Mesic grasslands
- E3 Seasonally wet and wet grasslands
- E4 Alpine and subalpine grasslands
- E5 Woodland fringes and clearings and tall forb stands
- E6 Inland salt steppes
- E7 Sparsely wooded grasslands

The assessment will be done on the extended window within the homogenous plot.

There are, however, other EUNIS types and some LUCAS points may lie e.g. in forest (e.g. wooded meadows), fen or heathland. In this case LUCAS will not classify them as grassland but as C = Woodland, D = Shrubland or H = Wetland.

Overview on possible EUNIS habitat types

D1 - Raised and blanket bogs
D2 - Valley mires, poor fens and transition mires
D3 - Aapa, palsa and polygon mires
D4 - Base-rich fens and calcareous spring mires
D5 - Sedge and reedbeds, normally without free-standing water
D6 - Inland saline and brackish marshes and reedbeds
E1 - Dry grasslands
E2 - Mesic grasslands
E3 - Seasonally wet and wet grasslands
E4 - Alpine and subalpine grasslands
E5 - Woodland fringes and clearings and tall forb stands
E6 - Inland salt steppes
E7 - Sparsely wooded grasslands: E7.1, E7.2, E7.3
F3 - Temperate and mediterranean-montane scrub
F4 - Temperate shrub heathland
F5 - Maquis, arborescent matorral and thermo-Mediterranean brushes
F6 - Garrigue
G1 - Broadleaved deciduous woodland
G2 - Broadleaved evergreen woodland
G3 - Coniferous woodland
G4 - Mixed deciduous and coniferous woodland
G5 - Lines of trees, small anthropogenic woodlands, recently felled woodland, early-stage woodland and coppice
I1 - Arable land and market gardens: I1.5 Bare tilled, fallow or recently abandoned arable land
XX - Other
Not Relevant

For the correct classification please use the habitat identification key in this annex 9.9 EUNIS Habitat Classification.



Comment:

A specific comment can be added

CODE	LABEL
GE02	Other extended grassland EUNIS comment

8.15.7 Photos and other extended grassland comments



Take photos as relevant: EXTGRASS_Other (multiple photos possible)

For any other comment on the extended grassland assessment, use the following code:



Comment needed:

CODE	LABEL
GE99	Extended grassland other comment

.

8.16 Specific module: Grassland

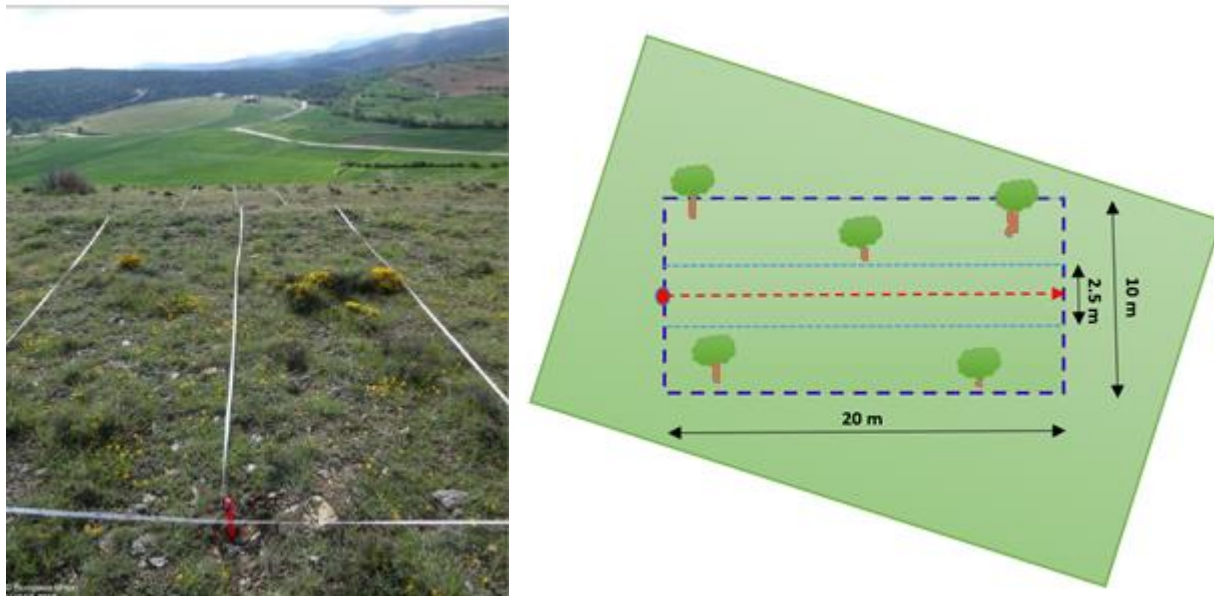
8.16.1 Areas of observation

There are three areas of observation:

Transect: a rectangle of 20 m by 2.5 m (50 m²), starting from the LUCAS point and proceeding towards the east.

Enlarged transect: expanding the area of the transect to a width of 10 m (20 m by 10 m).

Parcel: area of homogeneous land use, such as a fenced pasture. In the case of large or inhomogeneous parcels only the closer more or less homogeneous land is meant.



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Figure 8-58. Transect methodology: The core transect (20m * 2.5m) is marked on the photo; on the graphic can be also seen the enlarged transect (20m * 10m) and the parcel (green). Source of the figure: SUTCLIFFE, L.M.E., SCHRAML, A., EISELT, B., OPPERMANN, R. (2019): The LUCAS Grassland Module Pilot – qualitative monitoring of grassland in Europe. *Polarctic Grasslands* 40: 27-31.

8.16.2 Fields: grass point and region (H, I), survey possible: Can you do the grassland survey? (96)

Points that are part of the grassland survey have been selected based on the land cover information available in previous LUCAS surveys and in the LUCAS master. To qualify for the grassland sample a points needs to have at least 30% of grass-herb -moss/lichens (except for mosses/lichens on rocks) cover. This information is not available for all points. A modelling approach allows to estimate the grass cover for points where this information is not available.

If the point has been selected for the grassland sample fields H and I of the form will be prefilled as follows:

- Point is part of grassland module: Yes
- Grassland region of the point: Prefilled with relevant value

If the point has been selected for the grassland module and the following conditions are fulfilled in the field then the grassland survey shall be done:

- The point can be reached
- There is at least 30% grass-herb cover (i.e. non-woody vegetation)

If the above conditions are NOT fulfilled, then the grassland survey is NOT to be done and "No" has to be selected in the relevant field (96) "Grassland survey possible?". The remaining chapters of the grassland assessment will be skipped.



Comment needed:

CODE	LABEL
GN02	Grassland assesement not possible

8.16.3 Section: Grassland site (enlarged transect)

8.16.3.1 Field: Site orientation (100)

Orientation can be either "Flat" with no slope or indicate the direction of the slope: N = North, E = East, S = South, W = West; further differentiation: NE (Northeast), SE Southeast), SW (Southwest), NW (Northwest);

8.16.3.2 Field: Site slope (101)

The site slope is recorded with 8 classes from 0 (flat) to 7 (steep slope) according to the following classes, measured in %. The slope measured in % can be assessed by looking at the difference in height on a distance of 100 m. For example a slope of 8 % means that on 100 m in a specific direction, I lose or gain 8 m in height.

Please note:

A slope of 3% corresponds to 1.72 degrees, an inclination of 8 % corresponds to 4.5 °, 15 % = 8.5 °, 25 % = 14 °, 40 % = 22 °.

45 degrees corresponds to 100 % inclination!

0 = flat = 0%	0°
1 = > 0 – 3%	0 – 1.7°
2 = > 3 – 6%	1.7 – 3.5°
3 = > 6 -10%	3.5 – 5.5°
4 = >10 -15%	5.5 – 8.5°
5 = >15 -25%	8.5 – 14°
6 = >25 -40%	14 – 22°
7 = > 40%	>22°

8.16.3.3 Field: Site moisture (102)

There are six categories to judge the moisture of a site; please note that this refers to the **moisture of the soil** and not the moisture through current rainfall or the moisture of the vegetation.

Wet: Permanently wet through ground- or surface water or numerous floods per year with the water partly remaining on the surface. Reeds, sedges or rushes often dominate.

Moist: The grassland is frequently wet or moist throughout the year; reeds, sedges or rushes may occur frequently.

Mesic: The site has a regular water supply but it does not suffer heavily from flooding or permanently high ground water. Reeds, sedges, rushes do not dominate.

- Semi-dry: The site is partly dry during the growing season. This often occurs on south-facing slopes and free-draining calcareous or rocky soils.
- Dry: The site is permanently dry or, depending on the biogeographical region, in some seasons of the year (e.g. summer in the Mediterranean region).
- Mixed: There are also sites where both dryness and moisture occur. For example, some south-facing slopes (which tend to be dry) may contain soil layers that frequently carry water, which can be accessed by some plants but not others; thus, indicators of wet conditions (e.g. rushes like *Juncus effusus*) may occur alongside indicators of dry conditions (e.g. *Carlina acaulis*).



Figure 8-59. a and b: The moisture of a site is an important characteristic; good indicators are for example the *Juncus* species (photo of Swabian Jura, 05.05.2016).



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Figure 8-60. Here three types of moisture can be seen: in the middle the flooded plain comprise wet sites (throughout a large part of the year), in the plain but not under water are the moist sites and in the foreground at the edge of the plain there are mesic sites.



Figure 8-61. a, b, c, d, e and f: Site moisture – Examples (© Rainer Oppermann, 2016)

8.16.4 Field: Grassland fertilisation (106) (enlarged transect / field)

Fertilisation is not always easy to see. However, in some cases it is clear that there is fertilisation because you see for example the active spreading fertiliser or it lying on the surface. Please tick the boxes as follows:

- **no**: no fertilisation can be seen and the grassland seems to be unfertilised (sparse growth, grass may be yellowish or olive colour and not dark green;
- **probably**: there are no direct signs of fertilisation (no slurry, no injection tracks, no solid manure, no mineral fertiliser visible) but the grass is lush and dark green and the sward consists only of few species – thus it seems probable that the grassland is fertilised even if you don't see any direct signs of fertilisation.
- **sure**: you see direct signs of fertilisation (slurry, injection tracks for slurry, solid manure, obvious pasture dung (not sparse dung signs of occasional grazing of wild goats or other animals), or mineral fertiliser visible)
- **unclear**: there are neither positive signs nor negative signs of grassland fertilisation.

8.16.5 Field: Grassland type (768) (enlarged transect / field)

There are three main types of grasslands to be distinguished: meadows, pastures, and other grassland types. Some main types can be further distinguished based on the intensity of use. Select the most appropriate option.

Meadows:

Meadows are grasslands that are cut one or more times per year for harvesting hay, silage or biomass (e.g. for biogas). There is a very broad range of meadow types and uses. For example, in some meadows only the first spring growth is cut after which it is grazed, whilst others are mown only in autumn for harvesting litter (animal bedding) or are mown only every second year.

eC11	Very intensive meadow: mostly grass-dominated (few forbs), intense green colour
eC12	Intensive meadow: dense growth of grass and forbs but also with some flowers
eC13	Medium intensive to extensive meadow: a balance of grasses and forbs, often moderately dense flowers, less intensive green colour, less strong growth than eC11 and eC12
eC14	Extensive meadow: as eC13 but with less strong growth, but stronger growth than eC15
eC15	Very extensive meadow: delayed growth in spring, little biomass, often paler green colour
eC16	Meadow or Pasture: these grasslands are used for a first cut as meadow, but afterwards they are grazed (or vice versa). (only if obvious, e.g. when grassland is or will be mown but there are fences around the grassland parcel as well which point to grazing)

Pastures

Pastures can usually be easily distinguished from meadows e.g. by the presence of fences, by animals on the pasture, by animal tracks, by animal dung, by a water supply or shelters for animals, or by uneven height of the vegetation.

The following types of pastures are distinguished:

eC21	Intensive pasture (strip grazing, rotation grazing)
eC22	Medium pasture (regularly grazed at medium intensity)
eC23	Extensive pasture (regularly grazed but mostly only 1-3 times a year)
eC24	Occasionally grazed (very extensive, often with shrubs)
eC25	Pastured woodland: animals regularly graze the parcel and there is a tree cover

Other grassland

The following types of “other grassland” may occur:

eC31	Fallow or ruderal grassland: where vegetation has not been mown or grazed for at least 1 year, including areas that have self-seeded with grasses and forbs. In contrast to extensively used pastures or meadows, fallows have a large proportion of old vegetation and dead leaves from the previous year(s), and in some cases young trees and bushes.
eC41	Amenity grassland: areas with low, herbaceous vegetation not used primarily for agriculture in and around cities, villages and human settlements (e.g. parks, at the side of roads, football pitches, playgrounds, gardens etc.)
eC51	Unclear / Other kinds of grassland

8.16.6 Field: Growth status (enlarged transect / field) (769)

This parameter records the management stage at the time of survey.

There are three options:

- Grassland not yet mown or grazed this year – no cut edges visible in the vegetation, flowers are present: Recording is possible
- Grassland already mown or grazed a few days or weeks ago – vegetation is < 20 cm in height, no flowers, obvious signs of cutting (such as cut edges on leaves) or grazing (such as dung): Recording is possible but not relevant for some parameters.
- Grassland in regrowth after mowing / grazing – last cutting or grazing was several weeks ago and the vegetation has regrown so that it is > 20 cm in height and flowerheads are visible again: Parameters should be interpreted with caution.

The optimal time for carrying out the survey is before the first cut or grazing period. Once vegetation has been removed, some of the parameters can only be interpreted with caution, therefore it is important that information on the management stage is collected, in order to allow a correct analysis of the data.

8.16.7 Field: EUNIS Grassland Habitat Type (770) (enlarged transect)

EUNIS is an EU-wide habitat classification system. Please indicate the level 2 EUNIS habitat type according to the habitat identification key given in Annex 9.8 (98).

The most important habitat types for the grassland survey are E1 – E7, for their identification and description see Annex 9.8.

- E1 Dry grasslands
- E2 Mesic grasslands
- E3 Seasonally wet and wet grasslands
- E4 Alpine and subalpine grasslands
- E5 Woodland fringes and clearings and tall forb stands
- E6 Inland salt steppes
- E7 Sparsely wooded grasslands

There are, however, other EUNIS types and some LUCAS points may lie e.g. in forest (e.g. wooded meadows), fen or heathland. In this case LUCAS will not classify them as grassland but as C = Woodland, D = Shrubland or H = Wetland.

Overview on all possible EUNIS habitat types

D1 - Raised and blanket bogs
D2 - Valley mires, poor fens and transition mires
D3 - Aapa, palsa and polygon mires
D4 - Base-rich fens and calcareous spring mires
D5 - Sedge and reedbeds, normally without free-standing water
D6 - Inland saline and brackish marshes and reedbeds
E1 - Dry grasslands
E2 - Mesic grasslands
E3 - Seasonally wet and wet grasslands
E4 - Alpine and subalpine grasslands
E5 - Woodland fringes and clearings and tall forb stands
E6 - Inland salt steppes
E7 - Sparsely wooded grasslands: E7.1, E7.2, E7.3
F3 - Temperate and Mediterranean-montane scrub
F4 - Temperate shrub heathland
F5 - Maquis, arborescent matorral and thermo-Mediterranean brushes
F6 - Garrigue
G1 - Broadleaved deciduous woodland
G2 - Broadleaved evergreen woodland
G3 - Coniferous woodland
G4 - Mixed deciduous and coniferous woodland
G5 - Lines of trees, small anthropogenic woodlands, recently felled woodland, early-stage woodland and coppice
I1 - Arable land and market gardens: I1.5 Bare tilled, fallow or recently abandoned arable land
XX - Other
Not Relevant

For the correct classification please use the habitat identification key given in this annex9.9 EUNIS Habitat Classification.



Comment needed:

CODE	LABEL
GC99	Other grassland EUNIS comment

8.16.8 Field: Grassland age (127) (enlarged transect / field)

Grassland age is defined here as the length of time that the current vegetation has been growing without disturbance of the sward through ploughing or other cultivation. It is difficult to interpret this exactly in the field, but there are several signs that indicate if a grassland is more than or less than 5 years old:

Grassland 5 years old or younger (≤ 5 years):

- seeding rows of grassland are visible and / or
- only few grass and forb species occur (< 10 species)
- small patches of open ground

Grassland older than 5 years (> 5 years):

- species-rich composition (normally more than 10* herb and grass species present), **and**
- no seeding rows from a seeding drill visible, the sward is rather dense and the composition / distribution of the grass and herb species on the ground is balanced.

* Please note: very intensively managed meadows, with high slurry inputs after each of several cuts per year show less than 10 plant species and still are old in the sense of same use and no ploughing for many years.

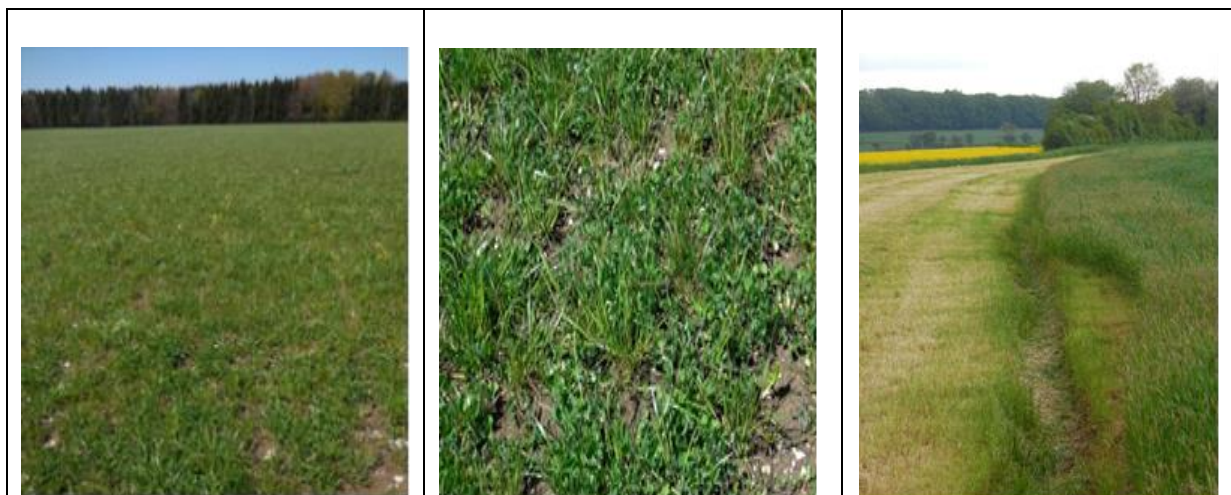


Comment and photo needed if grassland age unclear

If there are no clear signs for either grassland > 5 years or grassland ≤ 5 years, a comment is needed.

CODE	LABEL
GA99	Other grassland age comment

Take a photo of signs that could help to determine the grassland age: GRASS_UnclearAge



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Figure 8-62. a, b and c (from left to right): Seeded grassland can be *identified for example when the sward consists only of a few species (here Lolium perennis and Trifolium repens in (a) and (b)) and is partly open (Photos from 05.05.2016, Swabian Jura) or when the grassland is monostructured and the seed rows can be seen ((c), Photo from 10.05.2016, Luxembourg).*

8.16.9 Section: Grassland transect (130 – 140)

The grassland survey is based on a transect of 20 m length and 2.5 m width. Walking alongside the transect line, the surveyor should survey the vegetation 1.25 m to the left and 1.25 m to the right. In a few cases the observation window is a bit wider: these cases are indicated separately below (e.g. for orchards).

The starting point and the transect should be at least 5 m inside the grassland parcel (not at the edge bordering e.g. on another parcel a road, a forest, or a ditch, etc., as the information collected on the transect should be representative for the grassland type of the parcel and not that of the edge).

8.16.9.1 Fields: Grass transect direction/distance of grass transect shift

The starting point should be the LUCAS point (= exact coordinates are given). If it is not possible to complete a full transect of 20 * 2.5 m in any direction from this point (e.g. there are obstructions such as trees) or the vegetation is not representative for the parcel (e.g. the point falls directly on the edge of the parcel, or the sward has been disturbed by wild boar exactly here but nowhere else) then the point can be shifted. A shift should not be carried out unless absolutely necessary, and should be shifted as little as possible.

Please indicate if the transect starts directly on the point or was shifted (as the point falls on the edge). If a shift is necessary, please indicate the direction (N, NE, E, SE, S ...) and the distance of the shift (1-5 m).



Comment needed:

If there is a shift of the starting point, a comment on the reason for the shift is needed.

CODE	LABEL
GS99	Other grassland transect shift comment

8.16.9.2 Fields: Direction/ length of grass transect, start and end point

From the starting point, the transect should proceed in a strictly eastern direction. If it is not possible to walk a transect of at least 20 m to the east without coming within 5 m of the parcel edge, walk southeast. If this is not

possible, walk south, and if neither of these options is possible take the direction southwest, west etc. Finally, if none of these directions are possible take the direction of the longest possible transect walk within the parcel and indicate the direction of the transect.

If the longest possible transect is < 20 m length (extending until 1 m from the parcel edge) please indicate the length of the transect (e.g. 14 m).



Comment needed:

If the transect is shorter than 20m add a comment explaining why.

CODE	LABEL
GS01	Reason for shorter grassland transect

Register the coordinates of the starting point and of the end point.



Comment needed:

If there is a problem with the GPS and this has not been already reported, fill in field 14 "GPS coordinate system:" and make a comment (BExx) explaining the reason.

See 8.3.1 Fields: Access to point for possible comments.



Photos needed:

There are 5 photos mandatory on the transect.

1. Take a photo from the starting point of the transect in the direction of the end point of the transect(fields 659/661): GRASS_TransectStart.
2. Take one photo from 1.5 m height pointing the camera vertically downwards to a typical patch of vegetation for the transect (fields 662/664): GRASS_TransectAbove.
3. Take a photo from the end of transect point in the direction of the starting point of the transect (fields 665/667): GRASS_TransectEnd.
4. Take a photo of the flower posy (not necessary when there are no flowers at all).
5. Take a photo with the White linen tool (WLT).

8.16.10.1 Field: Vigour of vegetation (142)

Grassland vegetation can differ considerably in height and density (here termed “vigour”, roughly related to the biomass of plant material). Grasslands range from very dense (naturally or artificially fertilised) to very sparse (nutrient-poor and/or dry).

The vegetation on the transect should be assessed as one of five main categories of vigour. Intermediate stages (1.5, 2.5 etc.) should be assigned when the vigour is between two main categories. The height is not the only relevant factor for the type of vigour as the height depends mainly on the dominant species. The main characteristic is the density of the vegetation (i.e. the amount of biomass on a given area), especially to what extent sunlight can reach the low plant species near the ground. The type of vigour is assessed by looking down to the ground and checking to what extent the view (sunlight) can reach the different plant layers and the low plant species. Type 1 is a very sparse vegetation where the low plant species can be fully seen, type 3 is a light growth vegetation where the sunlight can reach almost all layers but with medium shadow effects given through higher plants and type 5 is very dense vegetation where only the tallest plants receive sunlight.

TYPE OF VIGOUR				
very meagre, sparse	meagre growth	light growth	dense growth	very dense, mass growth
				
1	2	3	4	5
	1.5	2.5	3.5	4.5

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Figure 8-63. Vigour of vegetation

However, difficulties may occur when the grassland is pastured or has already been mown. The type of vigour is indicated to show the full potential of the vegetation in spring; thus grazed pastures may be judged with the help of ungrazed patches remaining on the pasture. Meadows before the first cut should be judged according to the density that will be reached when most of the grass species are flowering. For example, a score of 4 or 5 can be expected if one or few highly productive species dominate, whilst the grassland will probably be 1, 2 or 3 if there is space between the tall grasses and forbs and sunlight reaches the ground.



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Figure 8-64. a, b and c: These three pictures show three types of vigour: (a) type 1.5 sparse, (b) type 3 (light/medium growth, sun light can still reach the ground) and (c) type 4.5 very dense growth



Photo needed:

In addition to noting the type of vigour, a photo of the vegetation shall be taken against a standardised white background (120*120°, as specified in 6.2, with height markings (provided to the surveyor): GRASS_Vigour.

The white linen tool:

The white linen tool (WLT) has the following characteristics:

Total weight: 1.5 kg.

Dimensions when packed: 10cm*10cm*130cm.

The following material is needed to produce the white linen tool:

- PVC-banner (120x120 cm) with hemstitch seam at the top and bottom,
- 2 wooden poles (130 cm x 2 cm Ø),
- 2 telescopic fishing pole holders with V-bracket (minimum high 130 cm) or a tripod
- 6 tent pegs + guy lines (for fixing the WLT to the ground in case of windy conditions)
- a bag for carrying the equipment

Preparing for the photo:

- *Choose a representative area of the vegetation on the transect*
- *Assemble the white linen tool at a 30-60° angle to the sun (if there is bright sunlight), so that the shadow of the photographer does not fall on it*
- *The white linen tool should be assembled from behind so as not to trample the vegetation in front of it*
- *Set up the tool and secure it in the ground using the ropes and pegs.*
- *In case of a slope, put the WLT alongside the slope (alongside the contourline) (not up or downwards) and take the photo upwards the hill.*

Assembling the white linen tool:

The PVC-banner of 120x120 cm has a hemstitch seam at the top and bottom. A 130 cm long wooden pole is inserted into both hemstitch seams with 5 cm protrude on the right and left side of the WLT. Starting from the bottom in the vegetation on the ground the WLT is rolled out.

For the grassland documentation the rolled WLT is placed in a location with representative grassland growth. A fishing pole holder is then inserted vertically into the ground to the left and right sides of the rolled WLT. The lower wooden pole rests on the ground while the upper wooden pole is pulled upwards (thus the WLT is rolled out) and placed in the V-brackets of the fishing rod holder. If necessary, pegs can fix the lower wooden pole on the ground and tent lines with pegs can stabilize the WLT against the wind.

On stony/rocky soils a tripod shall be used instead of a fishing rod holder.

Taking the photo:

- *The photo should be taken at a distance of 150 cm from the white linen tool, centrally, and at a 90° angle to the surface of the white linen tool. The camera should be horizontally, at a height of 70 cm, ensured by using a camera support.*
- *The whole white linen tool should be visible on the photo (Figure 4), also if the vegetation hides the lower layers of the vegetation.*
- *The photo ID must be noted.*

Further recommendations:

- *use a bag with shoulder strap for transport in the field*
- *take water and a cloth to clean the WLT*
- *take spare tent pegs, guy lines and runners*
- *attach colorful markers to the tent pegs to avoid them getting lost in the vegetation*

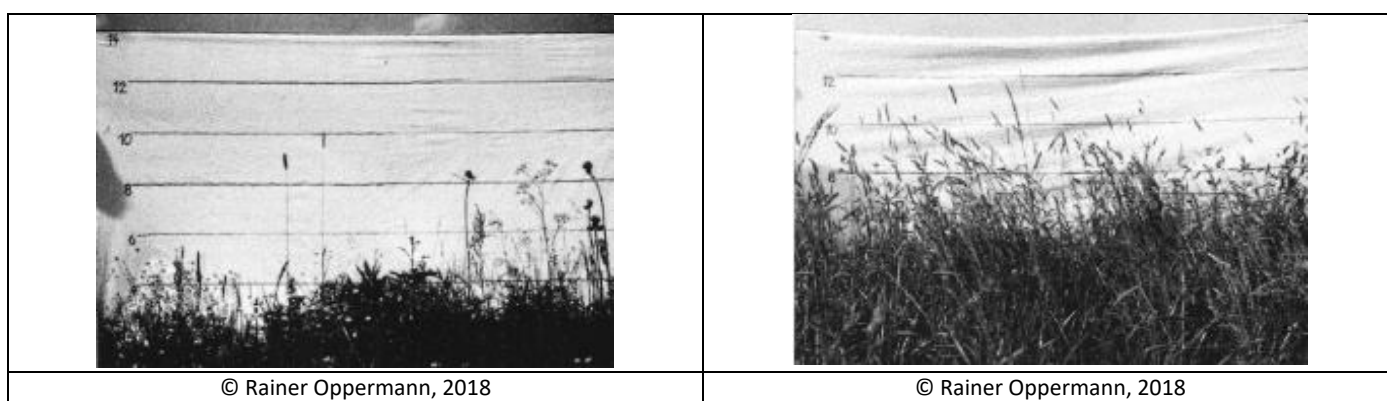


Figure 8-65: Photos of different vegetation types against the background of a white drapery with decimeter-lines.

8.16.10.2 Field: Grass-Forb ratio (771)

Within the herb layer, the ratio between the coverage of graminoids and forbs should be estimated, so that the sum of these two elements is 100. The aim is to determine which element dominates, or if the coverage is approximately equal.

- graminoid layer = all graminoid species including grasses, sedges, rushes and reeds;
- forb layer: all forb species (i.e. non-woody plants that are not grasses, sedges, rushes or reeds), as well as fern species;

In case of the presence of mosses, lichens etc. these shall not be considered in estimating the grass-forb-ratio.

Thus for a pure graminoid vegetation the ratio is 100:0;

For a pure forb vegetation (e.g. stinging nettles, also pure fern) the ratio is 0:100.

For most meadows and pastures there will be a mix of e.g. 90:10, 50:50, 40:60 or 20:80. The sum is always 100, also if the total coverage of this layer is only 50 % (and 50 % of bare layer or 50 % of bushes and dwarf shrub).

8.16.10.3 Fields: Layer components: Percentage of herbaceous, bare and woody layer on grass transect (143 – 145)

The horizontal coverage of the main layers (graminoid-herb layer, bare layer (open soil layer) and woody layer) is estimated in %.

Herbaceous layer + Bare layer + Woody layer = 100

The coverage of each main layer is estimated in predefined categories:

1	0 %
2	>0-3 %
3	>3-8 %
4	>8 -15 %
5	>15 -25 %
6	>25 -40 %
7	>40 -60 %
8	>60 -80 %
9	>80 -100%

Field: Herbaceous layer (143) (transect)

This layer comprises all areas covered by non-woody vegetation (grasses and broad-leaved plants), also mosses, fungi and lichens, when looking downwards from about 1.0 m height.

Field: Bare layer (144) (transect)

This layer relates to the open soil, thus earth, sand and stone soil as well as rocks, when looking downwards from about 1.5 m height.

Field: Woody layer (145) (transect)

This layer comprises all types of woody material, including dwarf shrubs, shrubs, trees and dead wood as a percentage of the woody vegetation touching the ground and below 1.0 m height, not the crown coverage.



Figure 8-66. On this photo,

all main types and many sub-types of layers occur: There is a woody plants layer (45%) with old trees (10%), with shrub (30%) and with dead wood (5%). Also a graminoid-herb layer with graminoids and forbs (ratio 40:60) can be seen.

The estimated percentage of the woody layer refers to the enlarged transect (20 m transect, 5m left and 5 m right), the herbaceous layer refers to the transect of 20m x 2,5 m.

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Fields: Woody layer components (lignified plants) (157 – 161) (enlarged transect)

The components of the woody layer are observed looking at the crown coverage from bird's eye perspective. This assessment is done on the enlarged transect (5 m left and 5 m right of the transect line – (thus 20 m x 10 m = 200 m²) as judging the tree canopy coverage requires a larger area to be considered than just the normal 2.5 m transect width.

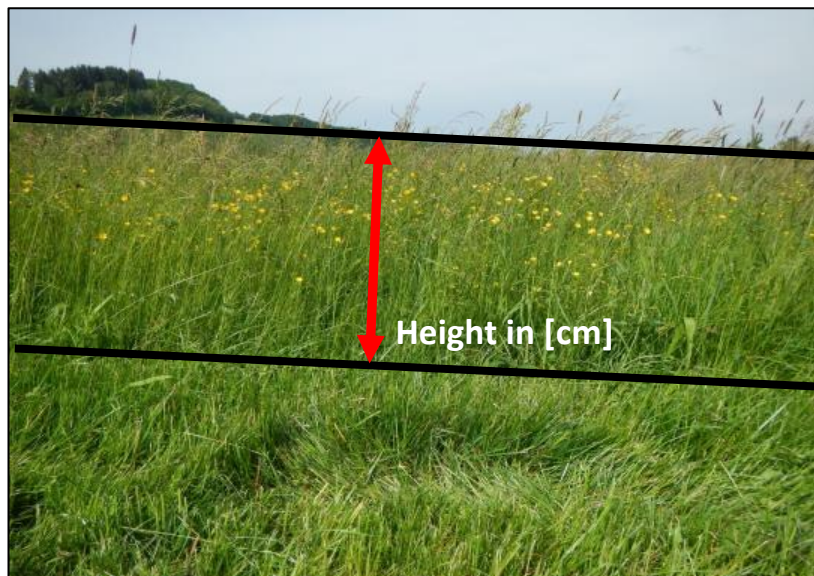
- orchard: all kind of fruit and nut trees (e.g. apple, pear, cherry, plum, walnut);
- old trees and trees > 5m: trees older than 20 years or higher than 5 m, often found in wood pastures, olive groves and dehesas; trees producing fruit or nuts older than 20 years are considered under orchard and not under old trees.
- Shrub and young/ low trees: all woody plants > 1 m and < 5 m height;
- dwarf shrub: small woody plants normally < 1 m in height, e.g. heath (the woody species of *Calluna* and *Erica*), blueberry or the hairy alpenrose in the Alps or the Phrygana and Garrigue vegetation in the Mediterranean countries (comprising woody plants with thorns or spines).
 - other woody species: all woody species not classified as shrub, dwarf shrub or heath, e.g. woody species like *Rubus* spec. or dead wood lying on the ground;

When estimating the percentage of the different woody layers, always consider the whole plant: for a shrub of 4 m height, also include the parts of the shrub lower than 1 m in the layer estimate thus looking down from bird's eye perspective; if there is dwarf shrub that is mainly below 1 m in height, also include the parts of the plants greater than 1 m in height.

The woody layer components are assessed from bird's eye view and the sum of all woody components reaches a maximum of 100%.

8.16.11 Field: Average grassland height (highest herb layer) (transect) (772)

Please measure the average height of the upper grass-herb layer. The upper layer of a grassland is determined by looking against the horizon and identifying the highest distinctive layer (often formed by grasses). Important: please note the average height, not the highest single plant!



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Photo of a grassland with the upper layer marked with the blue line. This can also be done on slopes as demonstrated on the photo (the foreground on the photo is mown).

Figure 8-67.

8.16.12 Section: Grassland flower (transect)

8.16.12.1 Field: Number of species of flowering forbs (173) and posy

Please count the number of currently flowering forb species (also the species which have well developed flower buds almost ready to flower, and species which have just finished flowering and the dead flowerheads can be seen). This applies only to insect-pollinated plant species (species with mostly coloured flowers, no grasses/sedges, no plants with inconspicuous flowers like *Plantago spec.*) but it applies to all flowering plants and not only to those indicated on the key species list.

Please count these flowering species exactly and please note also very small flowers (e.g. flower diameter 1-2 mm) and similar flowers (e.g. *Lotus corniculatus* and *Hippocrepis comosa*) (see Technical reference document C6 for example photos).

The exact number has to be counted and indicated (this will be a number between 0 and 33 – it's unlikely that there are more than 33 flowering species).

As you have to take a photo of a posy of the flowering forbs the best way to count the number of flowering forbs is to do the counting while by picking one flowering individual of each flowering plant species and afterwards taking the posy photo.



Photo needed:

Posy of flowering forbs: GRASS_Posy

In order to ease the counting of currently flowering forbs, please pick all different flowering species within the transect. For reasons of documentation, please take a photo of the posy.



Please make sure that the posy is sharply focused (not the background) or take a photo of the posy against a white background (e.g. put the flowers on the WLT).

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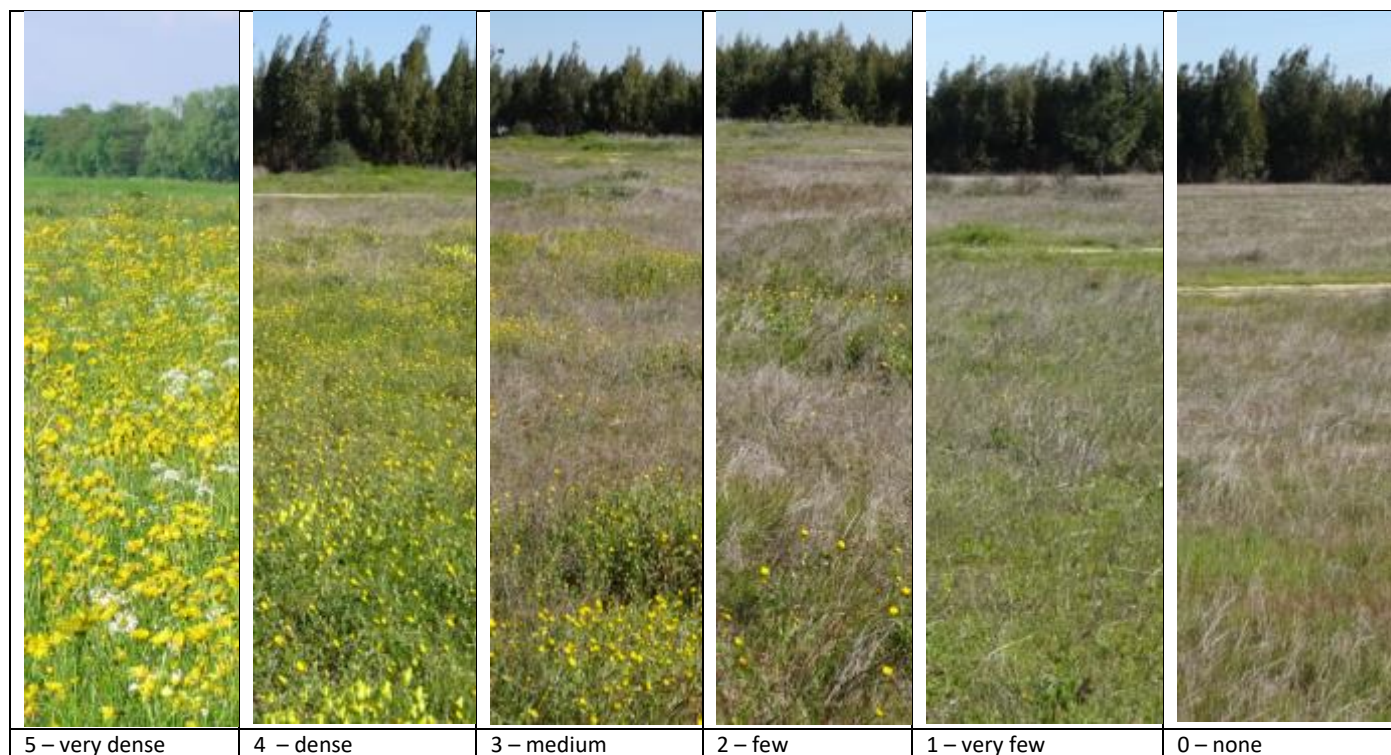
Figure 8-68. Photo with a posy of flowering forbs.

8.16.12.2 Field: Flower density (174)

Considering all flowers on the transect, estimate the flower density according to the scale below.

There are five main and four intermediate categories of flower density (plus the category 0). The surveyors shall indicate one of the main or intermediate categories.

0	none	no flowers in the transect
1	very few	hardly any flowers in the transect
1.5	very few to few	between hardly any and a few scattered flowers
2	few	a few scattered flowers
2.5	few to medium	between a few scattered and regular flowers
3	medium	regular flowering in the transect but not a dense flowering
3.5	medium to dense	between regular and many flowers
4	dense	many flowers occur, but not a dense “flower carpet”
4.5	dense to very dense	between many flowers and dense “flower carpet”
5	very dense	dense “flower carpet”



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Figure 8-69. 5, 4, 3, 2, 1 and 0: The flower density in these pictures reaches from (0) none to (5) very dense, passing by 4 other categories with different densities of flowers (1, 2, 3 and 4).

8.16.12.3 Fields: Colours of flowering forbs (773-782)

After having picked and counted all flowering species and after having taken a posy photo, indicate the colours of flowering forbs. There are 10 categories of flower colours including a category mixed colour. Mixed means that at least 20 % of the flower is of another colour than the dominating colour and there are at least 2 colours (e.g. a Daisy with yellow disc flowers in the centre and white ray flowers surrounding it)⁷:

- ☐ Brown
- ☐ Mixed*

Number of flower colours ticked: ...

(there will be an automatic counting of the number of the flower colours you have ticked).

8.16.13 Fields: Key species/species group richness ()

There are 12 key species groups for all European sub regions (only one list for whole Europe). The key species groups are species groups that in sum indicate species-rich or semi-natural grassland, as well as being easy to spot / identify, and neither too rare nor too common.

○ White	
○ Yellow	
○ Orange	
○ Red	
○ Magenta	
○ Violet	
○ Blue	
○ Green	
○ Brown	
○ Mixed*	Min 20% & min 2 colours

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The presence of these species should be recorded on the transect walk of 20 m length and 2.5 m width. While walking slowly along the transect line please tick the boxes of the key species that are present.

The 12 key species groups are the following:

	Apiaceae
	Campanula spp.
	Centaurea spp., Serratula tinctoria
	Cirsium spp., Carduus spp., Carlina spp.
	Juncus spp.
	Myosotis spp.
	Orchidaceae spp.
	Scabiosa spp., Knautia spp., Succisa spp.
	Silene spp., Lychnis flos-cuculi, Dianthus spp. (red flowering)
	Trifolium spp. (red flowering)
	Trifolium spp., Medicago spp., Lotus spp., Coronilla (yellow flowering)
	Vicia spp., Lathyrus spp., Astragalus spp. (blue or purple flowering)

If there are several species of one group (e.g. two *Campanula* species, for example *C. glomerata* and *C. patula*) the box is only ticked **once** and the abundance refers to **all the plants in the group** together. Attention: this is different for the number of flowering forbs (field 173), where all flowering species need to be counted (e.g. both species *Campanula glomerata* and *C. patula*).

After having ticked the presence of the key species groups please indicate the number of the flowering plant individuals in a rough scale of 4 classes:

0 = no individual flowering plant

1 = 1-2 individual flowering plants

2 = 3-10 individual flowering plants

3 = >10 individual flowering plants



Photo and comment needed:

If you encounter a rare or unknown species, take a photo GRASS_RareSpecies (multiple photos possible) and add a comment GO99 including the ID of the photo and the name of the species, if known.

CODE	LABEL
GO99	Other grassland other species comment

8.16.14 Fields: Structural species / species group: presence and % (enlarged transect)

Certain species tend to dominate under certain conditions. These may indicate positive conditions for grassland nature value (signs of extensive use) or negative conditions (e.g. abandonment or too intensive use of slurry).

There are 6 species groups which shall be indicated with their coverage:

	Shrub/ Brushwood/ Bramble
	Thistles (<i>Carduus</i> spp, and others)
	Docks (<i>Rumex obtusifolius</i> , and others)
	Rushes (<i>Juncus</i> spp.)
	Sedges (<i>Carex</i> spp.)
	Wood small-reed (<i>Calamagostis</i>)

Please tick the boxes if the species listed are present in the transect and record the classes of coverage (not the number of plants).

1	0 %
2	>0-3 %
3	>3-8 %
4	>8 -15 %
5	>15 -25 %
6	>25 -40 %
7	>40 -60 %
8	>60 -80 %
9	>80 -100%

Always indicate the coverage you estimate over the whole enlarged transect of 20m x 10m (200 m²).

All species of one species group are counted together (e.g. all *Juncus* species, both large and small species).

8.16.15 Field: Total cover of legumes (%) (transect) (236)

Record the total coverage of all herbaceous legume species, based on the following scale:

1	0 %
2	>0-3 %
3	>3-8 %
4	>8 -15 %
5	>15 -25 %
6	>25 -40 %
7	>40 -60 %
8	>60 -80 %
9	>80 -100%

Legumes are the following species groups:

Astragalus spec., *Coronilla spec.* *Onobrychis spec.*, *Hedysarum coronarium*

Medicago spec., *Lotus spec.*, *Coronilla spec.* – all species
Trifolium spec. – all species (red, yellow and white flowering)
Vicia spec., *Lathyrus spec.*, *Astragalus spec.* – all species
 other herbaceous species in the pea and bean family (Fabaceae; e.g. lucerne, lupin)



Figure 8-70. a and b: Legumes may form a considerable part of the herb layer as in (a) sainfoin (*Onobrychis viciifolia*); also legume species not listed in the key species list like white clover (b) have to be considered during the estimate of the coverage of the legume layer. For a more examples please have a look in the photo guidance.



Comment on legumes, if relevant:

CODE	LABEL
GL99	Other grassland legumes comments

8.16.16 Other grassland comments and photos



Other comments if relevant:

CODE	LABEL
GP01	Survey on small strips in orchards
GG99	Other grassland comments



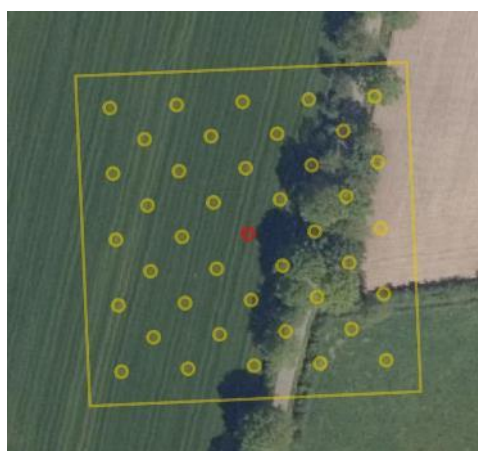
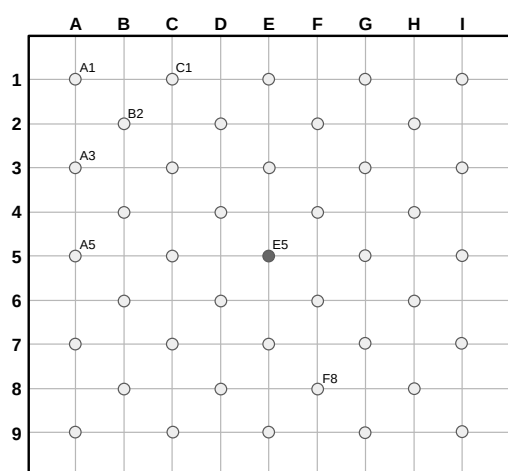
Take other photos as relevant: GRASS_Other (multiple photos possible)

8.17 Specific module: Landscape Features

The LUCAS Landscape Features (LF) module is a new module of the LUCAS survey 2022 and is carried-out on a sub-sample of the LUCAS points. For all LUCAS points that are part of the LF sample, the landscape feature assessment has to be done.

The purpose of this module is to survey all landscape features related to agricultural land. This means that the landscape features of interest in this context need to be placed on or adjacent to agricultural land. More detailed definitions of the landscape features of interest are given below.

The observation unit is a quadrat of 100x100m (1 ha), centered on a LUCAS sample point. Within each quadrat, a grid of 41 sub-points is to be evaluated according to a simple LF typology. Each sub-point shall be evaluated for the presence of landscape features of interest (defined in the sections below).



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Figure 8-71.: Grid of the 41 'sub-points' of the LUCAS LF module: the allocation of sub-point IDs (left), an example quadrat with the sub-points overlaid on an aerial ortho-photo centred on a LUCAS point (right: <https://www.pdok.nl/introductie/-/article/luchtfoto-pdok>). The sub-point in the middle (E5) always corresponds to the original LUCAS point.

The figure below shows the data structure to be recorded.

LUCAS point	Sub-point ID	LF1 PI	LF2-PI	LF PI comment	LF1 FS	LF2 FS	LF FS comment
375625	A1	W			W		
375625	C1						
375625	E1						
375625	G1						
375625	I1						
375625	B2						
375625	D2						
...	...	...			...		
375625	B4	W			W		
375625	D4						
...	...	...			...		

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Figure 8-72. Data structure to record landscape features on the grid of 41 sub-points

The evaluation of the sub-points is carried-out in 2 phases:

- Phase 1 is an office-based photo-interpretation (PI), where the sub points will be evaluated on very high resolution ortho-photos for the presence and type of LF).
- Phase 2 is the LUCAS field-survey (FS) of the same observation units, where the LF identified during Phase 1 will have to be confirmed or corrected in-situ.

The following section provides definitions for the landscape features and give instructions to their identification. These instructions are equally valid for both phases of the LUCAS LF work. The specificities of the office-based PI work (Phase 1) are described in Section 8.17.3, and field-work specific instructions are given in Section 8.17.4.

8.17.1 Identification of landscape features

The “landscape features” (LF) in the LUCAS LF module are small fragments of non-productive and (typically) semi-natural vegetation present in an agricultural context (embedded in or adjacent to agricultural land) which provide ecosystem services and support for biodiversity. The management and maintenance of these elements is also often connected to agricultural activities (e.g. cutting and harvesting the grass, pruning trees and shrubs etc...).

The definition above can be further elaborated using a LUCAS land use classification terminology. In order that a specific sub-point shall be considered as LF, it needs to correspond to a non-productive land use (e.g. U420, U413, U415, U315, U318, U319, or U321), while productive uses (U111, U113) as well as residential use (LU 360) are excluded. On the other hand, all LF are expected to be located on or adjacent to agricultural areas, with the exception of kitchen gardens (LU1 or LU2 = U111 or U112, but not U113). As a general rule, landscape features cover an area smaller than 0.5 ha or being narrower than 20 m. For some specific LF types, there is also a *minimum* width or size required, which aims to ensure that the LF is clearly identifiable and persistent.

The fulfilment of these criteria needs to be verified for each feature on each of the 41 sub-points.

The LF module aims at identifying LF located in or adjacent to agricultural land. More specifically, this includes all elements which are lying:

- within an agricultural field (i.e. typically within arable land, grassland or permanent crops);
- between agricultural fields;
- between an agricultural field and linear infrastructure (farm track, road, or railway);
- between an agricultural land and detached buildings (individual farmyards; agricultural buildings); and
- between an agricultural field and water bodies (rivers, lakes, reservoirs...).

On the other hand, no LF shall be recorded for features that are not in an agricultural context, or which are above or below the size limits⁸. Similarly, no LF should be recorded when a potential LF cannot be clearly distinguished from a neighbouring land parcel (e.g. grass along a ditch is not an LF if it is the same type and is pastured in the same way as the whole neighbouring grassland plot). More specifically, the following elements shall not be considered as LFs within the LUCAS LF module, while they are normally surveyed with specific codes in LUCAS core:

- elements within industrial or urban areas (including villages);
- elements within private gardens, parks that are connected to settlements;
- patches of grass, shrubs, trees (forest), their mosaic, or abandoned agricultural land, that are above the dimension limit for the LF elements;
- grass strips within permanent crops (e.g. between rows of fruit trees or vineyards);
- productive trees in agro-forestry systems or in orchards.

Terraced hillsides are complex landscapes that contain both productive and non-productive parts. In such landscapes only the non-productive elements (steep sections covered by permanent woody or grassy vegetation or stone walls) need to be recorded as (woody, grassy, or stony) LFs, but only if they meet all the criteria (e.g. they need to be embedded in an actively used agricultural landscape, and to meet all relevant geometric specifications).

The quantity of LF in agricultural land is estimated using a set of 41 'sub-points', which are marked on the ground documents in every LUCAS point selected for the LUCAS LF module. In those sub-points that fall on a landscape feature, the type of the LF shall be determined and recorded.⁹ Seven LF types are distinguished (see the following sections). An observation point can contain up to two landscape feature types (e.g. a hedgerow above a stone wall).

When there is more than one LF in one sub-point, two of them can be recorded: one as primary and one as secondary LF. Therefore, up to two landscape feature types for each sub-point are foreseen (e.g. an isolated tree on the bank of a ditch). The LF shall be recorded as primary and secondary LF, in the corresponding columns in the attribute table.

The allowed combinations are listed in the following figure:

⁸ If there are several LF types or artificial features (e.g. roads, buildings) next to each other in a single "non-cultivated island" embedded in agricultural land, then the maximum size limits (20m, 0.5 ha) need to be considered for the whole island. In other words, non-cultivated islands above these size limits are not considered to be part of the "agricultural context". Accordingly, for non-cultivated islands that meet the overall geometric criteria, all subpoints should be recorded as LF – except for subpoints that fall on artificial structures, like farm tracks, paved roads, buildings, dump sites, or mines.

⁹ To decide if a subpoint "falls on" a particular LF type or not, the following approach should be used

- (1) decide if the subpoint falls on a non-cultivated island (if not, it is not an LF – except for temporary herbaceous LF (T), which shall be recorded even within cultivated parcels);
- (2) determine the surface cover at the precise location of the point (centre of the map marker);
- (3) if the point falls exactly on a boundary, then apply the LUCAS "Look to the North/East" rule (i.e. select the LF type, that is to the North from the subpoint, or in the case of an exactly N-S divide, select the type to the East).

		LF2								
LF1 LF2		No LF	W	G	T	D	P	S	C	NA
LF1	No LF	2	0	0	0	0	0	0	0	0
	W	2	0	0	0	2	2	2	2	0
	G	2	0	0	0	0	0	0	2	0
	T	2	0	0	0	0	0	0	1	0
	D	2	0	0	0	0	0	0	0	0
	P	2	0	0	0	0	0	0	0	0
	S	2	0	0	0	0	0	0	1	0
	C	0	0	0	0	0	0	0	0	0
	NA	2	0	0	0	0	0	0	0	0

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Figure 8-73: Allowed combinations of LF1 & LF2. (W, G, T, D, P, S C: the main types of LF – W=Woody, G=Perm. herbaceous, T=Temporary herbaceous, D=Ditches, P=Ponds, S=Stone walls – rock outcrop, C=Cultural features, NA: no data (code only eligible in the PI phase); see Sections 8.17.1.1-8.17.1.7

2: possible/common combinations; 1: unlikely combinations; 0: impossible combinations).

Cultural features (C) can only be recorded as a secondary LF type (LF2). If woody landscape features are present, they are recorded as LF1.

The following sections give detailed instructions on the LF types, often making reference to the LU and LC types used in LUCAS core for better comprehension. Nevertheless, there is an important difference that should be kept in mind: in the context of LUCAS Landscape Features LC/LU always refer to the location of a sub-point, even if it falls on a patch narrower than 3m (no look to the North/East should be applied). Similarly, an extended window of observation should also never be applied.

8.17.1.1 Woody vegetation LFs (code: W)

This type includes isolated trees, trees in line, hedgerows, riparian woody vegetation (along water course), or any narrow strips (<20 m) of land covered by trees and shrubs within an agricultural context. This type can also include small groups of trees, field copses, or any small groups of woody semi-natural vegetation in an agricultural context. In case there is a grass (herb) layer under the woody vegetation, the woody feature is considered to incorporate the underlying grass layer too.

Woody LF can be linked to the woodland (LC: C10, C20, C30) and shrubland (LC: D10, D20) types from LUCAS core, nevertheless the type shall be determined at the subpoint, and no extended window of observation should be used.

In contrast with LUCAS Core, subpoints falling on the tree canopies of a woody LF overhanging a road or an arable field should also be recorded as W. Similarly, if a tree canopy of a woody LF overhangs a ditch (D), a pond (P), a stony (S), or a cultural LF (C), both features need to be registered (W as LF1, and the other one as LF2).

The maximum area for a woody LF is 0.5 ha. The minimum size of shrubs and young trees to be recognized is 1m². (Groups of) shrubs / young trees of <1m² that are surrounded by grassy (or wetland) vegetation are to be considered as part of the surrounding grassland (or wetland).



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Figure 8-74.



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Figure 8-83.

8.17.1.2 Permanent grass/herbaceous LFs (code: G)

This type consists of permanent semi-natural herbaceous vegetation (typically grass and/or perennial herbs) which are in the agricultural context, and which are not directly used for grazing, or fodder production (LU at the subpoint should not correspond to U111). This may include field margins, buffer strips (along ditches or ponds), or any other small pieces of semi-natural herbaceous vegetation (typically of LU U415 or U420) as long as they are between within or adjacent to arable or permanent crops fields. Grass margins next to meadows are counted as landscape features if they are not productive, as non-productivity is part of the definition of landscape features.

The minimum width of this type is 1m (for ensuring persistency). Nevertheless, this type of LF excludes parcels of actively managed grasslands (used for grazing or fodder production), and large patches of (semi-)natural grasslands (wider than 20 m or larger than 0.5 ha). Furthermore, farm tracks with grass, and grass strips between the rows of vineyards/orchards are also not grassy LF. Permanent grass/herbs landscape features do not include the grass layer under a woody feature, nor wet marsh vegetation (which are registered either with codes W or D / P respectively).



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Figure 8-84.



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Figure 8-86.



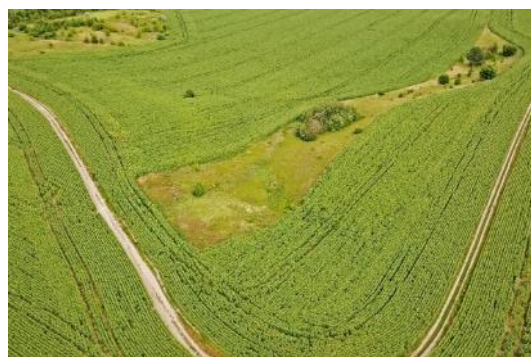
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Figure 8-87.



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Figure 8-88.



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Figure 8-89.

8.17.1.3 Temporary herbaceous LF (code: T)

This type consists of narrow strips of cropland planted with non-productive crops or flower-rich fallow (weed) vegetation inside arable land or permanent crops (typically along field margins), deliberately sown by the farmers to support biodiversity (U112). This is the only LF type that can occur within regularly cultivated agricultural parcels. The maximum width of this subtype is 20 m, and wider strips of flowers or fallows should not be considered in the LUCAS LF module. The minimum width of this feature type is 1m to ensure a consistent recognition.

Exclusions: Weedy spots resulting from the failure of arable crops.

If a grassland strip shows the characteristics of both a temporary (T) and a permanent (G) grassland strip, then G should be chosen.



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Figure 8-90.



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Figure 8-95.

8.17.1.4 Ditches and streams (code: D)

This type includes small water courses (G20) within an agricultural context, including the open water surface of streams, ditches, and small channels and an adjacent strip of marsh/wetland vegetation (H10) up to a maximum width of 20 m. Ditches that are dry at the moment of observation can also be registered as D if the vegetation reveals a regular presence of water.

Exclusions: Artificial constructions (channels with walls of concrete and subterranean constructions) are excluded. Shallow roadside ditches with neither permanent water nor wetland species should be considered either as grassy LF (G) or as "No LF". Similarly, if a sub-point falls on the dry parts of the banks of a small water bodies, it shall be recorded according to its nature, e.g. as grassy (G), or woody LF (W), if the other relevant criteria are satisfied.



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Figure 8-96.



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Figure 8-99.



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Figure 8-100.



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Figure 8-101.

8.17.1.5 Small ponds and small wetlands (code: P)

This type includes small patchy landscape features characterized by wetlands (H10) and water bodies (G10) in an agricultural context up to a maximum size of 0.5 ha. Small ponds and wetlands need to have a minimum area of 1 m² to be recorded as LF.

The type also includes accumulations of still water formed naturally (e.g. wetlands, lakes, natural lagoons, seepage areas) or artificially (e.g. pits and waterholes).

Small ponds can contain a core of open water and an adjacent wetland, characterized by marsh vegetation (e.g. reed or sedge beds) adapted to and dependent on the regular presence of surface water and high water levels.

Exclusions: reservoirs lined with concrete or plastic and depressions used as landfills.

If a sub-point falls on the grassy or woody banks along a small pond, with no standing water or apparent wetland vegetation, it shall be recorded either as grassy (G), or woody LF (W) (according to its nature), if it satisfies the relevant criteria.



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Figure 8-102.



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Figure 8-107.

8.17.1.6 Stone walls, stone terrace walls, cairns and rock outcrops (code: S)

This type includes piles of rock or stone (F10) in an agricultural context, and terraced agricultural landscapes. Such features may be natural (e.g. secular stones, small rock outcrops) or man-made, often historical, objects (e.g. dry stone walls, terrace support walls made of stone, clearance cairns), which may be in any condition (standing, collapsed, overgrown with vegetation). Stone walls and terrace walls made of artificial materials (e.g. concrete, brick and mortar) should not be considered. Terraced hillsides are anthropogenic structures created to reduce the risk of erosion, consisting of one or more “steps” (steep sections covered permanent woody or grassy vegetation or stone walls) and “land blocks” (flat sections that are used for agricultural production, separated by the steps).

Stone walls, stone terrace walls

As stony features typically consist of steep surfaces and many crevices, their horizontal area is significantly smaller than their ecologically active surface. This means that a simple point sampling in the horizontal space would significantly underrepresent the stony LF, possibly missing even major structures with large active surfaces. To correct for this, in the case of stone walls and terrace walls made of stone (and only these two subtypes!) a search range of 1.5 m is applied around each point. Practically, this means that if there is a stone wall within 1.5 m of the subpoint, then the subpoint shall be recorded as S, even if it does not fall directly “on” the (possibly vertical) wall.

If trees and shrubs (liana) cover the stone walls, both features shall be registered. Herbaceous vegetation, on the other hand, is considered to be an integral part of S.

Cairns and rock outcrops

In the case of cairns and natural rock outcrops the general geometric limits apply, i.e. only those subpoints can be registered as stony LF, which are embedded among agricultural fields in a “non-cultivated island” that is smaller than 0.5 ha or narrower than 20 m. Small isolated stones (e.g. in pastures) or cairns below an area of ~1m² should not be considered individually as stony LF (they should be considered as integral parts of the surrounding parcel / land cover / LF type). For cairns and rock outcrops there is no “search range” to be considered (similarly to all other LF types).

If trees and shrubs (liana) cover the cairns or rock outcrops, both features shall be registered. Herbaceous vegetation, on the other hand, is considered to be an integral part of S.



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Figure 8-108.



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Figure 8-111.



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Figure 8-112.



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Figure 8-113.

8.17.1.7 Cultural features (code: C)

Cultural features are local elements of cultural heritage that provide ecosystem services. This type includes historical mounds (round or elongated masses of earth protruding above a flat agricultural landscape, typically of (pre)historic origin and covered in semi-natural vegetation: e.g. burial mounds, prehistoric tells, Cumanian mounds, and historical earth banks). Historic mounds are covered either by permanent grass (G) or scrubs/trees (W), which shall be registered as the primary LF, and the LF code C can thus only be used as a secondary LF type.



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Figure 8-114.

<https://commons.wikimedia.org/w/index.php?curid=77731953>



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Figure 8-115.

<https://commons.wikimedia.org/w/index.php?curid=22865415>

8.17.2 Identifying LF during the PI and the FS phases: the decision process

To identify landscape features “embedded in or adjacent to agricultural land” we recommend that both the photo-interpreters (PI) and the field surveyors (FS) follow the following decision process:

- **Step 1: The delineation of agricultural land:** In the context of the LUCAS LF module, productive agricultural land (AL) is interpreted as the ensemble of all land that are actively used for agriculture (U111), including temporarily non-productive parcels (fallows), which are otherwise available for agricultural production (U112). Residential and leisure gardens, including kitchen gardens (U113) do **not** belong to this concept of AL.
- **Step 2: Analysis of gaps:** Landscape elements described in the above classes are considered landscape features if they are non-productive, they satisfy the size limits and the positional criteria (inside or adjacent/bordering to agricultural land). There can also be gaps in the agricultural land that are occupied by a combination of landscape elements: multiple linear and patchy objects (roads, tree alleys, ditches, hedges, grass strips, etc.). If the total size of this gap respects the size limits, all landscape elements are considered as landscape features.
- **Step 3: The identification “bordering features”:** Bordering landscape elements are to be considered landscape features if they respect the size limits, are adjacent to AL. Furthermore, narrow grass strips can be considered as parts of the AL, so the objects adjacent to them can also be seen as adjacent to the AL on their other side.
- **Step 4: Further possible LF:** Flower-rich fallow strips established by the farmers consciously within their arable fields (LF type T) are landscape features, even if their LU is classified as U112 (fallow land). Some types of agriculture (e.g. agroforestry areas: X06, X09) can also host significant amounts of non-productive elements which can be assessed as LF. To identify such LF, the decision depends on the productivity of the two components of the mosaic. If both components are productive, or neither of them are productive, then no LF should be identified in the mosaic considered. If just one component is productive, then the subpoints falling on the pieces of the non-productive component can be registered as LF – in case they also meet the remaining non-spatial criteria (permanent, semi-natural...).

8.17.3 Photo-interpretation (PI) in office (Phase 1: LF office PI)

To facilitate the work in the field and to establish an initial homogeneity of the visual interpretation, a preliminary PI of the LF points is performed. This preliminary PI shall be performed in an efficient GIS environment.

All sub-points of a quadrat shall be analysed, and LF codes should be assigned to the sub-points where a LF occurs (if it fulfils the definition of the section above) following the instructions and the typology presented above.

If no LF is present, the default value is to be left (No LF).

LF types T (temporary herbaceous LF) and C (cultural LF) should normally not be used during the PI phase, because these types typically cannot be reliably identified on the orthophotos. Temporary herbaceous LF, for example, are often involved in the crop rotation of the farmers, which means that even if a flower strip can be identified in an orthophoto from a previous year, it will probably not be present in the same location in the year of the survey.

However, if they can be clearly identified, they should be registered during PI.

If the observation of landscape features is technically not possible (e.g. because the ortho-photo is not available, or clouds/shadows make it impossible) the code “NA” should be used.

The photointerpretation of LF can be implemented directly within the DMT or the results may be uploaded in bulk to the DMT.



Comment needed:

- if the PI is entirely impossible (e.g. because the ortho-photo is not available, or clouds/shadows make it impossible) then a comment needs to be made (Code LF01, LF03) and the LF module is terminated for this LUCAS point.
- if the observation of landscape features is impossible for just some of the subpoints the LF classification code “NA” should be used, and the problem must also be described in a structured comment (LF04 + subpoint IDs).
- If any other problematic situation is encountered (e.g. the type of an LF cannot be determined based on the definitions provided in Section 8.17.1) add a comment LF99 + free text + and, if relevant add subpoint IDs + photo IDs.

CODE	LABEL
LF01	LF office PI not possible because appropriate ortho-photos are not available (LF office PI)
LF03	LF assessment is entirely impossible for any other reason + free text
LF04	LF type of subpoint is impossible to determine/verify + ID sub-point(s)
LF99	Other issue with LF identification LF + free text (+ ID sub-point(s) + photo(s) ID)

8.17.4 Field survey (Phase 2: LF field survey)

The main purpose of the field survey in the LUCAS LF module is to verify and update the LF types determined during the office-based PI (Phase 1). For this purpose, the results of the PI LF assessment have to be made available during the field survey, as well as the comments made during PI.

The surveyor has to confirm the result of the PI in case of agreement, or to change it if the field surveyor finds differences compared to the PI. If no LF is present, the value to be selected is “No LF”.

In addition, the field surveyor also needs to look out for flower strips (T) in the arable fields, as well as apparent historical mounds and earth banks (C, recorded as LF2), which typically can only be assessed in the field.

If the observation of landscape features is technically not possible (e.g. because the point is not visible and not reachable) the code “NA” is to be used.

After editing, confirm changes or close without changes.

In the LUCAS LF quadrats, the sub-points are relatively close to each other: the distance between the central point (E5) and the corner points is just 56 m. Accordingly, the surveyors are not expected to walk to all sub-points. Instead, the surveyors should pay close attention to the landscape features as they approach the LUCAS point, which lies exactly at the centre of the quadrat (sub-point E5). The existence of the LF and the correctness of their type assignments should be verified during this walk and from the centre of the quadrat, as much as possible. Additional walk should only be made, if absolutely needed to verify the sub-points (e.g. to look behind a group of trees, or to check the existence of suspected stonewall).



Comment needed:

- if the LF field survey is entirely impossible (e.g. the quadrat is inaccessible; Code LF02, LF03)
- if the LF type of some of the subpoints is impossible to determine/verify (e.g. they are inaccessible & not visible) then the LF classification code NA should be used and a comment needs to be made (LF04 + subpoint IDs + free text);
- if for some subpoints there is a change compared to the LF office PI assessment caused by the recent removal of an LF (LF05 + subpoint IDs);



Comment and photo needed;

- If for some subpoints there is an LF that is not visible on the orthophoto (e.g. LF too small to be identified on the orthophoto, newly established LF...)

Photo: LF_NotOnOrtho (multiple photos possible)

Comment: LF06 + subpoint IDs + photo ID s);

- If any other problematic situation is encountered (e.g. the type of an LF cannot be determined based on the definitions provided in Section 8.17.1

Photo: LF_Other

Comment LF99 + free text + and, if relevant add subpoint IDs + photo IDs.

CODE	LABEL
LF02	LF field survey not possible because the quadrat is inaccessible (field survey)
LF03	LF assessment is entirely impossible for any other reason + free text
LF04	LF type of subpoint is impossible to determine/verify + ID sub-point(s)
LF05	LF visible on the orto-photo is missing in the field due to removal + ID sub-point(s) (field survey)
LF06	LF not visible on orthophoto + ID sub-point(s) + photo ID
LF99	Other issue with LF identification LF + free text (+ ID sub-point(s) +photo(s) ID)

Additional information is to be provided in the following fields:

Office PI confirmed?

Orthophoto outdated within the quadrant?

Additional walking needed in the field?

8.18Specific module: soil

8.18.1 Purpose of the soil sampling in the LUCAS Survey

The soil component was incorporated to LUCAS in 2009 in order to create a harmonised and comparable dataset of soil properties and to improve the quality of soil modelling in the EU. LUCAS-Soil was developed as an expandable module, with the prospect of analysing new properties and add new sampling locations over successive surveys. The scope of LUCAS 2009 and 2015 was to assess the state and changes of soil physical and chemical properties ⁽¹⁰⁾ over time across different land uses/covers. Given the policy interests of the Commission ⁽¹¹⁾, analysis of biological properties and residues of Plant Protection Products (PPPs), and a visual assessment of soil erosion were included in the LUCAS survey of 2018. The action plan of the European Green Deal ⁽¹²⁾ has shown the importance of monitoring environmental performance of the initiatives outlined to achieve the Green Deal. As a result, in LUCAS 2022, analysis of physical, chemical, and biological properties have been extended to 41004 points, while gully erosion will be assessed on all LUCAS Core points. More than 63% of the soil points will be in cropland (Table 1).

Table 1. Distribution of points by land cover class in the soil component of LUCAS.

Land cover	LUCAS 2022
Artificial	20
Cropland	24283
Grassland	6826
Shrubland	43
Wetland	1230
Woodland	8602
Total	41004

For LUCAS 2022, we have reviewed the soil sampling instructions from 2018 to make room for new demands and improve some aspects of the soil survey.

- Increase of sampling depth to 30 cm
- Simplify the transport of soil samples for biological analysis
- Focus visual assessment to gully erosion

This document contains the following 2 manuals:

- **Manual 1:** spade sampling for points where physical (except bulk density) and chemical properties will be analysed
- **Manual 2:** Core sampling for points where bulk density will be measured

Surveyors will need to read carefully the manuals before going to the field.

8.18.2 Availability of results

Results of the various analysis and measurements will be published through the European Soil Data Centre (<http://esdac.jrc.ec.europa.eu/>). Should the land owner/land user ask to obtain a copy of the results of the soil analysis, it will be possible to do so, provided that full contact details are recorded in the Field Form.

⁽¹⁰⁾ Coarse fragments, sand, silt and clay contents, organic carbon content, nitrogen, phosphorus and potassium contents, cation exchange capacity, pH, electrical conductivity, and calcium carbonates content.

⁽¹¹⁾ Biodiversity strategy, Sustainable Use of Pesticides Directive, Sustainable Development Goal on Land degradation neutrality.

⁽¹²⁾ https://ec.europa.eu/info/strategy/priorities-2019-2024/european-green-deal_en

8.18.3 The soil sampling process

8.18.3.1 Different soil modules apply to different points

The present document is separated in chapters which relate to different soil collection modules. Only a selection of all LUCAS points is subject to soil collection (max. 41004). In 37000 points, samples will be taken with the standard sampling method. In the remaining 4000 points, samples will be taken with the core sampling method. It has to be noted that while the sampling methodology builds on previous LUCAS surveys, some modifications have been included in the instructions of both standard and core sampling for woodland points.

A maximum of 2000 out of the 4000 points where core sampling will be carried, an additional sample for the assessment of soil biodiversity will be collected. The explanation on how to collect the biodiversity sample is integrated in the same chapter of the core sampling. Note that special measures are needed for points where the biodiversity sample is to be collected (including careful cleaning of material and use of gloves by the surveyor).

In some cases, it will not be possible to collect the soil samples strictly according to the different instructions. Guidelines are given on how to collect the sample in special circumstances and in case of difficult access.

8.18.3.2 Can you do the soil assessment?

Soil samples should be collected only in points with observation type 1 (the point is observed at less than 100 m) and when the location reached by the surveyor is in the same land cover and use as the LUCAS point (ideally also in the same field).

If the LUCAS point cannot be reached (e.g. mature crops), the soil sample(s) may be collected from the same field provided the land cover and use are the same as the target location, being as close as possible to the LUCAS point, but not exceeding a distance of 300 m.

If you cannot do the soil assessment select “No” and add a comment explaining why (SA09).

8.18.3.3 Which soil samples to collect? (K, L, M, N, O, P)

If one or more specific modules on soil are to be assessed on the point this will be indicated with “yes” in the fields K – P:

- Soil point (K): standard soil sample is to be taken
- Bulk point (M, N, O): collect (and assess) bulk density
- BIO point (P): collect biodiversity sample



Comments and photos needed:

If it is not possible to take any one of these sub-modules a comment is needed (SA05, SA06, SA07, SA99).

In case the basic soil assessment or the bulk density cannot be done also a photo is needed: SOIL_NoSample.

If a sub-sample cannot be taken, a photo can document the reason: SOIL_NoSub

Any other important aspect and any exceptional arrangement for taking the soil sample needs to be reported in a comment: SB01, SB99, SC99, SA01, SA02, SA04, SA08, SA99.

If additional photos are needed use: SOIL_Other

If the relevant sub-modules are assessed a photo of the sampled area and the bags with the soil sample(s) (soil photo – SOIL_BagHoles: fields 626/628,) and a photo of each label is needed (biodiversity label – SOIL_BioLabel: fields 635/637, bulk label photo – SOIL_BulkLabel: 632/634, soil label photo – SOIL_StdLabel: fields 629/631).

Furthermore a photo of the litter surface in soil woodland points should be taken: SOIL_Litter.

When samples are collected this needs to be noted as well as the number of the label and the expedition date. The detailed process, including photos to be taken, is presented below.

Labels will be provided by the JRC. If labels are lost and there is a need to create own labels by the contractor or the surveyor, please use the point id as label code and include on stable paper within the sample bags (as described for the formal labels), but enter 000000 as the label code in the DMT.

Possible comments on soil are:

CODE	LABEL
SA01	Soil sample very wet
SA02	Stoned area in the point, stones removed
SA04	Soil sample collected at the edge. Approx. distance from the edge & reason:
SA05	Ground conditions unfavourable
SA06	Inaccessible point (e.g. fenced area)
SA07	Land owner refused the soil sample
SA08	Shallow soils – target soil depth (30 cm) not reached. Depth reached:
SA09	Soil assessment not possible
SA99	Other soil comments
SB01	Not all cores were collected
SB99	Other soil bulk comments
SC99	Other soil biodiversity comments

8.18.3.4 Location of soil site

Register latitude and longitude of the central soil point.



Comment needed:

If there is a problem with the GPS and this has not been already reported, fill in field 14 “GPS coordinate system:” and make a comment explaining the reason.

See 8.3.1 Fields: Access to point for possible comments.

8.18.3.5 Soil site (341)

In any case the percentage of stones on the surface has to be assessed.

8.18.3.6 Subsequent treatment

A number of subsequent steps are needed after the soil collection for the proper shipment, eventually drying and weighing of the sample. The treatments depend on the type of sample and are described in the respective chapters.

8.18.4 Standard/Spade sampling for points where physical and chemical properties will be analysed (except for bulk density)

This manual describes the spade sampling procedure for the analysis of physical (except bulk density) and chemical properties in cropland, grassland, and peatland (section 2.2.1) and woodland (section 2.2.2). Surveyors need to take a single sample of 30-cm depth from the soil surface in cropland, grassland, and peatland points

unless a physical barrier is met (e.g. hard rock, gravel layers, indurated horizon). In woodland points, surveyors need to take a sample of the litter layer, writing down in the corresponding field of the field form the depth of the layer, and a sample of the mineral topsoil unless a physical barrier is met.

Examples of shallow soils due to physical barriers (e.g. hard rock, gravel layers, indurated horizon)



© European Union, CC BY 4.0 photo by C. Kosmas

Figure 8-116.



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Figure 8-117.

https://esdac.jrc.ec.europa.eu/public_path/shared_folder/projects/DIS4ME/indicator_descriptions/indicator_description_images/soil_depth_1.jpg

https://content.ces.ncsu.edu/media/images/bedrock_6nnEvKq.jpg



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Figure 8-118.

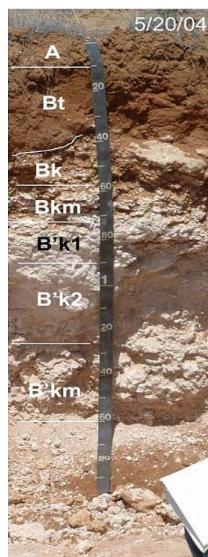


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Figure 8-119.

https://www.nrcs.usda.gov/wps/portal/nrcs/detail/soils/ref/?cid=nrcs142p2_054253

https://www.aswm.org/pdf_lib/soils_webinar/soil_texture_and_structure_rossi_071316.pdf



2004, Dr. H. Curtis Monger, CC BY-SA 3.0

https://commons.wikimedia.org/wiki/File:Jornada_G-IBPE_soil_pit_horizons.jpg

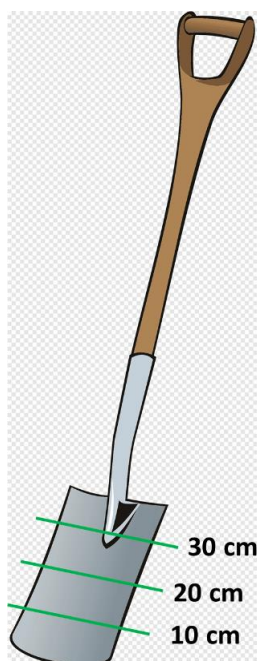
Figure 8-120.

Before the start of the survey, surveyors will receive from their coordinators the list of the specific points where spade sampling should be carried out.

Further aspects described concern measures to be taken before leaving the point and instructions for drying and shipping the sample.

8.18.4.1 Before going to the field

Surveyors need to mark the depth increments of 10, 20 and 30 cm on the blade of the spade. At each LUCAS point, before beginning the soil sampling, surveyors have to ensure that the marks are visible. If they are not visible, surveyors need to do the marks again before starting the sampling.



Picture source: <https://creazilla.com/nodes/28433-spade-clipart>

CC0 1.0 Universal

Figure 8-121: Mark the spade

For woodland points, surveyors need to tag 1 large plastic bag with the label “LITTER BAG”.

Calculate how many paces are required to reach a distance of 2 m.

8.18.4.2 Sampling in cropland, grassland, and peatland points

1 *The sample is taken only if the point is reached!*

Go to one of the LUCAS points of the list provided.

Ideal situation: you **reach** the exact point, so the samples are taken there. BUT if you are unable to reach the exact LUCAS point, then:

- The soil samples should be taken within a maximum of **100 m** distance from the LUCAS point, preferably in the same field where the LUCAS point is located. Soil samples will be taken always from the same land cover class as observed at the LUCAS point.
- In case the LUCAS point cannot be reached (e.g. mature crops) and it is not possible to take the samples within a maximum of 100 m distance to it, the soil sample may be collected from the same field provided the land cover and use are the same as the target location, being as close as possible to the LUCAS point, maintaining the same topographic setting (eg same contour on a slope), but not exceeding a distance of **300 m**.

If you cannot do the soil assessment check the relevant box and add a comment explaining why (SA09).

Avoid taking soil samples in the edge of fields (at least 5 m, ideally 10 m) or in between parcels.

If the "look to the North rule" applies, then move a maximum of 100 m distance from the LUCAS point.

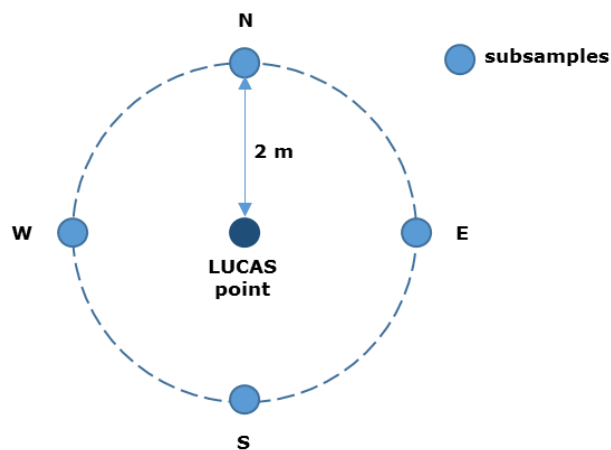
If it is not possible to take the soil sample in one LUCAS point due to inaccessibility problems (e.g. forbidden zone, wetland, impenetrable forest, barriers, access denied by landowner), then no soil sample will be taken for that point.



Take a photo SOIL_NoSample showing why the sample could not be taken.

In some cases, the soil being sampled is an organic soil or peat. In these situations, no mineral soil will be present and the boundary between litter and the soil is less evident. In these instances, if the mineral soil is not exposed within 30 cm of the surface then the entire organic layer (30 cm) can be sampled.

2 Take a composite sample of at least 500g of soil



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Figure 8-122.

A minimum of 500 gr ($\geq 0.5\text{kg}$) of topsoil should be collected.

The quantity of 500 g should be representative of a larger area. For this reason, the soil sample should be a composite one, resulting from the mixture of ideally 5 (at least 3) subsamples.

The first subsample will be taken in the LUCAS point, the other subsamples will be taken at a distance of 2 meters following the cardinal directions (North, East, South, West).

ATTENTION

Do not take soil samples in saturated peatland soils and organic-rich soils

In some cases, the soil being sampled will be an organic soil or peat. In these situations, no mineral soil will be present and the boundary between litter and the soil is less evident. In these instances, if the mineral soil is not exposed within 30 cm of the surface then the entire organic layer can be sampled (up to a maximum of 30 cm from the surface unless a hard surface such as bedrock is encountered).

Soil samples can only be taken in the mature surface layer of the peat soil under unsaturated conditions.

If the peat or the organic-rich soil is saturated with water do not take samples. We refer to a soil as water-saturated when it is inundated, and the surface is covered by water.

If you can stand without having your feet flooded (i.e. water does not come out to surface do to the pressure of your feet), then you can take the samples.



Take a photo SOIL_Other in case the sampling cannot be done due to saturation of peatland and organic-rich soils.



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Figure 8-123. Example of profile through a peat soil.

3 Remove residues and green vegetation from the surface

Remove vegetation residues, grass and litter, if any, from the surface using the spade and/or by hand. Some fine roots and brownish organic material from the upper part of the soil can remain in the sample as it is difficult to remove it completely.



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Figures 8-124. a



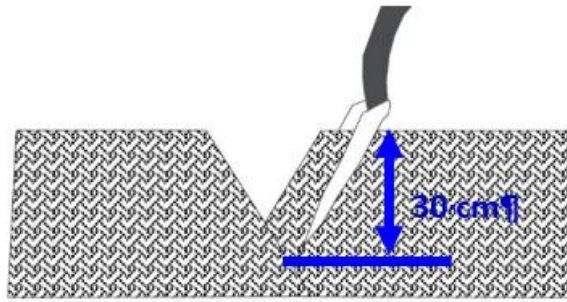
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Figures 8-124. b

4 Dig a V-shaped hole of 30-cm depth

Before beginning the sampling, make sure that the marks of depth increments are visible in the blade of the spade.

Use the spade to dig a (V-shaped) hole in the central point to a depth of 30 cm (the depth of the spade) as shown in the image.



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Figure 8-125.

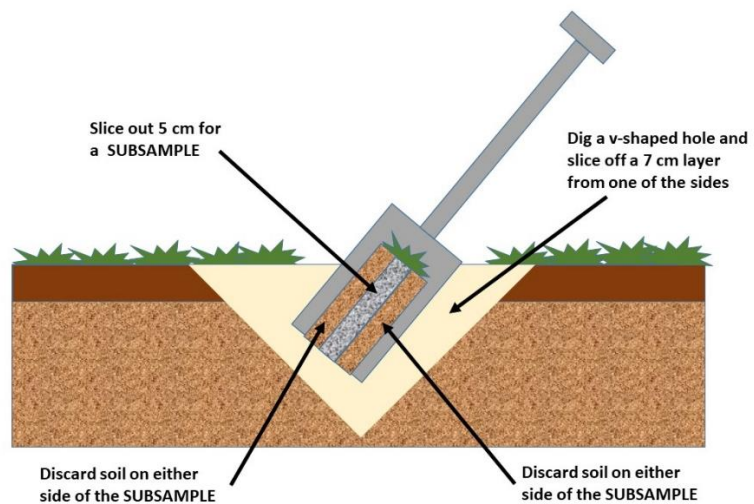
At some points, it may not be possible to sample to the nominated sampling depth (30 cm) due to the high density of roots (e.g. in grassland), presence of impermeable layers, stones or bedrock. In these cases, use your foot to help dig the hole. The actual sampling depth for each subsample taken must be recorded in the field form if the surveyor cannot dig to nominated depth (30 cm).



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Figure 8-126.

5 Cut a slice of soil, and trim the sides



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Figure 8-127.

Take a slice of soil with the spade. This slice is approximately 3 cm thick (a slightly thicker slice is also acceptable). Remove any remaining vegetation residues, stones, and litter.

Now trim the sides of the slice, leaving a 3 cm subsample on the spade.

6 Put soil in the bucket	© 2017, European Union, CC BY 4.0 Figure 8-128. Put the subsample in the bucket. Clean excess soil from the spade.
7 Repeat the process at least in 2 out of the 4 cardinal points (N, E, S, W)	The soil sample has to be a composite one, resulting from the mixture of ideally 5 (at least 3) subsamples. The first subsample is taken at the LUCAS point while the other subsamples are taken at a distance of 2 meters following the cardinal directions (North, East, South, West). From the central point, walk 2 meters to the North, take a subsample following steps 3 to 6 and put the subsample to the bucket. Clean excess soil from the spade and walk back to the central point. From the central point, walk 2 meters to the East, take a subsample following steps 3 to 6 and put the subsample to the bucket. Clean excess soil from the spade and walk back to the central point. From the central point, walk 2 meters to the South, take a subsample following steps 3 to 6, and put the subsample to the bucket, clean excess soil from the spade and walk back to the central point. From the central point, walk 2 meters to the West, take a subsample following steps 3 to 6 and put the subsample to the bucket. Clean excess soil from the spade and walk back to the central point.
8 Use the trowel to mix the sub-samples in the bucket	Use the trowel to mix the subsamples together. By mixing the subsamples you get the required composite soil sample. Also clods and lumps are broken up. Remove any extra vegetation residues, litter, and large stones (>6 cm) by hand.
9 Take at least 500g of soil and put it in the smaller plastic bag. Tag it and tie it	© 2017, European Union, CC BY 4.0 Figure 8-129. Take a minimum of 500 grams (about 5 heaped trowels full) of soil; place it in the plastic bag (no need to weigh the bag – a little more sample is not a problem). Put the first available plastic label with a progressive number and an “A” sign printed on it inside this bag (white label). Close the bag using a tie.
10 500g of dry soil are needed for the analysis	In case the soil is very wet, take more than 500 g to compensate for weight loss during the drying process.

11 Put the smaller plastic bag into the larger one. Tag again and tie it	 © 2017, European Union, CC BY 4.0 Figure 8-130. Put the plastic bag in a second one. Put a label with the same progressive number as in step 9, but with a “B” sign on it inside the outer bag (white label) facing outwards so that the bar code is visible. MAKE SURE THAT THE NUMBER ON THE ‘A’ AND ‘B’ LABELS MATCH. Close the bag using a tie. This is the sample to be shipped to the national coordinator.
12 Register the label number in the Field Form	Record the progressive label number in the appropriate Field Form.
13 Take the soil photo (LUCAS point and bag)	Put the bag next to the LUCAS point.  Take a photo of the LUCAS point with the bag (keep the label visible) and as many as possible of the 5 holes clearly visible: SOIL_BagHoles
14 Take additional photo of the label	 Take a photo of the label: SOIL_StdLabel. The label has to be clearly readable.
15 Complete the questionnaire in the Field Form	Complete the questions on the Field Form related to soil.
16 Percentage of surface covers (crops, residues, stones) is assessed on the 2 m radius area that serves as base for soil sampling	The surveyor has to estimate the percentage of the soil surface that is covered with standing crop, crop residues and stones. This assessment is made on the 2 m radius area that serves as base for the soil sampling.
17 Take additional pictures if needed	 If appropriate take any additional photo SOIL_NoSub, SOIL_Other (e.g. to justify that a subsample was not taken due to excessive amount of thorny shrubs).

18 Make any necessary comments	 If the procedure was not followed strictly, report any changes in the comments of the Field Form, explaining the reason (e.g. superficial sample as underground rock, hard due to tree roots). Add any other comment as needed. Possible comments: SA01, SA02, SA04, SA05, SA08, SA99 For free text comments the use of English is mandatory. Special characters are to be avoided.
19 If applicable take note of the land owner contacts	If the land owner/land user asks to obtain a copy of the results of the soil analysis, include his/her contact details (preferably e-mail address) in the Field Form.

Add a comment if relevant.

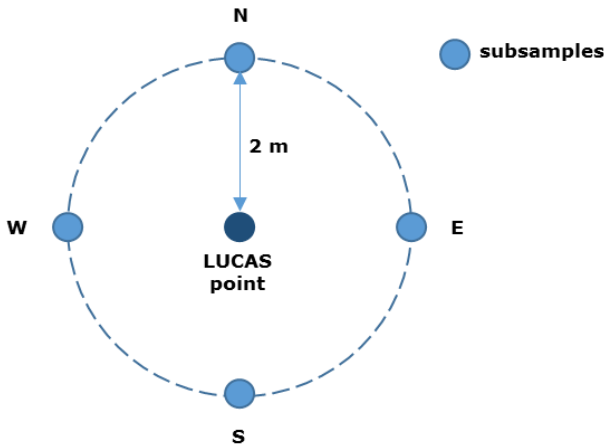


Before leaving the point

<i>Fill in the holes</i>	Using the excavated and excess soil, close the holes that you have dug.
<i>Clean and store the material</i>	Remove by hand the soil that could remain on the spade, the trowel and the bucket from the former sample. This is important in order to prevent cross-contamination.
<i>Don't forget to take all material and samples with you</i>	Be aware that you do not forget material or samples taken on the field before leaving the point. Put the closed bag in the box used for transporting the samples and put the box in the trunk of the car.

8.18.4.3 Sampling in woodland points

1 The sample is taken only if the point is reached!	Go to one of the LUCAS points of the list provided. Ideal situation: you reach the exact point, so the samples are taken there. BUT if you are unable to reach the exact LUCAS point, then: - The soil samples should be taken within a maximum of 100 m distance from the LUCAS point, preferably in the same field where the LUCAS point is located. Soil samples will be taken always from the same land cover class as observed at the LUCAS point. - In case the LUCAS point cannot be reached (e.g. mature crops) and it is not possible to take the samples within a maximum of 100 m distance to it, the soil sample may be collected from the same field provided the land cover and use are the same as the target location, being as close as possible to the LUCAS point, maintaining the same topographic setting (eg same contour on a slope), but not exceeding a distance of 300 m. If you cannot do the soil assessment check the relevant box and add a comment explaining why (SA09). Avoid taking soil samples: — in the edge of or in between parcels
---	---

	— within 0.5–1 m of tree trunks and disturbed areas such as wind-thrown trees, animal burrows and paths. If "look to the North rule" applies, then move a maximum of 100 m distance from the LUCAS point. If it is not possible to take the soil sample in one LUCAS point due to inaccessibility problems (e.g. forbidden zone, wetland, impenetrable forest, barriers, access denied by landowner), then no soil sample will be taken for that point.  Take a photo SOIL_NoSample showing why the sample could not be taken.
2 <i>Take composite samples of the litter layer and the mineral topsoil separately</i>	 © European Union, CC BY 4.0 Figure 8-131. A sample of the litter layer and a sample of the mineral topsoil should be taken at each point. Both samples should be representative of a larger area. For this reason, each sample should be a composite one, resulting from the mixture of ideally 5 (at least 3) subsamples. Just in the case the litter sample is deep enough, it may be enough material from a single subsample. A minimum of 500 gr ($\geq 0.5\text{kg}$) of mineral topsoil should be collected. The first subsamples of the litter layer and the mineral topsoil will be taken in the LUCAS point, the other subsamples will be taken at a distance of 2 meters following the cardinal directions (North, East, South, West).
2b <i>Take composite samples in organic or organic-rich soils</i>	In some cases, the soil being sampled will be an organic soil or peat. In these situations, no mineral soil will be present and the boundary between litter and the soil is less evident. In these instances, if the mineral soil is not exposed within 30 cm of the surface then the entire organic layer can be sampled (up to a maximum of 30 cm from the surface unless a hard surface such as bedrock is encountered). Do not take the samples, if the soil is completely saturated with water specially in organic-rich soils. We refer to a soil as water-saturated when it is inundated, and the surface is covered by water. If you can stand without having your feet flooded (i.e. water does not come out to surface do to the pressure of your feet), then you can take the samples.



Take a photo SOIL_NoSample in case the sampling cannot be done due to saturation of peatland and organic-rich soils.

3 *Collect the litter layer in a big plastic bag and tag it*



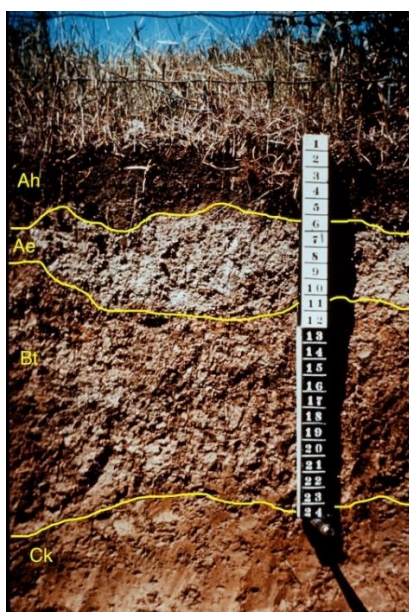
Take a photo of the surface of the litter layer; SOIL_Litter.

In woodlands, the mineral soil surface is normally covered by a litter layer (i.e. dead plant material), which consists of leaves (needles if the trees are conifers), pieces of bark, twigs, shoots and plant stems, in various stages of decay.



© Photo: Dr. Oliver van Straaten, University of Göttingen, Department of Soil Science of Tropical and Subtropical Ecosystems

Figure 8-132. Mineral soil with thin litter layer



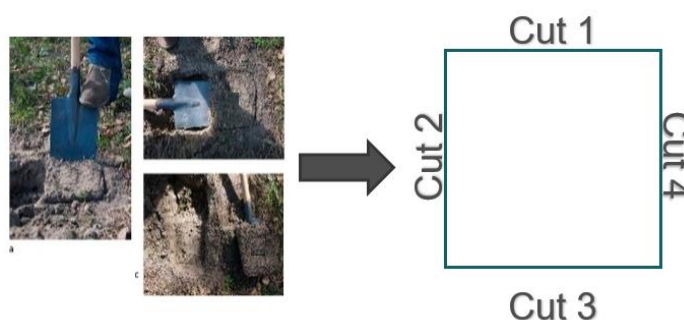
<https://soilsofcanada.ca/imagery-collection/luvisols.php#GrayBrownLuvisol>

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Figure 8-133. Mineral soil with thick litter layer

In some cases, the soil being sampled is an organic soil or peat. In these situations, no mineral soil will be present and the boundary between litter and the soil is less evident. In these instances, if the mineral soil is not exposed within 30 cm of the surface then the entire organic layer can be sampled.

Make 4 cuts with the spade down to 10 cm, avoid standing or stepping inside the square.



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Figure 8-134.

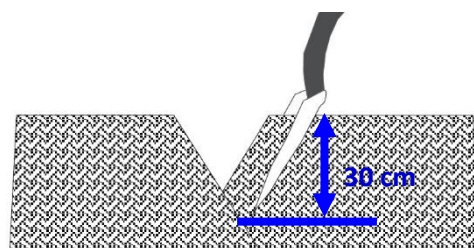
Take two or three handfuls of the the littler layer from within the square (including litter resting on the surface of the mineral soil) and place it in the big plastic bag tagged as “LITTER BAG”.

Carefully remove the remainder of the litter layer from the square to reveal the surface of the mineral soil.

4 Dig a V-shaped hole of 30-cm depth

Once the litter layer has been removed:

Use the spade to dig a (V-shaped) hole in the central point to a depth of 30 cm (the depth of the spade) as shown in the image.

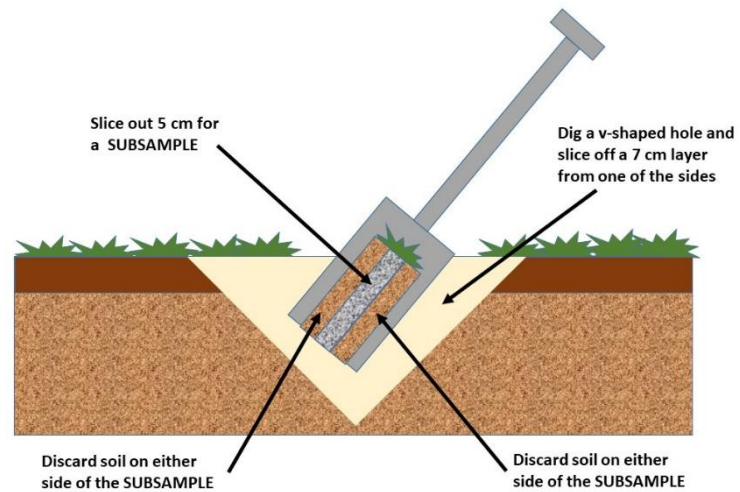


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Figure 8-135.

At some points, it may not be possible to sample to the nominated sampling depth (30 cm) due to the high density of roots, presence of impermeable layers, stones or bedrock. In this case, the actual sampling depth for each subsample taken must be recorded in the field form.

5 Cut a slice of soil, and trim the sides



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Figure 8-136.

Take a slice of soil with the spade. This slice is approximately 3 cm thick (a slightly thicker slice is also acceptable). Remove any remaining vegetation residues, stones, and litter.

Now trim the sides of the slice, leaving a 3 cm subsample on the spade.

6 Put soil in the bucket



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Figure 8-137.

Put the subsample in the bucket. Clean soil in excess from the spade.

7 Repeat the process at least in 2 out of the 4 cardinal points (N, E, S, W)

The samples of the litter layer and the mineral topsoil have to be composite, resulting from the mixture of ideally 5 (at least 3) subsamples.

The first subsample is taken at the LUCAS point then the other subsamples are taken at a distance of 2 meters following the cardinal directions (North, East, South, West).

From the central point, walk 2 meters to the North:

- take a subsample of the litter layer following step 3 and put it in the same plastic bag used in the central point
- take a subsample of the mineral topsoil following steps 4 to 6 and put the subsample to the bucket
- clean excess soil from the spade and walk back to the central point

Repeat the process walking 2 meters to the East, South and West from the central point.

8 Label and tie the plastic bag with the litter sample	Put the first white label with a progressive number and “LITTER” signed on it inside the bag. Close the bag using a tie.
9 Use the trowel to mix the sub-samples of mineral topsoil in the bucket	Use the trowel to mix the subsamples of mineral topsoil together. By mixing the subsamples you will get the required composite sample of the mineral topsoil. Also clods and lumps have to be broken up. Remove any extra vegetation residues, litter, and large stones (>6 cm) from the mineral soil sample.
10 Take at least 500g of soil and put it in the smaller plastic bag. Tag it and tie it	Five hundred grams of dry soil are needed for the analysis: — Take minimum 500 grams (about 5 heaped trowels full) of soil; place it in the plastic bag (no need to weigh the bag – a little more soil is not a problem). — In case the soil is too wet, take more than 500 g to compensate for weight loss during the drying process.  © 2017, European Union, CC BY 4.0 Figure 8-138. Put the white label with the same progressive number as for the bag of litter sample and an “A” sign printed on it inside this bag. Close the bag using a tie.
11 Put the smaller plastic bag into the larger one. Tag again and tie it	Put the plastic bag in a second one. Put the white label with the same progressive number as in step 10, but with a “B” sign on it inside the outer bag (white label) facing outwards so that the bar code is visible. MAKE SURE THAT THE NUMBER ON THE ‘A’ AND ‘B’ LABELS MATCH.  © 2017, European Union, CC BY 4.0 Figure 8-139. Close the bag using a tie. This is the sample to be shipped.
12 Register the label number in the Field Form	Record the progressive label number in the appropriate Field Form.

13 Take the soil photo (LUCAS point and bags)	 Put the bags of the litter and of the standard soil sample next to the LUCAS point. Take a photo of the LUCAS point with the bags (keep the labels visible) and as many as possible of the 5 holes clearly visible: SOIL_BagHoles
14 Take additional photo of the labels	 The labels of the bags of the litter sample and the standard sample have to be clearly readable.
15 Complete the questionnaire in the Field Form	Complete the questions on the Field Form related to soil.
16 Percentage of surface covers and stones is assessed on the 2 m radius area that serves as base for soil sampling	The surveyor has to estimate the percentage of the soil surface that is covered with standing crop, crop residues and stones. This assessment is made on the 2 m radius area that serves as base for the soil sampling. The presence of stones within the soil does not need to be estimated. As explained above, stones should be taken out of the subsamples before making a composite sample.
17 Take additional pictures if needed	 If appropriate take any additional pictures (e.g. to justify that a subsample was not taken due to excessive amount of thorny shrubs): SOIL_NoSub, SOIL_Other
18 Make any necessary comments	 If the procedure was not followed strictly, report any changes in the comments of the Field Form, explaining the reason (e.g. superficial sample as underground rock, hard due to tree roots). Add any other comment as needed. Possible comments are: SA01, SA02, SA04, SA05, SA08, SA99 For free text comments the use of English is mandatory. Special characters are to be avoided.
19 If applicable take note of the land owner contacts	If the land owner/land user asks to obtain a copy of the results of the soil analysis, include his/her contact details (preferably e-mail address) in the Field Form.



Before leaving the point

<i>Fill in the holes</i>	Using the excavated and excess soil, close the holes you have dug.
<i>Clean and store the material</i>	Remove by hand the soil that could remain on the spade, the trowel and the bucket from the former sample. This is important in order to prevent cross-contamination.

8.18.4.4 Air drying and shipping the standard soil sample

Before the shipment to the JRC, samples of litter layer and mineral topsoil should be air-dried. Wet or damp samples may give unreliable results, if inappropriately stored.

1 <i>Air-drying samples</i>	Air-dry samples of litter layer and mineral topsoil at the end of the day's work, when you have reached your night's location. Roll down the sides of both bags and open as much as possible to dry soil samples by exposure to the air in a well-ventilated room. <u>Do not</u> use an oven or exposure to heat source to dry the sample!
2 <i>Handle samples with care</i>	Make sure that samples do not fall over and contaminate other samples.
3 <i>Make sure that samples are well air-dried</i>	Crumble samples of mineral topsoil into small clods/peds by hand. Let samples stay to dry as needed and inspect daily to see whether samples have already dried. During air-drying, samples will become gradually lighter. At a certain point, no more moisture will evaporate from samples and their weight will be constant. This can take several days.
4 <i>Close the bags carefully</i>	Check that the labels are still inside the bags. Close the bags with their ties. Make sure all bags are properly closed in order to prevent contamination during transport when the boxes may not be kept upright all the time.
5 <i>Ship samples to the supervisor</i>	When samples from 5 to 10 samples points are taken, they should be shipped to the supervisor (e.g. via mail or by car). Tape the sides of the box for extra security. Use a new box for storing the next batch of samples.

8.18.5 Core sampling for points where bulk density, as well as physical and chemical properties, will be measured (with or without assessment of biodiversity)

This manual describes the procedure to sample soil cores for measuring bulk density, as well as physical and chemical properties, in a maximum of 4,000 points under cropland, grassland, and peatland (section 3.3.1) and woodland (section 3.3.2). Unlike the standard sampling, surveyors need to take separate soil samples from the 0-10 cm, 10-20 cm, and 20-30 cm depth increments in cropland, grassland, and peatland points unless a physical

barrier is met (e.g. hard rock, gravel layers, indurated horizon). In woodland points, surveyors need to take a sample of the litter layer and separate samples from the 0-10 cm, 10-20 cm, and 20-30 cm depth increments unless a physical barrier is met. Moreover, surveyors will take soil samples from 0 to 30-cm depth with the spade for analysing biological properties in a maximum of 2,000 out of the 4,000 points. Samples for analysing biological properties have to be kept fresh.

Examples of shallow soils due to physical barriers (e.g. hard rock, gravel layers, indurated horizon)



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Figure 8-140.



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Figure 8-141.

https://esdac.jrc.ec.europa.eu/public_path/shared_folder/projects/DIS4ME/indicator_descriptions/indicator_description_images/soil_depth_1.jpg

https://content.ces.ncsu.edu/media/images/bedrock_6nnEvKq.jpg



U.S. Department of Agriculture

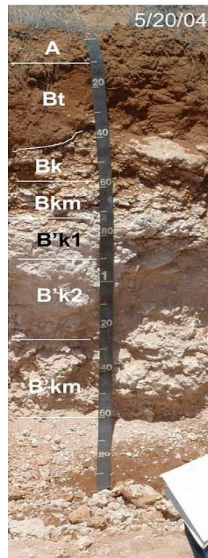
Figure 8-142.



U.S. Department of Agriculture

Figure 8-143.

https://www.nrcs.usda.gov/wps/portal/nrcs/detail/soils/ref/?cid=nrcs142p2_054253



2004, Dr. H. Curtis Monger, CC BY-SA 3.0

https://commons.wikimedia.org/wiki/File:Jornada_G-IBPE_soil_pit_horizons.jpg

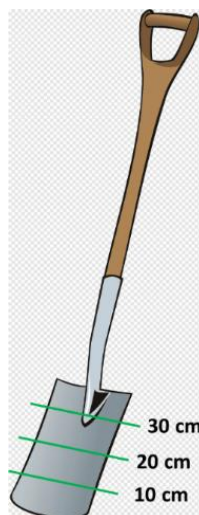
Figure 8-144.

Surveyors will receive from their coordinators the list of the specific points where they have to take soil cores and in which biological properties will be analysed.

Further aspects described concern measures to be taken before leaving the point and instructions for drying and shipping the sample.

8.18.5.1 Before going to the field

Surveyors need to mark in the blade of the spade the depth increments of 10, 20 and 30 cm. At each LUCAS point, before beginning the sampling, surveyors have to ensure that the marks are visible. If they are not visible, surveyors need to do the marks again before starting the sampling.



Picture source: <https://creazilla.com/nodes/28433-spade-clipart>

CC0 1.0 Universal

Figure 8-145. Mark the spade

For cropland, grassland and wetland points, surveyors will need to tag 3 small plastic bags per point with the following labels: Bag A 0-10 cm, Bag B 10-20 cm, and Bag C 20-30 cm.

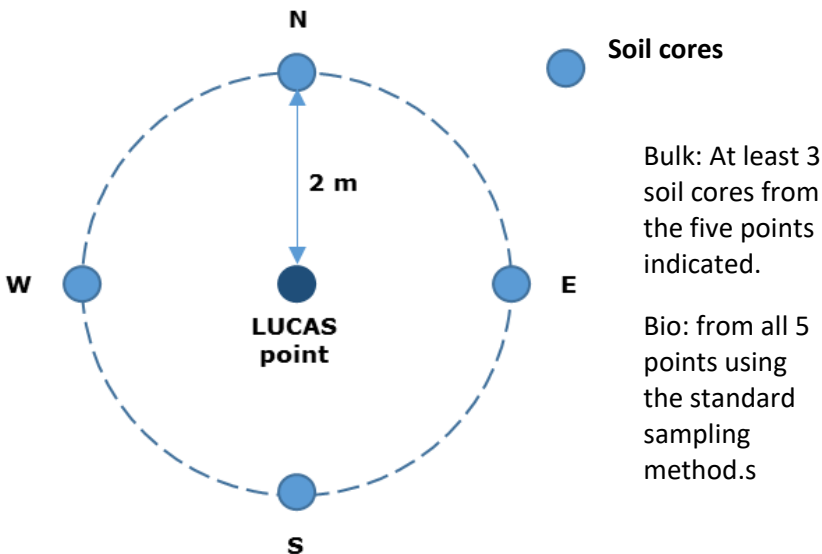
For woodland points, surveyors will need to tag 1 big plastic bag with the label “LITTER BAG” and 2 small plastic bags with the following labels: Bag A 0-10 cm, Bag B 10-20 cm, and Bag C 20-30 cm.

For those points where biological properties will be analysed, surveyors will need to:

1. At least 24/48 hours before the sampling, put the refrigerant/freezer packs that you will need for a sampling day into a freezer (remember that you will need 1 pack per point).
2. the morning before going to the field:
 - 2.1. place the refrigerant/freezer packs that you will need for the day into a polystyrene box/cool box.
 - 2.2. fill up the water bottles that you will need in the day (a bottle of water per point).

8.18.5.2 Sampling in cropland, grassland, and peatland points

1 <i>Samples are taken only if the point is reached!</i>	Go to one of the LUCAS points of the list provided. Ideal situation: you reach the exact point, so the samples are taken there. BUT if you are unable to reach the exact LUCAS point, then: - The soil samples should be taken within a maximum of 100 m distance from the LUCAS point, preferably in the same field where the LUCAS point is located. Soil samples will be taken always from the same land cover class as observed at the LUCAS point. - In case the LUCAS point cannot be reached (e.g. mature crops) and it is not possible to take the samples within a maximum of 100 m distance to it, the soil sample may be collected from the same field provided the land cover and use are the same as the target location, being as close as possible to the LUCAS point, maintaining the same topographic setting (eg same contour on a slope), but not exceeding a distance of 300 m. If you cannot do the soil assessment check the relevant box and add a comment explaining why (SA09). Avoid taking soil samples in the edge of or in between parcels. If "look to the North rule" applies, then move a maximum of 100 m distance from the LUCAS point. If it is not possible to take the soil sample in one LUCAS point due to inaccessibility problems (e.g. forbidden zone, wetland, impenetrable forest, barriers, access denied by landowner), then no soil sample will be taken for that point. Take a photo SOIL_NoSample showing why the sample could not be taken.
2 <i>Take composite samples</i>	For measuring bulk density, samples have to be taken with a metallic ring from 3 depth increments at each point: from 0–10 cm, 10-20 cm, and 20–30 cm. For assessing biodiversity, a single sample have to be taken from 0-30 cm depth with the spade. All samples should be representative of a larger area than the LUCAS point. For this reason, each sample should be a composite one, resulting from the mixture of: — Best case 5, ideally 4, at least 3 subsamples for measuring bulk density — 5 subsamples for biodiversity assessment Subsamples have to be taken based in the following sampling design. Note that a minimum of 500 g of soil should be taken in all cases.

	 Soil cores Bulk: At least 3 soil cores from the five points indicated. Bio: from all 5 points using the standard sampling method.s © European Union, CC BY 4.0 Figure 8-146. The first subsamples (both for measuring bulk density and assessing biodiversity) will be taken in the LUCAS point; the other subsamples will be taken at a distance of 2 meters following the cardinal directions (North, East, South, West). If, for any reason beyond your control, you cannot take one of the cores, indicate the reason and the number of subsamples finally taken by depth in the Field Form (see related questions in the Field Form).
ATTENTION <i>Do not take soil cores in saturated peatland soils and organic-rich soils</i>	In some cases, the soil being sampled will be an organic soil or peat. In these situations, no mineral soil will be present and the boundary between litter and the soil is less evident. In these instances, if the mineral soil is not exposed within 30 cm of the surface then the entire organic layer can be sampled (up to a maximum of 30 cm from the surface unless a hard surface such as bedrock is encountered). Soil samples can only be taken in the mature surface layer of the peat soil under unsaturated conditions. If the peat or the organic-rich soil is saturated with water do not take samples. We refer to a soil as water-saturated when it is inundated, and the surface is covered by water. If you can stand without having your feet flooded (i.e. water does not come out to surface do to the pressure of your feet), then you can take the samples.  Take a photo SOIL_NoSample in case the sampling cannot be done due to saturation of peatland and organic-rich soils.
Move to step 3 if you need to take a sample for analysing biological properties. Move to step 5 if you do not need to take a sample for analysing biological properties.	
3 <i>Use gloves if a sample for analysing biological properties will be taken</i>	Wear plastic gloves if you are in one of the 2,000 points where a sample for analysing biological properties will be taken.

4 Clean the sampling material	Ensure that the material for the sampling (spades, bucket) do not have soil residues from previous sampling. Firstly, clean the material using water to remove soil from previous sampling. Secondly, <u>clean the material using alcohol. This is especially important to prevent cross-contamination between samples.</u> Collect the material used to clean the equipment in a garbage bag. Do not forget to take the bag with you when leaving the point.
5 Remove residues and green vegetation from the surface	Remove vegetation residues, grass and litter, if any, from the surface using the spade and/or by hand. Some fine roots and brownish organic material from the upper part of the soil can remain in the sample, as it is difficult to remove it completely.  © 2017, European Union, CC BY 4.0  © 2017, European Union, CC BY 4.0 Figures 8-147. a Figures 8-147. b
6 Drive the ring into soil	Before beginning the sampling, make sure that the marks of depth increments are visible in the blade of the spade. Gently press the metallic ring, bevelled edge down, into the soil so that the top of the ring is flush with the soil surface. If needed, this can be done with the help of a mallet. In this case, place a block of wood on top of the metal ring and tap the wood with the mallet in order to avoid compaction of soil and protect the ring. Avoid pushing the ring in too far as the soil will compact.  © 2017, European Union, CC BY 4.0  © 2017, European Union, CC BY 4.0 Figure 8-148. a Figure 8-148. b Excavate a little bit around the outside edge of the ring (a knife can be useful for this). When the about 1 cm of the ring is exposed, put one of the plastic caps on the ring, as shown below.



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Figure 8-149. a



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Figure 8-149. b

7 Dig a square hole to a depth of 30 cm at a distance of 2-3 cm from the metallic ring

Dig a hole at a distance of 2-3 cm from the metallic ring.

Make 4 cuts with the spade to dig the hole, avoiding to stand/step where the ring is hammed.



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Figure 8-150. a



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Figure 8-150. b

Extract the resulting clump of soil with the spade. Take care that the hole remains clean.



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Figure 8-151.

	Keep the extracted soil close to the hole. Once the cores are taken, the extracted soil will be use later to close the hole.	
Move to step 8 if you need to take a sample for analysing biological properties.		
Move to step 10 if you do not need to take a sample for analysing biological properties.		
8 <i>Cut a slice of soil and trim the sides if a sample for analysing biological properties is needed</i>	Take a slice of soil to a DEPTH OF 20 CM with the spade from the face of the hole opposite to the ring. Remove any remaining vegetation residues, stones, and litter.	
		
	©2017, European Union, CC BY 4.0	©2017, European Union, CC BY 4.0
	Figure 8-152. a	Figure 8-152. b
	Trim the sides of the slice, leaving a 3 cm segment on the spade. Take care that the hole remains clean and that the trimmed material does not end up again in the hole.	
		
	© 2017, European Union, CC BY 4.0	© 2017, European Union, CC BY 4.0
	Figure 8-153. a	Figure 8-153. b
9 <i>Put collected soil in the bucket</i>	Put collected soil in the bucket. Clean soil residues from the spade.	



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Figure 8-154.

10 Remove the metallic ring

Make 3 cuts with the spade in the remaining sides around the ring, at a distance of 2-3 cm from the metallic ring.



8-155 a



b



c

Figure 8-155. a, b and c © 2017, European Union, CC BY 4.

Place the spade (or the trowel) underneath the ring in the fourth side and carefully remove the clump of soil with the ring intact.

Remove excess soil from the bottom of the ring and around the outside edge of the ring with the knife.



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Figure 8-156. a



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Figure 8-156. b



If the metallic ring is not completely filled with soil (less than 10% missing due to a stone or lost material when removing from soil), you can fill the ring with removed soil.



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Figure 8-157. a



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Figure 8-157. b



If the missing volume in the ring is greater, discard the soil core and take other on one of the other faces of the hole following steps 5, 6 and 10.

11 Put the soil core in the plastic bag tagged as “Bag A 0-10 cm”

Push out the soil in the core into the small plastic bag labelled as “Bag A 0-10 cm” using the knife.

Make sure the entire core is placed in the plastic bag.

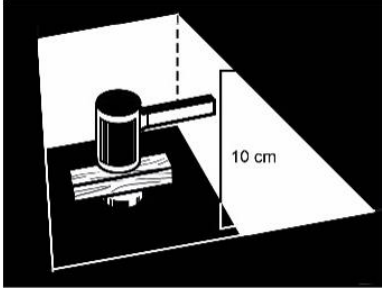


Figure 8-158 a



Figure 8-158 b

Figure 8-158. a and b © 2017, European Union, CC BY 4.0

12 Drive the ring into soil from 10-20 cm depth increment	Measure the depth of the hole where the first soil core was taken. If the depth of the hole is less than 10 cm, dig to reach at least 10 cm depth. Gently press the metallic ring, bevelled edge down, into the soil so that the top of the ring is flush with the soil surface at a depth of 10 cm. If needed, this can be done with the help of a mallet. In this case, place a block of wood on top of the metal ring and tap the wood with the mallet in order to avoid compaction of soil and protect the ring. As in the previous sample, excavate a little bit around the outside edge of the ring with a small, flat-bladed knife. Put the plastic cap on the ring.  Figure 8-159. a  Figure 8-159. b  Figure 8-159. c  Figure 8-159. d
13 Remove the metallic ring	Excavate more around the outside edge of the ring with a small, flat-bladed knife. Place the spade (or the trowel) underneath the ring and carefully remove it intact. Figure 8-126. a and b © 2017, European Union, CC BY 4.0  © 2017, European Union, CC BY 4.0 Figure 8-160 a  © 2017, European Union, CC BY 4.0 Figure 8-160 b Remove excess soil from the bottom of the ring and around the outside edge of the ring with the knife.



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Figure 8-161.



If the metallic ring is not completely filled with soil (less than 10% missing due to a stone or lost material when removing from soil), you can fill the ring with removed soil.



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Figure 8-162. a



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Figure 8-162. b



If the missing volume in the ring is greater, discard the soil core and take other on the opposite face of the hole. Dig at least to 10 cm depth and follow steps 12-13.

14 Put the soil core in the small plastic bag tagged as "Bag B 10-20 cm"

Push out the soil in the core into the small plastic bag labelled as "Bag B 10-20 cm" using the knife. Make sure the entire core is placed in the plastic bag.



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Figure 8-163. a

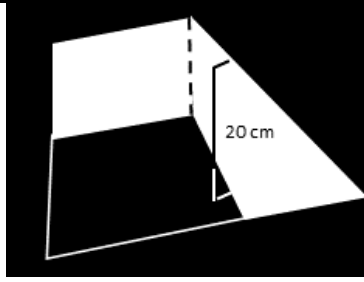


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Figure 8-163. b

15 Drive the ring into soil from 20-30-cm depth increment

Make sure that the depth of the hole, where the previous two soil cores were taken, is at least 20 cm.



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Figure 8-164. a



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Figure 8-164. b

Gently press the metallic ring, bevelled edge down, into the soil with the help of a mallet. Use a block of wood to push the ring with the mallet, in order to avoid compaction of soil and protect the ring. Avoid pushing the ring in too far or the soil will compact.

Excavate a little bit around the outside edge of the ring with a small, flat-bladed knife. Put the plastic cap to the ring.



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Figure 8-165. a



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Figure 8-165. b

16 Remove the metallic ring

Excavate more around the outside edge of the ring with a small, flat-bladed knife. Place the spade (or the trowel) underneath the ring and carefully remove it intact.



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Figure 8-166 a



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Figure 8-166 b

Remove excess soil from the bottom of the ring and around the outside edge of the ring with the knife.



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Figure 8-167



If the metallic ring is not completely filled with soil (less than 10% missing due to a stone or lost material when removing from soil), you can fill the ring with removed soil.



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Figure 8-168. a



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Figure 8-168. b



If the missing part in the ring is greater, discard the soil core and take other following steps 15-16.

17 Put the soil core in the small plastic bag tagged as "Bag C 20-30 cm"

Push out the soil in the core into the small plastic bag labelled as "Bag C 20-30 cm" using the knife. Make sure the entire core is placed in the plastic bag.

18 Repeat the process to take subsamples at the 4 cardinal directions (N, E, S, W)

Remember that at the best case 5, ideally 4 but at least 3 subsamples are needed for measuring bulk density, and 5 subsamples for assessing biodiversity.

From the central LUCAS point, walk 2 m to the North and repeat steps 5 to 17. When finished, clean the excess soil from the spade and walk back to the LUCAS point.

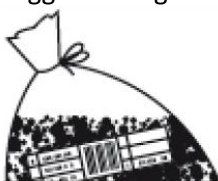
From the central LUCAS point, walk 2 m to the East and repeat steps 5 to 17. When finished, clean the excess soil from the spade and walk back to the LUCAS point.

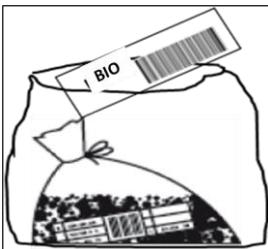
If time and conditions allow (optional)

From the central LUCAS point, walk 2 m to the South and repeat steps 5 to 17. When finished, clean the excess soil from the spade and walk back to the LUCAS point.

From the central LUCAS point, walk 2 m to the West and repeat steps 5 to 17. When finished, clean the excess soil from the spade and walk back to the LUCAS point.

If in any of the locations you can only collect the sample for biodiversity, then

	remove the surface residues and vegetation (step 5), dig a V-shaped hole to 20 cm, cut a slice of soil and trim the sides (step 8), and put the subsample in the bucket (step 9). Be aware that you take all material and samples with you when you move to a new location.
19 Label the small plastic bag (Bag A 0-10 cm) containing soil cores from 0-10-cm depth	Insert the first available orange plastic label with a progressive number and “BULK 0-10” sign printed on it inside the small bag (Bag A 0-10 cm). Close the bag with a tie.  European Union, CC BY 4.0 Figure 8-169.
20 Label the small plastic bag (Bag B 10-20 cm) containing soil cores from 10-20-cm depth	Insert the next available orange plastic label with the same progressive number as in step 19 and “BULK 10-20” sign printed on it inside the small bag tagged as Bag B 10-20 cm. Close the bag with a tie.  European Union, CC BY 4.0 Figure 8-170.
21 Label the small plastic bag (Bag C 20-30 cm) containing soil cores from 20-30 cm depth	Insert the next available orange plastic label with the same progressive number as in steps 19 and 20, and with “BULK 20-30” sign printed on it inside the small bag tagged as Bag C 20-30 cm. Close the bag with a tie.  European Union, CC BY 4.0 Figure 8-171.
22 Place the samples from the 3 depth increments (0-10, 10-20, and 20-30 cm depths) in the big plastic bag and label it	Put the small plastic bags containing the samples from the 3 depth increments (0-10, 10-20, and 20-30 cm depths) inside a big plastic bag. Insert the orange plastic label with the same progressive number as in steps 19, 20 and 21 and with “BULK” sign printed on it inside the big bag facing outwards so that the barcode is visible. MAKE SURE THAT THE NUMBER ON THE LABELS MATCH.  European Union, CC BY 4.0 Figure 8-172.
23 Move again to the central LUCAS point	Be aware that you take all material and samples with you when you move to a new location.
Move to step 24 if you need to take a sample for analysing biological properties. Move to step 28 if you do not need to take a sample for analysing biological properties.	

24 Mix the subsamples in the bucket if a sample for analysing biological properties have been taken	Use the trowel to mix subsamples in the bucket and to get a homogeneous composite sample. Clods and lumps should be broken up. Remove any extra vegetation residues, large stones (>6 cm), and litter.  ©European Union, CC BY 4.0 Figure 8-173.
25 Transfer 500 g of soil to the transparent plastic bag and label it	Take 500 g (about 5 handfuls) of the composite sample and place it in the small plastic bag. Insert the first available green plastic label with the same progressive number as in steps 19 to 22 and with “BIO A” sign printed on it in the plastic bag. Close the bag with the tie.  European Union, CC BY 4.0 Figure 8-174. Put the small plastic bag containing the sample inside the other plastic bag. Insert the green plastic label with the same progressive number and with “BIO B” sign printed on it inside the second bag facing outwards so that the bar code is visible. MAKE SURE THAT THE NUMBER ON THE ‘A’ AND ‘B’ LABELS MATCH. Close the bag with the tie.  European Union, CC BY 4.0 Figure 8-175.
26 Take a picture of the biodiversity sample with the LUCAS point	 Put the plastic bag next to the LUCAS points and take a picture of the point with the bag: SOIL_BagHoles_Bio. The green label in the bag should be clearly visible in the picture.

27 Take an additional photo of the label of the biodiversity sample	 Take a photo of the soil biodiversity label SOIL_BioLabel, the label should be clearly readable.
28 Place the bag with the sample inside the polystyrene box	Put the bag containing the sample inside the polystyrene box. Close the box. Box may contain up to 5 samples for soil biodiversity analysis and one freezer pack. This soil is taken for analysing biological properties. It should be shipped fresh to the country coordinator or JRC-Ispira depending on the shipping strategy adopted by the contractor.
29 Register the label numbers of the taken in the Field Form	Record the label numbers of the samples (core samples and biodiversity sample, if taken) in the appropriate section of the Field Form.
30 Take a photo of the LUCAS point with the bulk density sample soil photo	 Put the bag with core samples of the 3 depth increments next to the LUCAS point. Take a photo of the LUCAS point with the bag (keep the label visible) and as many as possible of the holes clearly visible: SOIL_BagHoles
31 Take additional photo of the label	 Take a photo of the bulk label SOIL_BulkLabel, the label should be clearly readable.
32 Complete the questionnaire on the Field Form	Complete the questions on the Field Form related to soil.
33 Percentage of surface covers and stones is assessed on the 2 m radius area that serves as base for soil sampling	The surveyor should estimate the percentage of the soil surface that is covered with standing crop, crop residues or biofilms and stones. This assessment is made on the 2 m radius area that serves as base for the soil sampling. The presence of stones within the soil does not need to be estimated. As explained above, stones should be taken out of the subsamples before making a composite sample.
34 Take additional pictures if needed	 If appropriate take any additional pictures (e.g. to justify that a soil core was not taken due to excessive amount of thorny shrubs): SOIL_Other.
35 Make any necessary comments	 If the procedure was not followed strictly, report any changes in the comments, explaining the reason (e.g. superficial sample as underground rock, hard due to tree roots). Add any other comment as needed.

	Possible comments: SA01, SA02, SA04, SA05, SA08, SB01, SB99+SC99 For free text comments the use of English is mandatory. Special characters are to be avoided.
36 <i>If applicable take note of the land owner contacts</i>	If the land owner/land user asks to obtain a copy of the results of the soil analysis, include his/her contact details (preferably e-mail address) in the Field Form.

Add a comment about the bulk density sample, if relevant.

Add a comment about the biodiversity sample, if relevant.



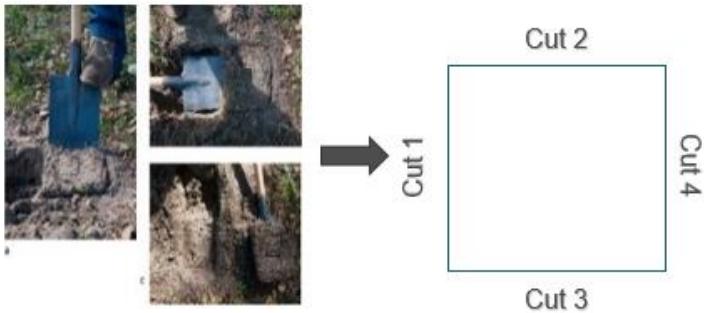
Before leaving the point

<i>Fill in the holes</i>	Using the excavated and excess soil, close the holes you have dug.
<i>Don't forget to take all material and samples with you</i>	Be aware that you do not forget material or samples taken on the field before leaving the point. Put the closed bag in the box used for shipping the samples and put the box in the trunk of the car.

8.18.5.3 Sampling in woodland points

1 <i>Samples are taken only if the point is reached!</i>	Go to one of the LUCAS points of the list provided. Ideal situation: you reach the exact point, so the samples are taken there. BUT if you are unable to reach the exact LUCAS point, then: - The soil samples should be taken within a maximum of 100 m distance from the LUCAS point, preferably in the same field where the LUCAS point is located. Soil samples will be taken always from the same land cover class as observed at the LUCAS point. - In case the LUCAS point cannot be reached (e.g. mature crops) and it is not possible to take the samples within a maximum of 100 m distance to it, the soil sample may be collected from the same field provided the land cover and use are the same as the target location, being as close as possible to the LUCAS point, maintaining the same topographic setting (eg same contour on a slope), but not exceeding a distance of 300 m. If you cannot do the soil assessment check the relevant box and add a comment explaining why (SA09). Avoid taking soil samples: — in the edge of or in between parcels — within 0.5–1 m of tree trunks and disturbed areas such as wind-thrown trees, animal burrows and paths If "look to the North rule" applies, then move a maximum of 100 m distance from the LUCAS point. If it is not possible to take the soil sample in one LUCAS point due to inaccessibility problems (e.g. forbidden zone, wetland, impenetrable forest, barriers, access denied by landowner), then no soil sample will be taken for that point.  Take a photo SOIL_NoSample showing why the sample could not be taken.
2 <i>Take composite samples</i>	For measuring bulk density, samples have to be taken with a metallic ring from 3 depth increments at each point: from 0–10 cm, 10–20 cm, and 20–30 cm. For assessing biodiversity, a single sample have to be taken from 0–30 cm depth with the spade. All samples should be representative of a larger area than the LUCAS point. For this reason, each sample should be a composite one, resulting from the mixture of: — Best case 5, ideally 4 (at least 3) subsamples for measuring bulk density — 5 subsamples for biodiversity assessment Subsamples have to be taken based in the following sampling design. Note that a minimum of 500 g of soil should be taken in all cases.

	© European Union, CC BY 4.0 Figure 8-176. The first subsamples of the litter layer and the mineral topsoil will be taken in the LUCAS point; the other subsamples will be taken at a distance of 2 meters following the cardinal directions (North, East, South, West). If, for any reason beyond your control, you cannot take subsamples in one of the locations, indicate the reason and the number of subsamples finally taken by depth in the Field Form (see related questions in the Field Form). Do not take the samples, if the soil is completely saturated with water specially in organic-rich soils. We refer to a soil as water-saturated when it is inundated, and the surface is covered by water. Take a photo SOIL_NoSample in case the sampling cannot be done due to saturation of peatland and organic-rich soils. Bulk: At least 3 soil cores from the five points indicated. Bio: from all 5 points using the standard sampling method.s
Move to step 3 if you need to take a sample for analysing biological properties.	
Move to step 5 if you do not need to take a sample for analysing biological properties.	
3 <i>Use gloves if a sample for analysing biological properties will be taken</i>	Wear plastic gloves if you are in one of the 2,000 points where a sample for analysing biological properties will be taken.
4 <i>Clean the sampling material</i>	Ensure that the material for the sampling (spades, bucket) do not have soil residues from previous sampling. Firstly, clean the material using water to remove soil from previous sampling. Secondly, <u>clean the material using alcohol. This is especially important to prevent cross-contamination between samples.</u> Collect the material used to clean the equipment in a garbage bag. Do not forget to take the bag with you when leaving the point.

5 <i>Collect the litter layer in the big plastic bag tagged as "LITTER BAG"</i>	Before beginning the sampling, make sure that the marks on depth increments are visible in the blade of the spade.  Take a photograph of the surface of the litter layer: SOIL_Litter Make 4 cuts with the spade down to 10 cm, avoiding standing or stepping inside the square.  © European Union, CC BY 4.0 Figure 8-177. The mineral soil surface is normally covered by a litter layer (i.e. dead plant material), which consists of leaves (needles if the trees are conifers), pieces of bark, twigs, shoots and plant stems, in various stages of decay. Take two or three handfuls of the litter layer from within the square (including material resting on the surface of the mineral soil) and place it in the big plastic bag tagged as "LITTER BAG". Carefully remove the litter layer from the square to reveal the surface of the mineral soil.
6 <i>Drive the ring into mineral topsoil</i>	Once the litter layer is taken: Gently press the metallic ring, bevelled edge down, into the mineral topsoil in the centre of the square designed in step 5 so that the top of the ring is flush with the soil surface. If needed, this can be done with the help of the mallet. In this case, place a block of wood on top of the ring and use the mallet to tap the ring into the soil, in order to avoid compaction of soil and protect the ring. Avoid pushing the ring in too far as the soil will compact.



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Figure 8-178. a



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Figure 8-178. b

Excavate a little bit around the outside edge of the ring (a knife can be useful for this). When the about 1 cm of the ring is exposed, put the plastic cap on the ring as shown below.



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Figure 8-179. a



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Figure 8-179. b

7 Dig a square hole to a depth of 30 cm at a distance of 2-3 cm from the metallic ring

Remove the litter layer from a wider area on one side of the metallic ring, using the spade and/or by hand.

Remove litter layer



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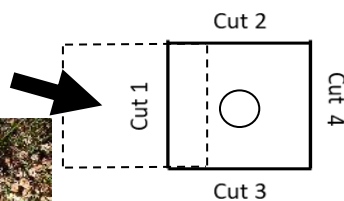


Figure 8-180.

Dig a hole at a distance of 2-3 cm from the metallic ring, where the litter layer has been removed.

Make 4 cuts with the spade to dig the hole, avoiding to stand/step where the ring is hammed.



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Figure 8-181. a



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Figure 8-181. b

Extract the resulting clump of soil with the spade. Take care that the hole remains clean.



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Figure 8-182.

Keep the extracted soil close to the hole. Once the cores are taken, the extracted soil will be use later to close the hole.

Move to step 8 if you need to take a sample for analysing biological properties.

Move to step 10 if you do not need to take a sample for analysing biological properties.

8 *Cut a slice of soil and trim the sides if a sample for analysing biological properties is needed*

Take a slice of soil with the spade from the face of the hole opposite to the ring. Remove any remaining vegetation residues, stones, and litter.

	 ©2017, European Union, CC BY 4.0  ©2017, European Union, CC BY 4.0 Figure 8-183. a Figure 8-183. b Trim the sides of the slice, leaving a 3 cm segment on the spade. Take care that the hole remains clean and that the trimmed material does not end up again in the hole.  © 2017, European Union, CC BY 4.0  © 2017, European Union, CC BY 4.0 Figure 8-184. a Figure 8-184. b
9 Put collected soil in the bucket	Put collected soil in the bucket. Clean soil residues from the spade.  © 2017, European Union, CC BY 4.0 Figure 8-185.
10 Remove the metallic ring	Make 3 cuts with the spade in the remaining sides around the ring, at a distance of 2-3 cm from the metallic ring. Place the spade (or the trowel) underneath the ring in the fourth side and carefully remove the clump of soil with the ring intact.



Figure 8-186. a



b



c

Figure 8-186. a, b and c © 2017, European Union, CC BY 4.

Remove excess soil from the bottom of the ring and around the outside edge of the ring with the knife.



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Figure 8-187. a



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Figure 8-187. b



If the metallic ring is not completely filled with soil (less than 10% missing due to a stone or lost material when removing from soil), you can fill the ring with removed soil.

	© 2017, European Union, CC BY 4.0 Figure 8-188. a © 2017, European Union, CC BY 4.0 Figure 8-188. b If the missing volume in the ring is greater, discard the soil core and take other on one of the other faces of the hole following steps 6 and 10.
11 Put the soil core in the plastic bag tagged as “Bag A 0-10 cm”	Push out the soil core into the small plastic bag labelled as “Bag A 0-10 cm” using the knife. Make sure the entire core is placed in the plastic bag. © European Union, CC BY 4.0 Figure 8-189. a © European Union, CC BY 4.0 Figure 8-189. b
12 Drive the ring into soil from 10-20 cm depth increment	Measure the depth of the hole where the first soil core was taken. If the depth of the hole is less than 10 cm, dig to reach at least 10 cm depth. Gently press the metallic ring, bevelled edge down, into the soil so that the top of the ring is flush with the soil surface at a depth of 10 cm. If needed, this can be done with the help of a mallet. In this case, place a block of wood on top of the metal ring and tap the wood with the mallet in order to avoid compaction of soil and protect the ring. As in the previous sample, excavate a little bit around the outside edge of the ring with a small, flat-bladed knife. Put the plastic cap to the ring. © European Union, CC BY 4.0 Figure 8-190. a © European Union, CC BY 4.0 Figure 8-190. b



Figure 8-190. c



Figure 8-190. d

13 Remove the metallic ring

Excavate more around the outside edge of the ring with a small, flat-bladed knife. Put the plastic cap on the ring.

Place the spade (or the trowel) underneath the ring and carefully remove it intact.



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Figure 8-191. a



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Figure 8-191. b

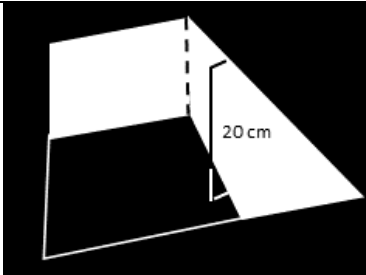
Remove excess soil from the bottom of the ring and around the outside edge of the ring with the knife.



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Figure 8-192.

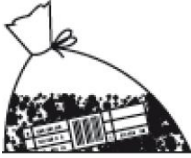


If the metallic ring is not completely filled with soil (less than 10% missing due to a stone or lost material when removing from soil), you can fill the ring with removed soil.

	 © 2017, European Union, CC BY 4.0 Figure 8-193. a	 © 2017, European Union, CC BY 4.0 Figure 8-193. b
14 Put the soil core in the small plastic bag tagged as “Bag B 10-20 cm”	Push out the soil in the core into the small plastic bag labelled as “Bag B 10-20 cm” using the knife. Make sure the entire core is placed in the plastic bag.  © European Union, CC BY 4.0 Figure 8-194. a  © European Union, CC BY 4.0 Figure 8-194. b	
15 Drive the ring into soil from 20-30-cm depth increment	Make sure that the depth of the hole, where the previous two soil cores were taken, is at least 20 cm.  © European Union, CC BY 4.0 Figure 8-195. a  © European Union, CC BY 4.0 Figure 8-195. b Gently press the metallic ring, bevelled edge down, into the soil with the help of a mallet. Use a block of wood to push the ring with the mallet, in order to avoid compaction of soil and protect the ring. Avoid pushing the ring in too far or the soil will compact. Excavate a little bit around the outside edge of the ring with a small, flat-bladed knife. Put the plastic cap to the ring.	

	 © European Union, CC BY 4.0 Figure 8-196. a	 © European Union, CC BY 4.0 Figure 8-196. b
16 Remove the metallic ring	Excavate more around the outside edge of the ring with a small, flat-bladed knife. Place the spade (or the trowel) underneath the ring and carefully remove it intact.  © European Union, CC BY 4.0 Figure 8-197. a  © European Union, CC BY 4.0 Figure 8-197. b Remove excess soil from the bottom of the ring and around the outside edge of the ring with the knife.  © European Union, CC BY 4.0 Figure 8-198.  If the metallic ring is not completely filled with soil (less than 10% missing due to a stone or lost material when removing from soil), you can fill the ring with removed soil.	

	 © 2017, European Union, CC BY 4.0 Figure 8-199. a	 © 2017, European Union, CC BY 4.0 Figure 8-199. b
	 If the missing part in the ring is greater, discard the soil core and take other following steps 15-16.	
17 Put the soil core in the small plastic bag tagged as “Bag C 20-30 cm”	Push out the soil in the core into the small plastic bag labelled as “Bag C 20-30 cm” using the knife. Make sure the entire core is placed in the plastic bag.	
18 Repeat the process at the 4 cardinal directions (N, E, S, W)	Remember that at the best case 5, ideally 4 but at least 3 subsamples are needed for measuring bulk density, and 5 subsamples for assessing biodiversity. From the central LUCAS point, walk 2 m to the North and repeat steps 5 to 17. When finished, clean the excess soil from the spade and walk back to the LUCAS point. From the central LUCAS point, walk 2 m to the East and repeat steps 5 to 17. When finished, clean the excess soil from the spade and walk back to the LUCAS point. If time and conditions allow (optional): From the central LUCAS point, walk 2 m to the South and repeat steps 5 to 17. When finished, clean the excess soil from the spade and walk back to the LUCAS point. From the central LUCAS point, walk 2 m to the West and repeat steps 5 to 17. When finished, clean the excess soil from the spade and walk back to the LUCAS point. If in any of the locations you can only collect the sample for biodiversity, then remove the surface residues and vegetation with the spade, dig a V-shaped hole to 20 cm, cut a slice of soil and trim the sides (step 8), and put the subsample in the bucket (step 9). Be aware that you take all material and samples with you when you move to a new location.	
19 Label the bag of the litter sample and tie it	Put the orange label with a progressive number and “LITTER” signed on it inside the bag. Close the bag using a tie.	

20 Label the small plastic bag (Bag A 0-10 cm) containing soil cores from 0-10-cm depth	Insert the first available orange plastic label with a progressive number and "BULK 0-10" sign printed on it inside the small bag (Bag A 0-10 cm). Close the bag with a tie.  © 2017, European Union, CC BY 4.0 Figure 8-200.
21 Label the small plastic bag (Bag B 10-20 cm) containing soil cores from 10-20-cm depth	Insert the next available orange plastic label with the same progressive number as in step 20 and "BULK 10-20" sign printed on it inside the small bag tagged as Bag B 10-20 cm. Close the bag with a tie.  © 2017, European Union, CC BY 4.0 Figure 8-201.
22 Label the small plastic bag (Bag C 20-30 cm) containing soil cores from 20-30 cm depth	Insert the next available orange plastic label with the same progressive number as in steps 20 and 21, and with "BULK 20-30" sign printed on it inside the small bag tagged as Bag C 20-30 cm. Close the bag with a tie.  © 2017, European Union, CC BY 4.0 Figure 8-202.
23 Place the samples from the 3 depth increments (0-10, 10-20, and 20-30 cm depths) in the big plastic bag and label it	Put the small plastic bags containing the samples from the 3 depth increments (0-10, 10-20, and 20-30 cm depths) inside a big plastic bag. Insert the orange plastic label with the same progressive number as in steps 20, 21 and 22 and with "BULK" sign printed on it inside the big bag facing outwards so that the barcode is visible. MAKE SURE THAT THE NUMBER ON THE LABELS MATCH.  © 2017, European Union, CC BY 4.0 Figure 8-203.
23 Move again to the central LUCAS point	Be aware that you take all material and samples with you when you move to a new location.

Move to step 24 if you need to take a sample for analysing biological properties.

Move to step 28 if you do not need to take a sample for analysing biological properties.

24 *Mix the subsamples in the bucket if a sample for analysing biological properties have been taken*

Use the trowel to mix subsamples in the bucket and to get a homogeneous composite sample. Clods and lumps should be broken up.

Remove any extra vegetation residues, stones, and litter.



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Figure 8-204.

25 *Transfer 500 g of soil to the transparent plastic bag and label it*

Take 500 g (about 5 handfuls) of the composite sample and place it in the small plastic bag.

Insert the first available **green plastic label** with the same progressive number as in steps 19 to 23 and with **"BIO A"** sign printed on it in the plastic bag.

Close the bag with the tie.



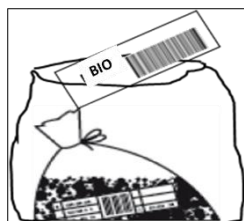
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Figure 8-205.

Put the small plastic bag containing the sample inside the other plastic bag. Insert the **green plastic label** with the same progressive number and with **"BIO B"** sign printed on it inside the second bag facing outwards so that the bar code is visible.

MAKE SURE THAT THE NUMBER ON THE 'A' AND 'B' LABELS MATCH.

Close the bag with the tie.



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Figure 8-206.

26 Take a photo of the biodiversity sample with the LUCAS point	 Put the plastic bag next to the LUCAS points and take a photo of the point with the bag: SOIL_BagHoles_Bio The green label in the bag should be clearly visible in the picture.
27 Take an additional photo of the biodiversity label	 Take a photo of the label: SOIL_BioLabel. The label should be clearly readable.
28 Place the bag with the sample inside the polystyrene box	Put the bag containing the sample inside the polystyrene box. Close the box. Box may contain up to 5 samples for soil biodiversity analysis and one freezer pack. This soil is taken for analysing biological properties. It should be kept cool, chilled and eventually shipped fresh to the country coordinator or JRC-Ispra depending on the shipping strategy adopted by the contractor.
29 Register the label numbers of the samples taken in the Field Form	Record the progressive label numbers of the samples taken (litter sample, core samples of the 3 depth increments, and biodiversity sample, if taken) in the appropriate section of the Field Form.
30 Take a photo of the LUCAS point with the bulk sample and the litter sample	 Put the bags next to the LUCAS point. Take a photo of the LUCAS point with the bags (keep the label visible) and as many as possible of the holes clearly visible: SOIL_BagHoles
31 Take additional photo of the labels of the bulk sample and the litter sample	 The label should be clearly readable: SOIL_BulkLabel
32 Complete the questionnaire on soil	Complete the questions on the Field Form related to soil.
33 Percentage of surface covers and stones is assessed on the 2 m radius area that serves as base for soil sampling	The surveyor should estimate the percentage of the soil surface that is covered with standing crop, crop residues or biofilms and stones. This assessment is made on the 2 m radius area that serves as base for the soil sampling. The presence of stones within the soil does not need to be estimated. As explained above, stones should be taken out of the subsamples before making a composite sample.

33 Take additional pictures if needed	 If appropriate take any additional pictures (e.g. to justify that a soil core was not taken due to excessive amount of thorny shrubs): SOIL_Other
34 Make any necessary comments	 If the procedure was not followed strictly, report any changes in the comments, explaining the reason (e.g. superficial sample as underground rock, hard due to tree roots). Add any other comment as needed. Possible comments: SA01, SA02, SA04, SA05, SA08, SB01, SB99+SC99 For free text comments the use of English is mandatory. Special characters are to be avoided.
35 If applicable take note of the land owner contacts	If the land owner/land user asks to obtain a copy of the results of the soil analysis, include his/her contact details (at least name, postal address and telephone number; if available, also e-mail address) in the Field Form.



Before leaving the point

<i>Close the holes</i>	Using the excavated and excess soil to close the holes you have dug.
<i>Don't forget to take all material and samples with you</i>	Be aware that you do not forget material or samples taken on the field before leaving the point. Put the closed bag in the box used for shipping the samples and put the box in the trunk of the car.

8.18.5.4 Shipment of fresh samples for the analysis of biological properties

1 Store the polystyrene box/cool box	Store the fresh sample in the polystyrene box/cool box until the end of the sampling day. At the end of the sampling day (within 10 hours of collection), , place the sample(s) in a fridge or freezer set to 4°C and keep it/them there until the shipment (samples can be grouped for shipment – we suggest a maximum of five) In situations where an overnight refrigerator is not available, the sample can be kept in the cool box by regularly replacing the ice pack with a new frozen version.
2 Prepare the polystyrene box for shipping	Surveyors can send a weekly batch of samples (maintaining the chilled status of the sample) by courier or express post using the supplied cool box to either a) the country coordination centre or b) the JRC. Place the plastic bags with the samples in the polystyrene box. Maximum 5 samples per box. Place 1 freezer pack (frozen before shipping) at the top/bottom. Close the box and seal it with the packaging tape to ensure the cover is secured. Attach the relevant address label on the box (JRC or national

	coordinator, as defined prior to the survey).
3 Ship the polystyrene box to the JRC-Ispira or national coordinator	The box must be shipped to the national coordinator or JRC-Ispira (depending on the shipping strategy adopted by the contractor) to arrive within 72 hours of shipment. Samples should be sent in such a way that they avoid spending the weekend in transit hubs. Opening hours of Customs at the JRC-Ispira: Monday to Thursday: 8.30-12.00 and 13.30-17.00 Friday: 8.30-12.00 and 13.30-16.00 The JRC will be unable to accept deliveries on April 15 & 18, May 9th, 26, June 2 & 6 and August 15 (including one working day either side of these dates). If sent to the national coordination centre, the box/sample must be either frozen immediately at -20°C or stored at 4°C for maximum two weeks, after which it must be stored at -20°C. Frozen samples can either a) be kept for the entire sampling period and can be sent in a single batch to the JRC at the end of the survey or b) sent to the JRC on a regular basis (e.g. monthly batches, maintaining the chilled status of the samples). Either the surveyor or contractor has to inform the JRC-Ispira of the shipment of each sample (requested details: sample ID, sampling date, shipping date, expected date arrival) through an <i>ad-hoc</i> online shared document (link will be provided by the JRC). If needed, the insulated box can be reused and returned to the surveyor after shipment to the JRC or national coordination centre (depending on the shipping strategy selected). If needed equivalent polystyrene boxes can be purchased.

8.18.5.5 Weighing and air-drying of the rest of the samples and shipping

Before shipping samples to the JRC-Ispira, the weight of samples must be recorded on the evening of collection (field-moist weight), and after air-drying (air-dry weight).

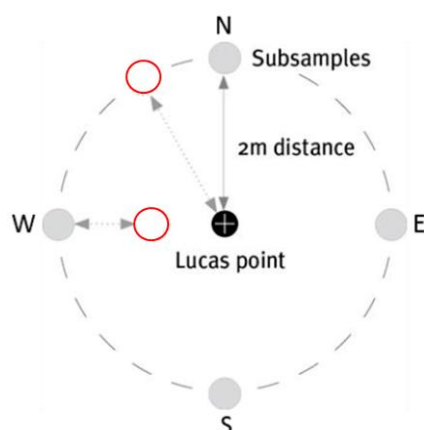
Do not air dry samples that have been taken for analysing biological properties! These samples need to be kept fresh and shipped to the JRC-Ispira refrigerated (see section 8.16.5.4).

1 Weigh soil samples from 0 to 20- and 20 to 30-cm depths	At the end of the day's work, when you have reached the night's location: Open the big plastic bag. Take out the small bags containing the samples from the big plastic bag. Open the small bags removing the ties. Weigh the samples within their bags. Record the weight of each sample (field-moist weight) in the Field Form.
--	--

2 <i>Air-dry samples</i>	Air-dry soil samples. Roll down the sides of bags and open them as much as possible to dry soil samples by exposure to the air in a well-ventilated room. <u>Do not</u> use an oven or exposure to heat source to dry the sample!
3 <i>Air-dry also the litter sample</i>	Open the bag removing the ties. Roll down the sides of bag and open them as much as possible to dry soil samples by exposure to the air in a well-ventilated room. <u>Do not</u> use an oven or exposure to heat source to dry the sample!
4 <i>Handle samples with care</i>	Make sure that samples do not fall over and contaminate other samples.
5 <i>Make sure that samples are well air-dried</i>	Crumble samples into small clods/peds by hand. Let the sample stay to dry as needed and inspect daily to see whether the samples have already dried. During air-drying, soil samples will become gradually lighter. At a certain point, no more moisture will evaporate from samples and the weight will be constant. This can take several days.
6 <i>Weigh air-dried samples of mineral topsoil</i>	Weigh air-dried samples of mineral topsoil within their bags. Record their weight (air-dried weight) in the Field Form.
7 <i>Close all plastic bags carefully</i>	For litter samples, close the bags containing the samples with the tie. Check that labels are still inside. For samples of mineral topsoil, close small bags containing samples with the ties and check that labels are still inside. Put them again in their corresponding big plastic bags. Make sure that the plastic label is still inside the big plastic bag. Close the bag with a tie. Make sure all bags are properly closed in order to prevent contamination during transport when the boxes may not be kept upright all the time.
8 <i>Ship the samples to the supervisor</i>	When 5 to 10 samples are taken (total weight around 2.5 kg to 5 kg), they should be shipped to the coordinator (e.g. via mail). Tape the sides of the box for extra security. Buy a new box that you will use for storing the samples to be taken until you will come again to a post office and that you will use for the next shipment.

8.18.6 Collecting the soil sample in special cases

If it is impossible to take the subsample in the appropriate location, find a suitable location within 2 m from the central point



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Figure 8-207.

If in one or more of the locations belonging to the LUCAS point (central, North, East, South or West) it is not possible to take a soil sample for any reason (accessibility problems, hardness of the ground, coverage – trees, houses etc.), walk along the dotted line until you find a suitable location or walk less than 2 m from central point to take the subsample (represented by the small red circles).

Remember that ideally 5 (at least 3) subsamples should be taken per LUCAS points.

A comment is needed: SA02, SA05, SA99, SB99, SC99

If a sample is very wet, treat it as a normal sample. It has to be air-dried in the same manner

What to do if the composite sample is very wet or comes from a peat bog? Treat these samples in the same way as any other sample unless it is inaccessible, or it is impossible to take soil.

Open the bag, roll down the sides and let the samples dry until you ship them.

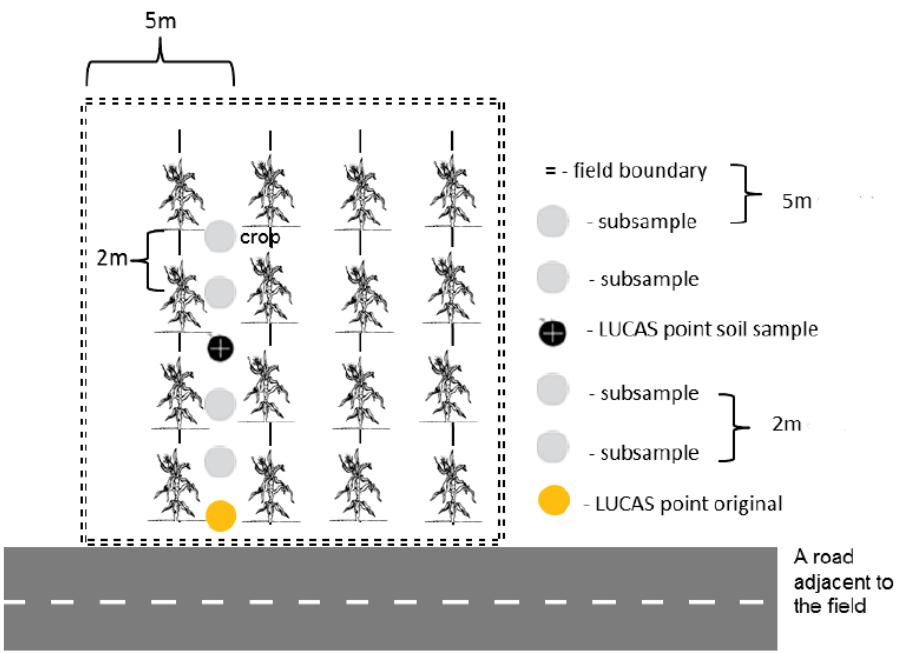
8.18.7 Collecting the soil sample in case of difficult access

If the access is difficult because of:

- a risk of damage to crops;
- harvesting of a crop;
- on-going fertiliser application;

then apply one of the following methods of soil collection. The methods are presented by order of preference.

Linear sample, at least 5 m from the edge of the field	© 2017, European Union, CC BY 4.0 Figure 8-208. The soil sampling is carried out inside the field, at least 5 m distance from the edge (while avoiding recently disturbed ground), in between the rows of the crops using a linear sampling pattern taking as the central one the LUCAS point (if not on the crop) or the one closest to the LUCAS point not on the crop (if LUCAS point on the crop). Remember that ideally 5 (at least 3) subsamples should be taken per LUCAS points.
Linear sample near the edge of the field	© Google Maps, Cadrezzate con Osmate, 2023 Figure 8-209. If it is not possible to take the soil sample in-between the rows of the crops, then the soil sample has to be taken near the edge inside the field as close as possible to where the original LUCAS point is located. Choose the edge closest to the LUCAS point and as far as possible from any artificial

	surface (e.g. road). The same linear sampling pattern, as previously described for sampling in-between rows, should be respected. Remember that ideally 5 (at least 3) subsamples should be taken per LUCAS points.
<i>Linear sample when point is close to a road</i>	 © 2017, European Union, CC BY 4.0 Figure 8-210. The same linear sampling schema should be applied if the LUCAS point is too near to the road. In this situation the soil sample can be taken between the rows or, if not possible, on the edge of the fields with the same approach shown in the example above. Remember that ideally 5 (at least 3) subsamples should be taken per LUCAS points.
<i>Use the comments field</i>	 The use of special arrangements for the collection of the soil sample in case of difficult access should be reported in the comments section of the Field Form. Possible comments: SA01, SA02, SA04, SA05, SA08, SA99, SB99, SC99

8.18.8 Impossibility to collect the soil sample

Eventually the ground conditions or the owner may prevent the surveyor from taking any soil sample in the LUCAS point. In this case a comment is needed.

CODE	LABEL
SA05	Ground conditions unfavourable
SA06	Inaccessible point (e.g. fenced area)
SA07	Land owner refused the soil sample
SA99	Other soil comments
SB99	Other soil bulk comments
SC99	Other soil biodiversity comments

<i>It is acceptable not to take a soil sample if certain conditions are met</i>	Ground conditions that prevent the soil collection include: - Abundance of rocks ($S \geq 50\%$) - Soil covered by water (flooding) - Excessive shrub cover that makes impossible to reach the soil (e.g. presence of thorny shrubs) - Frozen ground/thick snow cover If it is not possible to take soil, a comment must be added in the Field Form for the point explaining the reason.
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8.18.9 Summary of number of subsamples and photos

	Sampling			
	STANDARD / WOODLAND	BULK DENSITY	BIODIVERSITY	
N° subsamples 0-30 cm	5		5	
N° subsamples 0-10 cm		Ideally 5 (at least 3)		
N° subsamples 10-20 cm		Ideally 5 (at least 3)		
N° subsamples 20-30 cm		Ideally 5 (at least 3)		
Photo label	1	1	1	
Photo label + holes	1	1	1	

8.19 Photos

The surveyor has to document his observation by means of several photos.

The basic LUCAS photos are the point and Landscape photos in the 4 cardinal directions, taken on the point reached by the surveyor in the field.

What other photos have to be taken depends e.g. the land cover, presence or absence of water management, the need to collect a soil sample and the need to document conflict cases.

Photographs create a basis for a long-term monitoring of landscape changes. Moreover in the context of earth observation related mapping activities, LUCAS micro data and photos are used for production, verification and validation processes. It is thus important that they meet standard quality requirements.

8.19.1 How to take LUCAS photos

<i>Check settings before starting the survey</i>	REMEMBER to check the settings of the camera (date, time and the size of the pictures) before starting the survey! Be aware on the correct date format (regional settings of camera meta data).
<i>Pictures are to be delivered in JPG format</i>	In order to permit simple handling and archiving of the landscape photos, they have to be delivered in digital format (*.jpg file)
<i>EXIF</i>	The photos need to have EXIF data tags for coordinates. It is recommended to include the following information Gpslongitude, gpslongituderef, gpslatitude, gpslatituderef, gpsdatestamp, gpstimestamp, datetimeoriginal, datetimedigitized, datetime, gpsimgdirection, gpsimgdirectionref, imagesize, gpslatitudedd, gpslongitudedd, gpsgeom, gpsdatetimestampz
<i>All pictures are to be taken in landscape format</i>	The pictures are taken horizontally (landscape format).
<i>Dimensions required are 1600x1200 pixels</i>	The image dimension should be 1600 *1200 pixel
<i>Standard format is 4:3</i>	The standard format for LUCAS pictures is 4:3.
<i>Size of pictures between 1Mb to 1.5 Mb</i>	Good quality photos are expected to have a size between 1Mb and 1.5Mb. File sizes from 300Kb (compressed) up to a maximum of 2Mb are accepted.

Keep the camera levelled	 © European Union, CC BY 4.0 Figure 8-211. The camera should be kept level (i.e. not tilted). The photo above is unacceptable, since it is neither able to show the location of the point nor reflecting the surrounding landscape.
Use the 5/6 rule	In point and landscape photos the horizon must be about 5/6 of the way up the viewfinder (mark the respective line on the camera, or use the camera options).
Exposure	In point and landscape photos the exposure of the photo needs to be optimised for the landscape, not for the sky.
No date in the picture	Disable the date display setting on the pictures in the camera!
No strange elements on the picture	It is important to take care that the photographer's equipment, fingers, backpack, car, assistant, other people etc. are not visible in the picture.
Anonymization	The surveyor is requested to ensure that the photos do not display any recognisable feature (i.e. faces of persons, vehicle license plates). In case that photo-taking is not possible without recording such features, those elements in the photos need to be anonymized/blurred. The surveyor shall cover vehicle license plates and/or faces of persons using a dark grey box of adequate size, in a way that the feature(s) shall not be recognisable. The surveyor must do this before forwarding the photos to the Regional or Central office. The DMT includes a tool enabling to manipulate the photos. Points with at least one photo requiring anonymization which is not anonymised will be rejected.

EXIF data:

- Photos need to have exif data on coordinates.
- In addition the following exif data should be collected:

Gpslongitude, gpslongituderef, gpslatitude, gpslatituderef, gpsdatestamp, gpstimestamp, datetimeoriginal, datetimedigitized, datetime, gpsimgdirection, gpsimgdirectionref, imagesize, gpslatitudedd, gpslongitudedd, gpsgeom, gpsdatetimestampz.



Comment needed;

If a photo is required and cannot be taken (FA01 – FA4, FA99) or if there is a problem with a photo, (FA05 – FA08, FA22, FA99) a structured comment is required to describe the circumstances.

If there is a problem to anonymize or upload a photo, please add a comment (BH01, BH02, BH03).

The following comments can be relevant in this context:

CODE	LABEL
FA01	Camera's battery discharged - impossible to take photo(s)
FA02	Camera broken - impossible to take photo(s)
FA03	Memory Card Problem
FA04	Land owner refused photos
FA05	1/6 horizon rule not respected because an object in the front (house wall)
FA06	1/6 horizon rule not respected because photo taken in forest
FA07	Camera date/time set incorrectly
FA08	Compass malfunctioning
FA22	Landscape photo shifted (direction/degrees)
FA99	Other photo comments
BH01	Problem with data input: hardware
BH02	Problem with data input: software
BH03	Problem using the anonymization tool

8.19.2 Main types of photos

8.19.2.1 Point photo

The photos of the point should allow finding the exact location of the observed point in the next LUCAS survey.

<i>Land cover and land use observation refers to the point indicated on the orthophoto. So does the point photo.</i>	The information of GPS coordinates and precision are always referring to the position of the surveyor while doing the observation, but the information on land cover, land use and the photos of the point and of the crop/cover have to refer to the LUCAS point itself (the theoretical point), determined by the orthophoto, even if the point is further away from the real position of the surveyor.
<i>Find a stable field element to act as location reference</i>	The distance and the direction of the photo of the point are up to the surveyor's own judgement. The most important aspect is to have stable field elements which act as location references on the picture to ease the work in the next survey (e.g. house, barn, track or any other "quasi-stable" landmark).
<i>Using the flag</i>	A good point indicator is e.g. a bright paper flag in a stick (1 m long). The indicator shall be harmonised at country level, i.e. surveyors within one country shall use the same model. The flag (or point indicator) indicates whether the observation was made at the point (i.e. the point was reached). If a point is observed from a distance >100m it shall not be marked with the flag, even if in this case a point picture in the direction of the point is to be taken.
<i>The flag helps to</i>	When placing the flag take into consideration that it can be used to help the controllers

<i>assess the width of features</i>	assess the width of features (namely in case of the Look to the North/East rule).
<i>The point is located at the centre of the picture</i>	The photo has to be taken in landscape format (horizontally) with the point in the centre. If the indicator is not visible, then the centre of the picture is considered as the point location. Keep the 5/6 rule and orient the picture to a stable landmark.
<i>Show land cover on the point photo</i>	Point photos <u>need to show the registered land cover</u> . This is the reason why <i>in-situ</i> PI points do not need a point photo.
<i>Take meaningful pictures (especially in case of buildings)</i>	Since there is no need to take a “cover” photo in artificial areas, the point photo is often used to check the cover. Therefore, take care that the height of buildings at the point can be assessed: a picture of a wall is seldom helpful for quality controllers to check if a built-up area should be classified as A11 or A12.
<i>Example of a point picture (observation at 100m or less)</i>	 © European Union, CC BY 4.0 Figure 8-212. Good photo of a point, with point marker in the centre and a recognizable land mark. The 5/6 horizon rule is kept.

8.19.2.2 Crop/cover photo

The aim of the crop/cover photo is to enable the identification of the recorded land cover by means of the photo.

<i>If the point is not accessible the photo can be taken at the edge of the field</i>	If the point is not accessible but the surveyor can reach the border of the field parcel where the point is located and the point is visible, the crop/cover photo has to be taken from the edge of the field. The surveyor has to be sure that the crop is the same as at the point itself.
<i>Take the photo of the main land cover (LC1). An additional photo of the LC2 can also be taken.</i>	If there are multiple land covers registered (LC1 and LC2), the crop/cover photo of the main land cover has to be taken. An additional photo for LC2 can also be taken. For example, in case permanent crops (B7X, B8X) are registered, together with a second land cover, a zoomed photo of the permanent crop should be taken. The photo of LC2 is

	not mandatory in this case.
<i>Macro may be used for crops</i>	The zoomed picture should be taken at a close distance, so that the structure of leaves can be clearly seen, as well as flowers or fruits. If to correctly identify the crop, it has to be photographed from a distance of <2 m, use the camera's macro-mode (when available). Do not forget to switch off zoom and macro settings after each photo shot! They are not needed for the other pictures.
<i>Example: red cabbage</i>	 © 2019, Ralph (@Capri© 23Auto), Pixabay License Figure 8-213.
<i>Example: common wheat</i>	 © 2019, Osoian Marcel (@osoian-marcel), Pixabay License Figure 8-214. Fructification or leaves can be used as a means of crop identification.

Example: barley



© 2017, Sanjay Acharya, CC BY-SA 4.0
Figure 8-215.

Barley (*Hordeum vulgare* L.) seeds with husk (left) and with husk removed (right). Seeds and seedlings can be used as a means of identification.

Example: cherry



© 2016, (@Couleur), Pixabay License
Figure 8-216.

When possible, take a picture of leaves and fruit together.

Example: olive



© 2014, Hans, Pixabay License
Figure 8-217.

8.19.2.3 Landscape photos (N, E, S, W)

Landscape images illustrate the character of landscapes. With images it is possible to monitor changes in the landscapes, if the photographs are taken from the same place in different years. Keeping the 5/6 rule is very important in these pictures.

<i>Take the pictures from the place the point observation was made</i>	Landscape photos in digital format will be taken from the place where the visual observation was made. In case of application of the "Look to the North/East rule", the photos have to be taken from the original point, whether on a border or on a linear feature. In case the point is not accessible, the landscape photos have to be taken from the closest point reached (the same point where the GPS coordinates were recorded)
<i>Use a wide angle lens setting and no zoom. Focus at infinity.</i>	The landscape pictures should be taken using a wide-angle lens (equivalent to 35 mm or less). The objective is to photograph the landscape, so sharp foreground details are not necessary. The focus must be on the middle ground and background; this is obtained with the lens focused at infinity. Remember to set the zoom to minimum. Avoid automatic flash.
<i>4 landscape pictures are needed</i>	A photo must be taken in each of the 4 main compass directions (N, S, E, W). This applies even if there is a dense forest, or a wall close by.
<i>Take the pictures looking first to N and then clockwise to E, S and W</i>	The photograph sequence will be taken successively looking North, East, South and West (clockwise rotation). This order must be respected in order to archive the pictures without any misinterpretation.
<i>Use the magnetic compass to determine the correct direction</i>	The directions are determined with a handheld compass.
<i>If an error is made, delete and start again</i>	If an error is made during photo-taking, for example a picture is taken twice, incorrect order etc., the wrong ones should be deleted and the correct photo sequence has to be taken again.

Example: landscape photos in NESW sequence		
	© European Union, 2018	© European Union, 2018
	Figures 8-218.	Figures 8-219.
		
	© European Union, 2018	© European Union, 2018
	Figures 8-220.	Figures 8-221.
	LUCAS 2018 N, E, S, W photos	
Show land cover on the landscape photo if LTN was used	If the “Look to North/East” rule was applied, the N or (respectively) E photo must show the registered land cover	

8.19.2.4 Copernicus

The aim of the photo for the section on Copernicus is to document the land cover at the point the surveyor reaches. It is not a landscape photo. The surveyor step back from the point where he took the landscape photos (observation point), step back some 5m and then take a photo of the observation point and some 5m on each side.

8.19.2.5 Irrigation photos

If there is water management on the field, a photo of the water management system has to be taken.

<i>The irrigation photo may have to be taken away from the point of observation</i>	If the irrigation device is not at the point but on the same parcel, it might be necessary for the surveyor to come closer to the device for making an illustrative photo.
<i>If point is PI is irrigation photo needed?</i>	In case a point is PI in-situ, the irrigation photo is needed only if it can be seen from the location of the surveyor.
<i>Take a drainage photo</i>	If drainage is visible, take a drainage photo (as irrigation photo). If both elements (irrigation and drainage) are present take additional photos.

*Example: irrigation
photo (sprinkler)*



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Figure 8-222.

8.19.2.6 Soil photos

In case a soil sample (standard topsoil sample or bulk density sample) has been taken, take a photo of the LUCAS point with the bag, label and as many as possible of the 5 subsample holes clearly visible (3 in bulk density sites). For control purposes, please take also a picture of the label clearly readable. This will allow for an extra check to guarantee that the label code corresponds to the number filled in the field form. With geo tagged images a further control on location is also possible.

In case a soil biodiversity sample has been taken, please take a photo of the plastic jar with the label clearly visible.

8.19.2.7 Grassland

The aim of the photos for the grassland survey is to document the grassland, the transect, problems with the survey and important species that cannot be identified and/or that are rare.

Main grassland photos are as follows:

- Grassland with unclear age
- Posy
- Grass transect from above
- Grass transect start
- Grass transect end

Additional photos can concern legumes, rare species, dominant species.

8.19.3 Summary of main photos to be taken

Questionnaire	Value	Label	Point (P)	Cover (C)	North (N)	East (E)	South (S)	West (W)	Comment on photo
Observation type	1	Field Survey <= 100 m	Y	Y	Y	Y	Y	Y	
	2	Field Survey > 100 m	Y	Y	Y	Y	Y	Y	Take photo showing why point not reached
	3	Photo interpreted	N	N	Y	Y	Y	Y	Take photo showing why point not reached
	4	Point not visible. PI not possible	N	N	N	N	N	N	Take photo showing why point not reached
	5	Out of National territory	N	N	N	N	N	N	
	6	Out of EU 28	N	N	N	N	N	N	
	7	Photo-interpretation in the office	N	N	N	N	N	N	
Questionnaire		Value	Label	Comment on photo					
Direction of observation	2	North	Take photo supporting decision when "looking to the North"						
	3	East	Take photo supporting decision when "looking to the East"						

Questionnaire	Value	Land cover 2	Comment on photo
Land Cover		LC2 like 'B'	Take photo showing the crop in LC2 (if needed)
		LC2 like 'A', 'C', 'D', 'E', 'F', 'G' or 'H'	
	8	Not relevant	

Questionnaire	Value	Copernicus LC (K)	LC (K)	Comment on photo
Copernicus		LC	Y	Take photo showing the land cover on the point reached by the surveyor

Questionnaire	Value	Presence of water	Irrigation photo (I)	Comment on photo
- Presence of water management	1	Irrigation	Y	
	2	Potential irrigation	Y	
	3	Drainage	Y	
	4	Irrigation and drainage	Y	Take photo showing further irrigation/drainage elements.
	5	No visible water management	N	
	8	Not Relevant	N	

Questionnaire				
	Value	Soil biodiversity sample taken	Soil (L, Z)	Comment on photo
soil biodiversity sample taken	1	Yes	Y	Take a photo of the point, with the sample and as many holes as possible. Take a photo showing clearly the label code
	2	Not possible	N	Take photo showing the reason why soil not taken
	8	Not Relevant	N	

Questionnaire				
	Value	Soil bulk density sample taken	Soil (L, Y)	Comment on photo
soil bulk density sample taken?	1	Yes	Y	Take a photo of the point, with the sample and as many holes as possible. Take a photo showing clearly the label code
	2	Not possible	N	Take photo showing the reason why soil not taken
	8	Not Relevant	N	

Questionnaire	Value	Soil standard sample taken	Soil (L, X)	Comment on photo
soil standard sample taken ?	1	Yes	Y	Take a photo of the point, with the sample and as many holes as possible. Take a photo showing clearly the label code
	2	Not possible	N	Take photo showing the reason why soil not taken
	8	Not Relevant	N	

	Questionnaire	Value	Grassland age	Grassland age (A)	Comment on photo
Grass age	2	Unclear grassland age	Y	If the age of the grassland is unclear take a photo of the relevant characteristics present	
	Questionnaire	Value	Grassland transect	Grassland transect on map (M, O, Q, R)	Comment on photo
Grass transect					
			Transect start	Y (O)	Take a photo from the start of the transect in the direction of the transect end
			Transect end	Y (Q)	Take a photo from the end of transect point in the direction of the starting point of the transect
			Characteristic point from above	Y (R)	Take one photo from 1.5 m height pointing the camera vertically downwards to a typical patch of vegetation for the transect.
			Posy		
			Dominant species	N	If you find a dominant species not listed in the field form and that you cannot identify take a photo.

8.19.4 List of all photos

Chapter	Photo: content	Description
IDENTIFICATION AND ACCESS TO POINT	IDENTIFICATION AND ACCESS	
		IDACCESS_Other
POINT OBSERVATION	OBS	
8.3.3.1 Point is at more than 30 min walking distance	Photo (conflict) to document the impossibility to reach the point in the set time frame.	OBS_Conflict
8.3.3.2 Point not reached: restricted or inaccessible area, other difficulties	Photo (conflict) to document the impossibility to reach the point	OBS_Conflict
8.4.2.1 The point is visible (option 1 or 2)	Photo if the point is visible but the distance to the point is >100m the reason for not getting closer	OBS_Distance
8.4.2.1 The point is visible (option 1 or 2)	Photo of the point (fields 602, 604)	below
	and landscape photos (fields 608 – 619).	below
8.4.2.2 In situ photointerpretation (option 3)	Photo (fields 669 – 695)) why the LUCAS point was not reached	OBS_Conflict
8.4.2.3 Point not visible and photointerpretation not possible (option 4)	Additional photo (fields 669 – 695)) documenting why the LUCAS point was not reached has to be taken from the point reached.	OBS_Conflict
8.4.4 Field: Direction of observation (25)	Additional photo when "Look to the north" or the "Look to the east" rule is applied (fields 669 – 695). The photo needs to show the border or the linear feature.	OBS_LTNE
8.4.4 Field: Direction of observation (25)	Additional photos (type 2 – point observation) may be needed to show the land cover assessed (direction North or East).	OBS_LTNE_LC
		OBS_Other
LAND COVER AND LAND USE	LCLU	
8.4.6 Point and landscape photos	Photo of the Point (602)	LCLU_Point
8.4.6 Point and landscape photos	North (608)	LCLU_North
8.4.6 Point and landscape photos	East (611)	LCLU_East
8.4.6 Point and landscape photos	South (614)	LCLU_South

8.4.6 Point and landscape photos	West (617)	LCLU_West
8.6.4 Field: LC1 land cover 1 (28)	Crop/cover photo (fields 605/607) if the point is visible and LC1 is B, C, D or E	LCLU_CropLC1
8.6.5 Field: LC2 land cover 2 - Multiple registration of land cover (31)	Crop/cover photo (fields 605/607) or an additional photo (fields 660 – 695) if the point is visible and LC2 is B, C, D or E	LCLU_CropLC2
		LCLU_Other
INSPIRE	INSPIRE	
		INSPIRE_Other
FAO PARAMETERS	FAO	
		FAO_Other
LAND MANAGEMENT	LANDMNG	
8.9 Section: Land management	Land Management 1, 2, 3, 4 ...	LANDMNG_Other
SPECIAL REMARKS	REMARK	
8.10 Section: Special remarks	Special remark 1, 2, 3, 4, ...	REMARK_Other
GULLY EROSION	GULLY	
8.11.1 Field: Can you see signs of gully erosion?	Gully Erosion (before there was a general erosion photo)	GULLY_Biggest
		GULLY_Other
WATER MANAGEMENT	WATERMNG	
8.12 Section: Water management	Water management 1, 2, 3, 4 ...	WATERMNG_Other
COPERNICUS	COPERNICUS	
8.13 Specific module: Copernicus	Copernicus	COPERNICUS_LC
		COPERNICUS_Other
TREES WITH SECONDARY	TREEWITH	
		TREEWITH_Other
EXTENDED GRASSLAND	EXTGRASS	
		EXTGRASS_Other
GRASSLAND	GRASS	
		GRASS_UnclearAge
8.16.9.2 Fields: Direction/ length of grass transect, start and end point	TransectStart	GRASS_TransectStart
8.16.9.2 Fields: Direction/ length of grass transect, start and end point	TransectEnd	GRASS_TransectEnd

8.16.9.2 Fields: Direction/ length of grass transect, start and end point	TransectAbove	GRASS_TranssectAbove
8.16.10.1 Field: Vigour of vegetation	GrassVigour	GRASS_Vigour
8.16.12 Section: Grassland flower (transect)	Posy	GRASS_Posy
8.16.13 Fields: Key species/species group richness	RareSpecies	GRASS_RareSpecies
		GRASS_Other
LANDSCAPE FEATURES	LF	
8.17.3 Field survey (Phase 2: LF field survey)	LF not visible on orthophoto	LF_NotOnOrtho
8.17.3 Field survey (Phase 2: LF field survey)	Problem with LF identification	LF_Other
SOIL	SOIL	
8.18.3.3 Which soil samples to collect?	Photo if standard or bulk sample is not possible to take (additional photo "soil", fields 669 – 695).	SOIL_NoSample
8.18.3.3 Which soil samples to collect?	Photo of the sampled area if the relevant sub-modules are assessed (L-soil photo: fields 626/628,)	SOIL_StdBagHoles
8.18.3.3 Which soil samples to collect?	X - Soil label photo (629-631)	SOIL_StdLabel
8.18.3.3 Which soil samples to collect?	Y - Soil bulk label photo (632-634)	SOIL_BulkLabel
8.18.3.3 Which soil samples to collect?	Z - Soil biodiversity label photo (635-637)	SOIL_BioLabel
8.18.3.3 Which soil samples to collect?	Photo of the litter surface in soil woodland points	SOIL_Litter
8.18.4.2 Sampling in cropland, grassland, and peatland points	Photo of the LUCAS point with the bag (keep the label visible) and as many as possible of the 5 holes clearly visible.	SOIL_StdBagHoles
8.18.4.2 Sampling in cropland, grassland, and peatland points	Additional photo of the label which has to be clearly readable.	SOIL_StdLabel
8.18.4.2 Sampling in cropland, grassland, and peatland points	Additional pictures if appropriate (e.g. to justify that a subsample was not taken due to excessive amount of thorny shrubs).	SOIL_NoSub, SOIL_Other
8.18.4.3 Sampling in woodland points	Photo of the point with the bags of litter layer and mineral topsoil (keep the labels visible) and as many as possible of the 5 holes clearly visible.	SOIL_StdBagHoles
8.18.4.3 Sampling in woodland points	Photo of the labels of the bags of litter layer and mineral topsoil have to be clearly readable.	SOIL_StdLabel

8.18.4.3 Sampling in woodland points	Additional pictures if appropriate (e.g. to justify that a subsample was not taken due to excessive amount of thorny shrubs).	SOIL_NoSub, SOIL_Other
8.18.5.2 Sampling in cropland, grassland, and peatland points (bulk density)	Photo of the point with the biodiversity bag. The green label in the bag should be clearly visible.	SOIL_BagHoles_Bio
8.18.5.2 Sampling in cropland, grassland, and peatland points (bulk density)	Photo of the point with the bag with core samples of the 3 depth increments next to the point. keep the label visible and as many as possible of the holes clearly visible.	SOIL_BulkBagHoles
8.18.5.2 Sampling in cropland, grassland, and peatland points (bulk density)	Additional photo of the label	SOIL_BulkLabel
8.18.5.2 Sampling in cropland, grassland, and peatland points (bulk density)	Additional pictures (e.g. to justify that a soil core was not taken due to excessive amount of thorny shrubs).	SOIL_NoSub
		SOIL_Other
8.18.5.3 Sampling in woodland points (bulk)	Photo of the point with the bag of biodiversity sample. The green label in the bag should be clearly visible.	SOIL_BagHoles_Bio
8.18.5.3 Sampling in woodland points (bulk)	Photo of the point with the bags of litter layer and core samples of the 3 depth increments. keep the labels visible and as many as possible of the holes clearly visible.	SOIL_BulkBagHoles
8.18.5.3 Sampling in woodland points (bulk)	Additional photo of the labels of the bags of litter layer and soil cores	SOIL_BulkLabel
8.18.5.3 Sampling in woodland points (bulk)	Additional pictures (e.g. to justify that a soil core was not taken due to excessive amount of thorny shrubs).	SOIL_NoSub
		SOIL_Other

8.19.5 Naming of photos

The correct naming of the photos is essential. The identification is of high importance to enable correct archiving.

<i>Note the photo name.</i>	The camera will assign a name to the image. In order to ease renaming of the photos, the surveyor should write the photo name given by the camera on the field form (Photo ID). Together with the date and time of the field survey, it will be an unambiguous identification.
<i>DMT will rename the photos automatically</i>	LUCAS photos are named according to a convention. For each photo a unique identification code is to be given to each photo before delivery, composed of: Year PointId Description (see list of all photos for the possible descriptions) When uploading the images to DMT, they will be renamed automatically.
<i>Example: naming of a photo</i>	The name of the East photo for point 23456789 is : 202223456789ELCLU_East.jpg

9 Annexes

9.1 Field form - Record descriptor

This chapter gives an overview of the items, which have to be recorded in the field. The list below shows that beside the land cover and land use information of the point concerned a number of “meta information” has to be collected.

The explanations given follow the structure of the Field Form (LUCAS document C2) and provide a basic definition and description of each item. Moreover, references are given where to find more detailed information and explanations (as in LUCAS reference document C3, Land Use and Land Cover classification). Additional information can also be found in the following annexes.

9.1.1 Identification

Items to be filled in (including the item number in the field form)	Observed feature (including the code in the field form)	Short explanation / description
User ID (A)	Value (Char 8)	Unique identity code of surveyor. To be defined by contractors, according to rules.
Point ID (B)	Value (Char 8)	Unique code of the point as provided by Eurostat.
Point altitude (C)	In meters	Prefilled, based on the information on the sample
PI extension point (D)	Yes/No	Prefilled, based on the information on the sample
EX-ANTE point (E)	Yes/No	Prefilled, based on the information on the sample
Date (7)	YYY/MM/DD (e.g. 2022/05/23)	Date of observation. Valid date with YYYY=2022 or 2023, MM=01-12, DD=01..31
Start time (8)	HH/mm (e.g. 14:02)	Observation time starts when leaving the car. Valid time with HH=00..23 and mm=00..59
End time (9)	HH/mm (e.g. 14:50 h)	Observation time ends after returning to the car. Valid time with HH=00..23 and mm=00..59

9.1.2 Access to point

Car park latitude (11)	DD.ddddddd	GPS position (WGS84 latitude) of the location where the car was parked
Car park longitude (12, 13)	E/W DD.ddddddd	GPS position (WGS84 longitude) of the location where the car was parked
GPS coordinate system (14)	(1) WGS84	Normal functioning of GPS using "WGS 84" as coordinate system.
	(2) Problem with signal	No signal, or bad reception. The reason needs to be noted in the comments.
GPS Precision (15)	Value (in m)	Indication of average location error as given by GPS receiver (in meters)
GPS Elevation (16)	Value (in m)	GPS value of elevation of the location from which observation is done (in meters above sea level).
Point latitude (17)	DD.ddddddd	GPS position of the location from which observation is done (WGS84)
Point longitude (18, 19)	E/W DD.ddddddd	GPS position of the location from which observation is done (WGS84)
GPS distance to the point (20)	Value (in m)	Indication of the distance between observation location and the LUCAS point as provided by the GPS receiver (in meters).
GPS calculated distance to point (21)	Value (in m)	Distance calculated automatically by the DMT, based on the position of the surveyor according to the GPS coordinates
Way to the point (previous campaign (F))	Text	If there is information available from a previous campaign it is reported here.

9.1.2.1 Comments (on the way to the point)

In addition of the obligatory fields to be filled in, the surveyor has the opportunity - and in specific situations the obligation - to add specific comments and comments. Comments in the field form are essential for transparent reasoning of any decision taken by the surveyor and for all future LUCAS surveys.

Description of the way to the point (F, 500)	If the point can be accessed without any problem, a specific comment is not necessary. - Structured comments (to be preferred) - Free text comments
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9.1.3 Point observation

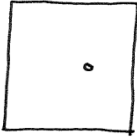
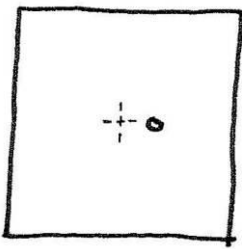
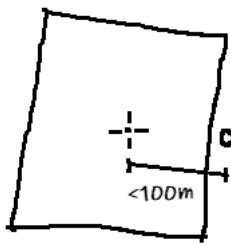
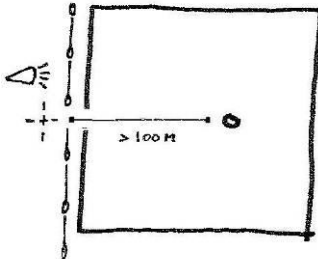
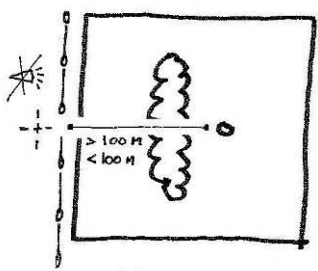
Type of observation (24)	(1) Field survey, Point visible, ≤ 100 m  Parcel with survey point  Surveyors position  Distance surveyor – survey point	Regular observation of the point.  If point is observed from a distance of > 50 m, thus not at the point directly, the reason needs to be noted in the comments (fence, high crop etc.). 
	(2) Field survey Point visible, >100 m  Point visible  Fence, barrier, high crop	If the <u>point</u> is not accessible in the field (e.g. wall, high crop), but still visible, observation from distance can be done in the field. The point has to be clearly visible from the closest reachable point and the LC and LU identifiable unambiguously.  Example: a point in a large rape field, ready for harvesting, in 300 m distance. The point can be seen in the rape field, but is not accessible (as it would destroy the crop). From 300 m distance and on condition that the topography allows seeing the point the surveyor can identify the crop (rape). For all these points the reason needs to be noted in the comments (fence, high crop, etc.).
	(3)	If the <u>point</u> is not accessible and not visible in

	Photo-interpretation in the field, Point not visible  Fence, barrier, high crop, hedge etc. impeding the point observation  Point not visible	the field (e.g. the point is located in a large inaccessible forest), an interpretation over the orthophoto has to be done in the field.  It is also important to indicate if the surveyor notices major differences (e.g. out dated orthophoto) between the land cover and the orthophotos. Those should be noted in the comments field. If point is photo interpreted, the reason needs to be noted in the comments (fence, high crop, etc.).
	(4) Point not visible. PI not possible Point not observed Photointerpretation not possible	If the point is not accessible in the field (e.g. located in forbidden zone) and the parcel with the point location in it is not visible, and no photointerpretation can be done (no orthophotos or very bad quality orthophoto; land cover/use – even using the simplified classification - cannot be recorded) the point is coded as not observed. The same applies for a point photointerpreted in the office, if the photointerpretation turns out to be impossible (no orthophotos or very bad quality orthophoto; land cover/use – even using the simplified classification - cannot be recorded) the point is coded as not observed. If the point is not observed because of inaccessibility, orthophoto unavailability or very bad quality, the reason needs to be noted in the comments.
	(5) Out of national territory	The point is located out of the national territory.
	(6) Out of EU 28	The point is located out of the national territory and out of the EU 28
	(7) Photo-interpretation in the office	The point has been assigned to photo-interpretation in the office a priori
Homogeneous plot fills extended window (?)	(1) Yes	The complete 40m diameter of the extended window is occupied by the land cover, which is assessed on the point (homogenous plot).

	(2) No	The extended window is not completely occupied by the homogeneous plot.
Direction of observation (25)	(1) On the point	Point regularly observed.
	(2) To North	"Look to the North" rule is applied if the point is located on a boundary/ edge or a small linear feature (<3 m wide).
	(3) To East	"Look to the East" rule is applied if the point is located on a boundary/edge or a small linear feature (<3 m wide) directed North/South.
	(8) N.R. (Not relevant)	If not applicable.
Reason for changing direction of observation (?)	(1) Border between (LC code) and (LC code)	In case the point is located on the boundary of two parcels, specify the LC codes on both sides of the border here.
	(2) Linear Feature (Transect feature code)	In case the point is located on a linear feature, specify the linear feature or small object here.

Comments about point observation (500)	Structured comments Free text comments	It is important to note all those facts related to the observation of the point, which pose specific difficulties. Possible comments are linked to: - Problems in the exact location of the point (radical changes in the field compared to the orthophoto, lack of adequate landmarks for orientation, loss of the GPS signal, wrong instructions from the previous survey, etc.), - Restricted access to the point, - The necessity to make the observation from far away, - Point is photo interpreted, - Problems in the coding of land cover or land use (e.g. crop recognition etc.). All such short comments should help to explain why the surveyor has taken a certain decision. For free text the use of English is mandatory. Special characters are to be avoided.
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9.1.4 Photo Interpretation (to be used if Type of observation is 3 or 7)

Poor Ortho-photo conditions (?)	(1) Outdated	The orthophoto is clearly outdated, landscape or constructions have changed significantly and the orthophoto does not present the actual surrounding of the LUCAS point anymore.
	(2) Detail quality Issue	The orthophoto's level of detail is too low to detect and assess the Land Cover, Land Use and other features correctly.
	(3) Outdated and detail quality Issue	Combination of both the before mentioned issues.
	(4) No	The orthophoto quality is fine to assess the point.
LC/LU assessment by using additional sources (?)	(1) For LC	Additional sources besides the orthophoto were needed and used to assess the Land Cover.
	(2) For LU	Additional sources besides the orthophoto were needed and used to assess the Land Use.
	(3) For LC/LU	Additional sources besides the orthophoto were needed and used to assess the Land Cover and Land Use.
	(4) Not used	The use of additional sources was not needed to assess the LUCAS point.
Additional sources used (?)	(1) Google Earth / Google Maps	As an additional source Google Earth / Google Maps was used
	(2) Previous campaign information	As an additional source information from the previous campaign (e.g. previous LC/LU, LTN-decision or previous photos) was used
	(3) Other additional sources	To be specified in a comment
	(4) No other suitable sources available	No additional sources were available, only a simplified photo-interpretation was possible. Please note that any information collected through photointerpretation

	(PI impossible)	needs to be reliable. If the orthophoto is so bad that no reliable information can be collected, then the photointerpretation is not possible. In such a case please go back to the field "Type of observation" and select option 4 "Point not visible and photointerpretation not possible".
Only assessment on lower detail level possible (?)	(1) Outdated	Land Cover can only be assessed on lower detail level
	(2) Detail quality Issue	Land Use can only be assessed on lower detail level
	(3) Outdated and detail quality Issue	Land Cover and Land Use can only be assessed on lower detail level
	(4) No	No assessment on lower detail level needed

9.1.5 Land cover and land use

While the information of GPS coordinates and precision are referring to the position of the surveyor while doing the observation, the **information on land cover, land use and the photos of the point and of the crop/cover have to refer to the LUCAS point itself** (the theoretical point), determined by the orthophoto, even if it is further away from the real position of the surveyor.

LC1 land cover 1 (28)	Land cover code	Coding of land cover according to LUCAS 2022 classification. BX1 or BX2 are only possible for points with observation type (24) = 3 (photo interpreted)
LC1 crop species (29)	BXXn	To be filled for crops (LC1=BXX) classified as “other” in LC1, and also for nurseries (see Annexes and Document C3 for code lists)
LC1 crop coverage (%) (30)	Value (0-100%) for land cover 1 (LC1)	Land cover 1 + 2 can sum up to more than 100%.
LC2 land cover 2 (31)	Land cover code	A second cover can be registered if necessary
LC2 crop species (32)	BXXn	To be filled for crops (LC1=BXX) classified as “other” in LC1
LC2 crop coverage (%) (33)	Value (0-100%) for land cover 2 (LC2)	Land cover 1 + 2 can sum up to more than 100%.
LU1 land use 1 (34)	Land use code	Coding of the land use according to LUCAS LU 2022 classification
LU1 land use type (35)	U22Xn U315n	To be filled for the secondary sector (LU1=U22X) and for transport via pipelines (LU1=U315)
LU1 coverage (%) (36)	Value (0-100%) for land use 1 (LU1)	Land use 1 + 2 can sum up to more than 100%.
LU2 land use 2 (37)	Land use code	A second land use can be registered if necessary.
LU2 land use type (38)	U22Xn U315n	To be filled for the secondary sector (LU2=U22X) and for transport via pipelines (LU2=U315)
LU2 coverage (%) (39)	Value (0-100%) for land use 2 (LU2)	Land use 1 + 2 can sum up to more than 100%.
Parcel area (in ha) (40)	(1) area < 0.1	Size of the observed parcel is smaller than 0.1 ha.
	(2) 0.1 ≤ area < 0.5	Size of the observed parcel ranges between 0.1 and 0.5 ha.
	(3)	Size of the observed parcel ranges between 0.5

	$0.5 \leq \text{area} < 1$	and 1 ha.
	(4) $1 \leq \text{area} < 10$	Size of the observed parcel ranges between 1 and 10 ha.
	(5) $\text{Area} \geq 10$	Size of the observed parcel is larger than 10 ha.

9.1.6 INSPIRE

To improve the mapping with other products some additional information is collected.

9.1.6.1 INSPIRE PLCC (Pure Land Cover Components)

The following fields are to assess INSPIRE pure land cover components. The assessment is done within **the extended window of observation (20m radius)**. Unlike what happens in LUCAS classes, where the sum of percentage of combined land cover can be more than 100%, the sum of PLCC must be 100%. Assessment of these percentages is made using the “birds-eye” view.

(81) Arable Land	Value (0-100%)	Assess the percentage of arable Land
(82) Artificial Constructions	Value (0-100%)	Assess the percentage of artificial constructions
(83) Coniferous forest trees	Value (0-100%)	Assess the percentage of coniferous trees
(84) Broadleaved forest trees	Value (0-100%)	Assess the percentage of broadleaved trees
(85) Shrubs	Value (0-100%)	Assess the percentage of shrubs
(86) Herbaceous plants	Value (0-100%)	Assess the percentage of herbaceous plants
(87) Lichens and mosses	Value (0-100%)	Assess the percentage of lichens and mosses
(88) Consolidated (bare) surface	Value (0-100%)	Assess the percentage of consolidated (bare) surface (e.g. rock outcrops)
(89) Unconsolidated (bare) surface	Value (0-100%)	Assess the percentage of unconsolidated (bare) surface (e.g. sand)
(91) Permanent woody and shrubby crops	Value (0-100%)	Assess the percentage of permanent woody and shrubby crops
(92) Wetlands and marshes	Value (0-100%)	Assess the percentage of wetlands and marshes

(93) Organic deposits (peatland)	Value (0-100%)	Assess the percentage of Organic deposits (peatland)
(94) Chemical deposits	Value (0-100%)	Assess the percentage of Chemical deposits
Intertidal flats	Value (0-100%)	Assess the percentage of intertidal flats
Fresh water course	Value (0-100%)	Assess the percentage of fresh water course

9.1.6.2 Urban areas and non vegetated areas

Point in urban area? (80)	(1) Yes	A good way to identify the “urban area” is to use the orthophoto. In general, the boundaries of an “urban area” are easily recognisable and marked by, for example, fences or walls around kitchen or allotment gardens, graveyards, residential areas or dwellings.
	(2) No	
Non-vegetated area (81)	Value (0-100%)	The degree of unvegetated area is the percentage (0 – 100%) of the area within the extended window that does not show vegetation. Previously to LUCAS 2022 it was named “Imperviousness”.

9.1.7 FAO parameters

The following fields are to be assessed when the point is located in an area with trees: in CX0, D10, E10 and for all points with crops in B70 or B80 (whatever the size of the parcel).

Height of trees at the moment of survey (42)	(1) < 5m	Applicable if the LC is CX0, D10, E10, B81, B82, B83 and trees in B70 (except B75, as this is a mixed class), assess the height of the trees at the moment of the survey.
	(2) ≥ 5m	
	(8) Not relevant	If not applicable.
Height of trees at maturity (43)	(1) < 5m	Applicable if the LC is CX0, D10, E10, B81, B82, B83 and trees in B70 (except B75, as this is a mixed class), assess the height of the trees at maturity.
	(2) ≥ 5m	

	(8) Not relevant	If not applicable.
Width of feature (44)	(1) < 20m	If LC is CX0, D10, E10, B81, B82, B83 and trees in B70 (except B75, as this is a mixed class), assess the width of the feature.
	(2) ≥ 20m	
	(8) Not relevant	If not applicable.

9.1.8 Land management

This information is to be collected for points with the following land cover: BXX, CXX, DXX, EXX, FXX, HXX, except standing vegetation and crop residues which are only to be assessed for BXX

Presence of stone walls (49)	(1) No	Based on visual assessment of the area around the point. Possible on all LC types
	(2) Stone wall, not maintained	Walls in poor state of repair. Collapse features or gaps evident. Would allow animals to pass from one field to another
	(3) Stone wall, well maintained	Walls in good state – act as a barrier to the passage of animals
	NR	If not applicable.
Presence of hedgerows (49)	Yes/ No N/R	Based on visual assessment of the area around the point. Possible on all LC types
Presence of grass margins (XX)	(1) No	Based on visual assessment of the field margin surrounding the point. Possible on all LC types
	(2) ≤ 1m width	Can be assessed visually or measured by pacing
	(3) > 1 m width	Can be assessed visually or measured by pacing
	NR	If not applicable
Presence of grass margins and hedgerows (XX)		Based on visual assessment of the field margin surrounding the point. Possible on all LC types
Standing crop	Yes/ No N/R	Indicate if crop present

	<25%	Low soil cover
	25% - 50%	Medium soil cover
	50% - 75%	High soil cover
	75% - 100%	Very high or total soil cover
Crop residue	Yes/ No N/R	Indicate if crop residue present
	<25%	Low soil cover
	25% - 50%	Medium soil cover
	50% - 75%	High soil cover
	75% - 100%	Very high or total soil cover

9.1.9 Special comments (on land cover/use)

Signs of grazing in the field (53)	(1) Visible signs of grazing	If signs of permanent or occasional grazing of the parcel can be found (e.g. animal tracks).
	(2) No signs of grazing	No signs of grazing can be found on the parcel. Note that this is the parcel where the area is assessed, not the homogeneous plot within the expended window.
	(8) N.R. Not relevant	If not applicable.
Special status (54)	(1) Protected	If the area is protected under a special regulation on nature conservation, (e.g. NATURA2000, national schemes). Either signs are visible or local knowledge of the surveyor has to be applied.
	(2) Hunting	If signs of hunting are visible (e.g. hunting reserve signals, signs of hunting, fences around forests, feeding of game, shooting towers, etc.)
	(3) Protected and Hunting	If signs of both special status are visible
	(4) No special status	No signs of neither a Protected Area nor Hunting can be found
Special comments on	(1)	The field has been harvested during the current

land cover / land use (55)	Crop not recognisable - Harvested Field	season and the crop is not recognisable by residuals. In that case, LC=F40 and LU= U111
	(2) Crop not recognisable - Tilled and/or sowed	If the parcel is tilled and/or sowed and the crop cannot be recognised: the surveyor has to re- visit the point
	(3) Clear Cut	If most of trees have been cut down uniformly. In this case LC = DXX or EXX or FXX and LU = U120
	(4) Burnt Area	Refers to a burnt area in any LC. Signs of fire need to be observed. The land cover observed is to be noted.
	(5) Fire Break	Man-made gaps in vegetation (cropland, woodland/forests, grassland, shrub land) in order to stop fires.
	(6) Nursery	Refers to nurseries under forestry use, normally found on forest areas that are classified as LC = CXX and LU = U120. Note that this is different from specialized forest nurseries (B83f).
	(7) Dump site	This special comment is used in the case the surveyor observes an area, which is used as a dump site. Open dump site LC1: A30
	(8) Currently dry	Applies to river beds and lakes which are temporarily dry (lower water level). Normally the limit of the water level should be visible.
	(9) Currently flooded	Applies to areas that are flooded at the time of the visit.
	(10) No comment	

9.1.10 Gully Erosion

For all points visited in the field.	Field	Content	Comment
Signs of gully erosion (257)		(1) Yes	Gully erosion within 500 m of LUCAS point
		(2) No	
		N.R. (not relevant)	
Type of gully erosion (Only is signs of gully erosion = 1) (Multiple choices possible)		(1) Ephemeral	
		(2) Permanent	
		(3) Badland	
		(4) No	
Gully erosion (within 500 m, if visible): direction and details of largest gully visible from LUCAS point			
Direction of largest gully from point		North (290)	
		East (291)	
		South (292)	
		West (293)	
Dimension of largest gully		(in m)	
Length		Value (in m)	
Width		Value (in m)	
Depth		Value (in m)	

9.1.11 Water management on the field

These fields are only relevant for points where LU = U111 or U112 or if the grassland module has been assessed. If more than one source of irrigation or delivery system exists, mark the most important source. Note if irrigation is visible from the way to the point or along the transect.

Presence of water management (251)	(1) Irrigation	Indicate if irrigation is present. Irrigation is the process of supplying water to crops by ditches, pipes, sprinklers, or other conduits and conveyances.
	(2) Potential irrigation	When the field is not irrigated but evidence exists that it will be irrigated in the present year, or of having been irrigated during at least the previous years.
	(3) Drainage	Drainage is the removal of excess surface water or groundwater from land by means of ditches, or subsurface drains (if recognizable). Drainage has only to be noted if the ditch bordering the field is linked to the drainage of the field itself and not e.g. only to a road situated next to the field.
	(4) Irrigation and drainage	Irrigation and drainage are present.
	(5) No visible water management	No visible signs of drainage or irrigation.
	(8) Not relevant	If not applicable.
Type of irrigation (252)	(1) Gravity	Water is delivered to the farm and/or field by canals or pipelines open to the atmosphere and water is distributed by the force of gravity down the field by: - Surface irrigation system (border, basin, furrow, corrugation, wild flooding, etc.) or - Subsurface irrigation pipelines or ditches.
	(2) Pressure: Sprinkle irrigation	Water is delivered to the farm and/or field in pump or elevation induced pressure pipelines and water is distributed across the field by sprinklers (centre pivot, linear move, traveling gun, side roll, hand move, big gun, or fixed set sprinklers).
	(3) Pressure: Micro-irrigation	Water is delivered to the farm and/or field by pressure and distributed across the field by micro-irrigation (drip emitters, continuous tube bubblers, micro spray, or micro sprinklers).

	(4) Gravity/Pressure	Farm delivery and field distribution of irrigation water are a combination of gravity and pressure facilities. For example, a valve is used to reduce pressurized water delivered to a farm or field for subsequent distribution by a gravity surface irrigation system.
	(5) Other/not identifiable	
	(8) Not relevant	If not applicable.
Source of irrigation (253)	(1) Well	A hole drilled or bored into the earth providing access to water.
	(2) Pond/Lake/Reservoir	Still water
	(3) Stream/Canal/Ditch	Running water
	(4) Lagoon/Wastewater	Wastewater
	(5) Other/not identifiable	
	(8) Not relevant	If not applicable.
Delivery system (254)	(1) Canal	An artificial waterway used for irrigation.
	(2) Ditch	A long, narrow trench or furrow dug in the ground, as for irrigation.
	(3) Pipeline	A conduit of pipe used for the conveyance of water.
	(5) Other/not identifiable	
	(8) Not relevant	If not applicable.
Signs of use of reclaimed water	(1) Yes	
	(2) No	

9.1.12 COPERNICUS

Only collect Copernicus if:

- the point is part of the Copernicus module (Copernicus point, G) (pre-filled),
- if the observation type (field 24) is 1, 2 or 3 and,
- If the point reached by the surveyor is not a linear feature < 3 m width, nor a small object with a diameter <3m

If at least one of the above requirements is not satisfied, the Copernicus assessment cannot be done.

Copernicus point (G)	Yes/No	prefilled
Can you do the Copernicus survey?	(1) Yes	Select yes, if the observation type (field 24) is (1), (2) or (3).
	(2) No	Select no, if the observation is made from a linear feature narrower than 3 m.
Copernicus LC (58)	Land cover level 3	
Extension of Copernicus LC in the 4 cardinal directions (in m)		
N (60)	Integer: 0 – 51	Extent in meters. If the extend is >= 50, put 51.
E (61)	Integer: 0 – 51	Extent in meters. If the extend is >= 50, put 51.
S (62)	Integer: 0 – 51	Extent in meters. If the extend is >= 50, put 51.
W (63)	Integer: 0 – 51	Extent in meters. If the extend is >= 50, put 51.

9.1.13 Trees with secondary crops, meadow or pasture

Trees with secondary crops, meadow or pasture (92) (previously to LUCAS2022 was called “EUNIS complex”)	(1) Cultivated trees with secondary crops, meadow or pasture (X06)	Crops, meadows or pastures developed under orchards or other cultivated tree plantations.
	(2) Forest trees with grazed grassland, heathland and/or woodland flora X09	Pasture woods (with a tree layer overlying pasture) with forest trees Large, open-grown or high forest trees (often pollards) at various densities, in a matrix of grazed grassland
	(3) NR	
Comments about Trees with secondary crops or grazing	- structured comment - free text	Optional

9.1.14 Extended Grassland

Possible if:

- The point belongs to the extended grassland module
- LC1 = EXX
- Same field than the LUCAS point reached
- OBS_TYPE = 1 (point visible, distance to the LUCAS point < 100 m).

But even if these conditions are satisfied, the extended grassland might not be possible, because the point is too far away.

The assessment will be done on the extended window within the homogenous plot.

9.1.14.1 Assessment possible?

Extended grassland point ()	Yes/No	Prefilled, based on the information on the sample
Extended grassland survey possible? ()	Yes/No	Is it possible to do the extended grassland assessment? If not, skip the complete part on extended grassland.
Grassland survey cannot be done Comment (500)	- structured comment - free text	Mandatory if extended grassland cannot be done.

9.1.14.2 Grassland composition (Extended window + homogenous plot)

Grassland composition	1) Graminoids dominated (>75%)	Grass like vegetation prevails
	2) Forbs dominated (>75%)	Herbaceous vegetation prevails
	3) Mixed graminoids / forbs (both > 25%)	Neither grass-like nor herbaceous vegetation prevails

9.1.14.3 Vegetation layers (Extended window + homogenous plot)

Grass-forb ratio	1) Grass : Forb = 100 : 0	
	2) Grass : Forb = 90 : 10	
	3) Grass : Forb = 80 : 20	
	4) Grass : Forb = 70 : 30	

	5) Grass : Forb = 60 : 40	
	6) Grass : Forb = 50 : 50	
	7) Grass : Forb = 40 : 60	
	8) Grass : Forb = 30 : 70	
	9) Grass : Forb = 20 : 80	
	10) Grass : Forb = 10 : 90	
	11) Grass : Forb = 0 : 100	
Herbaceous layer	1) 0 %	
	2) >0 - 3 %	
	3) >3 - 8 %	
	4) >8 - 15 %	
	5) >15 - 25 %	
	6) >25 - 40 %	
	7) >40 - 60 %	
	8) >60 - 80 %	
	9) >80 – 100 %	
Bare layer	1) 0 %	
	2) >0 - 3 %	
	3) >3 - 8 %	
	4) >8 - 15 %	
	5) >15 - 25 %	
	6) >25 - 40 %	
	7) >40 - 60 %	
	8) >60 - 80 %	
	9) >80 – 100 %	
Woody layer	1) 0 %	
	2) >0 - 3 %	
	3) >3 - 8 %	
	4) >8 - 15 %	
	5) >15 - 25 %	
	6) >25 - 40 %	

	7) >40 - 60 %	
	8) >60 - 80 %	
	9) >80 – 100 %	
Woody layer component - orchards	1) 0 %	
	2) >0 - 3 %	
	3) >3 - 8 %	
	4) >8 - 15 %	
	5) >15 - 25 %	
	6) >25 - 40 %	
	7) >40 - 60 %	
	8) >60 - 80 %	
	9) >80 – 100 %	
Woody layer component – old trees	1) 0 %	
	2) >0 - 3 %	
	3) >3 - 8 %	
	4) >8 - 15 %	
	5) >15 - 25 %	
	6) >25 - 40 %	
	7) >40 - 60 %	
	8) >60 - 80 %	
	9) >80 – 100 %	
Woody layer component - shrubs	1) 0 %	
	2) >0 - 3 %	
	3) >3 - 8 %	
	4) >8 - 15 %	
	5) >15 - 25 %	
	6) >25 - 40 %	
	7) >40 - 60 %	
	8) >60 - 80 %	

	9) >80 – 100 %	
Woody layer component – dwarf shrubs	1) 0 %	
	2) >0 - 3 %	
	3) >3 - 8 %	
	4) >8 - 15 %	
	5) >15 - 25 %	
	6) >25 - 40 %	
	7) >40 - 60 %	
	8) >60 - 80 %	
	9) >80 – 100 %	
Woody layer component – other wooded	1) 0 %	
	2) >0 - 3 %	
	3) >3 - 8 %	
	4) >8 - 15 %	
	5) >15 - 25 %	
	6) >25 - 40 %	
	7) >40 - 60 %	
	8) >60 - 80 %	
	9) >80 – 100 %	

9.1.14.4 Grassland type (Extended window + homogenous plot)

Grassland type	1) eC11	Very intensive meadow
	2) eC12	Intensive meadow
	3) eC13	Medium intensive meadow
	4) eC14	Extensive meadow
	5) eC15	Very extensive meadow
	6) eC16	Meadow or pasture (or both uses)
	7) eC21	Intensive pasture
	8) eC22	Medium intensive pasture
	9) eC23	Extensive pasture
	10) eC24	Only occasionally grazed (very extensive)
	11) eC25	Pastured woodland
	12) eC31	Ruderal / fallow grassland
	13) eC41	Amenity grassland
	14) eC51	Unclear / Other kinds of grassland

9.1.14.5 Growth status (Extended window + homogenous plot)

Growth status	1) Grassland not yet mown or pastured	
	2) Grassland mown or pastured a few days ago - (Cut /pastured grass / herbs, either lying on the ground or having been removed from plot, and vegetation ~ <20 cm – no inflorescences or blossoms)	
	3) Grassland in growth after mowing / grazing – (Vegetation > 20 cm)	
	4) NR	

9.1.14.6 EUNIS habitat type (Extended window + homogenous plot)

EUNIS Habitat Type	1) D1	Raised and blanket bogs
	2) D2	Valley mires, poor fens and transition mires
	3) D3	Aapa, palsa and polygon mires
	4) D4	Base-rich fens and calcareous spring mires
	5) D5	Sedge and reedbeds, normally without free-standing water
	6) D6	Inland saline and brackish marshes and reedbeds
	7) E1	Dry grasslands
	8) E2	Mesic grasslands
	9) E3	Seasonally wet and wet grasslands
	10) E4	Alpine and subalpine grasslands
	11) E5	Woodland fringes and clearings and tall forb stands
	12) E6	Inland salt steppes
	13) E7	Sparsely wooded grasslands: E7.1, E7.2, E7.3
	14) F3	Temperate and mediterranean-montane scrub
	15) F4	Temperate shrub heathland
	16) F5	Maquis, arborescent matorral and thermo-Mediterranean brushes
	17) F6	Garrigue
	18) G1	Broadleaved deciduous woodland
	19) G2	Broadleaved evergreen woodland
	20) G3	Coniferous woodland
	21) G4	Mixed deciduous and coniferous woodland
	22) G5	Lines of trees, small anthropogenic woodlands, recently felled woodland, early-stage woodland and coppice
	23) I1	Arable land and market gardens: I1.5 Bare tilled, fallow or recently abandoned arable land
	24) XX	Other
	25) NR	Not Relevant

9.1.15 Grassland

Possible if:

- If the point is part of the grassland module
- Grass cover is > 30% (grass, forbs as well as mosses/lichens if not on rocks)
- The point can be reached (obs_type = 1)

Grassland point (H)	Yes/No	Prefilled, based on the information on the sample
Grassland region (I)	Text	Prefilled, based on the information on the sample
Grassland survey possible? (96)	Yes/No	Is it possible to do the grassland assessment, including the transect, according to the rules? If not, skip the complete grassland module.
Grassland survey cannot be done Comment (500)	- structured comment - free text	Mandatory if survey cannot be done.

9.1.15.1 Grassland site (enlarged transect)

Site orientation (100)	(1) Flat	
	(2) N	North
	(3) NE	North-East
	(4) E	East
	(5) SE	South-East
	(6) S	South
	(7) SW	South-West
	(8) W	West
	(9) NW	North-West
	N.R.	Not relevant
Site slope (101)	0%	
	>0% - 3%	
	>3% - 8%	
	>8% – 15%	
	>15% - 25%	
	>25% – 40%	
	>40%	
Site moisture (102)	(1) Wet	

	(2) Moist	
	(3) Mesic	
	(4) Semi-dry	
	(5) Dry	
	(6) Mixed	
	N.R.	Not relevant

9.1.15.2 Grassland fertilisation (enlarged transect/field)

Fertilisation (106)	(1) No	
	(2) Probably	
	(3) For sure	
	(4) Unclear	
	N.R:	Not relevant

9.1.15.3 Grassland type (enlarged transect/field)

Grassland type	1) eC11	Very intensive meadow
	2) eC12	Intensive meadow
	3) eC13	Medium intensive meadow
	4) eC14	Extensive meadow
	5) eC15	Very extensive meadow
	6) eC16	Meadow or pasture (or both uses)
	7) eC21	Intensive pasture
	8) eC22	Medium intensive pasture
	9) eC23	Extensive pasture
	10) eC24	Only occasionally grazed (very extensive)
	11) eC25	Pastured woodland
	12) eC31	Ruderal / fallow grassland
	13) eC41	Amenity grassland
	14) eC51	Unclear / Other kinds of grassland

9.1.15.4 Growth status (enlarged transect)

Growth status	5) Grassland not yet mown or pastured	
	6) Grassland mown or pastured a few days ago - (Cut /pastured grass / herbs,	

	either lying on the ground or having been removed from plot, and vegetation ~ <20 cm – no inflorescences or blossoms)	
	7) Grassland in growth after mowing / grazing – (Vegetation > 20 cm)	
	8) NR	

9.1.15.5 EUNIS habitat type (enlarged transect)

EUNIS Habitat Type	1) D1	Raised and blanket bogs
	2) D2	Valley mires, poor fens and transition mires
	3) D3	Aapa, palsa and polygon mires
	4) D4	Base-rich fens and calcareous spring mires
	5) D5	Sedge and reedbeds, normally without free-standing water
	6) D6	Inland saline and brackish marshes and reedbeds
	7) E1	Dry grasslands
	8) E2	Mesic grasslands
	9) E3	Seasonally wet and wet grasslands
	10) E4	Alpine and subalpine grasslands
	11) E5	Woodland fringes and clearings and tall forb stands
	12) E6	Inland salt steppes
	13) E7	Sparsely wooded grasslands: E7.1, E7.2, E7.3
	14) F3	Temperate and mediterranean-montane scrub
	15) F4	Temperate shrub heathland
	16) F5	Maquis, arborescent matorral and thermo-Mediterranean brushes
	17) F6	Garrigue
	18) G1	Broadleaved deciduous woodland
	19) G2	Broadleaved evergreen woodland
	20) G3	Coniferous woodland
	21) G4	Mixed deciduous and coniferous woodland
	22) G5	Lines of trees, small anthropogenic woodlands, recently felled woodland,

		early-stage woodland and coppice
	23) I1	Arable land and market gardens: I1.5 Bare tilled, fallow or recently abandoned arable land
	24) XX	Other
	25) NR	Not Relevant

9.1.15.6 Grassland age (enlarged transect/field)

Grassland age (127)	(1) ≤5 years	
	(2) >5 years	
	(3) Unclear	
	N.R.	Not relevant
If visible signs are unclear Comment (500)	- structured comment - free text	

9.1.15.7 Grassland transect (transect)

Grassland transect starting point (130)	(1) On the point	
	(2) Shifted	
	N.R.	Not relevant
Direction of the grass transect shift (if shifted) (131)	(1) N	North
	(2) NE	North-East
	(3) E	East
	(4) SE	South-East
	(5) S	South
	(6) SW	South-West
	(7) W	West
	(8) NW	North-West
	N.R.	Not relevant
Distance of grass transect shift (if shifted) (132)	Value (in m)	
Reason of grass transect shift Comment (500)	- structured comment - free text	

Direction of grass transect (133)	(1) N	North
	(2) NE	North-East
	(3) E	East
	(4) SE	South-East
	(5) S	South
	(6) SW	South-West
	(7) W	West
	(8) NW	North-West
	N.R.	Not relevant
Length of grass transect (134)	(1) 20 m	
	(2) Shorter than 20 m	
Exact length, if shorter than 20 m	Value (in m)	
Start of grass transect latitude (135)	DD.ddddddd	GPS position (WGS84 latitude) of the start of grass transect latitude
Start of grass transect longitude (136, 137)	E/W DD.ddddddd	GPS position (WGS84 longitude) of the start of grass transect longitude
End of grass transect latitude (138)	DD.ddddddd	GPS position (WGS84 latitude) of the end of grass transect latitude
End of grass transect longitude (139, 140)	E/W DD.ddddddd	GPS position (WGS84 longitude) of the end of grass transect longitude

9.1.15.8 Grassland vegetation layers (transect, enlarged transect)

Vigour of vegetation (142)	1) 1	(transect)
	2) 1.5	
	3) 2	
	4) 2.5	
	5) 3	
	6) 3.5	
	7) 4	
	8) 4.5	
	9) 5	
	10) NR	Not relevant
Grass-forb ratio	1) Grass : Forb = 100 : 0	(transect)
	2) Grass : Forb = 90 : 10	
	3) Grass : Forb = 80 : 20	
	4) Grass : Forb = 70 : 30	
	5) Grass : Forb = 60 : 40	
	6) Grass : Forb = 50 : 50	
	7) Grass : Forb = 40 : 60	
	8) Grass : Forb = 30 : 70	
	9) Grass : Forb = 20 : 80	
	10) Grass : Forb = 10 : 90	
	11) Grass : Forb = 0 : 100	
Herbaceous layer	1) 0 %	(transect)
	2) >0 - 3 %	
	3) >3 - 8 %	
	4) >8 - 15 %	
	5) >15 - 25 %	
	6) >25 - 40 %	
	7) >40 - 60 %	
	8) >60 - 80 %	
	9) >80 - 100 %	

Bare layer	1) 0 %	(transect)
	2) >0 - 3 %	
	3) >3 - 8 %	
	4) >8 - 15 %	
	5) >15 - 25 %	
	6) >25 - 40 %	
	7) >40 - 60 %	
	8) >60 - 80 %	
	9) >80 – 100 %	
Woody layer	1) 0 %	(transect)
	2) >0 - 3 %	
	3) >3 - 8 %	
	4) >8 - 15 %	
	5) >15 - 25 %	
	6) >25 - 40 %	
	7) >40 - 60 %	
	8) >60 - 80 %	
	9) >80 – 100 %	
Woody layer component - orchards	1) 0 %	(enlarged transect)
	2) >0 - 3 %	
	3) >3 - 8 %	
	4) >8 - 15 %	
	5) >15 - 25 %	
	6) >25 - 40 %	
	7) >40 - 60 %	
	8) >60 - 80 %	
	9) >80 – 100 %	
Woody layer component – old trees	1) 0 %	(enlarged transect)
	2) >0 - 3 %	

	3) >3 - 8 %	
	4) >8 - 15 %	
	5) >15 - 25 %	
	6) >25 - 40 %	
	7) >40 - 60 %	
	8) >60 - 80 %	
	9) >80 – 100 %	
Woody layer component - shrubs	1) 0 %	(enlarged transect)
	2) >0 - 3 %	
	3) >3 - 8 %	
	4) >8 - 15 %	
	5) >15 - 25 %	
	6) >25 - 40 %	
	7) >40 - 60 %	
	8) >60 - 80 %	
	9) >80 – 100 %	
Woody layer component – dwarf shrubs	1) 0 %	(enlarged transect)
	2) >0 - 3 %	
	3) >3 - 8 %	
	4) >8 - 15 %	
	5) >15 - 25 %	
	6) >25 - 40 %	
	7) >40 - 60 %	
	8) >60 - 80 %	
	9) >80 – 100 %	
Woody layer component – other wooded	1) 0 %	(enlarged transect)
	2) >0 - 3 %	
	3) >3 - 8 %	
	4) >8 - 15 %	

	5) >15 - 25 %	
	6) >25 - 40 %	
	7) >40 - 60 %	
	8) >60 - 80 %	
	9) >80 – 100 %	

9.1.15.9 Grass height (transect)

Average height of the grassland	Value (in cm)	
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9.1.15.10 Grassland flowers (transect)

Number of species of flowering forbs (173)	1) 1	
	2) 2	
	33) 33	
	34) >33	
Flower density (174) (all flowers)	(1) 0 None	
	(2) 1 Very few	
	(3) 1.5 Very few to few	
	(4) 2 Few	
	(5) 2.5 Few to medium	
	(6) 3 Medium	
	(7) 3.5 Medium to dense	
	(8) 4 Dense	
	(9) 4.5 Dense to very dense	
	(10) 5 Very dense	
	N.R.	Not relevant

9.1.15.11 Flower colours (transect)

Flower colours	1) White	
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	2) Yellow	
	3) Orange	
	4) Red	
	5) Magenta	
	6) Violet	
	7) Blue	
	8) Green	
	9) Brown	
	10) Mixed	
Number of flower colours	Automatically filled	

9.1.15.12 Grassland key species / species group richness (transect)

Number of plants for each of the 12 key species group		
		Tick boxes Classification: 1) 0 2) 1-2 3) 3-10 4) >10
179	Apiaceae	
184	Campanula spp.	
185	Centaurea spp., Serratula tinctoria	
187	Cirsium spp., Carduus spp., Carlina spp.	
198	Juncus spp.	
201	Myosotis spp.	
202	Orchidaceae spp.	
209	Scabiosa spp., Knautia spp., Succisa spp.	
210	Silene spp., Lychnis flos-cuculi, Dianthus spp. (red flowering)	
214	Trifolium spp. (red flowering)	
215	Trifolium spp., Medicago spp., Lotus spp., Coronilla (yellow flowering)	
217	Vicia spp., Lathyrus spp., Astragalus spp. (blue or purple flowering)	

9.1.15.13 Grassland structural characterisation (transect)

Structural species / species group: (multiple choice)

- 1 ☐ 0 %
- 2 ☐ >0 - 3 %
- 3 ☐ >3 - 8 %
- 4 ☐ >8 - 15 %
- 5 ☐ >15 - 25 %
- 6 ☐ >25 - 40 %
- 7 ☐ >40 - 60 %
- 8 ☐ >60 - 80 %
- 9 ☐ >80 – 100 %

221	Shrub/ Brushwood/ Bramble	Tick boxes...
223	Thistles (Carduus spp, and others)	
225	Docks (Rumex obtusifolius, and others)	
227	Rushes (Juncus spp.)	
228	Sedges (Carex spp.)	
230	Wood small-reed (Calamagostis)	

9.1.15.14 Grassland legume cover (transect)

Total cover of legumes (%) (236)	1) 0 %	
	2) >0 - 3 %	
	3) >3 - 8 %	
	4) >8 - 15 %	
	5) >15 - 25 %	
	6) >25 - 40 %	
	7) >40 - 60 %	
	8) >60 - 80 %	
	9) >80 – 100 %	

9.1.16 Landscape features

9.1.16.1 Landscape feature module point (for LF PI and field assessment)

LF point?	Yes / No (pre-filled)	Indicates whether a point is to be considered for the LF module (Yes) or not (No)
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9.1.16.2 Landscape feature survey possible? (for LF PI and field assessment)

Landscape feature survey possible?	Yes	
	No	Go to the next module

9.1.16.3 Landscape feature types (should be filled in for all 41 subpoints in each LUCAS LF quadrat) (for LF PI and field assessment)

Sub-point ID	Value (Char 2)	Corresponds to the ID of the subpoint on the ground document (can take one of the following values: A1, A3, A5, A7, A9, B2, B4, B6, B8, C1, C3, C5, C7, C9, D2, D4, D6, D8, E1, E3, E5, E7, E9, F2, F4, F6, F8, G1, G3, G5, G7, G9, H2, H4, H6, H8, I1, I3, I5, I7, I9)
LF1 PI: primary LF type identified during the office PI	LF code (including "No LF", "NA")	Coding of LUCAS LF types according to section 8.15.1 (see also 9.2.8) Register a LF or "No LF" (default) or NA
LF2 PI: second LF type identified during the office PI	LF code (including "No LF", "NA")	Coding of LUCAS LF types according to section 8.15.1 (see also 9.2.8) A secondary LF type can be recorded if necessary Register a LF or "No LF" (default) or NA
LF1 FS: primary LF type identified during the field survey	LF code (including "No LF", "NA")	Coding of LUCAS LF types according to section 8.15.1 (see also 9.2.8) Register a LF or "No LF" (default) or NA
LF2 FS: second LF type identified	LF code (including "No LF", "NA")	Coding of LUCAS LF types according to section 8.15.1 (see also 9.2.8)

during the field survey	LF", "NA")	to section 8.15.1 (see also 9.2.8) A secondary LF type can be recorded if necessary Register a LF or "No LF" (default) or NA
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9.1.16.4 Landscape feature module: additional information (to be collected during the field assessment only)

Office PI confirmed?	1) Yes	Office PI is confirmed
	2) No, edit sub-points	Field situation different
If office PI is not confirmed: After editing	1) Confirm changes	
	2) Close without changes	
Orthophoto outdated within the quadrant?	1) Yes	
	2) No	
Additional walking needed?	1) Yes	
	2) No	

9.1.16.5 Landscape feature comments

Comments about the LF type assessment (500)	☐ Code XX1 LF office PI not possible because appropriate orto-photos are not available (LF office PI) ☐ Code XX2 LF field survey not possible because the quadrat is inaccessible (field survey) ☐ Code XX3 LF assessment is entirely impossible for any other reason + free text (LF office PI, field survey) ☐ Code YX: LF type of subpoint is impossible to determine/verify + ID sub-point(s) + free text (LF office PI, field survey) ☐ Code YY: LF present in the field is not visible on orthophoto + ID sub-point + photo ID (iterate if necessary) (field survey) ☐ Code YZ: LF visible on the orto-photo is missing in the field due to removal + ID sub-point(s) + free text (field survey) ☐ Code ZZ: Other issue with LF identification LF + free text (optional: ID sub-point(s) + photo ID (LF office PI, field survey) ☐	- structured comment - free text
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9.1.17 Soil

Only collect soil samples if observation distance ≤ 100 m and the point reached has the same land cover as the LUCAS point

9.1.17.1 Soil (Type of soil point)

Soil point? (K)	Yes/No (pre-filled)	Indicates whether a point is to be considered for soil collection (Yes) or not (No)
Bulk point (M) (0–10)	Yes/No (pre-filled)	Indicates whether a point is to be considered for collecting the bulk density sample (Yes) or not (No)
Bulk point (N) (10–20)	Yes/No (pre-filled)	Indicates whether a point is to be considered for collecting the bulk density sample (Yes) or not (No)
Bulk point (O) (20–30)	Yes/No (pre-filled)	Indicates whether a point is to be considered for collecting the bulk density sample (Yes) or not (No)
BIO point (P)	Yes/No (pre-filled)	Indicates whether a point is to be considered for collecting the biodiversity sample (Yes) or not (No)
Soil sample latitude	DD.ddddd	GPS position of the location from which the soil sample was taken (WGS84)
Soil sample longitude	E/W DD.ddddd	GPS position of the location from which the soil sample was taken (WGS84): - E/W - Coordinates

9.1.17.2 Litter sample

Litter sample taken	(1) Yes	If the point is a woodland point and the soil sample was taken.
	(2) Not possible	If the point is a woodland point and it is not possible to collect the litter sample (e.g. point not visible, or near a road).
	(8) Not relevant	If not applicable: the point is not a woodland soil point.
Litter label	Value (Char 6)	The number of the label placed on the plastic bag with the litter sample.
Structured comments Free text comments		For free text the use of English is mandatory. Special characters are to be avoided.

9.1.17.3 Soil site

Stones on the surface (%) (341)	1 - 100	Indicate the percentage of stones on the surface (does not include stones covered by soil)
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9.1.17.4 Soil biodiversity

If the point is a soil biodiversity point:

Soil biodiversity sample taken? (343)	Yes/No N.R.	
Biodiversity label (if sample is taken) (344)	Value (Char 6) N.R. = 888888	Corresponds to the number on the green plastic label placed in the plastic jar
Biodiversity expedition date (345)	YYY/MM/DD (e.g. 2022/05/23)	Date of observation. Valid date with YYYY=2022 or 2023, MM=01-12, DD=01..31
Comments about biodiversity (500)	Comments on special circumstances or problems: - Structured comments (to be preferred) - Free text comments	

9.1.17.5 Soil bulk density

If the point is a bulk density point:

Bulk (0–10) sample taken? (347)	Yes/No N.R.	
Bulk (0-10) label (if sample is taken) (348)	Value (Char 6) N.R. = 888888	Corresponds to the number on the yellow plastic label showing BULK 0-10
Bulk (10–20) sample taken? (349)	Yes/No N.R.	
Bulk (10-20) label (if sample is taken) (350)	Value (Char 6) N.R. = 888888	Corresponds to the number on the yellow plastic label showing BULK 10-20
Bulk (20–30) sample taken? (351)	Yes/No N.R.	
Bulk (20-30) label (if sample is taken) (352)	Value (Char 6) N.R. = 888888	Corresponds to the number on the yellow plastic label showing BULK 20-30
0-10 cores taken (multiple choice)		
Point (354)	Yes/No	
North (355)	Yes/No	
East (356)	Yes/No	
South (357)	Yes/No	
West (358)	Yes/No	
10-20 cores taken (multiple choice)		
Point (367)	Yes/No	
North (368)	Yes/No	
East (369)	Yes/No	
South (370)	Yes/No	
West (371)	Yes/No	
20-30 cores taken (multiple choice)		
Point (381)	Yes/No	
North (382)	Yes/No	
East (383)	Yes/No	

South (384)	Yes/No	
West (385)	Yes/No	
Comments about the bulk soil sample (500)	- structured comment - free text	
Weight of samples (field)		
Bulk 0-10 moist weight (395)	Value in g	
Bulk 10-20 moist weight (396)	Value in g	
Bulk 20-30 moist weight (397)	Value in g	
Weight of samples (air dried)		
Bulk 0-10 dried weight (399)	Value in g	
Bulk 10-20 dried weight (400)	Value in g	
Bulk 20-30 dried weight (401)	Value in g	

9.1.17.6 Standard soil sample

If the point is not a bulk density point.

Standard soil sample taken (403)	(1) Yes	If the point is a standard soil point and the soil sample was taken.
	(2) Not possible	If the point is a standard soil point and it is not possible to collect the soil sample (e.g. point not visible, or near a road).
	(8) Not relevant	If not applicable: the point is not a standard soil point.
Soil label (404)	Value (Char 6)	The number of the label placed on the plastic bag with the topsoil sample.
Comments about the soil sample (500)	Structured comments Free text comments	For free text the use of English is mandatory. Special characters are to be avoided.

9.1.17.7 Land owner details (soil sample results)

And owner requested soil results? (408)	Yes/No N.R:	
Land owner name (409)	CHAR(255)	
Land owner address (410)	CHAR(255)	(street/number/postal code/city/country)
Land owner phone (411)	CHAR(255)	
Land owner e-mail (412)	CHAR(255)	

9.1.18 Comments

Comments (500)	- structured comment - free text	See in the respective sections the fields numbered "500"
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9.1.19 Photos

It is mandatory that the surveyor does the anonymisation directly before sending the photos to the upper level (i.e. the Regional or the Central Office). According to the LUCAS 2022 tender, non-compliance to this rule is considered a breach of contract and will lead to legal consequences.

When a photo is taken, two fields have to be filled in:

Photo ID (P)	(Value) Photo ID	File name (in the camera). This information is for use as reference by the surveyor. The renaming of the photos by DMT is automatic.
To be anonymized	(Yes/No) To be anonymized	Tick the box if in the photo there are either people or vehicle license plates (including the plates of the surveyors' car!).

Photo of the Point	(1) Photo taken	The photo of the point is needed to find the point in the next survey. Therefore the photo should contain a recognisable and stable landmark. Use the flag when the point is reached. Never use the flag when the point is observed at more than 100m. The point photo is not to be taken when the point is PI.
	(2) Photo not taken	
	(8)	If not applicable. It is the case of a point not

	Not relevant	observed or photo interpreted.
Photo ID (P)	(Value) Photo ID	File name (in the camera). This information is for use as reference by the surveyor. The renaming of the photos by DMT is automatic.
To be anonymized	(Yes/No) To be anonymized	Tick the box if in the photo there are either people or vehicle license plates (including the plates of the surveyors' car!).
Photo of Crop	(1)Photo taken	Photo of the crop/cover should allow the identification of the crop and its phenological stage or the land cover. An adequate zoom (or macro) should be selected.
	(2) Photo not taken	
	(8) Not relevant	If not applicable.
Photo ID (C)	(Value) Photo ID	File name (in the camera). This information is for use as reference by the surveyor. The renaming of the photos by DMT is automatic.
To be anonymized	(Yes/No) To be anonymized	Tick the box if in the photo there are either people or vehicle license plates (including the plates of the surveyors' car!).
North East South West	(1) Photo taken	The landscape photos have to be taken in the four cardinal directions. The obligatory sequence (N-E-S-W) has to be respected.
	(2) Photo not taken	
	(8) Not relevant	If not applicable (e.g. in the case the point is not observed)
Photo ID (N, E, S, W)	(Value) Photo ID	File name (in the camera). This information is for use as reference by the surveyor. The renaming of the photos by DMT is automatic.
To be anonymized	(Yes/No) To be anonymized	Tick the box if in the photo there are either people or vehicle license plates (including the plates of the surveyors' car!).
Gully erosion (P, N, E, W, E)	(1) Photo taken	If there are signs of erosion a photo documenting these signs is needed

	(2) Photo not taken	
	(8) Not relevant	If not applicable (e.g. in the case the point is not observed)
Photo ID (J)	(Value) Photo ID	File name (in the camera). This information is for use as reference by the surveyor. The renaming of the photos by DMT is automatic.
To be anonymized	(Yes/No) To be anonymized	Tick the box if in the photo there are either people or vehicle license plates (including the plates of the surveyors' car!).
Copernicus photo	(1) Photo taken	For the Copernicus assessment a photo of the location the surveyor reached has to be taken
	(2) Photo not taken	
	(8) Not relevant	If not applicable (e.g. in the case the point is not observed)
Photo ID (K)	(Value) Photo ID	File name (in the camera). This information is for use as reference by the surveyor. The renaming of the photos by DMT is automatic.
To be anonymized	(Yes/No) To be anonymized	Tick the box if in the photo there are either people or vehicle license plates (including the plates of the surveyors' car!).
Photo of Irrigation	(1) Photo taken	Photo of the irrigation system should allow its identification.
	(2) Photo not taken	
	(8) Not relevant	If not applicable (e.g. in case point is not observed or LU is other than U111 or 112, or no irrigation equipment is visible from the observation point)
Photo ID (I)	(Value) Photo ID	File name (in the camera). This information is for use as reference by the surveyor. The renaming of the photos by DMT is automatic.
To be anonymized	(Yes/No) To be anonymized	Tick the box if in the photo there are either people or vehicle license plates (including the plates of the surveyors' car!).

Photo of soil standard	(1) Photo taken	In case a soil sample (standard topsoil sample or bulk density sample) has been taken, take a photo of the LUCAS point with the bag and as many as possible of the 5 holes clearly visible.
	(2) Photo not taken	
	(8) Not relevant	If not applicable.
Photo ID (L)	(Value) Photo ID	File name (in the camera). This information is for use as reference by the surveyor. The renaming of the photos by DMT is automatic.
To be anonymized	(Yes/No) To be anonymized	Tick the box if in the photo there are either people or vehicle license plates (including the plates of the surveyors' car!).
Photo of soil standard label	(1) Photo taken	In case a standard topsoil has been taken, take a photo of the label
	(2) Photo not taken	
	(8) Not relevant	If not applicable.
Photo ID (L)	(Value) Photo ID	File name (in the camera). This information is for use as reference by the surveyor. The renaming of the photos by DMT is automatic.
To be anonymized	(Yes/No) To be anonymized	Tick the box if in the photo there are either people or vehicle license plates (including the plates of the surveyors' car!).
Photo of bulk density label	(1) Photo taken	In case a bulk density sample has been taken, take a photo of the label
	(2) Photo not taken	
	(8) Not relevant	If not applicable.
Photo ID (Y)	(Value) Photo ID	File name (in the camera). This information is for use as reference by the surveyor. The renaming of the photos by DMT is automatic.
To be anonymized	(Yes/No) To be anonymized	Tick the box if in the photo there are either people or vehicle license plates (including the plates of the surveyors' car!).

Photo of biodiversity label	(1) Photo taken	In case a biodiversity sample has been taken, take a photo of the label
	(2) Photo not taken	
	(8) Not relevant	If not applicable.
Photo ID (Z)	(Value) Photo ID	File name (in the camera). This information is for use as reference by the surveyor. The renaming of the photos by DMT is automatic.
To be anonymized	(Yes/No) To be anonymized	Tick the box if in the photo there are either people or vehicle license plates (including the plates of the surveyors' car!).
	(2) Photo not taken	
	(8) Not relevant	If not applicable (e.g. in the case the point is not observed)
Photo ID (D, B, H, J, V)	(Value) Photo ID	File name (in the camera). This information is for use as reference by the surveyor. The renaming of the photos by DMT is automatic.
To be anonymized	(Yes/No) To be anonymized	Tick the box if in the photo there are either people or vehicle license plates (including the plates of the surveyors' car!).
Grassland with unclear age	(1) Photo taken	If the age of the grassland is unclear take a photo to document the available characteristics
	(2) Photo not taken	
	(8) Not relevant	If not applicable.
Photo ID (A)	(Value) Photo ID	File name (in the camera). This information is for use as reference by the surveyor. The renaming of the photos by DMT is automatic.
To be anonymized	(Yes/No) To be anonymized	Tick the box if in the photo there are either people or vehicle license plates (including the plates of the surveyors' car!).
Posy (xxx)	(1) Photo taken	Take a photo of a posy of flowering forbs found

	(2) Photo not taken	
	(8) Not relevant	If not applicable.
Photo ID (M)	(Value) Photo ID	File name (in the camera). This information is for use as reference by the surveyor. The renaming of the photos by DMT is automatic.
To be anonymized	(Yes/No) To be anonymized	Tick the box if in the photo there are either people or vehicle license plates (including the plates of the surveyors' car!).
Grassland transect start	(1) Photo taken	Take a photo from the start of the grassland transect looking towards the end of the grassland transect
	(2) Photo not taken	
	(8) Not relevant	If not applicable.
Photo ID (U)	(Value) Photo ID	File name (in the camera). This information is for use as reference by the surveyor. The renaming of the photos by DMT is automatic.
To be anonymized	(Yes/No) To be anonymized	Tick the box if in the photo there are either people or vehicle license plates (including the plates of the surveyors' car!).
Grassland from above	(1) Photo taken	Take a photo of a typical area of the grassland transect from above (by holding up the camera)
	(2) Photo not taken	
	(8) Not relevant	If not applicable.
Photo ID (R)	(Value) Photo ID	File name (in the camera). This information is for use as reference by the surveyor. The renaming of the photos by DMT is automatic.
To be anonymized	(Yes/No) To be anonymized	Tick the box if in the photo there are either people or vehicle license plates (including the plates of the surveyors' car!).
Grassland transect end	(1) Photo taken	Take a photo from the end of the grassland transect looking towards the start of the

		grassland transect
	(2) Photo not taken	
	(8) Not relevant	If not applicable.
Photo ID (Q)	(Value) Photo ID	File name (in the camera). This information is for use as reference by the surveyor. The renaming of the photos by DMT is automatic.
To be anonymized	(Yes/No) To be anonymized	Tick the box if in the photo there are either people or vehicle license plates (including the plates of the surveyors' car!).
Additional photos	Type of photo	These additional pictures can be used to illustrate conflict (why a point has not been reached) or complement the mandatory photos. Types can be: point access, point observation, land cover, land use, transect, soil, water management or other
Photo ID		File name (in the camera). This information is for use as reference by the surveyor. The renaming of the photos by DMT is automatic.
To be anonymized		Tick the box if in the photo there are either people or vehicle license plates (including the plates of the surveyors' car!).
Comments about photos Comment (500)	Structured comments Free text comments	Whenever there were problems in taking of photos the reasons have to be noted here. For free text the use of English is mandatory. Special characters are to be avoided.

9.1.20 GPS tracks and points

In addition to the data collected on the field form, and the pictures, the surveyor shall collect point and line data in the GPS.

9.2 Code lists

9.2.1 Code list for land cover (LUCAS LC 2022)

Land cover code	Land cover	Observation
8	Not relevant	Point (1.5m radius)
A11	Buildings with 1 to 3 floors	Point (1.5m radius)
A12	Buildings with more than 3 floors	Point (1.5m radius)
A13	Greenhouses	Point (1.5m radius)
A21	Non built-up area features	Point (1.5m radius)
A22	Non built-up linear features	Point (1.5m radius)
A30	Other artificial areas	Point (1.5m radius)
B11	Common wheat	Point (1.5m radius)
B12	Durum wheat	Point (1.5m radius)
B13	Barley	Point (1.5m radius)
B14	Rye	Point (1.5m radius)
B15	Oats	Point (1.5m radius)
B16	Maize	Point (1.5m radius)
B17	Rice	Point (1.5m radius)
B18	Triticale	Point (1.5m radius)
B19	Other cereals	Point (1.5m radius)
B21	Potatoes	Point (1.5m radius)
B22	Sugar beet	Point (1.5m radius)
B23	Other root crops	Point (1.5m radius)
B31	Sunflower	Point (1.5m radius)
B32	Rape and turnip rape	Point (1.5m radius)
B33	Soya	Point (1.5m radius)
B34	Cotton	Point (1.5m radius)
B35	Other fibre and oleaginous crops	Point (1.5m radius)
B36	Tobacco	Point (1.5m radius)
B37	Other non-permanent industrial crops	Point (1.5m radius)
B41	Dry pulses	Point (1.5m radius)
B42	Tomatoes	Point (1.5m radius)
B43	Other fresh vegetables	Point (1.5m radius)
B44	Floriculture and ornamental plants	Point (1.5m radius)
B45	Strawberries	Point (1.5m radius)
B51	Clovers	Point (1.5m radius)
B52	Lucerne	Point (1.5m radius)
B53	Other leguminous and mixtures for fodder	Point (1.5m radius)
B54	Mixed cereals for fodder	Point (1.5m radius)
B55	Temporary grasslands	Point (1.5m radius)
B71	Apple fruit	Extended window (20m radius)
B72	Pear fruit	Extended window (20m radius)
B73	Cherry fruit	Extended window (20m radius)
B74	Nuts trees	Extended window (20m radius)
B75	Other fruit trees and berries	Extended window (20m radius)
B76	Oranges	Extended window (20m radius)
B77	Other citrus fruit	Extended window (20m radius)

B81	Olive groves	Extended window (20m radius)
B82	Vineyards	Extended window (20m radius)
B83	Nurseries	Extended window (20m radius)
B84	Permanent industrial crops	Extended window (20m radius)
Bx1	Arable land (only PI)	Point (1.5m radius)
Bx2	Permanent crops (only PI)	Extended window (20m radius)
C10	Broadleaved woodland	Extended window (20m radius)
C21	Spruce dominated coniferous woodland	Extended window (20m radius)
C22	Pine dominated coniferous woodland	Extended window (20m radius)
C23	Other coniferous woodland	Extended window (20m radius)
C31	Spruce dominated mixed woodland	Extended window (20m radius)
C32	Pine dominated mixed woodland	Extended window (20m radius)
C33	Other mixed woodland	Extended window (20m radius)
D10	Shrubland with sparse tree cover	Extended window (20m radius)
D20	Shrubland without tree cover	Extended window (20m radius)
E10	Grassland with sparse tree/shrub cover	Extended window (20m radius)
E20	Grassland without tree/shrub cover	Extended window (20m radius)
E30	Spontaneously vegetated surfaces	Extended window (20m radius)
F10	Rocks and stones	Extended window (20m radius)
F20	Sand	Extended window (20m radius)
F30	Lichens and moss	Extended window (20m radius)
F40	Other bare soil	Extended window (20m radius)
G11	Inland fresh water bodies	Point (1.5m radius)
G12	Inland salty water bodies	Point (1.5m radius)
G21	Inland fresh running water	Point (1.5m radius)
G22	Inland salty running water	Point (1.5m radius)
G30	Transitional water bodies	Point (1.5m radius)
G40	Marine sea	Point (1.5m radius)
G50	Glaciers, permanent snow	Point (1.5m radius)
H11	Inland marshes	Extended window (20m radius)
H12	Peatbogs	Extended window (20m radius)
H21	Salt marshes	Extended window (20m radius)
H22	Salines and other chemical deposits	Extended window (20m radius)
H23	Intertidal flats	Extended window (20m radius)

9.2.2 Code list for land cover lower detail level for PI

Levels of Detail		
1st Level	2nd Level	3rd Level
Axx - Artificial	A1x Built-up artificial	A11 Buildings with 1 to 3 floors Point
		A12 Buildings with more than 3 floors Point
		A13 Greenhouses Point
	A2x Non-Built up artificial	A21 Non built-up area features Point
		A22 Non built-up linear features Point
		A30 Other artificial areas Point
Bxx - Cropland	B1x Field Crops (previously Bx1)	B11 Common wheat Point
		B12 Durum wheat Point
		B13 Barley Point
		B14 Rye Point
		B15 Oats Point
		B16 Maize Point
		B17 Rice Point
		B18 Triticale Point
		B19 Other cereals Point
		B21 Potatoes Point
		B22 Sugar beet Point
		B23 Other root crops Point
		B31 Sunflower Point
		B32 Rape and turnip rape Point
		B33 Soya Point
		B34 Cotton Point
		B35 Other fibre and oleaginous crops Point
		B36 Tobacco Point
		B37 Other non-permanent industrial crops Point
		B41 Dry pulses Point
		B42 Tomatoes Point
		B43 Other fresh vegetables Point
		B44 Floriculture and ornamental plants Point
		B45 Strawberries Point
		B51 Clovers Point
		B52 Lucerne Point
		B53 Other leguminous and mixtures for fodder Point
		B54 Mixed cereals for fodder Point
		B55 Temporary grasslands Point
	B7x Permanent Crops (previously Bx2)	B71 Apple fruit
		B72 Pear fruit
		B73 Cherry fruit
		B74 Nuts trees
		B75 Other fruit trees and berries
		B76 Oranges
		B77 Other citrus fruit
		B81 Olive groves
		B82 Vineyards
		B83 Nurseries
		B84 Permanent industrial crops

Levels of Detail		
1st Level	2nd Level	3rd Level
Cxx - Woodland	-	C10 Broadleaved woodland
	C2x - Spruce or Pine Dominated	C21 Spruce dominated coniferous woodland
		C22 Pine dominated coniferous woodland
		C23 Other coniferous woodland
	C3x - Spruce or Pine Dominated Mixed	C31 Spruce dominated mixed woodland
		C32 Pine dominated mixed woodland
		C33 Other mixed woodland
Dxx - Shrubland	-	D10 Shrubland with sparse tree cover
	-	D20 Shrubland without tree cover
Exx - Grassland	-	E10 Grassland with sparse tree/shrub cover
	-	E20 Grassland without tree/shrub cover
	-	E30 Spontaneously vegetated surfaces
Fxx - Bareland dominated	-	F10 Rocks and stones
	-	F20 Sand
	-	F30 Lichens and moss
	-	F40 Other bare soil
Gxx - Water	G1x - Inland water body	G11 Inland fresh water bodies Point
		G12 Inland salty water bodies Point
	G2x - Inland running water	G21 Inland fresh running water Point
		G22 Inland salty running water Point
	G3x - Sea Water	G30 Transitional water bodies Point
		G40 Marine sea Point
Hxx - Wetland	-	G50 Glaciers, permanent snow Point
	-	H11 Inland marshes
	-	H12 Peatbogs
	-	H21 Salt marshes
	-	H22 Salines and other chemical deposits
	-	H23 Intertidal flats

9.2.3 Code list for species: crops and nurseries

LC_SPECIES_CODE	LC_SPECIES_LABEL
8	Not relevant
B19a	Sorghum (<i>Sorghum bicolor</i>)
B19b	Buckwheat (<i>Fagopyrum esculentum</i> Mill.)
B19c	Common, golden or proso millet (<i>Panicum miliaceum</i> L.)
B19d	Canary grass (<i>Phalaris canariensis</i> L.)
B19f	Quinoa (<i>Chenopodium quinoa</i>)
B19g	Fonio (<i>Digitaria</i> spp.)
B19h	Other cereals not specified elsewhere
B23a	Fodder beet (roots of <i>Beta vulgaris</i>)
B23b	Fodder kale (<i>Brassica oleracea</i> L.)
B23c	Swedes (<i>Brassica napus</i> L. var. <i>napobrassica</i> (L.) Robb.)
B23d	Carrots (roots of <i>Daucus carota</i> L.)
B23e	Turnips (roots of <i>Brassica rapa</i> L.)
B23f	Jerusalem artichoke (<i>Helianthus tuberosus</i> L.)
B23g	Sweet potatoes (<i>Ipomoea batatas</i> (L.) Lam.)
B23h	Fodder parsnips (<i>Pastinaca sativa</i> L.)
B23i	Yam (<i>Discorea</i> sp.)
B23j	Manioc, Cassava (<i>Manihot esculenta</i> Crantz, syn. <i>M. utilissima</i>)
B23k	Horseradish (<i>Cochlearia armoracia</i>)
B23l	Tuberous chervil (<i>Chaerophyllum bulbosum</i>)
B23m	Other tropical tubers
B23n	Other root or tuber crops not specified elsewhere
B35a	Flax (<i>Linum usitatissimum</i> straw)
B35b	Hemp (<i>Cannabis sativa</i> straw)
B35c	Poppy seeds (<i>Papaver somniferum</i> L.)
B35d	Mustard seeds (<i>Sinapis alba</i> L., <i>Sinapis nigra</i>)
B35e	Safflower (<i>Carthamus tinctorius</i> L.)
B35f	Earth almond (<i>Cyperus esculentus</i> L.)
B35g	Sesame (<i>Sesamum indicum</i>)
B35h	Jute (<i>Corchorus</i> spp.)
B35i	Ramie (<i>Boehmeria nivea</i>)
B35j	Sisal (<i>Agave sisalana</i>)
B35k	Kenaf (<i>Hibiscus cannabinus</i>)
B35l	Bird rape (<i>Brassica rapa</i> var. <i>silvestris</i>)
B35m	Castor beans (<i>Ricinus communis</i>)
B35n	Other fibre and oleaginous crops not elsewhere specified
B37a	Chicory roots (<i>Cichorium intybus</i> var. <i>sativum</i>)
B37b	Hops (<i>Humulus lupulus</i>)
B37c	Medicinal plants (see C3)
B37d	Spices and aromatic plants (see C3)
B37e	Sugar cane (<i>Saccharum officinarum</i>)
B37f	Other non permanent industrial crops not specified elsewhere

B43a	Brassicas
B43b	Leaf and stalked vegetables other than brassicas (see C3)
B43c	Leguminous vegetables, incl. production for tins
B43e	Vegetables cultivated for fruit other than tomatoes
B43f	Cultivated mushrooms
B43g	Wild products (e.g.truffle, water chesnut)
B43h	Other fresh vegetables not specified elsewhere
B53a	Sainfoin (<i>Onobrychis viciifolia</i> Scop.)
B53b	Sweet clover or Trefoil (<i>Lotus corniculatus</i> L.)
B53c	Chickling vetch (<i>Lathyrus sativus</i>)
B53d	Black medick (<i>Medicago lupulina</i> L.)
B53e	Melilot (<i>Melilotus alba</i> Lam.)
B53f	Serradella (<i>Ornithopus sativus</i>)
B53g	Fenugreek (<i>Trigonella foenum-graecum</i>)
B53h	Sulla (<i>Hedysarium coronarium</i>)
B53i	Sweet lupins (<i>Lupinus albus</i> , <i>Lupinus angustifolius</i> , <i>Lupinus luteus</i>)
B53j	California bluebell (<i>Phacelia tanacetifolia</i>)
B53k	Other leguminous and mixtures for fodder not specified elsewhere
B75a	Apricot (<i>Prunus armeniaca</i>)
B75b	Cornel (<i>Cornus mas</i>)
B75c	Figs (<i>Ficus carica</i>)
B75d	Juneberry (<i>Amelanchier vulgaris</i>)
B75e	Kiwi (<i>Actinidia chinensis</i>)
B75f	Loquat (<i>Eriobotrya japonica</i>)
B75g	Medlar (<i>Mespilus germanica</i>)
B75h	Mirabellas (<i>Prunus insititia</i> var. <i>syriaca</i>)
B75i	Peach, nectarine and platerine (<i>Prunus persica</i> ; <i>Amygdalus persica</i> ; <i>Persica laevis</i>)
B75j	Persimons (<i>Diospyros kaki</i>)
B75k	Plums (<i>Prunus domestica</i>)
B75l	Pomegranates (<i>Punica granatum</i>)
B75m	Oleaster (<i>Eleagnus</i> sp.)
B75n	Quinces (<i>Cydonia oblonga</i>)
B75o	Sloes (<i>Prunus spinosa</i>)
B75p	Small fruits and berries (see C3)
B75q	Tropical fruits (see C3)
B75r	Other fruit trees and berries not specified elsewhere
B77a	Bergamot (<i>Citrus bergamia</i>)
B77b	Citron (<i>Citrus. medica</i> var. <i>cedrata</i>)
B77c	Grapefruit and pomelo (<i>Citrus maxima</i> , <i>C. grandis</i> , <i>C. paradisea</i>)
B77d	Lemons (<i>Citrus limon</i>)
B77e	Limes (<i>Citrus aurantifolia</i> , <i>Citrus limetta</i>)
B77f	Satsumas (<i>Citrus unshiu</i>)
B77g	Mandarines (<i>Citrus deliciosa</i> , <i>C. nobilis</i>)
B77h	Clementines (<i>C. clementina</i>)
B77i	Kumquats (<i>Fortunella</i> spp.)

B77k	Other citrus fruit not specified elsewhere
B83a	Horticultural nurseries
B83b	Vine and root-stock nurseries
B83c	Fruit tree nurseries
B83d	Ornamental plants nurseries
B83e	Tree and bushes nurseries (for planting in gardens, parks, embankments)
B83f	Forest tree nurseries
B83g	Other, non-specialized nurseries
B84a	Reeds, canary grass (<i>Phalaris arundinacea</i>) when cultivated and not natural
B84b	Other plants for plaiting (see C3)
B84c	Coffee
B84d	Tea
B84e	Other industrial plants for beverages (see C3)
B84f	Industrial plants for oil (see C3)
B84g	Permanent aromatic plants (see C3)
B84h	Permanent textile plants (see C3)
B84i	Plants for rubber and tannin (see C3)
B84j	Rose trees cultivated for rose-oil production (not as ornamental plants)
B84k	Other permanent industrial crops (see C3)
B84l	Miscanthus (<i>Miscanthus giganteus</i>)
B84m	Willow

9.2.4 Code list for land use (LUCAS LU 2022)

LU_CODE	LU_LABEL
8	Not relevant
U111	Agriculture (excluding fallow land and kitchen gardens)
U112	Fallow land
U113	Kitchen garden
U120	Forestry
U130	Aquaculture and fishing
U140	Mining and quarrying
U150	Other primary production
U210	Energy production
U221	Manufacturing of food, beverages and tobacco products
U222	Manufacturing of textile products
U223	Coal, oil and metal processing
U224	Production of non-metal mineral goods
U225	Chemical and allied industries and manufacturing
U226	Machinery and equipment
U227	Wood based products
U228	Printing and reproduction
U311	Railway transport
U312	Road transport
U313	Water transport
U314	Air transport
U315	Transport via pipelines
U316	Telecommunication
U317	Logistics and storage
U318	Protection infrastructures
U319	Electricity, gas and thermal power distribution
U321	Water supply and treatment
U322	Waste treatment
U330	Construction
U341	Commerce
U342	Financial, professional and information services
U350	Community services
U361	Amenities, museums, leisure
U362	Sport
U370	Residential
U411	Abandoned industrial areas
U412	Abandoned commercial areas
U413	Abandoned transport areas
U414	Abandoned residential areas
U415	Other abandoned areas
U420	Semi-natural and natural areas not in use

9.2.5 Code list for land use: lower detail level for PI

Levels of Detail	
2nd Level	3rd Level
U1x - Agricultural Use	U111 Agriculture (excluding fallow land and kitchen gardens)
	U112 Fallow land
	U113 Kitchen garden
-	U120 Forestry
-	U130 Aquaculture and fishing
-	U140 Mining and quarrying
-	U150 Other primary production
U2x - Industrial Use	U210 Energy production
	U221 Manufacturing of food, beverages and tobacco
	U222 Manufacturing of textile products
	U223 Coal, oil and metal processing
	U224 Production of non-metal mineral goods
	U225 Chemical and allied industries and manufacturing
	U226 Machinery and equipment
	U227 Wood based products
U3x - Transport Use	U228 Printing and reproduction
	U311 Railway transport
	U312 Road transport
	U313 Water transport
	U314 Air transport
	U315 Transport via pipelines
-	U316 Telecommunication
-	U317 Logistics and storage
-	U318 Protection infrastructures
-	U319 Electricity, gas and thermal power distribution
-	U321 Water supply and treatment
-	U322 Waste treatment
-	U330 Construction
-	U341 Commerce
-	U342 Financial, professional and information services
-	U350 Community services
-	U361 Amenities, museums, leisure
-	U362 Sport
-	U370 Residential
U4x - Abandoned Use	U411 Abandoned industrial areas
	U412 Abandoned commercial areas
	U413 Abandoned transport areas
	U414 Abandoned residential areas
	U415 Other abandoned areas
	U420 Semi-natural and natural areas not in use

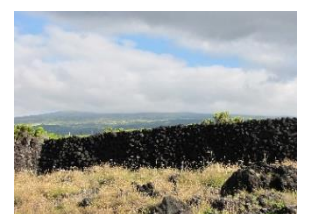
9.2.6 Code list for land use types

LU_TYPE_CODE	LU_TYPE_LABEL
U221L	Manufacture of food beverages or tobacco products (light end)
U222R	Manufacturing of textile products (raw)
U222L	Manufacturing of textile products (light end)
U223R	Coal, oil and metal processing (raw)
U223H	Coal, oil and metal processing (heavy end)
U224R	Production of non-metal mineral goods (raw)
U225R	Chemical and allied industries and manufacturing (raw)
U225L	Chemical and allied industries and manufacturing (light end)
U226H	Machinery and equipment (heavy end)
U226L	Machinery and equipment (light end)
U227R	Wood based products (raw)
U228L	Printing and reproduction (light end)
U315W	Transport of water via pipelines
U315O	Transport of other material via pipelines

9.2.7 Code list for linear features

Find below the codes to be used in the case of crossing with a linear feature.

Type	<3m	≥3m		
<u>Grass margins</u> Strip of mainly uncultivated (not agriculturally used) vegetation, dominated by grasses, grass-like plants or herbs. Often located at the edge of fields, between cropped areas (beetle banks) or bordering roads and tracks (roadside verge) as well as associated with water courses.	01	E10 E20 E30	 © 2013, Kay Harrington (@Kay), Pixabay License.	 © 2014, Manfred Antranias Zimmer (@Antranias), Pixabay License.
			Figure 9-1	Figure 9-2
<u>Heath/Shrub, tall herb fringes</u>	02	D10 D20 E30	 © Allenpaul1000, adobestock.com.	
			Figure 9-3	
<u>Single bushes/trees</u>	10		 © 2020, DEZALB, Pixabay License.	 © 2018, Joe (@Jplenio), Pixabay License.
			Figure 9-4	Figure 9-5
<u>Avenue trees or other lines of trees</u> Refers to one line of trees, not clustered trees. Two lines of trees (avenue trees) are normally separated by a road.	11		 © 2013, Anna (@AnnaER), Pixabay License	 © 2014, Manfred Antranias Zimmer (@Antranias), Pixabay License.
			Figure 9-6	Figure 9-7

<u>Conifer hedges</u>	12	C2X D10 D20 E30	 © 2019, Peter H (@tama66), Pixabay License.	 © 2017, Helga Loessl (@helgaloessl1), Pixabay License.
			Figure 9-8	Figure 9-9
<u>Managed bush or tree hedges or coppices</u> They should be visibly managed (e.g. pollarded). Generally they are < 5 m height	13	C10 C2X C3X	 © 2017, O K (@islandworks), Pixabay License.	 © 2018, Mabel Amber (@MabelAmber), Pixabay License.
			Figure 9-10	Figure 9-11
<u>Not managed bush or tree hedges</u> They can have single trees or shrubs, deriving from abandonment. Shrub or wood margins are found as field boundaries within agricultural land or alongside roads or water courses.	14	C10 C2X C3X D10 D20	 © 2019, Franck Barske (@braskefrank), Pixabay License.	 © 2020, obenbleibenstuttgart, Pixabay License.
			Figure 9-12	Figure 9-13
<u>Grove/Woodland margins (if no hedgerow)</u>	15	C10 C2X C3X	 © 2017, Catkin, Pixabay License.	 © 2012, PublicDomainPictures, Pixabay License.
			Figure 9-14	Figure 9-15
<u>Dry stone walls</u> Also includes stone heaps which were collected by the farmer on the field even though not in a linear form.	21		 © 2017, John Mounsey (@mounsey), Pixabay License.	 © 2017, MB1975, Pixabay License.
			Figure 9-16	Figure 9-17

<u>Artificial constructions (other than dry stone walls)</u> Various man-made built structures (e.g. walls, noise walls, pipelines, wind turbines, dams or terraces etc.) of different material such as metal, bricks or concrete	22	A	 © 2019, Jim Black (@jdblack), Pixabay License.	 © 2018, Andreas Senftleben (@AS_PHOTOGRAPHICS), Pixabay License.
			Figure 9-18	Figure 9-19
<u>Fences</u>	23		 © 2012, Tina Shaskus (@minx©267), Pixabay License.	 © 2020, Armin Foster (@ArminEP), Pixabay License.
			Figure 9-20	Figure 9-21
<u>Electric lines and other aerial cables</u>	24	24	 © 2015, Walter Bichler (@Silberfuchs), Pixabay License.	 © 2014, Jan Mallander (@Jan-Mallander), Pixabay License.
			Figure 9-22	Figure 9-23
<u>Ditches, channels</u> Artificial drainage or irrigation line, usually straight, temporarily or permanently wet. Ditches are frequently found in agricultural land for lower the water table or drainage. They are often associated with roadside verges used to drain the runoff from the associated road. Ditches are to be recorded independently from their width and inside artificial areas (A). Edges or banks along the small water body are to be recorded separately	31	G21 G22	 © 2019, Kobus van Leer (@Kobus-van-Leer), Pixabay License.	 © 2018, Mabel Amber (@MabelAmber), Pixabay License.
			Figure 9-24	Figure 9-25

<u>Rivers and streams</u> A linear body of water, often flowing in its naturally shaped bed through the land into a body of water such another stream, a lake or the ocean. Banks or edges (riverside vegetation) have to be recorded separately as grass, shrub or wood margin. Rivers and streams are collected even if within artificial areas (A)	32	G21 G22	 © 2012, Larisa Koshkina (@Larisa-K), Pixabay License.	 © 2020, Polina Andreeva (@mystraysoul), Pixabay License.
			Figure 9-26	Figure 9-27
<u>Ponds, wetlands</u>	41	G11 H1X H2X	 © 2020, Annette Meyer (@Nennieinzweidrei), Pixabay License.	 © 2017, Marisa04, Pixabay License.
			Figure 9-28	Figure 9-29
<u>Rock outcrops with some natural vegetation</u>	51	F10	 © 2021, Julita (@pasja1000), Pixabay License.	 © 2017, Demetrius (@NaturePhoto), Pixabay License.
			Figure 9-30	Figure 9-31
<u>Tracks</u> Usually rough tracks, mainly used to access agricultural land or forests, in most cases unpaved. They are not part of the public road network thus often closed for public transport. This category includes all type of paths. Roadside vegetation has to be recorded separately.	61	61	 © 2014, Manfred Antranas Zimmer (@Antranas), Pixabay License.	 © 2014, alohamalakov, Pixabay License.
			Figure 9-32	Figure 9-33

<u>Paved roads, cycling and walking lanes</u> Mainly part of the official traffic road network composed of roads of different levels (urban streets to highways). Roadside vegetation has to be recorded separately if it is wider than 1m. If a road and a walking or cycling lane are separated by a vegetation stripe wider than 1 m all elements need to be coded separately. If there is no vegetation stripe wider than 1 m one code is enough.	62	62	 © 2014, Ryan McGuire (@RyanMcGuire), Pixabay License.	 © 2013, Kay Harrington (@Kay), Pixabay License.
			Figure 9-34	Figure 9-35
<u>Railways</u> A set of rails on which trains run. Green linear features bordering the railway track are to be recorded separately.	63	63	 © 2019, Jerzy Górecki (@JerzyGorecki), Pixabay License.	 © 2016, Joshua_Wilson, Pixabay License.
			Figure 9-36	Figure 9-37
<u>Other structural elements</u> such as firebreaks.	71	71		

9.2.8 Code list for landscape features

Code list for the LF fields (LF1 PI, LF2 PI, LF1 FS, LF2 FS):

LF CODE	LF label
W	Woody vegetation LF (trees, shrubs, hedgerows)
G	Permanent grass/herbaceous vegetation LF
T	Temporary herbaceous LF (flower strips on arable land)
D	Ditches, channels, streams
P	Small ponds, small wetlands
S	Stone walls, terraces
C	Cultural features
No LF	There is no LF at the sub-point
NA	The LF of the sub-point could not be determined (comment needed!)

9.2.9 Code lists for structured comments

9.2.9.1 Type of comments for field work

CODE	TYPE OF COMMENT (Field work)
AA	Description of the way to the point
AB	Reasons for not reaching the point
AC	Vehicle
BA	Quality of background information
BD	Land cover
BE	GPS
BF	Exceptional survey time
BH	Data entry
BJ	Point marker
BP	Panel points
FA	Photos
GA	Grassland age
GC	Grassland EUNIS
GE	Extended Grassland
GL	Grassland legumes
GN	Grassland assessment not possible
GO	Grassland other species
GP	Grassland specific site characteristics
GS	Grassland transect
GG	Grassland – other comments
HA	Trees with secondary crops/meadow/pasture
LF	Landscape features
LL	Land Use - other comments
PA	Copernicus
QA	Land management
RA	Gully Erosion
SA	Soil
SB	Soil bulk
SC	Soil biodiversity
TA	INSPIRE Non vegetated area
XX	General comment
ZZ	RO/CO awaiting dry soil sample weight

9.2.9.2 Type of comments for quality checks

CODE	TYPE OF COMMENT (Quality checks)
EA	Overall quality
EB	Observation problems
EC	Land cover problems
ED	Land use problems
EF	Additional information problems
EG	Grassland problems
EH	Trees with secondary crops/meadow/pasture
EI	Photos missing
EJ	Corrupt photos

EK	Bad quality photos
EL	Non-anonymized photos
EM	Missing geo-tags in photos
EN	Missing GPS tracks
EO	Other comments
EP	Copernicus problems
EQ	Land management problems
ER	Erosion problems
ES	Soil problems
EW	Water management problems

9.2.9.3 Code list for field work comments

CODE	LABEL
AA01	Long walk
AA02	Water body between the road and the point
AA03	Easiest access from North
AA04	Easiest access from South
AA05	Easiest access from East
AA06	Easiest access from West
AA07	Information requested by land owner
AA08	Point revisited (insert date of previous visit)
AA99	Other description of the way comments
AB01	Fence, wall, locked gate/door, natural obstacle
AB02	Access forbidden sign found
AB03	Landowner refused the entrance
AB04	Adverse meteorological conditions (storm, flooding, snow ice etc.)
AB05	Dangerous animals (cows/bulls, dogs, wild animals etc.)
AB06	Impenetrable vegetation / poisonous plants
AB07	Point in water
AB08	Point in wetland
AB09	Shortest possible walking time (from car to point and back without time for data collection) more than 1 hour
AB10	Military or official area (e.g. naval, airfield)
AB11	Steep slope
AB12	High crop on the ground
AB13	Non-drivable road/track
AB14	Access forbidden due to natural protected area
AB15	Restricted access due to fire hazard
AB16	Restricted access due to hunting
AB17	Restricted access due to fitosanitary treatment
AB18	Restricted access. Payment needed
AB19	Need to cross by boat to access the point
AB20	GPS problem : impossible to reach the point
AB99	Other reasons for not reaching the point comments
AC01	Car damaged
BA01	Ground documents not correct
BA02	Orthophoto not available
BA03	Not updated orthophoto
BA04	PI reference other than GD ortho
BA99	Other quality of background information comments
BD01	Dehesa

BD02	Layered crop
BD03	Local knowledge
BD04	Tilled arable land not sown during current season
BD05	Management of migratory animals
BD07	Windthrow
BD08	LC classified with the help of other information
BD99	Other land cover comments
BE01	GPS battery discharged - track to the point not saved
BE02	GPS battery discharged - point location not saved
BE03	GPS battery discharged - track and point location not saved
BE04	GPS not working - track to the point not saved
BE05	GPS not working - point location not saved
BE06	GPS not working - track and point location not saved
BE07	GPS signal weak, lost or high positional error (forest, dense urban,...)
BE08	GPS wrong settings (different ellipsoid/coordinate system,...)
BE20	GPS problem : impossible to reach the point
BE21	Ground document quality problem : impossible to reach the point
BE22	GPS & Ground document quality problems : impossible to reach the point
BE99	Other GPS comments
BF04	Survey time < 10' because it was easy to assess the LC/LU
BF06	Survey time > 1h 15' because it was very difficult to assess the LC/LU
BF07	Survey time < 10' because LC/LU photo interpreted
BF08	Survey time > 1h 15' because it was very difficult to reach the point
BF10	Survey time < 10' because it was impossible to reach the point
BF99	Other survey time comments
BH01	Problem with data input: hardware
BH02	Problem with data input: software
BH03	Problem using the anonymization tool
BJ01	Point marker/flag missing
BJ02	Point marker/flag replaced by other object
BJ03	Point marker/flag placed by mistake or not on point
BJ99	Other point marker comments
BP01	Previous data probably erroneous or wrongly located
BP02	Different location from previous survey/visit: better GD
BP03	Different location from previous survey/visit: point not reachable in present campaign
BP04	Different location from previous survey/visit: closer or better view to the point
BP05	Different Look to N/E decision than in previous survey/visit. Explain why.
BP06	Shift in Orthophotos of different years
BP07	Shift between Orthophoto and GPS
BP08	Different decision on LC: real change
BP09	Different decision on LU: real change
BP11	Different decision than in previous survey/visit: rules changed
BP12	Different decision than in previous survey/visit: PI in previous survey
BP13	No real change. Reason not clear
BP14	Different decision on LC/LU because of shift in orthophotos of different years
BP15	No real change: Other reason
BP99	Other panel points comments
FA01	Camera's battery discharged - impossible to take photo(s)
FA02	Camera broken - impossible to take photo(s)
FA03	Memory Card Problem
FA04	Land owner refused photos
FA05	1/6 horizon rule not respected because an object in the front (house wall)
FA06	1/6 horizon rule not respected because photo taken in forest
FA07	Camera date/time set incorrectly
FA08	Compass malfunctioning
FA22	Landscape photo shifted (m, direction)

FA99	Other photo comments
GA99	Other grassland age comments
GC99	Other grassland EUNIS comments
GE01	Extended grassland not possible
GE02	Other extended grassland EUNIS comment
GE99	Extended grassland other comments
GL99	Other grassland legumes comments
GO99	Other grassland other species comments
GP01	Survey on small strips in orchards
GS01	Reason for shorter grassland transect
GS99	Other grassland transect shift comments
GG99	Other grassland comments
GN02	Grassland assessment not possible
HA99	Other Trees with secondary crops/meadow/pasture comments
LF01	LF office PI not possible because appropriate ortho-photos are not available (Office PI)
LF02	LF field survey not possible because the quadrat is inaccessible (Field survey)
LF03	LF assessment is entirely impossible for any other reason + free text
LF04	LF type of subpoint is impossible to determine/verify + ID sub-point(s)
LF05	LF visible on the orto-photo is missing in the field due to removal + ID sub-point(s) (Field survey)
LF06	LF not visible on orthophoto + ID sub-point(s) + photo ID (Field survey)
LF99	Other issue with LF identification LF + free text (+ ID sub-point(s) +photo(s) ID)
LL99	Land use - Other comment
PA01	LC change hidden by a feature <3 m or the terrain
PA02	LC change existing but not located within the area covered by the photo
PA99	Other Copernicus comments
QA99	Other land management comments
RA99	Other gully erosion comments
SA01	Soil sample very wet
SA02	Many stones in the area, stones removed
SA04	Soil sample collected at the edge. Approx. distance from the edge & reason:
SA05	Ground conditions unfavourable
SA06	Inaccessible point (e.g. fenced area)
SA07	Land owner refused the soil sample
SA08	Shallow soils - target soil depth (30 cm) not reached. Depth reached:
SA99	Other soil comments
SB01	Not all cores were collected
SB99	Other soil bulk comments
SC99	Other soil biodiversity comments
TA01	Unvegetated area partly assessed through PI
XX99	Any other comment on the survey
ZZ01	RO/CO awaiting dry soil sample weight

9.2.9.4 Code list for quality checks comments

CODE	LABEL
EA01	Point ID and location do not match
EA02	Data entry not complete
EA03	Observed data do not belong to the point
EA04	Point not correctly surveyed – to be visited again
EA05	Point corrected after DR (double rejection)
EA06	Point not correctable after DR (double rejection)
EA99	Other overall quality problems
EB01	Incorrect date

EB02	Incorrect start/end time
EB03	Observation time rare - missing explanation
EB04	Point not accessible questionable
EB05	Observation type questionable
EB06	PI reasons not justifiable
EB07	Observation direction questionable/wrong
EB08	Coordinates or distance to the point wrong
EB09	Distance to point unreasonably long
EB10	Track record and lat/long do not match
EB11	Missing track records
EB12	Incorrect application of look to the north/east rule
EB13	Comments about special circumstances missing
EB99	Other observation problems
EC01	LC 1 questionable/not corresponding to the plant photo
EC02	LC 2 questionable/not corresponding to the plant photo
EC03	LC 1 questionable/not corresponding to the orthophoto
EC04	LC 2 questionable/not corresponding to the orthophoto
EC05	LC 1 questionable/not corresponding to previous data
EC06	LC 2 questionable/not corresponding to previous data
EC07	LC 2 missing
EC08	Percentage of land cover 1 wrong
EC09	Percentage of land cover 2 wrong
EC99	Other land cover problems
ED01	LU 1 questionable/not corresponding to the plant photo
ED02	LU 2 questionable/not corresponding to the plant photo
ED03	LU 1 questionable/not corresponding to the orthophoto
ED04	LU 2 questionable/not corresponding to the orthophoto
ED05	LU 1 questionable/not corresponding to previous data
ED06	LU 2 questionable/not corresponding to previous data
ED07	LU 2 missing
ED08	LC & LU unlikely combination
ED99	Other land use problems
EF01	Height of trees questionable
EF02	Parcel area questionable
EF03	Width of feature missing/wrong
EF04	Plant species missing/wrong
EF05	Questionable grazing code
EF06	Wrong special comment
EF99	Other additional information problems
EG99	Other grassland problems
EH99	Other EUNIS Habitat Complex problems
EI01	Point photo missing
EI02	Crop/cover photo missing
EI03	North photo missing
EI04	East photo missing
EI05	South photo missing
EI06	West photo missing
EI08	Irrigation photo missing
EI09	Soil photo missing
EI10	Additional photo missing
EI11	Crop/cover photo missing due to change in LC (ex-post)
EI99	Other photo missing problems
EJ01	Point photo corrupt
EJ02	Crop/cover photo corrupt
EJ03	North photo corrupt
EJ04	East photo corrupt

EJ05	South photo corrupt
EJ06	West photo corrupt
EJ07	Transect photo corrupt
EJ08	Irrigation photo corrupt
EJ09	Soil photo corrupt
EJ10	Additional picture corrupt
EJ99	Other corrupt photo problems
EK01	Point photo of bad quality
EK02	Crop/cover photo of bad quality
EK03	North photo of bad quality
EK04	East photo of bad quality
EK05	South photo of bad quality
EK06	West photo of bad quality
EK07	Transect photo of bad quality
EK08	Irrigation photo of bad quality
EK09	Soil photo of bad quality
EK10	Additional photo of bad quality
EK99	Other bad quality photo problems
EL01	Point photo not anonymised
EL02	Crop/cover photo not anonymised
EL03	North photo not anonymised
EL04	East photo not anonymised
EL05	South photo not anonymised
EL06	West photo not anonymised
EL07	Transect photo not anonymised
EL08	Irrigation photo not anonymised
EL09	Soil photo not anonymised
EL10	Additional photo not anonymised
EM01	Point photo with wrong/missing geo-tag
EM02	Crop/cover photo with wrong/missing geo-tag
EM03	North photo with wrong/missing geo-tag
EM04	East photo with wrong/missing geo-tag
EM05	South photo with wrong/missing geo-tag
EM06	West photo with wrong/missing geo-tag
EM07	Transect photo with wrong/missing geo-tag
EM08	Irrigation photo with wrong/missing geo-tag
EM09	Soil photo with wrong/missing geo-tag
EM10	Additional photo with wrong/missing geo-tag
EN01	GPS Point missing
EN02	GPS Point and Track missing
EN99	GPS Track missing
EO01	Requires Eurostat correction in previous campaign
EO02	Quality check at CO was bypassed via DMT Client
EP99	Other Copernicus problems
EQ99	Other land management problems
ER99	Other erosion problems
ES99	Other soil problems
EW01	Missing irrigation data (seen on the photo)
EW02	No source of irrigation identified
EW03	Type of irrigation questionable
EW04	Water delivery system questionable
EW99	Other water management problems

9.3 Combinations of LC1 x LC2

C1_Annexes_9.3-9.7_CrossTables_LUCAS2022.xlsx

9.4 Combinations of LU1 x LU2

C1_Annexes_9.3-9.7_CrossTables_LUCAS2022.xlsx

9.5 Combinations of LC x LU

C1_Annexes_9.3-9.7_CrossTables_LUCAS2022.xlsx

9.6 Unlikely land cover changes

C1_Annexes_9.3-9.7_CrossTables_LUCAS2022.xlsx

9.7 Unlikely land use changes

C1_Annexes_9.3-9.7_CrossTables_LUCAS2022.xlsx

9.8 Cross table LF1 * LF2

The allowed combinations of LF1 and LF2 are the following:

		LF2								
LF1 LF2		No LF	W	G	T	D	P	S	C	NA
LF1	No LF	2	0	0	0	0	0	0	0	0
	W	2	0	0	0	2	2	2	2	0
	G	2	0	0	0	0	0	0	2	0
	T	2	0	0	0	0	0	0	1	0
	D	2	0	0	0	0	0	0	0	0
	P	2	0	0	0	0	0	0	0	0
	S	2	0	0	0	0	0	0	1	0
	C	0	0	0	0	0	0	0	0	0
	NA	0	0	0	0	0	0	0	0	2

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With the following meaning:

2 = possible/common combinations

1 = unlikely combinations

0 = impossible combinations

9.9 EUNIS Habitat Classification

9.9.1 Introduction

EUNIS (European nature information system) is a classification of vegetation types based on vegetation composition, therefore is significantly different from the LU/LC types in LUCAS. For harmonisation with other environmental and nature conservation databases, a simplified version of EUNIS was added as a parameter in the LUCAS grassland module. The following EUNIS types were selected for recording in the LUCAS grassland module.

D1 - Raised and blanket bogs

D2 - Valley mires, poor fens and transition mires

D3 - Aapa, palsa and polygon mires

D4 - Base-rich fens and calcareous spring mires

D5 - Sedge and reedbeds, normally without free-standing water

D6 - Inland saline and brackish marshes and reedbeds

E1 - Dry grasslands

E2 - Mesic grasslands

E3 - Seasonally wet and wet grasslands

E4 - Alpine and subalpine grasslands

E5 - Woodland fringes and clearings and tall forb stands

E6 - Inland salt steppes

E7 - Sparsely wooded grasslands: E7.1, E7.2, E7.3

F3 - Temperate and mediterranean-montane scrub

F4 - Temperate shrub heathland

F5 - Maquis, arborescent matorral and thermo-Mediterranean brushes

F6 - Garrigue

G1 - Broadleaved deciduous woodland

G2 - Broadleaved evergreen woodland

G3 - Coniferous woodland

G4 - Mixed deciduous and coniferous woodland

G5 - Lines of trees, small anthropogenic woodlands, recently felled woodland, early-stage woodland and coppice

I1 - Arable land and market gardens: I1.5 Bare tilled, fallow or recently abandoned arable land

XX - Other

Not Relevant

9.9.2 EUNIS descriptions of the most frequent habitat types in the LUCAS grassland survey to Level 2

Several broad habitat types have similar descriptions for both LUCAS and **EUNIS** systems, for example:

- C10 Broadleaved woodland and **G1 Broadleaved deciduous woodland or G2 Broadleaved evergreen woodland**. Similarly, the definitions of
- E10 Grassland with sparse tree/shrub cover and **E7 Sparsely wooded grassland** match quite closely (note that both LUCAS and EUNIS define woodland as areas with a canopy cover of >10 %).

However, this is not always the case and particular care should be taken when assigning habitat types in the following categories:

- Inland wetlands (LUCAS H10) are differentiated into at least seven categories in EUNIS (C3, D1-6)
- Grassland without tree/shrub cover (LUCAS E20) can belong to at least 6 EUNIS types, including specialised habitats ranging **from subalpine grasslands to salt steppes** (E1-6).
- Shrubland with sparse tree cover (LUCAS D10) or Shrubland without tree cover (D20) could correspond to 10 EUNIS types (F2-9, FA or FB).

Please note, that for reasons of simplicity a few EUNIS-types which occur very seldom in grasslands have not been listed in the following list, e.g. C3, FA, FB; thus types which do not at all meet the descriptions of the following types have to be recorded as **EUNIS type XX** in the form.

The following definitions of the EUNIS codes are taken from the official EUNIS guidance (Version 2017: <https://www.eea.europa.eu/data-and-maps/data/eunis-habitat-classification>, accessed 14/01/2021) – for more information, please consult the original definitions which are free to download. Please note that this guidance is written for whole Europe, so some types can be immediately excluded for specific regions (e.g. Mediterranean Maquis doesn't apply for the Boreal zone). Additional notes relevant for the application of the EUNIS codes are written in *italics*.

D Mires, bogs and fens

D1 Raised and blanket bogs

Peatlands formed by ombrotrophic acid peat, which is (or was while actively growing) capable of growth fed by rainfall rather than by the inflow of water from higher ground in the vicinity.

D2 Valley mires, poor fens and transition mires

Weakly to strongly acid peatlands, flushes and vegetated rafts formed in situations where they receive water from the surrounding landscape or are intermediate between land and water. Included are quaking bogs and vegetated non-calcareous springs. Excluded are calcareous fens (D4), and reedbeds (C3, D5).

D3 Aapa, palsa and polygon mires

Patterned mire complexes of the arctic, subarctic and northern boreal zones.

D4 Base-rich fens and calcareous spring mires

Peatlands, flushes and vegetated springs with calcareous or eutrophic ground water, within river valleys, alluvial plains, or on hillsides. As in poor fens, the water level is at or near the surface of the substratum and peat formation depends on a permanently high watertable. Excluded are reedbeds (C3, D5).

D5 Sedge and reedbeds, normally without free-standing water

Sedge and reedbeds forming terrestrial mire habitats, not closely associated with open water. Excluded are reedbeds and sedges where they form emergent or fringing vegetation beside water bodies (C3.2).

D6 Inland saline and brackish marshes and reedbeds

Saline wetlands, with closed or open vegetation, which are the non-coastal analogue of coastal saltmarshes and saline reedbeds (A2.5). Drier saline habitats are classified as inland salt steppe (E6) or saline scrubland (F6.8).

E Grasslands and lands dominated by forbs, mosses or lichens

E1 Dry grasslands

Well-drained or dry lands dominated by grass or herbs, mostly not fertilized and with low productivity. Included are [Artemisia] steppes. Excluded are dry mediterranean lands with shrubs of other genera where the shrub cover exceeds 10%; these are listed as garrigue (F6).

E2 Mesic grasslands

Lowland and montane mesotrophic and eutrophic pastures and hay meadows of the boreal, nemoral, warm-temperate humid and mediterranean zones. They are generally more fertile than dry grasslands (E1), and include sports fields and agriculturally improved and reseeded pastures.

E3 Seasonally wet and wet grasslands

Unimproved or lightly improved wet meadows and tall herb communities of the boreal, nemoral, warm-temperate humid, steppic and mediterranean zones.

E4 Alpine and subalpine grasslands

Primary and secondary grass- or sedge- dominated formations of the alpine and subalpine levels of boreal, nemoral, mediterranean, warm-temperate humid and Anatolian mountains.

E5 Woodland fringes and clearings and tall forb stands

Stands of tall herbs or ferns, occurring on disused urban or agricultural land, by watercourses, at the edge of woods, or invading pastures. Stands of shorter herbs forming a distinct zone (seam) at the edge of woods.

E6 Inland salt steppes

Saline land with dominant salt-tolerant grasses and herbs. Excludes saline scrubland, listed under F6.8 xero-halophile scrubs.

E7 Sparsely wooded grasslands

Grasslands with a wooded overstorey that normally has less than 10% cover.

This category E7 is restricted to three subcategories – please use E7 only for these types:

E 7.1: Atlantic parkland: Extensive surfaces of Atlantic regions of nemoral Europe occupied by grassland dotted with widely planted trees, characteristic of the British Isles, where they are usually enclosed, used for cattle or deer grazing.

E7.2: Sub-continental parkland: Grassland dotted with widely planted trees, to the east of the Atlantic zone of nemoral Europe.

E7.3: Dehesa: A characteristic landscape of the southwestern quadrant of the Iberian peninsula in which crops, pasture land or Mediterranean scrub, in juxtaposition or rotation, are shaded by a fairly closed to very open canopy of native oaks, *Quercus suber*, *Quercus rotundifolia*, *Quercus pyrenaica*, *Quercus faginea*.

F Heathland, scrub and tundra

(in this definition, "shrubs" are <5 m in height)

F3 Temperate and mediterranean-montane scrub

Shrub communities of nemoral affinities. They include deciduous and evergreen scrubs of the nemoral zone, and deciduous scrubs of the submediterranean and supramediterranean zones. Excluded are heathlands with dominant [Ericaceae] F4, and the typically mediterranean maquis F5, garrigue F6 and phrygana F7.

F4 Temperate shrub heathland

Shrub communities of nemoral affinities, in which [Ericaceae] are dominant or at least prominent. Such heaths are best developed on acid soils in the Atlantic zone and also in sub-Atlantic Europe.

F5 Maquis, arborescent matorral and thermo-Mediterranean brushes

Evergreen sclerophyllous or lauriphyllous shrub vegetation, with a closed or nearly closed canopy structure, having nearly 100% cover of shrubs, with few annuals and some vernal geophytes; trees are nearly always present, some of which may be in shrub form. Shrubs, sometimes tall, of [Arbutus], [Cistus], [Cytisus], [Erica], [Genista], [Lavandula], [Myrtus], [Phillyrea], [Pistacia], [Quercus] and [Spartium] are typical. Included is pseudomaquis, in which the dominants are mixed deciduous and evergreen shrubs.

F6 Garrigue

Evergreen sclerophyllous or lauriphyllous shrub vegetation, with an open canopy structure and some bare ground, usually with many winter annuals and vernal geophytes. Low shrubs of [Cistus], [Lavandula], [Rosmarinus] and [Stoechas] are usually present, and there may be some larger shrubs and scattered trees. Garrigue is found mostly in the Mediterranean, Macaronesian and Pontic regions, where it typically derives from degradation or regrowth of broad-leaved evergreen forests (G2), but it extends into deciduous forest areas in the supra-Mediterranean zone and sub-Mediterranean zones and into steppe areas in Anatolia. Includes scrubby land with mainly herbaceous vegetation and a large component of unpalatable non-vernal monocots ([Asphodelus], [Urginea]) and thistles, provided that shrub cover exceeds 10%.

G Woodland, forest and other wooded land

G1 Broadleaved deciduous woodland

Woodland, forest and plantations dominated by summer-green non-coniferous trees that lose their leaves in winter. Includes woodland with mixed evergreen and deciduous broadleaved trees, provided that the deciduous cover exceeds that of evergreens. Excludes mixed forests (G4) where the proportion of conifers exceeds 25%.

G2 Broadleaved evergreen woodland

Temperate forests dominated by broad-leaved sclerophyllous or lauriphyllous evergreen trees, or by palms. They are characteristic of the Mediterranean and warm-temperate humid zones.

G3 Coniferous woodland Woodland, forest and plantations dominated by coniferous trees, mainly evergreen ([Abies], [Cedrus], [Picea], [Pinus], [Taxus], Cupressaceae) but also deciduous [Larix]. Excludes mixed forests (G4) where the proportion of broadleaved trees exceeds 25%.

G4 Mixed deciduous and coniferous woodland

Forest and woodland of mixed broad-leaved deciduous or evergreen and coniferous trees of the nemoral, boreal, warm-temperate humid and mediterranean zones. They are mostly characteristic of the boreonemoral transition zone between taiga and temperate lowland deciduous forests, and of the montane level of the major mountain

ranges to the south. Neither coniferous, nor broadleaved species account for more than 75% of the crown cover. Deciduous forests with an understorey of conifers or with a small admixture of conifers in the dominant layer are included in unit G1. Conifer forests with an understorey of deciduous trees or with a small admixture of deciduous trees in the dominant layer are included in unit G3.

G5 Lines of trees, small anthropogenic woodlands, recently felled woodland, early-stage woodland and coppice

Stands of trees greater than 5 m in height or with the potential to achieve this height, either in more or less continuous narrow strips or in small (less than about 0.5 ha) plantations or small (less than about 0.5 ha) intensively-managed woods. Woodland and coppice that is temporarily in a successional or non-woodland stage but which can be expected to develop into woodland in the future. Excludes parkland (E7.1, E7.2).

I Regularly or recently cultivated agricultural, horticultural and domestic habitats

I1 Arable land and market gardens

For the purposes of LUCAS, I1 Arable land and market gardens is typified by the subtype I1.5 Bare tilled, fallow or recently abandoned arable land:

Fields abandoned or left to rest, and other interstitial spaces on disturbed ground. Set-aside or abandoned arable land with forbs planted for purposes of soil protection, stabilization, fertilisation or reclamation. Abandoned fields are colonised by numerous pioneering, introduced or nitrophilous plants. They sometimes provide habitats that can be used by animals of open spaces.

9.9.3 Summary of EUNIS and LUCAS types in grassland-relevant habitats

Broad habitat categories	Potential habitat types	
	EUNIS	LUCAS
Grasslands	E1-E7, FA, FB	E10, E20, E30, B55, B70-B77
Grassland-relevant habitats other than lowland agricultural or semi-natural grassland		
Seashore	A2, B1	H20
Lakeshore	C3	H10
Mires, bogs and swamps	D1-D6, F9	H10
Riverbanks	C3, E3, E5, F9	E10, E20, E30, H10
Subalpine and alpine	E4, E5, F2	B55, D10, D20, E10, E20, E30
Heath and scrub	F2- F9	D10, D20, E10
Woodland/scattered trees	E5, E7, G1-G5	B70, B80, C10, C20, C30, D10, E10
Anthropogenic	I1, I2	A30, B50, B55, E10, E20, E30

Steppe	E1, E6, E7	C10, E10, E20
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The following key to the EUNIS types frequently encountered in the LUCAS survey is based on the principle that first those habitat situations which are very specific and only apply to a limited number of points are excluded (e.g. if there is a direct marine or coastal influence, a littoral zone of an inland surface waterbody or woodland).

As this key is developed for the whole of the European LUCAS grassland survey the surveyors may exclude some of the options presented in this key for their survey zone.

9.9.4 Key to the EUNIS types frequently encountered in the LUCAS survey

Please note that this is a shortened summary of the official EUNIS key, adapted to show the most common grassland types that may be encountered during the LUCAS survey.

Sub-key D Mires, bogs and fens

1. Does the habitat contain large areas of tall grasses and reeds normally without free-standing water?

Yes – go to 2

No – go to 3

2. Is the habitat influenced by saline or brackish water?

Yes - D6 Inland saline and brackish marshes and reedbeds

No - D5 Sedge and reedbeds, normally without free-standing water

3. Is the habitat in the arctic, subarctic or northern boreal zone with conspicuous patterning caused by frozen mounds or ridges interspersed with hollows?

Yes – D3 Aapa, palsa and polygon mires

No – go to 4

4. Is the area dominated by relatively species-rich herbaceous vegetation with little sphagnum moss visible?

Yes - D4 Base-rich fens and calcareous spring mires

No – go to 5

5. Is the vegetation dominated by sedges and/or has a convex form (valley or basin)?

Yes – D2 Valley mires, poor fens and transition mires

No - D1 Raised and blanket bogs

Sub-key E Grasslands and lands dominated by forbs, mosses or lichens

1. Is there significant tree presence (but <10% canopy cover) scattered over the survey area

Yes - E7 Sparsely wooded grasslands

No – go to 2

2. Is the habitat adjacent to a woodland area, and obviously affected by the shading of the canopy (e.g. with tall forbs)?

Yes - E5 Woodland fringes and clearings and tall forb stands

No – go to 3

3. Is the habitat in the alpine or subalpine zone (from around 1200 m a.s.l. upwards)?

Yes - E4 Alpine and subalpine grasslands

No – go to 4

4. Does the habitat contain aturally sparse or low-growing herbaceous vegetation?

Yes – go to 5

No – go to 6

5. Is the habitat saline (e.g. with succulent plants, crystalline salt on the surface)

Yes – E6 Inland salt steppes

No – E1 Dry grasslands

6. Is there generally dense herbaceous vegetation with signs of excess water supply (e.g. standing water after recent rain, or patches of rushes, club-rushes or sedges)

Yes - E3 Seasonally wet and wet grasslands

No – E2 Mesic grasslands

Sub-key **F Heathland, scrub and tundra**

1. Is the habitat in the Mediterranean zone (below the montane belt)?

Yes – go to 2

No – go to 4

2. Is the habitat dominated by low-growing, spiny shrubs?

Yes – F7 Spiny Mediterranean heaths (phrygana, hedgehog-heaths and related coastal cliff vegetation)

No – go to 3

3. Does the vegetation have an open structure, usually with some open ground and patches of shrubs (e.g. Cistus, Lavendula and Rosmarinus)?

Yes – F6 Garrigue

No (closed structure, mainly consisting of shrubs with scattered trees) - F5 Maquis, arborescent matorral and thermo-Mediterranean brushes

4. Is the habitat in the Canary islands and Madeira, on generally dry slopes and rich in succulents?

Yes – F8 Thermo-Atlantic xerophytic scrub

No – go to 5

5. Is the habitat dominated by heather?

Yes – F4 Temperate shrub heathland

No – F3 Temperate and mediterranean-montane scrub

Sub-key **G Woodland, forest and other wooded land**

1. Is there a strong anthropogenic influence?

Yes - **G5 Lines of trees, small anthropogenic woodlands, recently felled woodland, early-stage woodland and coppice**

No – go to 2

2. > 75% of the stand is dominated by...

Broadleaved deciduous trees - **G1 Broadleaved deciduous woodland**

Broadleaved evergreen trees - **G2 Broadleaved evergreen woodland**

Conifers - **G3 Coniferous woodland**

Mixture of deciduous and conifer trees - **G4 Mixed deciduous and coniferous woodland**

9.10 LUCAS Survey – Decision Tree

C1_Annex_9.10_DecisionTree_LUCAS2022_V5.2.vsd

C1_Annex_9.10_DecisionTree_LUCAS2022_V5.2.pdf

1) IDENTIFICATION AND ACCESS TO THE POINT

Beginning of the Field survey
Prefilled information: User ID (A), Point ID (B), Point
Attitude (C)

Point is part
of PI extension?
(Field D)

No

Yes

Point is
EX-ANTE?

Yes

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9.11 INSPIRE Pure Land Cover Components

C1_Annex_9.11_INSPIRE-PLCC_v3_20210122.docx

Regional statistics and
Geographic Information
Author: E4.LUCAS (ESTAT)



eurostat

Technical Documents
2022

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